

Notes on Measure-Theoretic Probability

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2022

1 Measure spaces

Definition 1.1. A set S contains s if $s \in S$. A set S includes F if $F \subseteq S$.

Definition 1.2. An algebra Σ_0 on a set S is a set of subsets of S such that

- $S \in \Sigma_0$,
- If $F \in \Sigma_0$, then $F^c \in \Sigma_0$, where $F^c = S \setminus F$,
- If $F, G \in \Sigma_0$, then $F \cup G \in \Sigma_0$.

Proposition 1.1. If Σ_0 is an algebra on S ,

- $\emptyset \in \Sigma_0$,
- If $F, G \in \Sigma_0$, then $F \cap G \in \Sigma_0$.

Definition 1.3. A trivial algebra on S is given by $\{\emptyset, S\}$.

Definition 1.4. A σ -algebra Σ on S is an algebra on S such that

$$\bigcup_{n \in \mathbb{N}} F_n \in \Sigma$$

for any sequence $(F_n \in \Sigma \mid n \in \mathbb{N})$, which also implies

$$\bigcap_{n \in \mathbb{N}} F_n \in \Sigma.$$

Definition 1.5. A measurable space (S, Σ) is a pair composed of a set S and a σ -algebra Σ on S . An element of Σ is called a Σ -measurable subset of S .

Definition 1.6. Let $\mathcal{C}$ be a set of subsets of S . The σ -algebra $\sigma(\mathcal{C})$ generated by $\mathcal{C}$ is the smallest σ -algebra Σ on S such that $\mathcal{C} \subseteq \Sigma$. The σ -algebra $\sigma(\mathcal{C})$ is the intersection of all the σ -algebras on S that include $\mathcal{C}$.

Note that the set $\mathcal{P}(S)$ of all subsets of S is a σ -algebra on S that includes any set of subsets $\mathcal{C}$.

Definition 1.7. The Borel $\mathcal{B}(\mathbb{R})$ σ -algebra is the σ -algebra on $\mathbb{R}$ generated by the set of open sets of real numbers.

Proposition 1.2. Let $\pi(\mathbb{R}) = \{(-\infty, x] \mid x \in \mathbb{R}\}$ be the set that contains every interval that contains every real number smaller or equal to every real number $x \in \mathbb{R}$. The σ -algebra generated by $\pi(\mathbb{R})$ is $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$.

Proof. First, recall that $(-\infty, x] = \bigcap_{n \in \mathbb{N}^+} (-\infty, x + n^{-1})$. Because $\mathcal{B}(\mathbb{R})$ is a σ -algebra on $\mathbb{R}$ that contains every $(-\infty, x + n^{-1})$, we have $(-\infty, x] \in \mathcal{B}(\mathbb{R})$. Because $\mathcal{B}(\mathbb{R})$ is a σ -algebra on $\mathbb{R}$ that includes $\pi(\mathbb{R})$ and $\sigma(\pi(\mathbb{R}))$ is the smallest σ -algebra on $\mathbb{R}$ that includes $\pi(\mathbb{R})$, we have $\sigma(\pi(\mathbb{R})) \subseteq \mathcal{B}(\mathbb{R})$.

Second, recall that every open set of real numbers is a countable union of open intervals. Because $\sigma(\pi(\mathbb{R}))$ is a σ -algebra on $\mathbb{R}$, if it contains every open interval, then it contains every open set of real numbers. This would also imply that $\mathcal{B}(\mathbb{R}) \subseteq \sigma(\pi(\mathbb{R}))$, since $\sigma(\pi(\mathbb{R}))$ is a σ -algebra on $\mathbb{R}$ and $\mathcal{B}(\mathbb{R})$ is the smallest σ -algebra on $\mathbb{R}$ that contains every open set of real numbers. In order to show that $\sigma(\pi(\mathbb{R}))$ contains every open interval, first note that $(a, u] = (-\infty, u] \cap (-\infty, a]^c \in \sigma(\pi(\mathbb{R}))$ for any $u > a$ and then note that $(a, b) = \bigcup_{n \in \mathbb{N}^+} (a, b - \epsilon n^{-1}]$ for $\epsilon = (b - a)/2$. $\square$

Definition 1.8. Consider an algebra Σ_0 on a set S . A function $\mu_0 : \Sigma_0 \rightarrow [0, \infty]$ is called additive if $\mu_0(\emptyset) = 0$ and, for any $F, G \in \Sigma_0$ such that $F \cap G = \emptyset$,

$$\mu_0(F \cup G) = \mu_0(F) + \mu_0(G).$$

Definition 1.9. A function $\mu_0 : \Sigma_0 \rightarrow [0, \infty]$ is called countably additive if $\mu_0(\emptyset) = 0$ and, for any sequence $(F_n \in \Sigma_0 \mid n \in \mathbb{N})$ such that $F_n \cap F_m = \emptyset$ for $n \neq m$,

$$\mu_0 \left(\bigcup_{n \in \mathbb{N}} F_n \right) = \sum_{n \in \mathbb{N}} \mu_0(F_n)$$

whenever $\bigcup_{n \in \mathbb{N}} F_n \in \Sigma_0$. This last requirement is always met when Σ_0 is a σ -algebra.

Definition 1.10. Let (S, Σ) be a measurable space. A countably additive function $\mu : \Sigma \rightarrow [0, \infty]$ is called a measure on (S, Σ) . The triple (S, Σ, μ) is called a measure space.

Proposition 1.3. A measure space (S, Σ, μ) has the following properties:

- If $\mu(S) < \infty$ and $A, B \in \Sigma$, then $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$,
- If $A, B \in \Sigma$, then $\mu(A \cup B) \leq \mu(A) + \mu(B)$,
- $\mu \left(\bigcup_{n \in \mathbb{N}} F_n \right) \leq \sum_{n \in \mathbb{N}} \mu(F_n)$ for any sequence $(F_n \in \Sigma \mid n \in \mathbb{N})$.

Definition 1.11. A measure μ on the measurable space (S, Σ) is called finite if $\mu(S) < \infty$.

Definition 1.12. A measure μ on the measurable space (S, Σ) is called σ -finite if there is a sequence $(S_n \in \Sigma \mid n \in \mathbb{N})$ such that $\mu(S_n) < \infty$ and $\bigcup_{n \in \mathbb{N}} S_n = S$.

Definition 1.13. A measure μ on the measurable space (S, Σ) is called a probability measure if $\mu(S) = 1$. The triple (S, Σ, μ) is then called a probability triple. A set $F \in \Sigma$ is called μ -null if $\mu(F) = 0$. If a statement is false only for elements of a μ -null set $F \in \Sigma$, then the statement is said to be true almost everywhere.

Definition 1.14. A π -system $\mathcal{I}$ on S is a set of subsets of S such that if $I_1, I_2 \in \mathcal{I}$, then $I_1 \cap I_2 \in \mathcal{I}$.

Definition 1.15. A d -system $\mathcal{D}$ on S is a set of subsets of S such that

- $S \in \mathcal{D}$;
- If $A, B \in \mathcal{D}$ and $A \subseteq B$, then $B \setminus A \in \mathcal{D}$;
- For any sequence $(A_n \in \mathcal{D} \mid n \in \mathbb{N})$, if $A_n \subseteq A_{n+1}$ for every $n \in \mathbb{N}$, then $\bigcup_n A_n \in \mathcal{D}$.

Definition 1.16. Let $\mathcal{C}$ be a set of subsets of S . The d -system $d(\mathcal{C})$ generated by $\mathcal{C}$ is the smallest d -system on S such that $\mathcal{C} \subseteq d(\mathcal{C})$. The d -system $d(\mathcal{C})$ is the intersection of all the d -systems on S that include $\mathcal{C}$.

Proposition 1.4. A set Σ of subsets of S is a σ -algebra on S if and only if Σ is a π -system and a d -system on S .

Proof. If Σ is a σ -algebra on S , then:

- If $E, F \in \Sigma$, then $E \cap F \in \Sigma$;
- $S \in \Sigma$;
- If $E, F \in \Sigma$ and $E \subseteq F$, then $F \setminus E \in \Sigma$, since $F \setminus E = F \cap E^c$;
- For any sequence $(F_n \in \Sigma \mid n \in \mathbb{N})$, if $F_n \subseteq F_{n+1}$ for every $n \in \mathbb{N}$, then $\bigcup_n F_n \in \Sigma$.

If Σ is a π -system and a d -system on S , then:

- $S \in \Sigma$;
- If $E \in \Sigma$, then $E^c \in \Sigma$, since $E \subseteq S$ and $S \setminus E = E^c$;
- If $E, F \in \Sigma$, then $E \cup F \in \Sigma$, since $(E^c \cap F^c)^c = E \cup F$;
- If $(E_n \in \Sigma \mid n \in \mathbb{N})$ is a sequence, then $\bigcup_n E_n \in \Sigma$. In order to see this, let $G_k = \bigcup_{n \leq k} E_n$ for every $k \in \mathbb{N}$. Since $G_k \in \Sigma$ and $G_k \subseteq G_{k+1}$ for every $k \in \mathbb{N}$, we know that $\bigcup_k G_k \in \Sigma$.

□

Lemma 1.1 (Dynkin's lemma). If $\mathcal{I}$ is a π -system on S , then $d(\mathcal{I}) = \sigma(\mathcal{I})$.

Proof. We will show that $d(\mathcal{I})$ is a π -system on S , so that $d(\mathcal{I})$ is a σ -algebra on S and $\sigma(\mathcal{I}) \subseteq d(\mathcal{I})$. Because $\sigma(\mathcal{I})$ is a d -system on S that includes $\mathcal{I}$, we know that $d(\mathcal{I}) \subseteq \sigma(\mathcal{I})$, which will show that $d(\mathcal{I}) = \sigma(\mathcal{I})$.

Let $\mathcal{D}_1 = \{B \in d(\mathcal{I}) \mid B \cap C \in d(\mathcal{I}) \text{ for every } C \in \mathcal{I}\}$. For every $B, C \in \mathcal{I}$, we know that $B \cap C \in \mathcal{I}$. Because $\mathcal{I} \subseteq d(\mathcal{I})$, we also know that $\mathcal{I} \subseteq \mathcal{D}_1$. Furthermore, $\mathcal{D}_1$ is a d -system on S :

- $S \in \mathcal{D}_1$, since $S \in d(\mathcal{I})$ and $S \cap C = C$ and $C \in d(\mathcal{I})$ for every $C \in \mathcal{I}$.
- If $B_1, B_2 \in \mathcal{D}_1$ and $B_1 \subseteq B_2$, then $B_2 \setminus B_1 \in \mathcal{D}_1$. In order to see this, note that, for every $C \in \mathcal{I}$,

$$(B_2 \cap C) \setminus (B_1 \cap C) = (B_2 \cap C) \cap (B_1^c \cup C^c) = B_2 \cap (C \cap (B_1^c \cup C^c)) = B_2 \cap (B_1^c \cap C) = (B_2 \setminus B_1) \cap C.$$

Since $(B_1 \cap C) \in d(\mathcal{I})$ and $(B_2 \cap C) \in d(\mathcal{I})$ and $(B_1 \cap C) \subseteq (B_2 \cap C)$, we know that $(B_2 \cap C) \setminus (B_1 \cap C) \in d(\mathcal{I})$. Therefore, $B_2 \setminus B_1 \in d(\mathcal{I})$ and $(B_2 \setminus B_1) \cap C \in d(\mathcal{I})$ for every $C \in \mathcal{I}$, so that $B_2 \setminus B_1 \in \mathcal{D}_1$.

- For any sequence $(B_n \in \mathcal{D}_1 \mid n \in \mathbb{N})$, if $B_n \subseteq B_{n+1}$ for every $n \in \mathbb{N}$, then $\cup_n B_n \in \mathcal{D}_1$. In order to see this, note that, for every $n \in \mathbb{N}$ and $C \in \mathcal{I}$, we have $B_n \cap C \in d(\mathcal{I})$ and $B_n \cap C \subseteq B_{n+1} \cap C$. Since $\cup_n (B_n \cap C) = (\cup_n B_n) \cap C$, we know that $\cup_n B_n \in d(\mathcal{I})$ and $(\cup_n B_n) \cap C \in d(\mathcal{I})$ for every $C \in \mathcal{I}$, so that $\cup_n B_n \in \mathcal{D}_1$.

Because $\mathcal{I} \subseteq \mathcal{D}_1$, we know that $d(\mathcal{I}) \subseteq \mathcal{D}_1$. By definition, we know that $\mathcal{D}_1 \subseteq d(\mathcal{I})$, so that $\mathcal{D}_1 = d(\mathcal{I})$. Therefore, for every $B \in d(\mathcal{I})$ and $C \in \mathcal{I}$, we know that $B \cap C \in d(\mathcal{I})$.

Let $\mathcal{D}_2 = \{A \in d(\mathcal{I}) \mid A \cap B \in d(\mathcal{I}) \text{ for every } B \in d(\mathcal{I})\}$. From the previous result, we know that $\mathcal{I} \subseteq \mathcal{D}_2$. Furthermore, $\mathcal{D}_2$ is a d -system on S :

- $S \in \mathcal{D}_2$, since $S \in d(\mathcal{I})$ and $S \cap B = B$ and $B \in d(\mathcal{I})$ for every $B \in d(\mathcal{I})$.
- If $A_1, A_2 \in \mathcal{D}_2$ and $A_1 \subseteq A_2$, then $A_2 \setminus A_1 \in \mathcal{D}_2$. In order to see this, note that, for every $B \in d(\mathcal{I})$,

$$(A_2 \cap B) \setminus (A_1 \cap B) = (A_2 \cap B) \cap (A_1^c \cup B^c) = A_2 \cap (B \cap (A_1^c \cup B^c)) = A_2 \cap (A_1^c \cap B) = (A_2 \setminus A_1) \cap B.$$

Since $(A_1 \cap B) \in d(\mathcal{I})$ and $(A_2 \cap B) \in d(\mathcal{I})$ and $(A_1 \cap B) \subseteq (A_2 \cap B)$, we know that $(A_2 \cap B) \setminus (A_1 \cap B) \in d(\mathcal{I})$. Therefore, $A_2 \setminus A_1 \in d(\mathcal{I})$ and $(A_2 \setminus A_1) \cap B \in d(\mathcal{I})$ for every $B \in d(\mathcal{I})$, so that $A_2 \setminus A_1 \in \mathcal{D}_2$.

- For any sequence $(A_n \in \mathcal{D}_2 \mid n \in \mathbb{N})$, if $A_n \subseteq A_{n+1}$ for every $n \in \mathbb{N}$, then $\cup_n A_n \in \mathcal{D}_2$. In order to see this, note that, for every $n \in \mathbb{N}$ and $B \in d(\mathcal{I})$, we have $A_n \cap B \in d(\mathcal{I})$ and $A_n \cap B \subseteq A_{n+1} \cap B$. Since $\cup_n (A_n \cap B) = (\cup_n A_n) \cap B$, we know that $\cup_n A_n \in d(\mathcal{I})$ and $(\cup_n A_n) \cap B \in d(\mathcal{I})$ for every $B \in d(\mathcal{I})$, so that $\cup_n A_n \in \mathcal{D}_2$.

Because $\mathcal{I} \subseteq \mathcal{D}_2$, we know that $d(\mathcal{I}) \subseteq \mathcal{D}_2$. By definition, we know that $\mathcal{D}_2 \subseteq d(\mathcal{I})$, so that $\mathcal{D}_2 = d(\mathcal{I})$. Therefore, for every $A \in d(\mathcal{I})$ and $B \in d(\mathcal{I})$, we have $A \cap B \in d(\mathcal{I})$, which shows that $d(\mathcal{I})$ is a π -system on S . $\square$

Proposition 1.5. If $\mathcal{I}$ is a π -system on S and $\mathcal{D}$ is a d -system on S such that $\mathcal{I} \subseteq \mathcal{D}$, then $\sigma(\mathcal{I}) \subseteq \mathcal{D}$.

Proof. Since $d(\mathcal{I}) \subseteq \mathcal{D}$, this is a direct consequence of the previous lemma. $\square$

Proposition 1.6. Let $\Sigma = \sigma(\mathcal{I})$ be the σ -algebra generated by a π -system $\mathcal{I}$. If μ_1 and μ_2 are measures on the measurable space (S, Σ) such that $\mu_1(S) = \mu_2(S) < \infty$ and $\mu_1(I) = \mu_2(I)$ for any $I \in \mathcal{I}$, then $\mu_1(F) = \mu_2(F)$ for any $F \in \Sigma$. Therefore, if two probability measures agree on a π -system, then they agree on the σ -algebra generated by that π -system.

Theorem 1.1 (Carathéodory's extension theorem). If Σ_0 is an algebra on S and $\Sigma = \sigma(\Sigma_0)$ is the σ -algebra generated by Σ_0 and $\mu_0 : \Sigma_0 \rightarrow [0, \infty]$ is a countably additive function, then there exists a measure μ on the measurable space (S, Σ) such that $\mu(F) = \mu_0(F)$ for any $F \in \Sigma_0$. If $\mu_0(S) < \infty$, then μ is unique, since an algebra is a π -system.

Definition 1.17. Let Σ_0 be the algebra on the set $S = (0, 1]$ that contains every F such that

$$F = \bigcup_{k=1}^r (a_k, b_k],$$

where $r \in \mathbb{N}$ and $0 \leq a_1 \leq b_1 \leq \dots \leq a_r \leq b_r \leq 1$.

Let $\mu_0 : \Sigma_0 \rightarrow [0, 1]$ denote the countably additive function given by

$$\mu_0(F) = \sum_{k=1}^r (b_k - a_k).$$

Let $\mathcal{B}((0, 1]) = \sigma(\Sigma_0)$ be the σ -algebra generated by Σ_0 . The unique measure $\mu : \mathcal{B}((0, 1]) \rightarrow [0, 1]$ on the measurable space $((0, 1], \mathcal{B}((0, 1]))$ that agrees with μ_0 on the algebra Σ_0 is called the Lebesgue measure Leb on $((0, 1], \mathcal{B}((0, 1]))$. The σ -finite Lebesgue measure Leb on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is similarly defined.

Intuitively, a Lebesgue measure Leb assigns *lengths*.

Definition 1.18. Let $a_n \uparrow a$ denote that a sequence of real numbers $(a_n \mid n \in \mathbb{N})$ is such that $a_n \leq a_{n+1}$ and $a = \lim_{n \rightarrow \infty} a_n$. Similarly, let $a_n \downarrow a$ denote that a sequence of real numbers $(a_n \mid n \in \mathbb{N})$ is such that $a_{n+1} \leq a_n$ and $a = \lim_{n \rightarrow \infty} a_n$.

Definition 1.19. Let $A_n \uparrow A$ denote that a sequence of sets $(A_n \mid n \in \mathbb{N})$ is such that $A_n \subseteq A_{n+1}$ and $A = \bigcup_{n \in \mathbb{N}} A_n$. Similarly, let $A_n \downarrow A$ denote that a sequence of sets $(A_n \mid n \in \mathbb{N})$ is such that $A_{n+1} \subseteq A_n$ and $A = \bigcap_{n \in \mathbb{N}} A_n$.

Proposition 1.7 (Monotone-convergence property of measure). Consider the measure space (S, Σ, μ) . For a sequence $(F_n \in \Sigma \mid n \in \mathbb{N})$, if $F_n \uparrow F$, then $\mu(F_n) \uparrow \mu(F)$. Similarly, for a sequence $(G_n \in \Sigma \mid n \in \mathbb{N})$, if $G_n \downarrow G$ and $\mu(G_k) < \infty$ for some k , then $\mu(G_n) \downarrow \mu(G)$.

2 Events

Definition 2.1. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. An element $\omega \in \Omega$ is called an outcome. The set Ω is called an outcome space. A set of outcomes $F \in \mathcal{F}$ is called an event. The probability measure $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is defined on a σ -algebra $\mathcal{F}$ on the outcome space Ω .

A probability $\mathbb{P}(F)$ assigns a degree of belief to the statement that the outcome $\omega \in \Omega$ of an experiment belongs to the event $F \in \mathcal{F}$. For instance, a probability $\mathbb{P}(F) = 1$ indicates that $\omega \in F$ almost surely, while a probability $\mathbb{P}(F) = 0$ indicates that $\omega \notin F$ almost surely. In general, a statement about an outcome is said to be true almost surely if $\mathbb{P}(F) = 1$, where $F \in \mathcal{F}$ is the event that contains every outcome $\omega \in \Omega$ for which the statement is true.

Example 2.1. Consider an experiment where a coin is tossed twice. Let $H = 0$ represent heads and $T = 1$ represent tails. The outcome space Ω may be defined as $\Omega = \{(H, H), (H, T), (T, H), (T, T)\}$. The σ -algebra $\mathcal{F}$ on the outcome space Ω may be defined as the set of all subsets of Ω , which is denoted by $\mathcal{F} = \mathcal{P}(\Omega)$. The event F where at least one head is observed is then given by $F = \{(H, H), (H, T), (T, H)\}$.

Example 2.2. Consider an experiment where a coin is tossed infinitely often. The outcome space Ω may be defined as the set of infinite binary sequences $\Omega = \{H, T\}^{\mathbb{N}}$. In order to at least assign probabilities to every event $F = \{\omega \in \Omega \mid \omega_n = W\}$ where $n \in \mathbb{N}$ and $W \in \{H, T\}$, the σ -algebra $\mathcal{F}$ on the outcome space Ω may be generated as $\mathcal{F} = \sigma(\{\{\omega \in \Omega \mid \omega_n = W\} \mid n \in \mathbb{N}, W \in \{H, T\}\})$.

Proposition 2.1. Consider a sequence of events $(F_n \in \mathcal{F} \mid n \in \mathbb{N})$. If $\mathbb{P}(F_n) = 1$ for every $n \in \mathbb{N}$, then $\mathbb{P}(\bigcap_{n \in \mathbb{N}} F_n) = 1$.

Definition 2.2. The infimum $\inf_n x_n$ of a sequence of real numbers $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ is the largest $r \in [-\infty, \infty]$ such that $r \leq x_n$ for every $n \in \mathbb{N}$. The supremum $\sup_n x_n$ of a sequence of real numbers $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ is the smallest $r \in [-\infty, \infty]$ such that $r \geq x_n$ for every $n \in \mathbb{N}$.

Definition 2.3. The limit inferior of a sequence of real numbers $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ is defined by

$$\liminf_{n \rightarrow \infty} x_n = \sup_m \inf_{n \geq m} x_n = \lim_{m \rightarrow \infty} \inf_{n \geq m} x_n.$$

Note that the sequence $(\inf_{n \geq m} x_n \mid m \in \mathbb{N})$ is non-decreasing. Let $z \in [-\infty, \infty]$. If $z < \liminf_{n \rightarrow \infty} x_n$, then $z < x_n$ for all sufficiently large $n \in \mathbb{N}$. If $z > \liminf_{n \rightarrow \infty} x_n$, then $z > x_n$ for infinitely many $n \in \mathbb{N}$.

Definition 2.4. The limit superior of a sequence of real numbers $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ is defined by

$$\limsup_{n \rightarrow \infty} x_n = \inf_m \sup_{n \geq m} x_n = \lim_{m \rightarrow \infty} \sup_{n \geq m} x_n.$$

Note that the sequence $(\sup_{n \geq m} x_n \mid m \in \mathbb{N})$ is non-increasing. Let $z \in [-\infty, \infty]$. If $z > \limsup_{n \rightarrow \infty} x_n$, then $z > x_n$ for all sufficiently large $n \in \mathbb{N}$. If $z < \limsup_{n \rightarrow \infty} x_n$, then $z < x_n$ for infinitely many $n \in \mathbb{N}$.

Proposition 2.2. For any sequence $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$, the limit inferior and the limit superior are related by

$$-\liminf_{n \rightarrow \infty} x_n = \lim_{m \rightarrow \infty} -\inf_{n \geq m} x_n = \lim_{m \rightarrow \infty} \sup_{n \geq m} -x_n = \limsup_{n \rightarrow \infty} -x_n.$$

Definition 2.5. A sequence of real numbers $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ is said to converge in $[-\infty, \infty]$ if and only if

$$\liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_n.$$

Definition 2.6. The limit inferior of a sequence of sets $(E_n \mid n \in \mathbb{N})$ is defined by

$$\liminf_{n \rightarrow \infty} E_n = \bigcup_{m \in \mathbb{N}} \bigcap_{n \geq m} E_n.$$

Let $F_m = \bigcap_{n \geq m} E_n$. Note that $F_m \subseteq F_{m+1}$. Furthermore, $\omega \in \liminf_{n \rightarrow \infty} E_n$ if and only if $\omega \in E_n$ for all sufficiently large $n \in \mathbb{N}$.

Definition 2.7. The limit superior of a sequence of sets $(E_n \mid n \in \mathbb{N})$ is defined by

$$\limsup_{n \rightarrow \infty} E_n = \bigcap_{m \in \mathbb{N}} \bigcup_{n \geq m} E_n.$$

Let $F_m = \bigcup_{n \geq m} E_n$. Note that $F_m \supseteq F_{m+1}$. Furthermore, $\omega \in \limsup_{n \rightarrow \infty} E_n$ if and only if $\omega \in E_n$ for infinitely many $n \in \mathbb{N}$.

Proposition 2.3. For any sequence of sets $(E_n \subseteq \Omega \mid n \in \mathbb{N})$, the limit inferior and the limit superior are related by

$$\left(\liminf_{n \rightarrow \infty} E_n \right)^C = \limsup_{n \rightarrow \infty} E_n^C.$$

Definition 2.8. Consider a measurable space $(\Omega, \mathcal{F})$. The indicator function $\mathbb{I}_F : \Omega \rightarrow \{0, 1\}$ of an event $F \in \mathcal{F}$ is defined by

$$\mathbb{I}_F(\omega) = \begin{cases} 1, & \text{if } \omega \in F, \\ 0, & \text{if } \omega \notin F. \end{cases}$$

Proposition 2.4. For any outcome $\omega \in \Omega$ and sequence of events $(E_n \in \mathcal{F} \mid n \in \mathbb{N})$,

$$\begin{aligned} \mathbb{I}_{\liminf_{n \rightarrow \infty} E_n}(\omega) &= \liminf_{n \rightarrow \infty} \mathbb{I}_{E_n}(\omega), \\ \mathbb{I}_{\limsup_{n \rightarrow \infty} E_n}(\omega) &= \limsup_{n \rightarrow \infty} \mathbb{I}_{E_n}(\omega). \end{aligned}$$

Lemma 2.1 (Reverse Fatou Lemma). For a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence $(E_n \in \mathcal{F} \mid n \in \mathbb{N})$,

$$\mathbb{P} \left(\limsup_{n \rightarrow \infty} E_n \right) \geq \limsup_{n \rightarrow \infty} \mathbb{P}(E_n).$$

Proof. Let $F_m = \bigcup_{n \geq m} E_n$ such that $F_m \supseteq F_{m+1}$. By definition, $F_m \downarrow \limsup_{n \rightarrow \infty} E_n$, which implies $\mathbb{P}(F_m) \downarrow \mathbb{P}(\limsup_{n \rightarrow \infty} E_n)$. Because $A \subseteq (B \cup A)$ implies $\mathbb{P}(A) \leq \mathbb{P}(B \cup A)$ for any events $A, B \in \mathcal{F}$,

$$\mathbb{P}(F_m) = \mathbb{P} \left(\bigcup_{n \geq m} E_n \right) \geq \sup_{n \geq m} \mathbb{P}(E_n).$$

By taking the limit of both sides of the equation above when $m \rightarrow \infty$,

$$\lim_{m \rightarrow \infty} \mathbb{P}(F_m) = \mathbb{P} \left(\limsup_{n \rightarrow \infty} E_n \right) \geq \lim_{m \rightarrow \infty} \sup_{n \geq m} \mathbb{P}(E_n) = \limsup_{n \rightarrow \infty} \mathbb{P}(E_n).$$

□

Lemma 2.2 (Fatou Lemma for sets). For a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence $(E_n \in \mathcal{F} \mid n \in \mathbb{N})$,

$$\mathbb{P}\left(\liminf_{n \rightarrow \infty} E_n\right) \leq \liminf_{n \rightarrow \infty} \mathbb{P}(E_n).$$

Proof. Let $F_m = \bigcap_{n \geq m} E_n$ such that $F_m \subseteq F_{m+1}$. By definition, $F_m \uparrow \liminf_{n \rightarrow \infty} E_n$, which implies $\mathbb{P}(F_m) \uparrow \mathbb{P}(\liminf_{n \rightarrow \infty} E_n)$. Because $(A \cap B) \subseteq B$ implies $\mathbb{P}(A \cap B) \leq \mathbb{P}(B)$ for any events $A, B \in \mathcal{F}$,

$$\mathbb{P}(F_m) = \mathbb{P}\left(\bigcap_{n \geq m} E_n\right) \leq \inf_{n \geq m} \mathbb{P}(E_n).$$

By taking the limit of both sides of the equation above when $m \rightarrow \infty$,

$$\lim_{m \rightarrow \infty} \mathbb{P}(F_m) = \mathbb{P}\left(\liminf_{n \rightarrow \infty} E_n\right) \leq \lim_{m \rightarrow \infty} \inf_{n \geq m} \mathbb{P}(E_n) = \liminf_{n \rightarrow \infty} \mathbb{P}(E_n).$$

□

Lemma 2.3 (First Borel-Cantelli Lemma). For a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence of events $(E_n \in \mathcal{F} \mid n \in \mathbb{N})$ such that $\sum_{n=0}^{\infty} \mathbb{P}(E_n) < \infty$,

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} E_n\right) = 0.$$

Proof. Let $F_m = \bigcup_{n \geq m} E_n$ such that $F_m \supseteq F_{m+1}$. By definition, $F_m \downarrow \limsup_{n \rightarrow \infty} E_n$, which implies $\mathbb{P}(F_m) \downarrow \mathbb{P}(\limsup_{n \rightarrow \infty} E_n)$. Because $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$ for any events $A, B \in \mathcal{F}$,

$$\mathbb{P}(F_m) = \mathbb{P}\left(\bigcup_{n \geq m} E_n\right) \leq \sum_{n \geq m} \mathbb{P}(E_n).$$

By taking the limit of both sides of the equation above when $m \rightarrow \infty$,

$$\lim_{m \rightarrow \infty} \mathbb{P}(F_m) = \mathbb{P}\left(\limsup_{n \rightarrow \infty} E_n\right) \leq \lim_{m \rightarrow \infty} \sum_{n \geq m} \mathbb{P}(E_n) = 0,$$

where the last equality comes from the fact that, for any $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that, for all $m - 1 \geq N$,

$$\epsilon > \left| \sum_{n=0}^{\infty} \mathbb{P}(E_n) - \sum_{n=0}^{m-1} \mathbb{P}(E_n) \right| = \sum_{n \geq m} \mathbb{P}(E_n).$$

□

3 Random variables

Definition 3.1. Consider a measurable space (S, Σ) and a function $h : S \rightarrow \mathbb{R}$. The function h^{-1} is defined as

$$h^{-1}(A) = \{s \in S \mid h(s) \in A\}$$

for any $A \subseteq \mathbb{R}$. The function h is called Σ -measurable if $h^{-1}(A) \in \Sigma$ for every $A \in \mathcal{B}(\mathbb{R})$. In an extended definition, a function $h : S \rightarrow [-\infty, \infty]$ is called Σ -measurable if $h^{-1}(A) \in \Sigma$ for every $A \in \mathcal{B}([-\infty, \infty])$.

Definition 3.2. A $\mathcal{B}(\mathbb{R})$ -measurable function $h : \mathbb{R} \rightarrow \mathbb{R}$ is said to be Borel.

Definition 3.3. The set of Σ -measurable functions on S is denoted by $\text{m}\Sigma$. The set of non-negative Σ -measurable functions on S is denoted by $(\text{m}\Sigma)^+$. The set of bounded Σ -measurable functions on S is denoted by $\text{b}\Sigma$.

Proposition 3.1. Consider a function $h : S \rightarrow \mathbb{R}$. For any set $A \subseteq \mathbb{R}$,

$$h^{-1}(A^c) = \{s \in S \mid h(s) \in A^c\} = \{s \in S \mid h(s) \in A\}^c = (h^{-1}(A))^c.$$

Proposition 3.2. Consider a function $h : S \rightarrow \mathbb{R}$. For any sequence of sets $(A_n \subseteq \mathbb{R} \mid n \in \mathbb{N})$,

$$h^{-1}\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \{s \in S \mid h(s) \in \bigcup_{n \in \mathbb{N}} A_n\} = \bigcup_{n \in \mathbb{N}} \{s \in S \mid h(s) \in A_n\} = \bigcup_{n \in \mathbb{N}} h^{-1}(A_n).$$

Similarly,

$$h^{-1}\left(\bigcap_{n \in \mathbb{N}} A_n\right) = \{s \in S \mid h(s) \in \bigcap_{n \in \mathbb{N}} A_n\} = \bigcap_{n \in \mathbb{N}} \{s \in S \mid h(s) \in A_n\} = \bigcap_{n \in \mathbb{N}} h^{-1}(A_n).$$

Proposition 3.3. Consider a measurable space (S, Σ) and a function $h : S \rightarrow \mathbb{R}$. The set $\mathcal{E} = \{B \in \mathcal{B}(\mathbb{R}) \mid h^{-1}(B) \in \Sigma\}$ is a σ -algebra on $\mathbb{R}$.

Proof. First, note that $h^{-1}(\mathbb{R}) = \{s \in S \mid h(s) \in \mathbb{R}\} = S$ and $S \in \Sigma$. Therefore, $\mathbb{R} \in \mathcal{E}$. Consider an element $B \in \mathcal{E}$. In that case, $h^{-1}(B) \in \Sigma$, which implies $(h^{-1}(B))^c = h^{-1}(B^c) \in \Sigma$. Therefore, $B^c \in \mathcal{E}$. Finally, consider a sequence $(B_n \in \mathcal{E} \mid n \in \mathbb{N})$. In that case, $h^{-1}(B_n) \in \Sigma$ for every $n \in \mathbb{N}$, which implies $\bigcup_n h^{-1}(B_n) \in \Sigma$. Therefore, $h^{-1}(\bigcup_n B_n) \in \Sigma$ and $\bigcup_n B_n \in \mathcal{E}$. $\square$

Proposition 3.4. Consider a measurable space (S, Σ) , a function $h : S \rightarrow \mathbb{R}$, and a set $\mathcal{C}$ of subsets of $\mathbb{R}$. If $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$ and $h^{-1}(C) \in \Sigma$ for every $C \in \mathcal{C}$, then h is Σ -measurable.

Proof. Note that the set $\mathcal{E} = \{B \in \mathcal{B}(\mathbb{R}) \mid h^{-1}(B) \in \Sigma\}$ is a σ -algebra on $\mathbb{R}$. Because $\mathcal{C} \subseteq \mathcal{E}$, $\mathcal{E} \subseteq \mathcal{B}(\mathbb{R})$, and $\mathcal{B}(\mathbb{R})$ is the smallest σ -algebra that includes $\mathcal{C}$, we know that $\mathcal{E} = \mathcal{B}(\mathbb{R})$, which implies that $h^{-1}(B) \in \Sigma$ for every $B \in \mathcal{B}(\mathbb{R})$. $\square$

Proposition 3.5. If a function $h : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then it is Borel.

Proof. First, consider the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and let $\mathcal{C}$ be the set of open sets of real numbers. Recall that $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C})$. Second, recall that a function h is continuous if $h^{-1}(A) \in \mathcal{C}$ is an open set for every open set $A \in \mathcal{C}$. Using the previous result, $h^{-1}(B) \in \mathcal{B}(\mathbb{R})$ for every $B \in \mathcal{B}(\mathbb{R})$. $\square$

Proposition 3.6. Consider a measurable space (S, Σ) and a function $h : S \rightarrow \mathbb{R}$. For any $c \in \mathbb{R}$, define

$$\{h \leq c\} = h^{-1}((-\infty, c]) = \{s \in S \mid h(s) \leq c\}.$$

If $\{h \leq c\} \in \Sigma$ for every $c \in \mathbb{R}$, then h is Σ -measurable.

Proof. First, let $\mathcal{C} = \{(-\infty, x] \mid x \in \mathbb{R}\}$ be the set that contains every interval that contains every real number smaller or equal to every real number $x \in \mathbb{R}$. Recall that $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C})$. By assumption, $h^{-1}(C) \in \Sigma$ for every $C \in \mathcal{C}$, and so h is Σ -measurable. $\square$

Note that analogous results apply for $\{h \geq c\}$, $\{h < c\}$, and $\{h > c\}$.

Proposition 3.7. Consider a measurable space (S, Σ) . Let $h : S \rightarrow \mathbb{R}$, $h_1 : S \rightarrow \mathbb{R}$, and $h_2 : S \rightarrow \mathbb{R}$ be Σ -measurable functions and let $\lambda \in \mathbb{R}$ be a constant. In that case, $h_1 + h_2$ is a Σ -measurable function, $h_1 h_2$ is a Σ -measurable function, and λh is a Σ -measurable function.

Proof. We will only show the first of these statements. Based on the previous result, if $\{h_1 + h_2 > c\} = \{s \in S \mid h_1(s) + h_2(s) > c\} \in \Sigma$ for every $c \in \mathbb{R}$, then $h_1 + h_2$ is Σ -measurable. Recall that $h_1(s) + h_2(s) > c$ if and only if there is a rational $q \in \mathbb{Q}$ such that $h_1(s) > q > c - h_2(s)$. Therefore,

$$\{h_1 + h_2 > c\} = \{s \in S \mid h_1(s) > q \text{ and } q > c - h_2(s) \text{ for some } q \in \mathbb{Q}\} = \bigcup_{q \in \mathbb{Q}} \{s \in S \mid h_1(s) > q \text{ and } q > c - h_2(s)\},$$

which is a countable union of elements of Σ given by

$$\{h_1 + h_2 > c\} = \bigcup_{q \in \mathbb{Q}} \{s \in S \mid h_1(s) > q\} \cap \{s \in S \mid q > c - h_2(s)\} = \bigcup_{q \in \mathbb{Q}} \{h_1 > q\} \cap \{h_2 > c - q\}.$$

$\square$

Proposition 3.8. Consider a measurable space (S, Σ) and a Σ -measurable function $h : S \rightarrow \mathbb{R}$. Consider also the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and a $\mathcal{B}(\mathbb{R})$ -measurable function $f : \mathbb{R} \rightarrow \mathbb{R}$. For all $s \in S$, let $(f \circ h)(s) = f(h(s))$. For any $A \subseteq \mathbb{R}$,

$$(f \circ h)^{-1}(A) = \{s \in S \mid (f \circ h)(s) \in A\} = \{s \in S \mid f(h(s)) \in A\}.$$

Note that $f^{-1}(A) \subseteq \mathbb{R}$ for any $A \subseteq \mathbb{R}$, since $f^{-1}(A) = \{r \in \mathbb{R} \mid f(r) \in A\}$. Therefore,

$$(h^{-1} \circ f^{-1})(A) = h^{-1}(f^{-1}(A)) = \{s \in S \mid h(s) \in f^{-1}(A)\} = \{s \in S \mid f(h(s)) \in A\} = (f \circ h)^{-1}(A),$$

where we used the fact that $f(h(s)) \in A$ if and only if $h(s) \in f^{-1}(A)$, for all $s \in S$ and $A \subseteq \mathbb{R}$. Furthermore, since $f^{-1}(A) \in \mathcal{B}(\mathbb{R})$ for any $A \in \mathcal{B}(\mathbb{R})$ and $h^{-1}(f^{-1}(A)) \in \Sigma$ for any $f^{-1}(A) \in \mathcal{B}(\mathbb{R})$, the function $f \circ h$ is Σ -measurable.

Definition 3.4. Consider the measurable spaces (S_1, Σ_1) and (S_2, Σ_2) . A function $h : S_1 \rightarrow S_2$ is called Σ_1/Σ_2 -measurable if $h^{-1}(A) \in \Sigma_1$ for every $A \in \Sigma_2$. Therefore, a function on a measurable space (S, Σ) is Σ -measurable if it is $\Sigma/\mathcal{B}(\mathbb{R})$ -measurable.

Proposition 3.9. Consider the measurable spaces (S_1, Σ_1) and (S_2, Σ_2) and a π -system $\mathcal{I}$ on S_2 such that $\sigma(\mathcal{I}) = \Sigma_2$. If $h : S_1 \rightarrow S_2$ is a function such that $h^{-1}(A) \in \Sigma_1$ for every $A \in \mathcal{I}$, then h is Σ_1/Σ_2 -measurable.

Proof. Note that the set $\mathcal{E} = \{A \in \Sigma_2 \mid h^{-1}(A) \in \Sigma_1\}$ is a σ -algebra on S_2 :

- $S_2 \in \mathcal{E}$, since $S_2 \in \Sigma_2$ and $h^{-1}(S_2) = S_1$ and $S_1 \in \Sigma_1$.
- If $A \in \mathcal{E}$, then $A^c \in \mathcal{E}$. In order to see this, note that $A \in \Sigma_2$ and $h^{-1}(A) \in \Sigma_1$. Because $A^c \in \Sigma_2$ and $h^{-1}(A^c) = h^{-1}(A)^c$ and $h^{-1}(A)^c \in \Sigma_1$, we know that $h^{-1}(A^c) \in \Sigma_1$.
- If $(A_n \in \mathcal{E} \mid n \in \mathbb{N})$ is a sequence, then $\cup_n A_n \in \mathcal{E}$. In order to see this, note that $\cup_n A_n \in \Sigma_2$ and $h^{-1}(\cup_n A_n) = \cup_n h^{-1}(A_n)$. Because $h^{-1}(A_n) \in \Sigma_1$ for every $n \in \mathbb{N}$, we know that $h^{-1}(\cup_n A_n) \in \Sigma_1$.

Because $\mathcal{I} \subseteq \mathcal{E}$ and $\sigma(\mathcal{I}) = \Sigma_2$, we know that $\Sigma_2 \subseteq \mathcal{E}$. Because $\mathcal{E} \subseteq \Sigma_2$, we know that $\mathcal{E} = \Sigma_2$. Therefore, $h^{-1}(A) \in \Sigma_1$ for every $A \in \Sigma_2$, so that h is Σ_1/Σ_2 -measurable. $\square$

Definition 3.5. Consider the measurable spaces (S_1, Σ_1) and (S_2, Σ_2) and a function $h : S_1 \rightarrow S_2$. The σ -algebra $\sigma(h)$ on S_1 generated by h is given by $\sigma(h) = \{h^{-1}(A) \mid A \in \Sigma_2\}$.

Proposition 3.10. Consider the measurable spaces (S_1, Σ_1) and (S_2, Σ_2) . Suppose that the function $h_1 : S_1 \rightarrow S_2$ has the inverse $h_2 : S_2 \rightarrow S_1$. If $\sigma(h_1) = \Sigma_1$, then $\sigma(h_2) = \Sigma_2$.

Proof. Consider the function $(h_1 \circ h_2) : S_2 \rightarrow S_2$ given by $(h_1 \circ h_2)(s_2) = h_1(h_2(s_2)) = s_2$. For every $A_2 \in \Sigma_2$, note that $(h_1 \circ h_2)^{-1}(A_2) = h_2^{-1}(h_1^{-1}(A_2)) = A_2$. Let $\mathcal{E} = \{A_1 \in \Sigma_1 \mid h_2^{-1}(A_1) \in \Sigma_2\}$. Since $h_1^{-1}(A_2) \in \mathcal{E}$ for every $A_2 \in \Sigma_2$, note that $\sigma(h_1) = \Sigma_1 \subseteq \mathcal{E}$, so that $\mathcal{E} = \Sigma_1$ and $\sigma(h_2) = \{h_2^{-1}(A_1) \mid A_1 \in \Sigma_1\} \subseteq \Sigma_2$. For every $A_2 \in \Sigma_2$, let $A_1 = h_1^{-1}(A_2)$, so that $A_1 \in \Sigma_1$. Since $A_2 = h_2^{-1}(A_1)$, note that $A_2 \in \sigma(h_2)$. Therefore, $\Sigma_2 \subseteq \sigma(h_2)$. $\square$

Proposition 3.11. Consider the measure space (S_1, Σ_1, μ_1) and the measurable space (S_2, Σ_2) . If $h : S_1 \rightarrow S_2$ is Σ_1/Σ_2 -measurable, then the function $\mu_2 : \Sigma_2 \rightarrow [0, \infty]$ given by $\mu_2(A) = \mu_1(h^{-1}(A))$ is a measure on (S_2, Σ_2) .

Proof. Note that $\mu_2(\emptyset) = \mu_1(h^{-1}(\emptyset)) = \mu_1(\emptyset) = 0$ and $\mu_2(S_2) = \mu_1(h^{-1}(S_2)) = \mu_1(S_1)$.

For any sequence $(A_n \in \Sigma_2 \mid n \in \mathbb{N})$ such that $A_n \cap A_m = \emptyset$ for $n \neq m$,

$$\mu_2\left(\bigcup_n A_n\right) = \mu_1\left(h^{-1}\left(\bigcup_n A_n\right)\right) = \mu_1\left(\bigcup_n h^{-1}(A_n)\right) = \sum_n \mu_1(h^{-1}(A_n)) = \sum_n \mu_2(A_n),$$

where we used the fact that $h^{-1}(A_n) \cap h^{-1}(A_m) = h^{-1}(A_n \cap A_m) = h^{-1}(\emptyset) = \emptyset$ for $n \neq m$. $\square$

Consider a measurable space (S, Σ) and a sequence of $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable functions $(h_n \mid n \in \mathbb{N})$.

Definition 3.6. For any $s \in S$, the function $\inf_n h_n : S \rightarrow [-\infty, \infty]$ is given by

$$\left(\inf_n h_n\right)(s) = \inf_n h_n(s).$$

Proposition 3.12. The function $\inf_n h_n$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable.

Proof. Note that if $\{\inf_n h_n \geq c\} \in \Sigma$ for every $c \in \mathbb{R}$, then $\inf_n h_n$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable. For every $c \in \mathbb{R}$,

$$\{\inf_n h_n \geq c\} = \{s \in S \mid \inf_n h_n(s) \geq c\} = \{s \in S \mid h_n(s) \geq c \text{ for all } n \in \mathbb{N}\},$$

where we used the fact that $\inf_n h_n(s) \geq c$ if and only if $h_n(s) \geq c$ for all $n \in \mathbb{N}$, for all $s \in S$ and $c \in \mathbb{R}$. Therefore,

$$\{\inf_n h_n \geq c\} = \bigcap_{n \in \mathbb{N}} \{s \in S \mid h_n(s) \geq c\} = \bigcap_{n \in \mathbb{N}} \{h_n \geq c\},$$

which is a countable intersection of elements of Σ . $\square$

Definition 3.7. For any $s \in S$, the function $\sup_n h_n : S \rightarrow [-\infty, \infty]$ is given by

$$\left(\sup_n h_n\right)(s) = \sup_n h_n(s).$$

Proposition 3.13. The function $\sup_n h_n$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable.

Proof. Note that if $\{\sup_n h_n \leq c\} \in \Sigma$ for every $c \in \mathbb{R}$, then $\sup_n h_n$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable. For every $c \in \mathbb{R}$,

$$\{\sup_n h_n \leq c\} = \{s \in S \mid \sup_n h_n(s) \leq c\} = \{s \in S \mid h_n(s) \leq c \text{ for all } n \in \mathbb{N}\},$$

where we used the fact that $\sup_n h_n(s) \leq c$ if and only if $h_n(s) \leq c$ for all $n \in \mathbb{N}$, for all $s \in S$ and $c \in \mathbb{R}$. Therefore,

$$\{\sup_n h_n \leq c\} = \bigcap_{n \in \mathbb{N}} \{s \in S \mid h_n(s) \leq c\} = \bigcap_{n \in \mathbb{N}} \{h_n \leq c\},$$

which is a countable intersection of elements of Σ . $\square$

Definition 3.8. For any $s \in S$, the function $\liminf_{n \rightarrow \infty} h_n : S \rightarrow [-\infty, \infty]$ is given by

$$\left(\liminf_{n \rightarrow \infty} h_n\right)(s) = \liminf_{n \rightarrow \infty} h_n(s).$$

Proposition 3.14. The function $\liminf_{n \rightarrow \infty} h_n$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable.

Proof. Each function in the sequence $(L_n = \inf_{r \geq n} h_r \mid n \in \mathbb{N})$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable, which implies that $\sup_n L_n$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable. Also,

$$\left(\liminf_{n \rightarrow \infty} h_n\right)(s) = \liminf_{n \rightarrow \infty} h_n(s) = \sup_n \inf_{r \geq n} h_r(s) = \sup_n \left(\inf_{r \geq n} h_r\right)(s) = \sup_n L_n(s) = \left(\sup_n L_n\right)(s).$$

Definition 3.9. For any $s \in S$, the function $\limsup_{n \rightarrow \infty} h_n : S \rightarrow [-\infty, \infty]$ is given by

$$\left(\limsup_{n \rightarrow \infty} h_n\right)(s) = \limsup_{n \rightarrow \infty} h_n(s).$$

Proposition 3.15. The function $\limsup_{n \rightarrow \infty} h_n$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable.

Proof. Each function in the sequence $(L_n = \sup_{r \geq n} h_r \mid n \in \mathbb{N})$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable, which implies that $\inf_n L_n$ is $\Sigma/\mathcal{B}([-\infty, \infty])$ -measurable. Also,

$$\left(\limsup_{n \rightarrow \infty} h_n\right)(s) = \limsup_{n \rightarrow \infty} h_n(s) = \inf_n \sup_{r \geq n} h_r(s) = \inf_n \left(\sup_{r \geq n} h_r\right)(s) = \inf_n L_n(s) = \left(\inf_n L_n\right)(s).$$

Proposition 3.16. Consider the set $F = \{s \in S \mid \lim_{n \rightarrow \infty} h_n(s) \text{ exists in } \mathbb{R}\}$. Recall that $\lim_{n \rightarrow \infty} h_n(s)$ exists in $\mathbb{R}$ if and only if

$$-\infty < \liminf_{n \rightarrow \infty} h_n(s) = \limsup_{n \rightarrow \infty} h_n(s) < \infty.$$

Therefore, $F \in \Sigma$, since F is an intersection of elements of Σ :

$$F = \{s \in S \mid \liminf_{n \rightarrow \infty} h_n(s) > -\infty\} \cap \{s \in S \mid \limsup_{n \rightarrow \infty} h_n(s) < \infty\} \cap \{s \in S \mid \left(\limsup_{n \rightarrow \infty} h_n\right)(s) \leq \left(\liminf_{n \rightarrow \infty} h_n\right)(s)\}.$$

Definition 3.10. Consider a measurable space $(\Omega, \mathcal{F})$. An $\mathcal{F}$ -measurable function $X : \Omega \rightarrow \mathbb{R}$ is a random variable. By definition, for any $B \in \mathcal{B}(\mathbb{R})$, $X^{-1}(B) \in \mathcal{F}$.

Proposition 3.17. The indicator function $\mathbb{I}_F : \Omega \rightarrow \{0, 1\}$ of any event $F \in \mathcal{F}$ is a random variable.

Proof. The function $\mathbb{I}_F$ is defined by

$$\mathbb{I}_F(\omega) = \begin{cases} 1, & \text{if } \omega \in F, \\ 0, & \text{if } \omega \notin F. \end{cases}$$

Recall that if $\{\omega \in \Omega \mid \mathbb{I}_F(\omega) \leq c\} \in \mathcal{F}$ for every $c \in \mathbb{R}$, then $\mathbb{I}_F$ is $\mathcal{F}$ -measurable. For every $c < 0$, we have $\{\omega \in \Omega \mid \mathbb{I}_F(\omega) \leq c\} = \emptyset$. For every $0 \leq c < 1$, we have $\{\omega \in \Omega \mid \mathbb{I}_F(\omega) \leq c\} = \{\omega \in \Omega \mid \omega \notin F\} = F^c$. For every $c \geq 1$, we have $\{\omega \in \Omega \mid \mathbb{I}_F(\omega) \leq c\} = \Omega$. In every case, the set belongs to $\mathcal{F}$. $\square$

Example 3.1. Once again consider an experiment where a coin is tossed infinitely often. Let $H = 0$ represent heads and $T = 1$ represent tails. The outcome space Ω may be defined as the set of infinite binary sequences $\Omega = \{H, T\}^{\mathbb{N}^+}$. Let $F_{n,W} = \{\omega \in \Omega \mid \omega_n = W\}$ be the set of infinite binary sequences whose n -th element is W . The σ -algebra $\mathcal{F}$ on the outcome space Ω may be generated as $\mathcal{F} = \sigma(\{F_{n,W} \mid n \in \mathbb{N}^+, W \in \{H, T\}\})$. Note that $\mathbb{I}_{F_{n,W}}$ is a random variable, since $F_{n,W} \in \mathcal{F}$. Therefore, for any $n \in \mathbb{N}^+$, the function $A_{n,W}$ given by

$$A_{n,W}(\omega) = \left(n^{-1} \sum_{i=1}^n \mathbb{I}_{F_{i,W}} \right) (\omega) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{F_{i,W}}(\omega)$$

is also a random variable. For a given sequence $\omega \in \Omega$, $A_{n,W}(\omega)$ is the fraction of the first n tosses resulting in W .

For a given $p \in [0, 1]$, consider the set $\Lambda_W = \{\omega \in \Omega \mid \lim_{n \rightarrow \infty} A_{n,W}(\omega) = p\}$. Clearly,

$$\Lambda_W = \{\omega \in \Omega \mid \liminf_{n \rightarrow \infty} A_{n,W}(\omega) = p\} \cap \{\omega \in \Omega \mid \limsup_{n \rightarrow \infty} A_{n,W}(\omega) = p\},$$

which can be rewritten as

$$\Lambda_W = \left(\liminf_{n \rightarrow \infty} A_{n,W} \right)^{-1}(\{p\}) \cap \left(\limsup_{n \rightarrow \infty} A_{n,W} \right)^{-1}(\{p\}).$$

Note that $\Lambda_W \in \mathcal{F}$, since both the limit inferior and the limit superior of the sequence of $\mathcal{F}$ -measurable functions $(A_{n,W} \mid n \in \mathbb{N}^+)$ are $\mathcal{F}$ -measurable functions. Therefore, a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ would define the probability $\mathbb{P}(\Lambda_W)$ that the fraction of tosses with result W tends to a given $p \in [0, 1]$.

Definition 3.11. Consider a function $X : \Omega \rightarrow \mathbb{R}$. The σ -algebra $\sigma(X)$ on Ω is defined as $\sigma(X) = \sigma(\{X^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R})\})$.

Note that if X is a random variable on a measurable space $(\Omega, \mathcal{F})$, then $\sigma(X) \subseteq \mathcal{F}$.

Definition 3.12. Consider a set of functions $\{Y_\gamma \mid \gamma \in \mathcal{C}\}$ where $Y_\gamma : \Omega \rightarrow \mathbb{R}$. The σ -algebra $\sigma(\{Y_\gamma \mid \gamma \in \mathcal{C}\})$ is defined by

$$\sigma(\{Y_\gamma \mid \gamma \in \mathcal{C}\}) = \sigma(\{Y_\gamma^{-1}(B) \mid \gamma \in \mathcal{C}, B \in \mathcal{B}(\mathbb{R})\}).$$

Note that if $Y_\gamma : \Omega \rightarrow \mathbb{R}$ is a random variable on a measurable space $(\Omega, \mathcal{F})$ for every γ , then $\sigma(\{Y_\gamma \mid \gamma \in \mathcal{C}\}) \subseteq \mathcal{F}$.

Proposition 3.18. Consider a measurable space $(\Omega, \mathcal{F})$ and a random variable $Y : \Omega \rightarrow \mathbb{R}$. For a set $\mathcal{E}$ of subsets of $\mathbb{R}$, let $Y^{-1}(\mathcal{E}) = \{Y^{-1}(E) \mid E \in \mathcal{E}\}$. By definition, $\sigma(Y) = \sigma(Y^{-1}(\mathcal{B}(\mathbb{R})))$. In that case, $\sigma(Y) = Y^{-1}(\mathcal{B}(\mathbb{R}))$.

Proof. By definition, $Y^{-1}(\mathcal{B}(\mathbb{R})) = \{Y^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R})\}$. Because $\mathbb{R} \in \mathcal{B}(\mathbb{R})$, $Y^{-1}(\mathbb{R}) \in Y^{-1}(\mathcal{B}(\mathbb{R}))$, where $Y^{-1}(\mathbb{R}) = \Omega$. Consider an element $Y^{-1}(B) \in Y^{-1}(\mathcal{B}(\mathbb{R}))$. Because $B^c \in \mathcal{B}(\mathbb{R})$, $Y^{-1}(B^c) \in Y^{-1}(\mathcal{B}(\mathbb{R}))$, where $Y^{-1}(B^c) = (Y^{-1}(B))^c$. Finally, consider a sequence $(Y^{-1}(B_n) \in Y^{-1}(\mathcal{B}(\mathbb{R})) \mid n \in \mathbb{N})$. Because $\cup_n B_n \in \mathcal{B}(\mathbb{R})$, $Y^{-1}(\cup_n B_n) \in Y^{-1}(\mathcal{B}(\mathbb{R}))$, where $Y^{-1}(\cup_n B_n) = \cup_n Y^{-1}(B_n)$. Therefore, $Y^{-1}(\mathcal{B}(\mathbb{R}))$ is a σ -algebra on Ω . Because $\sigma(Y)$ is the smallest σ -algebra on Ω that includes $Y^{-1}(\mathcal{B}(\mathbb{R}))$, we know that $\sigma(Y) = Y^{-1}(\mathcal{B}(\mathbb{R}))$. $\square$

Proposition 3.19. Additionally, consider the π -system $\pi(\mathbb{R}) = \{(-\infty, x] \mid x \in \mathbb{R}\}$ and let $\pi(Y) = Y^{-1}(\pi(\mathbb{R}))$. In that case, $\sigma(Y) = \sigma(\pi(Y))$.

Proof. By definition, $\sigma(\pi(Y)) = \sigma(\{Y^{-1}((-\infty, x]) \mid (-\infty, x] \in \pi(\mathbb{R})\})$. Clearly, $\pi(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R})$ implies $\sigma(\pi(Y)) \subseteq \sigma(Y)$, since $\sigma(Y) = \sigma(\{Y^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R})\})$. Because $\{Y \leq x\} \in \sigma(\pi(Y))$ for every $x \in \mathbb{R}$, Y is $\sigma(\pi(Y))$ -measurable. Therefore, $\sigma(Y) \subseteq \sigma(\pi(Y))$. $\square$

Proposition 3.20. If $Y : \Omega \rightarrow \mathbb{R}$, then $Z : \Omega \rightarrow \mathbb{R}$ is a $\sigma(Y)$ -measurable function if and only if there is a Borel function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $Z = f \circ Y$.

Proposition 3.21. If $Y_1, Y_2, \dots, Y_n$ are functions from Ω to $\mathbb{R}$, then a function $Z : \Omega \rightarrow \mathbb{R}$ is $\sigma(\{Y_1, Y_2, \dots, Y_n\})$ -measurable if and only if there is a Borel function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $Z(\omega) = f(Y_1(\omega), Y_2(\omega), \dots, Y_n(\omega))$ for every $\omega \in \Omega$.

Definition 3.13. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable $X : \Omega \rightarrow \mathbb{R}$. For any $B \in \mathcal{B}(\mathbb{R})$, $X^{-1}(B) \in \sigma(X)$, $\sigma(X) \subseteq \mathcal{F}$, and $\mathbb{P}(X^{-1}(B)) \in [0, 1]$. For any $B \in \mathcal{B}(\mathbb{R})$, this allows defining the law $\mathcal{L}_X : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$ of X as

$$\mathcal{L}_X(B) = \mathbb{P}(X^{-1}(B)).$$

Proposition 3.22. The law $\mathcal{L}_X$ is a probability measure on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Proof. First, note that

$$\begin{aligned}\mathcal{L}_X(\mathbb{R}) &= \mathbb{P}(X^{-1}(\mathbb{R})) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) \in \mathbb{R}\}) = \mathbb{P}(\Omega) = 1, \\ \mathcal{L}_X(\emptyset) &= \mathbb{P}(X^{-1}(\emptyset)) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) \in \emptyset\}) = \mathbb{P}(\emptyset) = 0.\end{aligned}$$

Second, consider a sequence of sets $(B_n \in \mathcal{B}(\mathbb{R}) \mid n \in \mathbb{N})$ such that $B_n \cap B_m = \emptyset$ for $n \neq m$ and note that

$$\mathcal{L}_X\left(\bigcup_{n \in \mathbb{N}} B_n\right) = \mathbb{P}\left(X^{-1}\left(\bigcup_{n \in \mathbb{N}} B_n\right)\right) = \mathbb{P}\left(\bigcup_{n \in \mathbb{N}} X^{-1}(B_n)\right) = \sum_{n \in \mathbb{N}} \mathbb{P}(X^{-1}(B_n)) = \sum_{n \in \mathbb{N}} \mathcal{L}_X(B_n),$$

where we used the fact that $X^{-1}(B_n) \cap X^{-1}(B_m) = X^{-1}(B_n \cap B_m) = X^{-1}(\emptyset) = \emptyset$ for $n \neq m$. $\square$

Definition 3.14. The (cumulative) distribution function $F_X : \mathbb{R} \rightarrow [0, 1]$ of the random variable X is defined by

$$F_X(c) = \mathcal{L}_X((-\infty, c]) = \mathbb{P}(X^{-1}((-\infty, c])) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) \leq c\}) = \mathbb{P}(\{X \leq c\}).$$

Proposition 3.23. Recall that the σ -algebra generated by $\pi(\mathbb{R}) = \{(-\infty, x] \mid x \in \mathbb{R}\}$ is $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$. Consider a probability measure μ on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\mu((-\infty, c]) = F_X(c) = \mathcal{L}_X((-\infty, c])$ for every $c \in \mathbb{R}$. Because μ and $\mathcal{L}_X$ agree on the π -system $\pi(\mathbb{R})$, we have $\mu = \mathcal{L}_X$. Therefore, F_X fully determines the law $\mathcal{L}_X$ of X .

Consider a random variable $X : \Omega \rightarrow \mathbb{R}$ carried by a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the distribution function $F_X : \mathbb{R} \rightarrow [0, 1]$.

Proposition 3.24. If $a \leq b$, then $F_X(a) \leq F_X(b)$.

Proof. Clearly, $\{X \leq a\} \subseteq \{X \leq b\}$, which implies $\mathbb{P}(\{X \leq a\}) \leq \mathbb{P}(\{X \leq b\})$. $\square$

Proposition 3.25. $\lim_{x \rightarrow -\infty} F_X(x) = 0$.

Proof. Recall that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $\lim_{x \rightarrow -\infty} f(x) = L$ for some $L \in \mathbb{R}$ if and only if $\lim_{n \rightarrow \infty} f(x_n) = L$ for all non-increasing sequences $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ such that $\lim_{n \rightarrow \infty} x_n = -\infty$.

Consider a non-increasing sequence $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ such that $\lim_{n \rightarrow \infty} x_n = -\infty$ and the sequence of sets $(A_n = (-\infty, x_n] \mid n \in \mathbb{N})$. Because $A_n \downarrow \emptyset$, $\mathcal{L}_X(A_n) \downarrow 0$. Therefore, $\lim_{n \rightarrow \infty} \mathcal{L}_X((-\infty, x_n]) = 0$, which implies

$$\lim_{x \rightarrow -\infty} F_X(x) = \lim_{x \rightarrow -\infty} \mathcal{L}_X((-\infty, x]) = 0.$$

Proposition 3.26. $\lim_{x \rightarrow \infty} F_X(x) = 1$.

Proof. Recall that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $\lim_{x \rightarrow \infty} f(x) = L$ for some $L \in \mathbb{R}$ if and only if $\lim_{n \rightarrow \infty} f(x_n) = L$ for all non-decreasing sequences $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ such that $\lim_{n \rightarrow \infty} x_n = +\infty$.

Consider a non-decreasing sequence $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ such that $\lim_{n \rightarrow \infty} x_n = +\infty$ and the sequence of sets $(A_n = (-\infty, x_n] \mid n \in \mathbb{N})$. Because $A_n \uparrow \mathbb{R}$, $\mathcal{L}_X(A_n) \uparrow 1$. Therefore, $\lim_{n \rightarrow \infty} \mathcal{L}_X((-\infty, x_n]) = 1$, which implies

$$\lim_{x \rightarrow \infty} F_X(x) = \lim_{x \rightarrow \infty} \mathcal{L}_X((-\infty, x]) = 1.$$

□

Proposition 3.27. The function F_X is right-continuous.

Proof. Recall that $f : \mathbb{R} \rightarrow \mathbb{R}$ is right continuous if and only if $\lim_{n \rightarrow \infty} f(x_n) = f(x)$ for every $x \in \mathbb{R}$ and every non-increasing sequence $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ such that $\lim_{n \rightarrow \infty} x_n = x$ and $x_n > x$ for every $n \in \mathbb{N}$.

Consider $x \in \mathbb{R}$ and a non-increasing sequence $(x_n \in \mathbb{R} \mid n \in \mathbb{N})$ such that $\lim_{n \rightarrow \infty} x_n = x$ and $x_n > x$ for every $n \in \mathbb{N}$. Consider also the sequence of sets $(A_n = (-\infty, x_n] \mid n \in \mathbb{N})$. Because $A_n \downarrow (-\infty, x]$, $\mathcal{L}_X((-\infty, x_n]) \downarrow \mathcal{L}_X((-\infty, x])$. Therefore, $\lim_{n \rightarrow \infty} \mathcal{L}_X((-\infty, x_n]) = \mathcal{L}_X((-\infty, x])$, which implies

$$\lim_{n \rightarrow \infty} F_X(x_n) = \lim_{n \rightarrow \infty} \mathcal{L}_X((-\infty, x_n]) = \mathcal{L}_X((-\infty, x]) = F_X(x).$$

□

Proposition 3.28. Consider a right-continuous function $F : \mathbb{R} \rightarrow [0, 1]$ such that if $a \leq b$, then $F(a) \leq F(b)$; $\lim_{x \rightarrow -\infty} F(x) = 0$; and $\lim_{x \rightarrow \infty} F(x) = 1$. There is a unique probability measure $\mathcal{L}$ on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\mathcal{L}((-\infty, x]) = F(x)$ for every $x \in \mathbb{R}$.

Proof. Consider the probability triple $((0, 1), \mathcal{B}((0, 1)), \text{Leb})$, where Leb denotes the restriction to $(0, 1)$ of the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, and a function $X^- : (0, 1) \rightarrow \mathbb{R}$ given by

$$X^-(\omega) = \inf\{z \in \mathbb{R} \mid F(z) \geq \omega\}.$$

In words, $X^-(\omega)$ is the infimum $z \in \mathbb{R}$ such that $F(z)$ reaches $\omega \in (0, 1)$.

First, note that $\omega \leq F(c)$ if and only if $X^-(\omega) \leq c$ for every $c \in \mathbb{R}$. Clearly, if $\omega \leq F(c)$, then $X^-(\omega) \leq c$. Now suppose $X^-(\omega) \leq c$. Because F is non-decreasing, $F(X^-(\omega)) \leq F(c)$. Because F is also right-continuous, $F(X^-(\omega)) \geq \omega$. Therefore, $\omega \leq F(c)$. This also implies that X^- is a random variable since, for every $c \in \mathbb{R}$,

$$\{X^- \leq c\} = \{\omega \in (0, 1) \mid X^-(\omega) \leq c\} = \{\omega \in (0, 1) \mid \omega \leq F(c)\} = (0, F(c)] \cap (0, 1).$$

For every $c \in \mathbb{R}$, the distribution function F_{X^-} on the probability triple $((0, 1), \mathcal{B}((0, 1)), \text{Leb})$ is given by

$$F_{X^-}(c) = \mathcal{L}_{X^-}((-\infty, c]) = \text{Leb}(\{X^- \leq c\}) = \text{Leb}((0, F(c)] \cap (0, 1)) = F(c).$$

Finally, recall that the distribution function F_{X^-} fully determines the law $\mathcal{L}_{X^-}$ of X^- , which is the desired unique probability measure on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\mathcal{L}_{X^-}((-\infty, x]) = F(x)$ for every $x \in \mathbb{R}$. □

Theorem 3.1 (Monotone-class theorem). If

- $\mathcal{H}$ is a set of bounded functions from a set S into $\mathbb{R}$,
- $\mathcal{H}$ is a vector space over $\mathbb{R}$,
- The constant function 1 is an element of $\mathcal{H}$,
- If $(f_n \in \mathcal{H} \mid n \in \mathbb{N})$ is a sequence of non-negative functions in $\mathcal{H}$ such that $f_n \uparrow f$, where f is a bounded function on S , then $f \in \mathcal{H}$,
- $\mathcal{H}$ contains the indicator function of every set in some π -system $\mathcal{I}$,

then $\mathcal{H}$ contains every bounded $\sigma(\mathcal{I})$ -measurable function on S .

4 Independence

Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 4.1. The sub- σ -algebras $\mathcal{G}_1, \mathcal{G}_2, \dots$ of $\mathcal{F}$ are called independent if, for every choice of distinct indices $i_1, i_2, \dots, i_n$ and events $G_{i_1}, G_{i_2}, \dots, G_{i_n}$ such that $G_{i_k} \in \mathcal{G}_{i_k}$ for every i_k ,

$$\mathbb{P}\left(\bigcap_{k=1}^n G_{i_k}\right) = \prod_{k=1}^n \mathbb{P}(G_{i_k}).$$

Definition 4.2. The random variables $X_1, X_2, \dots$ are called independent if the σ -algebras $\sigma(X_1), \sigma(X_2), \dots$ are independent.

Definition 4.3. The events $E_1, E_2, \dots$ are called independent if the σ -algebras $\mathcal{E}_1, \mathcal{E}_2, \dots$ are independent, where $\mathcal{E}_k = \{\emptyset, E_k, E_k^c, \Omega\}$.

Proposition 4.1. The events $E_1, E_2, \dots$ are independent if and only if the random variables $\mathbb{I}_{E_1}, \mathbb{I}_{E_2}, \dots$ are independent.

Proof. We have already shown that each indicator function $\mathbb{I}_{E_k}$ is $\mathcal{E}_k$ -measurable. Since $\mathbb{I}_{E_k}^{-1}(\{1\}) = E_k$, we know that $E_k \in \sigma(\mathbb{I}_{E_k})$, which implies $\mathcal{E}_k = \sigma(\mathbb{I}_{E_k})$. $\square$

Proposition 4.2. The events $E_1, E_2, \dots$ are independent if and only if, for every choice of distinct indices $i_1, i_2, \dots, i_n$,

$$\mathbb{P}\left(\bigcap_{k=1}^n E_{i_k}\right) = \prod_{k=1}^n \mathbb{P}(E_{i_k}).$$

Proposition 4.3. If $X_1, X_2, \dots$ are independent random variables, then the events $\{X_1 \leq x_1\}, \{X_2 \leq x_2\}, \dots$ are independent for every $x_1, x_2, \dots \in \mathbb{R}$, since $X_n^{-1}((-\infty, x_n]) \in \sigma(X_n)$ for every $n \in \mathbb{N}^+$.

Proposition 4.4. Suppose that $\mathcal{G}$ and $\mathcal{H}$ are sub- σ -algebras of $\mathcal{F}$. Furthermore, let $\mathcal{I}$ and $\mathcal{J}$ be π -systems such that $\sigma(\mathcal{I}) = \mathcal{G}$ and $\sigma(\mathcal{J}) = \mathcal{H}$. If $\mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J)$ for every $I \in \mathcal{I}$ and $J \in \mathcal{J}$, we say that $\mathcal{I}$ and $\mathcal{J}$ are independent. The sub- σ -algebras $\mathcal{G}$ and $\mathcal{H}$ are independent if and only if $\mathcal{I}$ and $\mathcal{J}$ are independent.

Proof. Suppose that $\mathcal{G}$ and $\mathcal{H}$ are independent. In that case, $\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H)$ for every $G \in \mathcal{G}$ and $H \in \mathcal{H}$. Since $\mathcal{I} \subseteq \mathcal{G}$ and $\mathcal{J} \subseteq \mathcal{H}$, $\mathcal{I}$ and $\mathcal{J}$ are independent.

Suppose that $\mathcal{I}$ and $\mathcal{J}$ are independent. For every $I \in \mathcal{I}$ and $H \in \mathcal{H}$, let $\mu_I(H) = \mathbb{P}(I \cap H)$ and $\eta_I(H) = \mathbb{P}(I)\mathbb{P}(H)$. Clearly, $\mu_I(\emptyset) = 0 = \eta_I(\emptyset)$. Also, $\mu_I(\Omega) = \mathbb{P}(I) = \eta_I(\Omega)$. Finally, if $(H_n \in \mathcal{H} \mid n \in \mathbb{N})$ is a sequence of events such that $H_n \cap H_m = \emptyset$ for $n \neq m$,

$$\begin{aligned} \mu_I\left(\bigcup_n H_n\right) &= \mathbb{P}\left(I \cap \left(\bigcup_n H_n\right)\right) = \mathbb{P}\left(\bigcup_n (I \cap H_n)\right) = \sum_n \mathbb{P}(I \cap H_n) = \sum_n \mu_I(H_n), \\ \eta_I\left(\bigcup_n H_n\right) &= \mathbb{P}(I)\mathbb{P}\left(\bigcup_n H_n\right) = \mathbb{P}(I) \sum_n \mathbb{P}(H_n) = \sum_n \mathbb{P}(I)\mathbb{P}(H_n) = \sum_n \eta_I(H_n). \end{aligned}$$

Considered together, these results imply that μ_I and η_I are finite measures on $(\Omega, \mathcal{H})$. By assumption, $\mu_I(J) = \mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J) = \eta_I(J)$ for every $I \in \mathcal{I}$ and $J \in \mathcal{J}$. Therefore, μ_I and η_I agree on the π -system $\mathcal{J}$, which implies that they agree on the σ -algebra $\sigma(\mathcal{J}) = \mathcal{H}$. In other words, for every $I \in \mathcal{I}$ and $H \in \mathcal{H}$, we have $\mathbb{P}(I \cap H) = \mu_I(H) = \eta_I(H) = \mathbb{P}(I)\mathbb{P}(H)$.

For every $H \in \mathcal{H}$ and $G \in \mathcal{G}$, let $\mu'_H(G) = \mathbb{P}(H \cap G)$ and $\eta'_H(G) = \mathbb{P}(H)\mathbb{P}(G)$. Analogously, μ'_H and η'_H are finite measures on $(\Omega, \mathcal{G})$. From our previous result, for every $I \in \mathcal{I}$ and $H \in \mathcal{H}$, we have $\mathbb{P}(I \cap H) = \mu'_H(I) = \eta'_H(I) = \mathbb{P}(I)\mathbb{P}(H)$. Therefore, μ'_H and η'_H agree on the π -system $\mathcal{I}$, which implies that they agree on the σ -algebra $\sigma(\mathcal{I}) = \mathcal{G}$. In other words, for every $G \in \mathcal{G}$ and $H \in \mathcal{H}$, we have $\mathbb{P}(G \cap H) = \mu'_H(G) = \eta'_H(G) = \mathbb{P}(G)\mathbb{P}(H)$. $\square$

Proposition 4.5. Consider the random variables X and Y on the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. For every $A \in \mathcal{B}(\mathbb{R})$ and $B \in \mathcal{B}(\mathbb{R})$ such that $\mathbb{P}(Y^{-1}(B)) > 0$, let $\mathbb{P}(X^{-1}(A) \mid Y^{-1}(B)) = \mathbb{P}(X^{-1}(A) \cap Y^{-1}(B)) / \mathbb{P}(Y^{-1}(B))$. If X and Y are independent, then $\mathbb{P}(X^{-1}(A) \mid Y^{-1}(B)) = \mathbb{P}(X^{-1}(A))$, since $X^{-1}(A) \in \sigma(X)$ and $Y^{-1}(B) \in \sigma(Y)$.

In what follows, we will employ a common abuse of notation. Consider the random variables X and Y on the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. For every $x \in \mathbb{R}$, we will let $\mathbb{P}(X \leq x)$ denote $\mathbb{P}(\{X \leq x\})$. Furthermore, for every $x, y \in \mathbb{R}$, we will let $\mathbb{P}(X \leq x, Y \leq y)$ denote $\mathbb{P}(\{X \leq x\} \cap \{Y \leq y\})$. We will employ analogous notation when there are more random variables and different predicates.

Proposition 4.6. Consider the random variables X and Y on the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that, for every $x, y \in \mathbb{R}$, $\mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x)\mathbb{P}(Y \leq y)$. The random variables X and Y are independent.

Proof. Recall that $\pi(\mathbb{R}) = \{(-\infty, x] \mid x \in \mathbb{R}\}$ and $\pi(X) = \{X^{-1}((-\infty, x]) \mid (-\infty, x] \in \pi(\mathbb{R})\} = \{X \leq x \mid x \in \mathbb{R}\}$. Note that $\pi(X)$ is a π -system on Ω : for any $x_1, x_2 \in \mathbb{R}$, if $\{X \leq x_1\} \in \pi(X)$ and $\{X \leq x_2\} \in \pi(X)$, then $\{X \leq x_1\} \cap \{X \leq x_2\} = \{\omega \in \Omega \mid X(\omega) \leq x_1 \text{ and } X(\omega) \leq x_2\} = \{\omega \in \Omega \mid X(\omega) \leq \min(x_1, x_2)\} = \{X \leq \min(x_1, x_2)\}$. By assumption, $\mathbb{P}(\{X \leq x\} \cap \{Y \leq y\}) = \mathbb{P}(\{X \leq x\})\mathbb{P}(\{Y \leq y\})$ for any $\{X \leq x\} \in \pi(X)$ and $\{Y \leq y\} \in \pi(Y)$. By definition, the π -systems $\pi(X)$ and $\pi(Y)$ are independent. Therefore, $\sigma(\pi(X))$ and $\sigma(\pi(Y))$ are independent. Based on a previous result, we know that $\sigma(\pi(X)) = \sigma(X)$ and $\sigma(\pi(Y)) = \sigma(Y)$. $\square$

Proposition 4.7. In general, the random variables $X_1, X_2, \dots, X_n$ are independent if and only if, for every $x_1, x_2, \dots, x_n \in \mathbb{R}$,

$$\mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) = \mathbb{P}\left(\bigcap_{k=1}^n \{X_k \leq x_k\}\right) = \prod_{k=1}^n \mathbb{P}(X_k \leq x_k).$$

Lemma 4.1 (Second Borel-Cantelli Lemma). Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence of independent events $(E_n \in \mathcal{F} \mid n \in \mathbb{N})$ such that $\sum_{n=0}^{\infty} \mathbb{P}(E_n) = \infty$. In that case,

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} E_n\right) = 1.$$

Proof. Because the events are independent, for any $m, r \in \mathbb{N}$ such that $m \leq r$,

$$\mathbb{P}\left(\bigcap_{m \leq n \leq r} E_n^c\right) = \prod_{m \leq n \leq r} \mathbb{P}(E_n^c) = \prod_{m \leq n \leq r} (1 - \mathbb{P}(E_n)).$$

Let e denote Euler's number. For any $x \geq 0$, recall that $1 - x \leq e^{-x}$. Therefore,

$$\mathbb{P}\left(\bigcap_{m \leq n \leq r} E_n^c\right) \leq \prod_{m \leq n \leq r} e^{-\mathbb{P}(E_n)} = e^{-\sum_{m \leq n \leq r} \mathbb{P}(E_n)}.$$

Because both sides of the inequation above are non-increasing with respect to r , we may take the limit of both sides when $r \rightarrow \infty$ and use the fact that $\sum_{n=0}^{\infty} \mathbb{P}(E_n) = \infty$ to conclude that

$$\lim_{r \rightarrow \infty} \mathbb{P}\left(\bigcap_{m \leq n \leq r} E_n^c\right) = \mathbb{P}\left(\bigcap_{n \geq m} E_n^c\right) \leq \lim_{r \rightarrow \infty} e^{-\sum_{m \leq n \leq r} \mathbb{P}(E_n)} = 0.$$

Using the relationship between the limit superior and the limit inferior,

$$\mathbb{P}\left(\left(\limsup_{n \rightarrow \infty} E_n\right)^c\right) = \mathbb{P}\left(\liminf_{n \rightarrow \infty} E_n^c\right) = \mathbb{P}\left(\bigcup_m \bigcap_{n \geq m} E_n^c\right) \leq \sum_m \mathbb{P}\left(\bigcap_{n \geq m} E_n^c\right) = 0.$$

$\square$

Definition 4.4. A valid distribution function $F : \mathbb{R} \rightarrow [0, 1]$ is a right-continuous function such that if $a \leq b$, then $F(a) \leq F(b)$; $\lim_{x \rightarrow -\infty} F(x) = 0$; and $\lim_{x \rightarrow \infty} F(x) = 1$.

Proposition 4.8. For any sequence of valid distribution functions $(F_n \mid n \in \mathbb{N})$, there is a sequence of independent random variables $(X_n \mid n \in \mathbb{N})$ on the probability triple $([0, 1], \mathcal{B}([0, 1]), \text{Leb})$ such that F_n is the distribution function of X_n .

Definition 4.5. Let $(X_n \mid n \in \mathbb{N})$ be a sequence of independent random variables on the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. If $\mathbb{P}(X_n \leq x) = F(x)$ for every $n \in \mathbb{N}$, $x \in \mathbb{R}$, and a distribution function $F : \mathbb{R} \rightarrow [0, 1]$, then the random variables are considered independent and identically distributed.

Example 4.1. As an application of the Borel-Cantelli lemmas, consider a sequence of independent random variables $(X_n \mid n \in \mathbb{N}^+)$ on the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that each random variable X_n is exponentially distributed with rate 1 such that $\mathbb{P}(X_n > x_n) = 1 - \mathbb{P}(X_n \leq x_n) = e^{-x_n}$ for every $x_n \geq 0$. If $x_n = \alpha \log n$ for some $\alpha > 0$, then

$$\mathbb{P}(X_n > \alpha \log n) = e^{-\alpha \log n} = (e^{\log n})^{-\alpha} = \frac{1}{n^\alpha}.$$

For some $\alpha > 0$, consider the sequence of independent events $(\{X_n > \alpha \log n\} \in \mathcal{F} \mid n \in \mathbb{N}^+)$ and recall that

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n > \alpha \log n) = \sum_{n=1}^{\infty} \frac{1}{n^\alpha} < \infty$$

if and only if $\alpha > 1$. Using the Borel-Cantelli lemmas,

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} \{X_n > \alpha \log n\}\right) = \begin{cases} 0, & \text{if } \alpha > 1, \\ 1, & \text{if } \alpha \leq 1. \end{cases}$$

Recall that $\omega \in \limsup_{n \rightarrow \infty} \{X_n > \alpha \log n\}$ if and only if $X_n(\omega) > \alpha \log n$ for infinitely many $n \in \mathbb{N}^+$.

In particular, if $\omega \in \limsup_{n \rightarrow \infty} \{X_n > \log n\}$, then for every $m \in \mathbb{N}^+$ there is an $n > m$ such that $X_n(\omega) > \log n$ and $X_n(\omega)/\log n > 1$. In that case,

$$\limsup_{n \rightarrow \infty} \frac{X_n(\omega)}{\log n} = \lim_{m \rightarrow \infty} \sup_{n \geq m} \frac{X_n(\omega)}{\log n} \geq 1,$$

so that $\omega \in \left\{ \limsup_{n \rightarrow \infty} \frac{X_n}{\log n} \geq 1 \right\}$ and

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} \frac{X_n}{\log n} \geq 1\right) \geq \mathbb{P}\left(\limsup_{n \rightarrow \infty} \{X_n > \log n\}\right) = 1.$$

For every $k \in \mathbb{N}^+$, if $\omega \in \left\{ \limsup_{n \rightarrow \infty} \frac{X_n}{\log n} > 1 + 2k^{-1} \right\}$, then for every $m \in \mathbb{N}^+$ there is an $n > m$ such that $\frac{X_n(\omega)}{\log n} > 1 + 2k^{-1}$ and $X_n(\omega) > (1 + 2k^{-1}) \log n$. In that case, $\omega \in \limsup_{n \rightarrow \infty} \{X_n > (1 + 2k^{-1}) \log n\}$ and

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} \frac{X_n}{\log n} > 1\right) = \mathbb{P}\left(\bigcup_{k \in \mathbb{N}^+} \left\{ \limsup_{n \rightarrow \infty} \frac{X_n}{\log n} > 1 + 2k^{-1} \right\}\right) \leq \sum_{k \in \mathbb{N}^+} \mathbb{P}\left(\limsup_{n \rightarrow \infty} \{X_n > (1 + 2k^{-1}) \log n\}\right) = 0.$$

By combining the previous results,

$$\mathbb{P}\left(\limsup_{n \rightarrow \infty} \frac{X_n}{\log n} = 1\right) = 1.$$

Definition 4.6. For any set $\mathcal{C}$, a set (or sequence) of random variables $Y = (Y_\gamma \mid \gamma \in \mathcal{C})$ on a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a stochastic process parameterized by $\mathcal{C}$.

Proposition 4.9. Consider a measurable space $(\Omega, \mathcal{F})$ and a function $X : \Omega \rightarrow C$, where $C \subseteq \mathbb{N}$. If $\{X = c\} \in \mathcal{F}$ for every $c \in C$, then X is $\mathcal{F}$ -measurable.

Proof. For any $B \in \mathcal{B}(\mathbb{R})$, let $A = B \cap C$ and note that

$$X^{-1}(B) = \{\omega \in \Omega \mid X(\omega) \in B\} = \{\omega \in \Omega \mid X(\omega) \in B \text{ and } X(\omega) \in C\} = X^{-1}(B \cap C) = X^{-1}(A).$$

Furthermore, note that

$$X^{-1}(A) = X^{-1}\left(\bigcup_{a \in A} \{a\}\right) = \bigcup_{a \in A} X^{-1}(\{a\}) = \bigcup_{a \in A} \{X = a\}.$$

Because $A \subseteq C$, we have $\{X = a\} \in \mathcal{F}$ for every $a \in A$. Because $\mathcal{F}$ is a σ -algebra, we have $X^{-1}(A) \in \mathcal{F}$. Therefore, for every $B \in \mathcal{B}(\mathbb{R})$, we have $X^{-1}(B) \in \mathcal{F}$. □

Definition 4.7. Consider a set $E \subseteq \mathbb{N}$. Let P be a stochastic matrix whose (i, j) -th element is given by $p_{i,j} \geq 0$ and suppose that $\sum_{k \in E} p_{i,k} = 1$ for every $i \in E$. Let μ be a probability measure on the measurable space $(E, \mathcal{P}(E))$, where $\mathcal{P}(E)$ is the set of all subsets of E , and let μ_i denote $\mu(\{i\})$ for every $i \in E$. A time-homogeneous Markov chain $Z = (Z_n \mid n \in \mathbb{N})$ on E with initial distribution μ and 1-step transition matrix P is a stochastic process parameterized by $\mathbb{N}$ such that, for every $n \in \mathbb{N}^+$ and $i_0, i_1, \dots, i_n \in E$,

$$\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \mu_{i_0} p_{i_0, i_1} \cdots p_{i_{n-1}, i_n} = \mu_{i_0} \prod_{k=1}^n p_{i_{k-1}, i_k}.$$

Proposition 4.10. A probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ carrying the aforementioned stochastic process Z exists.

Proof. First, for any set of valid distribution functions $\{F_n \mid n \in \mathbb{N}\}$, recall that there is a set of independent random variables $\{X_n \mid n \in \mathbb{N}\}$ on a certain probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ such that F_n is the distribution function of X_n . Using this result, for every $i, j \in E$ and $n \in \mathbb{N}^+$, let $Z_0 : \Omega \rightarrow E$ and $Y_{i,n} : \Omega \rightarrow E$ be independent random variables on a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\mathbb{P}(Z_0 = i) = \mu_i$ and $\mathbb{P}(Y_{i,n} = j) = p_{i,j}$.

For every $\omega \in \Omega$ and $n \in \mathbb{N}^+$, let $Z_n(\omega) = Y_{Z_{n-1}(\omega), n}(\omega)$. Using induction, we will show that the function $Z_n : \Omega \rightarrow E$ is a random variable for every $n \in \mathbb{N}$. We already know that Z_0 is a random variable. Suppose that Z_{n-1} is a random variable. We will show that $\{Z_n = i_n\} \in \mathcal{F}$ for every $i_n \in E$. By definition,

$$\{Z_n = i_n\} = \{\omega \in \Omega \mid Z_n(\omega) = i_n\} = \{\omega \in \Omega \mid Y_{Z_{n-1}(\omega), n}(\omega) = i_n\} = \bigcup_{i \in E} \{\omega \in \Omega \mid Z_{n-1}(\omega) = i \text{ and } Y_{i,n}(\omega) = i_n\},$$

which implies

$$\{Z_n = i_n\} = \bigcup_{i \in E} \{Z_{n-1} = i\} \cap \{Y_{i,n} = i_n\}.$$

Because Z_{n-1} and $Y_{i,n}$ are random variables for every $i \in E$, $\{Z_n = i_n\} \in \mathcal{F}$, as we wanted to show.

Using induction, we will now show that, for every $n \in \mathbb{N}$ and $i_0, \dots, i_n \in E$,

$$\bigcap_{k=0}^n \{Z_k = i_k\} = \{Z_0 = i_0\} \cap \bigcap_{k=1}^n \{Y_{i_{k-1}, k} = i_k\}.$$

The statement above is true when $n = 0$, so suppose it is true for some $n - 1 \in \mathbb{N}$. Using a previous result,

$$\bigcap_{k=0}^n \{Z_k = i_k\} = \left(\bigcap_{k=0}^{n-1} \{Z_k = i_k\} \right) \cap \{Z_n = i_n\} = \left(\bigcap_{k=0}^{n-1} \{Z_k = i_k\} \right) \cap \left(\bigcup_{i \in E} \{Z_{n-1} = i\} \cap \{Y_{i,n} = i_n\} \right).$$

By distributing the intersection over the union,

$$\bigcap_{k=0}^n \{Z_k = i_k\} = \bigcup_{i \in E} \left(\bigcap_{k=0}^{n-1} \{Z_k = i_k\} \right) \cap \{Z_{n-1} = i\} \cap \{Y_{i,n} = i_n\}.$$

Because $\{Z_{n-1} = i_{n-1}\} \cap \{Z_{n-1} = i\} = \emptyset$ whenever $i \neq i_{n-1}$,

$$\bigcap_{k=0}^n \{Z_k = i_k\} = \left(\bigcap_{k=0}^{n-1} \{Z_k = i_k\} \right) \cap \{Y_{i_{n-1}, n} = i_n\} = \{Z_0 = i_0\} \cap \bigcap_{k=1}^n \{Y_{i_{k-1}, k} = i_k\},$$

where the last equation follows from the inductive hypothesis.

The event above is the intersection of events from the σ -algebras of independent random variables, which implies

$$\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \mathbb{P}\left(\bigcap_{k=0}^n \{Z_k = i_k\}\right) = \mathbb{P}(Z_0 = i_0) \prod_{k=1}^n \mathbb{P}(Y_{i_{k-1}, k} = i_k) = \mu_{i_0} \prod_{k=1}^n p_{i_{k-1}, i_k}.$$

□

Example 4.2. Consider a time-homogeneous Markov chain $Z = (Z_n \mid n \in \mathbb{N})$ on E with initial distribution μ and 1-step transition matrix P . Consider also a finite sequence of elements of E given by $I = i_0, i_1, \dots, i_n$. We say that the sequence I appears in outcome $\omega \in \Omega$ at time t if $Z_{t+k}(\omega) = i_k$ for every $k \leq n$. We will now show how several interesting events related to the appearance of the sequence I may be defined.

The event M_t composed of outcomes where the sequence I appears at time t is given by

$$M_t = \bigcap_{k=0}^n \{Z_{t+k} = i_k\} = \bigcap_{k=0}^n \{\omega \in \Omega \mid Z_{t+k}(\omega) = i_k\}.$$

The event S_t composed of outcomes where the sequence I appears at least once at or after time t is given by

$$S_t = \bigcup_{t' \geq t} M_{t'}.$$

The event $L_{t,m}$ composed of outcomes where the sequence I appears at least m times up to time t is given by

$$L_{t,m} = \bigcup_{l_1, \dots, l_m} \bigcap_{k=1}^m M_{l_k},$$

where the union ranges over every finite sequence $l_1, \dots, l_m$ of distinct times such that $l_k \in \mathbb{N}$ and $l_k \leq t$ for every $k \leq m$.

The event L_m composed of outcomes where I appears at least m times is given by $L_m = \bigcup_{t \in \mathbb{N}} L_{t,m}$, since $L_{t,m} \subseteq L_{t+1,m}$ for every $t \in \mathbb{N}$.

The event M_∞ composed of outcomes where the sequence I appears infinitely many times is given by

$$M_\infty = \limsup_{t \rightarrow \infty} M_t.$$

5 Integration

Consider a measure space (S, Σ, μ) . The integral with respect to μ of a Σ -measurable function $f : S \rightarrow \mathbb{R}$ is denoted by $\mu(f)$.

Definition 5.1. For any set $A \in \Sigma$, the integral with respect to μ of the indicator function $\mathbb{I}_A : S \rightarrow \{0, 1\}$ is defined as

$$\mu(\mathbb{I}_A) = \mu(A).$$

Definition 5.2. A simple function is a Σ -measurable function $f : S \rightarrow [0, \infty]$ that can be written as

$$f(s) = \sum_{k=1}^m a_k \mathbb{I}_{A_k}(s)$$

for every $s \in S$, for some fixed $a_1, a_2, \dots, a_m \in [0, \infty]$ and $A_1, A_2, \dots, A_m \in \Sigma$.

Intuitively, when $A_1, A_2, \dots, A_m$ partition S , each set A_k is assigned a value a_k .

Definition 5.3. The integral with respect to μ of the simple function $f : S \rightarrow [0, \infty]$ as written above is defined as

$$\mu(f) = \sum_{k=1}^m a_k \mu(A_k).$$

It is possible to show that the right side of the equation above is equivalent for every choice of sets and constants used to write the simple function f . Therefore, the integral $\mu(f)$ with respect to μ of a simple function f is well-defined. Intuitively, when $A_1, A_2, \dots, A_m$ partition S , the integral with respect to μ accumulates the measure $\mu(A_k)$ of each set A_k multiplied by the value a_k assigned to it.

Proposition 5.1. If $f : S \rightarrow [0, \infty]$ and $g : S \rightarrow [0, \infty]$ are simple functions, then

- $f + g$ is a simple function and $\mu(f + g) = \mu(f) + \mu(g)$,

- if $c \geq 0$, then cf is a simple function and $\mu(cf) = c\mu(f)$,
- if $\mu(f \neq g) = \mu(\{s \in S \mid f(s) \neq g(s)\}) = 0$, then $\mu(f) = \mu(g)$,
- if $f \leq g$ such that $f(s) \leq g(s)$ for every $s \in S$, then $\mu(f) \leq \mu(g)$,
- if $h = \min(f, g)$ such that $h(s) = \min(f(s), g(s))$ for every $s \in S$, then h is a simple function,
- if $h = \max(f, g)$ such that $h(s) = \max(f(s), g(s))$ for every $s \in S$, then h is a simple function.

Definition 5.4. The integral with respect to μ of a Σ -measurable function $f : S \rightarrow [0, \infty]$ is defined as

$$\mu(f) = \sup\{\mu(h) \mid h \text{ is simple and } h \leq f\}.$$

Proposition 5.2. Consider a Σ -measurable function $f : S \rightarrow [0, \infty]$. If $\mu(f) = 0$, then $\mu(\{f > 0\}) = 0$.

Proof. Because the measure μ is non-negative, this is equivalent to showing that if $\mu(\{f > 0\}) > 0$, then $\mu(f) > 0$.

For every $n \in \mathbb{N}^+$, let $A_n = \{f > n^{-1}\} = \{s \in S \mid f(s) > n^{-1}\}$ and note that

$$\{f > 0\} = \{s \in S \mid f(s) > 0\} = \bigcup_{n \in \mathbb{N}^+} \{s \in S \mid f(s) > n^{-1}\} = \bigcup_{n \in \mathbb{N}^+} A_n.$$

For every $s \in S$ and $n \in \mathbb{N}^+$, if $f(s) > n^{-1}$, then $f(s) > (n+1)^{-1}$. Therefore, $A_n \subseteq A_{n+1}$ and $A_n \uparrow \{f > 0\}$. Furthermore, the monotone-convergence property of measure guarantees that $\mu(A_n) \uparrow \mu(\{f > 0\})$.

Suppose that $\mu(\{f > 0\}) > 0$. In that case, there is an $n \in \mathbb{N}^+$ such that

$$\mu(\mathbb{I}_{\{f > n^{-1}\}}) = \mu(\{f > n^{-1}\}) = \mu(A_n) > 0.$$

For such an $n \in \mathbb{N}^+$, consider now the simple function $g = n^{-1}\mathbb{I}_{\{f > n^{-1}\}}$ given by

$$g(s) = n^{-1}\mathbb{I}_{\{f > n^{-1}\}}(s) = \begin{cases} n^{-1} & f(s) > n^{-1}, \\ 0 & f(s) \leq n^{-1}. \end{cases}$$

The fact that $f \geq g$ implies that $\mu(f) \geq \mu(g)$ even if f is not simple. Therefore,

$$\mu(f) \geq \mu(g) = \mu(n^{-1}\mathbb{I}_{\{f > n^{-1}\}}) = n^{-1}\mu(\mathbb{I}_{\{f > n^{-1}\}}) > 0,$$

where the last inequality follows from the fact that $n^{-1} > 0$. □

Definition 5.5. Let $f_n \uparrow f$ denote that a sequence of functions $(f_n : S \rightarrow \mathbb{R} \mid n \in \mathbb{N})$ is such that $f_n(s) \uparrow f(s)$ for every $s \in S$. Similarly, let $f_n \downarrow f$ denote that a sequence of functions $(f_n : S \rightarrow \mathbb{R} \mid n \in \mathbb{N})$ is such that $f_n(s) \downarrow f(s)$ for every $s \in S$.

Theorem 5.1 (Monotone-convergence theorem). If $(f_n : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$ is a sequence of Σ -measurable functions such that $f_n \uparrow f$, then $\mu(f_n) \uparrow \mu(f)$.

Before showing how the integral with respect to μ of a given Σ -measurable function is the limit of a sequence of integrals with respect to μ of simple functions, it is convenient to introduce staircase functions.

Definition 5.6. Let $\alpha_n : [0, \infty] \rightarrow [0, n]$ denote the n -th staircase function given by $\alpha_n(x) = \min(n, \lfloor 2^n x \rfloor / 2^n)$ for every $n \in \mathbb{N}$ and $x \in [0, \infty]$.

Intuitively, the n -th staircase function partitions its domain into a sequence of intervals of length $1/2^n$. The value assigned to the first interval is zero, and the value of each following interval is $1/2^n$ plus the value of the previous interval, with values truncated at n .

Proposition 5.3. Let $h : [0, \infty] \rightarrow [0, \infty]$ denote the identity function given by $h(x) = x$ for every $x \in [0, \infty]$. In that case, $\alpha_n \uparrow h$.

Proof. We will start by showing that $\min(n, \lfloor 2^n x \rfloor / 2^n) = \alpha_n(x) \leq \alpha_{n+1}(x) = \min(n+1, \lfloor 2^{n+1} x \rfloor / 2^{n+1})$, for every $n \in \mathbb{N}$ and $x \in [0, \infty]$. When $x = \infty$, we have $\alpha_n(x) = n \leq n+1 = \alpha_{n+1}(x)$. When $x < \infty$, the fact that $n \leq n+1$ implies that we only need to show that $\lfloor 2^n x \rfloor / 2^n \leq \lfloor 2^{n+1} x \rfloor / 2^{n+1}$. Note that $\lfloor 2^n x \rfloor \leq 2^n x$, which implies $2\lfloor 2^n x \rfloor \leq 2^{n+1} x$. By the monotonicity of the floor function, $\lfloor 2\lfloor 2^n x \rfloor \rfloor \leq \lfloor 2^{n+1} x \rfloor$. Because the floor of an integer is itself an integer, $2\lfloor 2^n x \rfloor \leq \lfloor 2^{n+1} x \rfloor$. Dividing both sides of the previous inequation by 2^{n+1} completes the proof.

In order to show that $\alpha_n \uparrow h$, it remains to show that, for every $x \in [0, \infty]$,

$$\lim_{n \rightarrow \infty} \alpha_n(x) = x.$$

The case where $x = \infty$ is trivial, since $\alpha_n(x) = n$. When $x < \infty$, note that $2^n x \geq \lfloor 2^n x \rfloor$ implies $x \geq \lfloor 2^n x \rfloor / 2^n$, and so $n > x$ implies $n > \lfloor 2^n x \rfloor / 2^n$. Therefore, for every sufficiently large $n \in \mathbb{N}$, we know that $\alpha_n(x) = \lfloor 2^n x \rfloor / 2^n$ when $x < \infty$. It remains to show that $\lim_{n \rightarrow \infty} \lfloor 2^n x \rfloor / 2^n = x$. By noting that $2^n x - 1 \leq \lfloor 2^n x \rfloor \leq 2^n x$ and dividing each term by 2^n ,

$$x - \frac{1}{2^n} = \frac{2^n x - 1}{2^n} \leq \frac{\lfloor 2^n x \rfloor}{2^n} \leq \frac{2^n x}{2^n} = x.$$

Using the squeeze theorem with $n \rightarrow \infty$ completes the proof that $\alpha_n \uparrow h$. □

Proposition 5.4. Consider a Σ -measurable function $f : S \rightarrow [0, \infty]$. For every $n \in \mathbb{N}$, consider $f_n : S \rightarrow [0, n]$ such that

$$f_n(s) = \alpha_n(f(s)) = \sum_{k=1}^m a_k \mathbb{I}_{\{f_n = a_k\}}(s),$$

where $a_1, \dots, a_m \in [0, n]$ are the (distinct) elements of the (finite) image of f_n . In that case, $\mu(f_n) \uparrow \mu(f)$.

Proof. Because f is Σ -measurable and α_n is $\mathcal{B}([0, \infty])$ -measurable, we know that $f_n = \alpha_n \circ f$ is Σ -measurable, which implies that f_n is also simple. For every $s \in S$, we have $f(s) \in [0, \infty]$ and $(\alpha_n \circ f)(s) \uparrow f(s)$. Therefore, $f_n \uparrow f$. From the monotone-convergence theorem, $\mu(f_n) \uparrow \mu(f)$. □

In other words, the integral with respect to μ of a given Σ -measurable function $f : S \rightarrow [0, \infty]$ is the limit of a sequence of integrals with respect to μ of simple functions $(f_n : S \rightarrow [0, n] \mid n \in \mathbb{N})$.

Proposition 5.5. Let $f : S \rightarrow [0, \infty]$ and $g : S \rightarrow [0, \infty]$ be Σ -measurable functions. If $\mu(\{f \neq g\}) = 0$, then $\mu(f) = \mu(g)$.

Proof. Recall that we already have the analogous result for simple functions. For any $n \in \mathbb{N}$, let $f_n = \alpha_n \circ f$ and $g_n = \alpha_n \circ g$, where α_n is the n -th staircase function. Note that

$$\{f_n \neq g_n\} = \{s \in S \mid f_n(s) \neq g_n(s)\} \subseteq \{s \in S \mid f(s) \neq g(s)\} = \{f \neq g\},$$

which implies $\mu(\{f_n \neq g_n\}) \leq \mu(\{f \neq g\}) = 0$. Because f_n and g_n are simple functions such that $\mu(\{f_n \neq g_n\}) = 0$, we know that $\mu(f_n) = \mu(g_n)$. From the monotone-convergence theorem, $\mu(f_n) \uparrow \mu(f)$ and $\mu(g_n) \uparrow \mu(g)$, so

$$\mu(f) = \lim_{n \rightarrow \infty} \mu(f_n) = \lim_{n \rightarrow \infty} \mu(g_n) = \mu(g).$$

Proposition 5.6. Consider a Σ -measurable function $f : S \rightarrow [0, \infty]$ and a sequence of Σ -measurable functions $(f_n : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$ such that $f_n(s) \uparrow f(s)$ for every $s \in S \setminus N$ for some μ -null set $N \subseteq S$. In that case, $\mu(f_n) \uparrow \mu(f)$.

Proof. Consider the Σ -measurable function $f \mathbb{I}_{S \setminus N}$ such that $(f \mathbb{I}_{S \setminus N})(s) = f(s) \mathbb{I}_{S \setminus N}(s)$ for every $s \in S$. Clearly, $\{f \mathbb{I}_{S \setminus N} \neq f\} \subseteq N$. Therefore, $\mu(\{f \mathbb{I}_{S \setminus N} \neq f\}) \leq \mu(N) = 0$ and $\mu(f \mathbb{I}_{S \setminus N}) = \mu(f)$.

Analogously, consider the Σ -measurable function $f_n \mathbb{I}_{S \setminus N}$ such that $(f_n \mathbb{I}_{S \setminus N})(s) = f_n(s) \mathbb{I}_{S \setminus N}(s)$ for every $s \in S$ and $n \in \mathbb{N}$. Clearly, $\{f_n \mathbb{I}_{S \setminus N} \neq f_n\} \subseteq N$. Therefore, $\mu(\{f_n \mathbb{I}_{S \setminus N} \neq f_n\}) \leq \mu(N) = 0$ and $\mu(f_n \mathbb{I}_{S \setminus N}) = \mu(f_n)$.

Note that $(f_n \mathbb{I}_{S \setminus N})(s) \uparrow (f \mathbb{I}_{S \setminus N})(s)$, whether $s \in N$ or $s \in S \setminus N$. Therefore, $\mu(f_n \mathbb{I}_{S \setminus N}) \uparrow \mu(f \mathbb{I}_{S \setminus N})$, which implies $\mu(f_n) \uparrow \mu(f)$. □

Lemma 5.1 (Fatou lemma). For a sequence of Σ -measurable functions $(f_n : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$,

$$\mu \left(\liminf_{n \rightarrow \infty} f_n \right) \leq \liminf_{n \rightarrow \infty} \mu(f_n).$$

Proof. For any $m \in \mathbb{N}$, consider the function $g_m = \inf_{n \geq m} f_n$ such that

$$\liminf_{n \rightarrow \infty} f_n = \lim_{m \rightarrow \infty} \inf_{n \geq m} f_n = \lim_{m \rightarrow \infty} g_m.$$

Because $g_{m+1} \geq g_m$ for every $m \in \mathbb{N}$, we have that $g_m \uparrow \liminf_{n \rightarrow \infty} f_n$. Because $g_m : S \rightarrow [0, \infty]$ is also Σ -measurable for every $m \in \mathbb{N}$, the monotone-convergence theorem guarantees that $\mu(g_m) \uparrow \mu(\liminf_{n \rightarrow \infty} f_n)$.

For any $n \geq m$, note that $g_m \leq f_n$ and $\mu(g_m) \leq \mu(f_n)$, which also implies $\mu(g_m) \leq \inf_{n \geq m} \mu(f_n)$. By taking the limit of both sides of the previous inequation when $m \rightarrow \infty$,

$$\mu \left(\liminf_{n \rightarrow \infty} f_n \right) = \lim_{m \rightarrow \infty} \mu(g_m) \leq \lim_{m \rightarrow \infty} \inf_{n \geq m} \mu(f_n) = \liminf_{n \rightarrow \infty} \mu(f_n).$$

□

Proposition 5.7. For a Σ -measurable function $f : S \rightarrow [0, \infty]$ and a constant $c \geq 0$, we have $\mu(cf) = c\mu(f)$.

Proof. Recall that we already have the analogous result for simple functions. For any $n \in \mathbb{N}$, let $f_n = \alpha_n \circ f$, where α_n is the n -th staircase function. Because $f_n \uparrow f$, we know that $cf_n \uparrow cf$. Because cf_n is Σ -measurable for every $n \in \mathbb{N}$, the monotone-convergence theorem guarantees that $\mu(cf_n) \uparrow \mu(cf)$. Because $\mu(cf_n) = c\mu(f_n)$, we have $c\mu(f_n) \uparrow \mu(cf)$. Because $c\mu(f_n) \uparrow c\mu(f)$, we have $\mu(cf) = c\mu(f)$. □

Proposition 5.8. Consider a Σ -measurable function $f : S \rightarrow [0, \infty]$ and a Σ -measurable function $g : S \rightarrow [0, \infty]$. In that case, $\mu(f + g) = \mu(f) + \mu(g)$.

Proof. Recall that we already have the analogous result for simple functions. For any $n \in \mathbb{N}$, let $f_n = \alpha_n \circ f$ and $g_n = \alpha_n \circ g$, where α_n is the n -th staircase function. Because $f_n \uparrow f$ and $g_n \uparrow g$, we know that $f_n + g_n \uparrow f + g$. Because $f_n + g_n$ is Σ -measurable for every $n \in \mathbb{N}$, the monotone-convergence theorem guarantees that $\mu(f_n + g_n) \uparrow \mu(f + g)$. Because $\mu(f_n + g_n) \uparrow \mu(f) + \mu(g)$, we have $\mu(f + g) = \mu(f) + \mu(g)$. □

Lemma 5.2 (Reverse Fatou lemma). Consider a sequence of Σ -measurable functions $(f_n : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$ such that $f_n \leq g$ for every $n \in \mathbb{N}$ and some Σ -measurable function $g : S \rightarrow [0, \infty]$ such that $\mu(g) < \infty$. In that case,

$$\mu \left(\limsup_{n \rightarrow \infty} f_n \right) \geq \limsup_{n \rightarrow \infty} \mu(f_n).$$

Proof. For every $n \in \mathbb{N}$, consider the function $h_n = g - f_n$. Because g and f_n are Σ -measurable and $f_n \leq g$, we know that $h_n : S \rightarrow [0, \infty]$ is Σ -measurable. From the Fatou lemma,

$$\mu \left(\liminf_{n \rightarrow \infty} (g - f_n) \right) \leq \liminf_{n \rightarrow \infty} \mu(g - f_n).$$

By using the fact that $\mu(g) = \mu(g - f_n) + \mu(f_n)$ and moving g and $\mu(g)$ outside the corresponding limits,

$$\mu \left(g + \liminf_{n \rightarrow \infty} -f_n \right) \leq \mu(g) + \liminf_{n \rightarrow \infty} -\mu(f_n).$$

By using the relationship between limit inferior and limit superior,

$$\mu \left(g - \limsup_{n \rightarrow \infty} f_n \right) \leq \mu(g) - \limsup_{n \rightarrow \infty} \mu(f_n).$$

By using the fact that $\mu(g) = \mu(g - \limsup_{n \rightarrow \infty} f_n) + \mu(\limsup_{n \rightarrow \infty} f_n)$,

$$\mu(g) - \mu \left(\limsup_{n \rightarrow \infty} f_n \right) \leq \mu(g) - \limsup_{n \rightarrow \infty} \mu(f_n).$$

The proof is completed by reorganizing terms in the inequation above. □

Definition 5.7. For a Σ -measurable function $f : S \rightarrow \mathbb{R}$, the Σ -measurable function $f^+ : S \rightarrow [0, \infty]$ is given by

$$f^+(s) = \max(f(s), 0) = \begin{cases} f(s), & \text{if } f(s) > 0, \\ 0, & \text{if } f(s) \leq 0. \end{cases}$$

Definition 5.8. For a Σ -measurable function $f : S \rightarrow \mathbb{R}$, the Σ -measurable function $f^- : S \rightarrow [0, \infty]$ is given by

$$f^-(s) = \max(-f(s), 0) = \begin{cases} 0, & \text{if } f(s) > 0, \\ -f(s), & \text{if } f(s) \leq 0. \end{cases}$$

Proposition 5.9. For a Σ -measurable function $f : S \rightarrow \mathbb{R}$, whether $f(s) > 0$ or $f(s) \leq 0$,

$$f(s) = f^+(s) - f^-(s).$$

Furthermore, whether $f(s) > 0$ or $f(s) \leq 0$,

$$|f(s)| = f^+(s) + f^-(s).$$

In other words, $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

Definition 5.9. A function $f : S \rightarrow \mathbb{R}$ is μ -integrable if it is Σ -measurable and $\mu(|f|) = \mu(f^+ + f^-) = \mu(f^+) + \mu(f^-) < \infty$.

Definition 5.10. The set of all μ -integrable functions in the measure space (S, Σ, μ) is denoted by $\mathcal{L}^1(S, \Sigma, \mu)$. The set of all non-negative μ -integrable functions in the measure space (S, Σ, μ) is denoted by $\mathcal{L}^1(S, \Sigma, \mu)^+$.

Definition 5.11. The integral $\mu(f)$ with respect to μ of a μ -integrable function $f : S \rightarrow \mathbb{R}$ is defined as

$$\mu(f) = \mu(f^+) - \mu(f^-).$$

Alternatively, the integral $\mu(f)$ with respect to μ of a μ -integrable function $f : S \rightarrow \mathbb{R}$ is denoted by

$$\int_S f d\mu = \int_S f(s) \mu(ds) = \mu(f).$$

Proposition 5.10. If a function $f : S \rightarrow \mathbb{R}$ is μ -integrable, then $\mu(f^+) < \infty$ and $\mu(f^-) < \infty$. By the triangle inequality,

$$|\mu(f)| = |\mu(f^+) + (-\mu(f^-))| \leq |\mu(f^+)| + |-\mu(f^-)| = \mu(f^+) + \mu(f^-) = \mu(|f|).$$

Proposition 5.11. Consider a μ -integrable function $f : S \rightarrow \mathbb{R}$. Because $-f : S \rightarrow \mathbb{R}$ is Σ -measurable and $\mu(|-f|) = \mu(|f|) < \infty$, we know that $-f$ is μ -integrable. Furthermore, $\mu(-f) = -\mu(f)$.

Proof. For every $s \in S$, $(-f)^+(s) = \max(-f(s), 0) = f^-(s)$ and $(-f)^-(s) = \max(f(s), 0) = f^+(s)$. Therefore,

$$\mu(-f) = \mu((-f)^+) - \mu((-f)^-) = -(\mu((-f)^-) - \mu((-f)^+)) = -(\mu(f^+) - \mu(f^-)) = -\mu(f).$$

□

Proposition 5.12. Consider a μ -integrable function $f : S \rightarrow \mathbb{R}$ and a constant $c \in \mathbb{R}$. Because $cf : S \rightarrow \mathbb{R}$ is Σ -measurable and $\mu(|cf|) = \mu(|c||f|) = |c|\mu(|f|) < \infty$, we know that cf is μ -integrable. Furthermore, $\mu(cf) = c\mu(f)$.

Proof. Because $f = f^+ - f^-$, we know that $cf = cf^+ - cf^-$. Furthermore, $(cf) = (cf)^+ - (cf)^-$. Therefore,

$$(cf)^+ - (cf)^- = cf^+ - cf^-.$$

By rearranging negative terms,

$$(cf)^+ + cf^- = (cf)^- + cf^+.$$

We will now consider the case where $c \geq 0$. By the linearity of the integral of non-negative functions,

$$\mu((cf)^+) + \mu(cf^-) = \mu((cf)^-) + \mu(cf^+).$$

By rearranging terms,

$$\mu((cf)^+) - \mu((cf)^-) = \mu(cf^+) - \mu(cf^-).$$

Because cf is μ -integrable and by the linearity of the integral of non-negative functions,

$$\mu(cf) = c\mu(f^+) - c\mu(f^-) = c(\mu(f^+) - \mu(f^-)) = c\mu(f).$$

When $c < 0$, note that $\mu(cf) = \mu(-|c|f) = |c|\mu(-f) = -|c|\mu(f) = c\mu(f)$. □

Proposition 5.13. Consider a μ -integrable function $f : S \rightarrow \mathbb{R}$ and a μ -integrable function $g : S \rightarrow \mathbb{R}$. Because $f + g : S \rightarrow \mathbb{R}$ is Σ -measurable and $|f + g| \leq |f| + |g|$ implies $\mu(|f + g|) \leq \mu(|f|) + \mu(|g|) < \infty$, we know that $f + g$ is μ -integrable. Furthermore, $\mu(f + g) = \mu(f) + \mu(g)$.

Proof. We know that $f + g = (f^+ - f^-) + (g^+ - g^-)$. Furthermore, $(f + g) = (f + g)^+ - (f + g)^-$. Therefore,

$$(f + g)^+ - (f + g)^- = (f^+ - f^-) + (g^+ - g^-).$$

By rearranging negative terms,

$$(f + g)^+ + f^- + g^- = (f + g)^- + f^+ + g^+.$$

By the linearity of the integral of non-negative functions,

$$\mu((f + g)^+) + \mu(f^-) + \mu(g^-) = \mu((f + g)^-) + \mu(f^+) + \mu(g^+).$$

By rearranging terms,

$$\mu((f + g)^+) - \mu((f + g)^-) = (\mu(f^+) - \mu(f^-)) + (\mu(g^+) - \mu(g^-))$$

Because $f + g$ is μ -integrable,

$$\mu(f + g) = \mu(f) + \mu(g).$$
□

Proposition 5.14. Let $f : S \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$ be μ -integrable functions. If $\mu(\{f \neq g\}) = 0$, then $\mu(f) = \mu(g)$.

Proof. Recall that we already have the analogous result for non-negative Σ -measurable functions. First, note that if $f^+(s) \neq g^+(s)$ or $f^-(s) \neq g^-(s)$ for some $s \in S$, then $f(s) \neq g(s)$. Therefore,

$$\{s \in S \mid f^+(s) \neq g^+(s)\} \cup \{s \in S \mid f^-(s) \neq g^-(s)\} \subseteq \{s \in S \mid f(s) \neq g(s)\},$$

so that $\mu(\{f^+ \neq g^+\}) \leq \mu(\{f \neq g\})$ and $\mu(\{f^- \neq g^-\}) \leq \mu(\{f \neq g\})$. Because $\mu(\{f \neq g\}) = 0$, we know that $\mu(\{f^+ \neq g^+\}) = 0$ and $\mu(\{f^- \neq g^-\}) = 0$. Because f^+, f^-, g^+ , and g^- are non-negative Σ -measurable functions, we know that $\mu(f^+) = \mu(g^+)$ and $\mu(f^-) = \mu(g^-)$. Therefore,

$$\mu(f) = \mu(f^+) - \mu(f^-) = \mu(g^+) - \mu(g^-) = \mu(g).$$
□

Definition 5.12. The integral with respect to μ of a μ -integrable function $f : S \rightarrow \mathbb{R}$ over the set $A \in \Sigma$ is defined as

$$\mu(f; A) = \mu(f\mathbb{I}_A).$$

Because $f\mathbb{I}_A$ is Σ -measurable and $|f\mathbb{I}_A| \leq |f|$ implies $\mu(|f\mathbb{I}_A|) \leq \mu(|f|) < \infty$, we know that $f\mathbb{I}_A$ is μ -integrable. Alternatively, the integral $\mu(f; A)$ with respect to μ of f over the set $A \in \Sigma$ is denoted by

$$\int_A f d\mu = \int_A f(s) \mu(ds) = \mu(f; A).$$

Proposition 5.15. Consider a sequence of real numbers $(x_n \mid n \in \mathbb{N})$ and the measure space $(\mathbb{N}, \mathcal{P}(\mathbb{N}), \mu)$, where $\mu(\{n\}) = 1$ for every $n \in \mathbb{N}$. Furthermore, consider a function $f : \mathbb{N} \rightarrow \mathbb{R}$ such that $f(n) = x_n$. In that case, f is μ -integrable if and only if $\sum_n |x_n| < \infty$. Also, if f is μ -integrable, then $\mu(f) = \sum_n x_n$.

Proof. Suppose that $f(n) \geq 0$ for every $n \in \mathbb{N}$. For every $k \in \mathbb{N}$, consider the function $f_k : \mathbb{N} \rightarrow [0, \infty]$ given by

$$f_k(n) = \sum_{i=0}^k f(i) \mathbb{I}_{\{i\}}(n) = \begin{cases} f(n), & \text{if } n \leq k, \\ 0, & \text{if } n > k. \end{cases}$$

Clearly, if $k \rightarrow \infty$, then $f_k \rightarrow f$. Because f_k is a simple function,

$$\mu(f_k) = \sum_{i=0}^k f(i) \mu(\{i\}) = \sum_{i=0}^k f(i) = \sum_{i=0}^k x_i.$$

Because $f_k \leq f_{k+1}$, we have $f_k \uparrow f$. By the monotone-convergence theorem, $\mu(f_k) \uparrow \mu(f)$. Therefore,

$$\mu(f) = \lim_{k \rightarrow \infty} \sum_{i=0}^k x_i = \sum_n x_n.$$

Now suppose $f(n) \in \mathbb{R}$ for every $n \in \mathbb{N}$. Based on our previous result,

$$\mu(|f|) = \mu(f^+) + \mu(f^-) = \sum_n (\max(x_n, 0) + \max(-x_n, 0)) = \sum_n |x_n|.$$

By definition, f is integrable if and only if $\mu(|f|) = \sum_n |x_n| < \infty$, in which case

$$\mu(f) = \mu(f^+) - \mu(f^-) = \sum_n (\max(x_n, 0) - \max(-x_n, 0)) = \sum_n x_n.$$

□

Theorem 5.2 (Dominated convergence theorem). Consider a sequence of Σ -measurable functions $(f_n : S \rightarrow \mathbb{R} \mid n \in \mathbb{N})$ and a Σ -measurable function $f : S \rightarrow \mathbb{R}$ such that $\lim_{n \rightarrow \infty} f_n = f$. Furthermore, suppose there is a μ -integrable non-negative function $g \in \mathcal{L}^1(S, \Sigma, \mu)^+$ that dominates this sequence of functions such that $|f_n| \leq g$ for every $n \in \mathbb{N}$. In that case, f is μ -integrable and $\lim_{n \rightarrow \infty} \mu(f_n) = \mu(f)$.

Proof. Because g is μ -integrable and non-negative, $\mu(g) = \mu(|g|) < \infty$. Because $|f_n| \leq g$ for every $n \in \mathbb{N}$, we know that $\mu(|f_n|) \leq \mu(g) < \infty$, which implies that f_n is μ -integrable. Because the function $|\cdot|$ is continuous, we know that $\lim_{n \rightarrow \infty} |f_n| = |f|$, which implies $|f| \leq g$. Because $\mu(|f|) \leq \mu(g) < \infty$, we know that f is μ -integrable.

Because $|f_n| \leq g$ and $|f| \leq g$, we know that $|f_n| + |f| \leq 2g$. By the triangle inequality,

$$|f_n - f| = |f_n + (-f)| \leq |f_n| + |f| \leq 2g.$$

Because $|f_n - f| : S \rightarrow [0, \infty]$ is a Σ -measurable function and $|f_n - f| \leq 2g$ for every $n \in \mathbb{N}$, where $2g : S \rightarrow [0, \infty)$ is a Σ -measurable function such that $\mu(2g) = 2\mu(g) < \infty$, the reverse Fatou lemma states that

$$\mu \left(\limsup_{n \rightarrow \infty} |f_n - f| \right) \geq \limsup_{n \rightarrow \infty} \mu(|f_n - f|).$$

Since the function $|\cdot|$ is continuous, we know that $\lim_{n \rightarrow \infty} |f_n - f| = 0$, where 0 is the zero function. Therefore,

$$\limsup_{n \rightarrow \infty} |f_n - f| = \liminf_{n \rightarrow \infty} |f_n - f| = \lim_{n \rightarrow \infty} |f_n - f| = 0.$$

By taking the integral with respect to μ of these non-negative functions,

$$\mu \left(\limsup_{n \rightarrow \infty} |f_n - f| \right) = \mu \left(\liminf_{n \rightarrow \infty} |f_n - f| \right) = \mu \left(\lim_{n \rightarrow \infty} |f_n - f| \right) = \mu(0) = 0.$$

Because $f_n - f$ is μ -integrable for every $n \in \mathbb{N}$ and $|\mu(f_n - f)| \leq \mu(|f_n - f|)$,

$$0 \geq \limsup_{n \rightarrow \infty} \mu(|f_n - f|) \geq \limsup_{n \rightarrow \infty} |\mu(f_n - f)| \geq \liminf_{n \rightarrow \infty} |\mu(f_n - f)| \geq 0.$$

Because the limit superior and limit inferior in the inequation above must be equal to zero, we know that $\lim_{n \rightarrow \infty} |\mu(f_n - f)| = 0$, which implies $\lim_{n \rightarrow \infty} \mu(f_n - f) = 0$. By the linearity of the integral with respect to μ ,

$$\lim_{n \rightarrow \infty} \mu(f_n) = \mu(f).$$

□

Lemma 5.3 (Scheffé's lemma for non-negative functions). Consider a sequence of μ -integrable non-negative functions $(f_n : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$ and a μ -integrable non-negative function $f : S \rightarrow [0, \infty]$ such that $\lim_{n \rightarrow \infty} f_n = f$ (almost everywhere). In that case,

$$\lim_{n \rightarrow \infty} \mu(|f_n - f|) = 0 \text{ if and only if } \lim_{n \rightarrow \infty} \mu(f_n) = \mu(f).$$

Proof. First, suppose $\lim_{n \rightarrow \infty} \mu(|f_n - f|) = 0$. Since $0 \leq |\mu(f_n - f)| \leq \mu(|f_n - f|)$, the squeeze theorem implies that $\lim_{n \rightarrow \infty} |\mu(f_n - f)| = 0$, which also implies that $\lim_{n \rightarrow \infty} \mu(f_n - f) = 0$. By the linearity of the integral with respect to μ , we conclude that $\lim_{n \rightarrow \infty} \mu(f_n) = \mu(f)$.

Now suppose $\lim_{n \rightarrow \infty} \mu(f_n) = \mu(f)$ and consider the function $(f_n - f)^- : S \rightarrow [0, \infty]$ given by

$$(f_n - f)^-(s) = \max(-(f_n - f)(s), 0) = \max((f - f_n)(s), 0) = (f - f_n)^+(s) = \begin{cases} f(s) - f_n(s), & \text{if } f(s) > f_n(s), \\ 0, & \text{if } f(s) \leq f_n(s). \end{cases}$$

Note that $(f_n - f)^- \leq f$ for every $n \in \mathbb{N}$. Because $\lim_{n \rightarrow \infty} f_n = f$, we know that for every $s \in S$ and $\epsilon > 0$ there is an $N \in \mathbb{N}$ such that $n > N$ guarantees that $|f(s) - f_n(s)| < \epsilon$. Note that, for every $n > N$, if $f(s) > f_n(s)$, then $|(f_n - f)^-(s)| = |f(s) - f_n(s)| < \epsilon$. If $f(s) \leq f_n(s)$, then $|(f_n - f)^-(s)| = 0 < \epsilon$. Therefore, for every $s \in S$ and $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that $n > N$ guarantees that $|(f_n - f)^-(s)| < \epsilon$. By definition, $\lim_{n \rightarrow \infty} (f_n - f)^- = 0$, where 0 denotes the zero function.

Consider the sequence of Σ -measurable functions $((f_n - f)^- : S \rightarrow \mathbb{R} \mid n \in \mathbb{N})$ and the Σ -measurable function $0 : S \rightarrow \mathbb{R}$ such that $\lim_{n \rightarrow \infty} (f_n - f)^- = 0$. Furthermore, consider the μ -integrable non-negative function $f \in \mathcal{L}^1(S, \Sigma, \mu)^+$ such that $|(f_n - f)^-| = (f_n - f)^- \leq f$ for every $n \in \mathbb{N}$. By the dominated convergence theorem, we know that $\lim_{n \rightarrow \infty} \mu((f_n - f)^-) = \mu(0) = 0$.

For every $n \in \mathbb{N}$, recall that $(f_n - f)^+ = (f_n - f) + (f_n - f)^-$. By the linearity of the integral with respect to μ ,

$$\lim_{n \rightarrow \infty} \mu((f_n - f)^+) = \lim_{n \rightarrow \infty} (\mu(f_n) - \mu(f) + \mu((f_n - f)^-)) = \mu(f) - \mu(f) + \lim_{n \rightarrow \infty} \mu((f_n - f)^-) = 0.$$

For every $n \in \mathbb{N}$, recall that $|f_n - f| = (f_n - f)^+ + (f_n - f)^-$. By the linearity of the integral with respect to μ ,

$$\lim_{n \rightarrow \infty} \mu(|f_n - f|) = \lim_{n \rightarrow \infty} (\mu((f_n - f)^+) + \mu((f_n - f)^-)) = 0.$$

□

Lemma 5.4 (Scheffé's lemma). Consider a sequence of μ -integrable functions $(f_n : S \rightarrow \mathbb{R} \mid n \in \mathbb{N})$ and a μ -integrable function $f : S \rightarrow \mathbb{R}$ such that $\lim_{n \rightarrow \infty} f_n = f$ (almost everywhere). In that case,

$$\lim_{n \rightarrow \infty} \mu(|f_n - f|) = 0 \text{ if and only if } \lim_{n \rightarrow \infty} \mu(|f_n|) = \mu(|f|).$$

Proof. First, suppose $\lim_{n \rightarrow \infty} \mu(|f_n - f|) = 0$. By the triangle inequality,

$$\begin{aligned} |f_n| &= |(f_n - f) + f| \leq |f_n - f| + |f|, \\ |f| &= |(f - f_n) + f_n| \leq |f_n - f| + |f_n|. \end{aligned}$$

Because the integral with respect to μ is non-decreasing and linear,

$$\begin{aligned} \mu(|f_n - f|) &\geq \mu(|f_n|) - \mu(|f|), \\ \mu(|f_n - f|) &\geq \mu(|f|) - \mu(|f_n|). \end{aligned}$$

Because $\mu(|f_n - f|) \geq a$ and $\mu(|f_n - f|) \geq -a$ for $a = \mu(|f_n|) - \mu(|f|)$,

$$\mu(|f_n - f|) \geq |\mu(|f_n|) - \mu(|f|)| \geq 0.$$

By the squeeze theorem, $\lim_{n \rightarrow \infty} |\mu(|f_n|) - \mu(|f|)| = 0$, which implies $\lim_{n \rightarrow \infty} \mu(|f_n|) - \mu(|f|) = 0$. By the linearity of the integral with respect to μ , we conclude that $\lim_{n \rightarrow \infty} \mu(|f_n|) = \mu(|f|)$.

Now suppose $\lim_{n \rightarrow \infty} \mu(|f_n|) = \mu(|f|)$. Because the function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = \max(x, 0)$ is continuous,

$$\begin{aligned}\lim_{n \rightarrow \infty} f_n^+(s) &= \lim_{n \rightarrow \infty} \max(f_n(s), 0) = \max(f(s), 0) = f^+(s), \\ \lim_{n \rightarrow \infty} f_n^-(s) &= \lim_{n \rightarrow \infty} \max(-f_n(s), 0) = \max(-f(s), 0) = f^-(s).\end{aligned}$$

Because $(f_n^+ : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$ and $(f_n^- : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$ are sequences of Σ -measurable functions, the Fatou lemma guarantees that

$$\begin{aligned}\mu(f^+) &= \mu\left(\lim_{n \rightarrow \infty} f_n^+\right) = \mu\left(\liminf_{n \rightarrow \infty} f_n^+\right) \leq \liminf_{n \rightarrow \infty} \mu(f_n^+), \\ \mu(f^-) &= \mu\left(\lim_{n \rightarrow \infty} f_n^-\right) = \mu\left(\liminf_{n \rightarrow \infty} f_n^-\right) \leq \liminf_{n \rightarrow \infty} \mu(f_n^-).\end{aligned}$$

Consider the integrals $\mu(f_n^+)$ and $\mu(f_n^-)$ written as

$$\begin{aligned}\mu(f_n^+) &= \mu(f_n^+) + \mu(f_n^-) - \mu(f_n^-), \\ \mu(f_n^-) &= \mu(f_n^-) + \mu(f_n^+) - \mu(f_n^+).\end{aligned}$$

By taking the limit superior of both sides,

$$\begin{aligned}\limsup_{n \rightarrow \infty} \mu(f_n^+) &= \limsup_{n \rightarrow \infty} (\mu(f_n^+) + \mu(f_n^-) - \mu(f_n^-)), \\ \limsup_{n \rightarrow \infty} \mu(f_n^-) &= \limsup_{n \rightarrow \infty} (\mu(f_n^-) + \mu(f_n^+) - \mu(f_n^+)).\end{aligned}$$

By the subadditivity of the limit superior,

$$\begin{aligned}\limsup_{n \rightarrow \infty} \mu(f_n^+) &\leq \limsup_{n \rightarrow \infty} (\mu(f_n^+) + \mu(f_n^-)) + \limsup_{n \rightarrow \infty} -\mu(f_n^-) \\ \limsup_{n \rightarrow \infty} \mu(f_n^-) &\leq \limsup_{n \rightarrow \infty} (\mu(f_n^-) + \mu(f_n^+)) + \limsup_{n \rightarrow \infty} -\mu(f_n^+).\end{aligned}$$

From our assumption that $\lim_{n \rightarrow \infty} \mu(|f_n|) = \mu(|f|)$,

$$\limsup_{n \rightarrow \infty} (\mu(f_n^+) + \mu(f_n^-)) = \limsup_{n \rightarrow \infty} (\mu(f_n^-) + \mu(f_n^+)) = \limsup_{n \rightarrow \infty} \mu(|f_n|) = \lim_{n \rightarrow \infty} \mu(|f_n|) = \mu(|f|).$$

Therefore, by the relationship between the limit inferior and the limit superior,

$$\begin{aligned}\limsup_{n \rightarrow \infty} \mu(f_n^+) &\leq \mu(|f|) - \liminf_{n \rightarrow \infty} \mu(f_n^-), \\ \limsup_{n \rightarrow \infty} \mu(f_n^-) &\leq \mu(|f|) - \liminf_{n \rightarrow \infty} \mu(f_n^+).\end{aligned}$$

By non-decreasing the right sides of the previous inequations using our previous result,

$$\begin{aligned}\limsup_{n \rightarrow \infty} \mu(f_n^+) &\leq \mu(|f|) - \mu(f^-) = \mu(f^+) + \mu(f^-) - \mu(f^-) = \mu(f^+), \\ \limsup_{n \rightarrow \infty} \mu(f_n^-) &\leq \mu(|f|) - \mu(f^+) = \mu(f^+) + \mu(f^-) - \mu(f^+) = \mu(f^-).\end{aligned}$$

By noting that the limit superior is at least as large as the limit inferior and combining the previous results,

$$\begin{aligned}\mu(f^+) &\leq \liminf_{n \rightarrow \infty} \mu(f_n^+) \leq \limsup_{n \rightarrow \infty} \mu(f_n^+) \leq \mu(f^+), \\ \mu(f^-) &\leq \liminf_{n \rightarrow \infty} \mu(f_n^-) \leq \limsup_{n \rightarrow \infty} \mu(f_n^-) \leq \mu(f^-).\end{aligned}$$

Because the previous inequations imply that the limits must match,

$$\begin{aligned}\lim_{n \rightarrow \infty} \mu(f_n^+) &= \mu(f^+), \\ \lim_{n \rightarrow \infty} \mu(f_n^-) &= \mu(f^-).\end{aligned}$$

Because $(f_n^+ : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$ and $(f_n^- : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$ are sequences of μ -integrable non-negative functions and $f^+ : S \rightarrow [0, \infty]$ and $f^- : S \rightarrow [0, \infty]$ are μ -integrable non-negative functions such that $\lim_{n \rightarrow \infty} f_n^+ = f^+$ and $\lim_{n \rightarrow \infty} f_n^- = f^-$, Scheffé's lemma for non-negative functions guarantees that

$$\begin{aligned}\lim_{n \rightarrow \infty} \mu(|f_n^+ - f^+|) &= 0, \\ \lim_{n \rightarrow \infty} \mu(|f_n^- - f^-|) &= 0.\end{aligned}$$

By the triangle inequality,

$$|f_n - f| = |(f_n^+ - f_n^-) - (f^+ - f^-)| = |(f_n^+ - f^+) + (f^- - f_n^-)| \leq |f_n^+ - f^+| + |f_n^- - f^-|.$$

Because the integral with respect to μ is non-negative for non-negative functions, non-decreasing, and linear,

$$0 \leq \mu(|f_n - f|) \leq \mu(|f_n^+ - f^+|) + \mu(|f_n^- - f^-|).$$

By the squeeze theorem, and as we wanted to show,

$$\lim_{n \rightarrow \infty} \mu(|f_n - f|) = 0.$$

□

Proposition 5.16. Consider the measure space (S, Σ, μ) . For a set $A \in \Sigma$, consider the triple (A, Σ_A, μ_A) such that $\Sigma_A = \{B \in \Sigma \mid B \subseteq A\}$ and $\mu_A(B) = \mu(B)$ for every $B \in \Sigma_A$. In that case, (A, Σ_A, μ_A) is a measure space restricted to A .

Proof. First, we will show that Σ_A is a σ -algebra on A . Because $A \in \Sigma$ and $A \subseteq A$, we have $A \in \Sigma_A$. If $B \in \Sigma_A$, then $B \in \Sigma$ and $A \cap B^c \in \Sigma$. Because $A \cap B^c \subseteq A$, we have $A \setminus B \in \Sigma_A$. For any sequence $(B_n \in \Sigma_A \mid n \in \mathbb{N})$, the fact that $B_n \in \Sigma$ guarantees that $\cup_n B_n \in \Sigma$. Because $B_n \subseteq A$ for every $n \in \mathbb{N}$, we know that $\cup_n B_n \subseteq A$, which implies $\cup_n B_n \in \Sigma_A$.

Second, we will show that the non-negative function $\mu_A : \Sigma_A \rightarrow [0, \infty]$ is a measure on the measurable space (A, Σ_A) . Because $\emptyset \in \Sigma$ and $\emptyset \in \Sigma_A$, we know that $\mu_A(\emptyset) = \mu(\emptyset) = 0$. For any sequence $(B_n \in \Sigma_A \mid n \in \mathbb{N})$ such that $B_n \cap B_m = \emptyset$ for every $n \neq m$, we know that $\cup_n B_n \in \Sigma$ and $\cup_n B_n \in \Sigma_A$ and

$$\mu_A\left(\bigcup_n B_n\right) = \mu\left(\bigcup_n B_n\right) = \sum_n \mu(B_n) = \sum_n \mu_A(B_n).$$

□

Proposition 5.17. Consider the measure space (S, Σ, μ) and a Σ -measurable function $f : S \rightarrow \mathbb{R}$. Consider also the measure space (A, Σ_A, μ_A) restricted to $A \in \Sigma$ and the function $f|_A : A \rightarrow \mathbb{R}$ restricted to A given by $f|_A(a) = f(a)$ for every $a \in A$. The function $f|_A$ is Σ_A -measurable because, for every $B \in \mathcal{B}(\mathbb{R})$,

$$(f|_A)^{-1}(B) = \{a \in A \mid f(a) \in B\} = \{s \in S \mid f(s) \in B\} \cap A = f^{-1}(B) \cap A.$$

Proposition 5.18. Consider the measure space (S, Σ, μ) , a Σ -measurable function $f : S \rightarrow \mathbb{R}$, and a set $A \in \Sigma$. Then $f|_A$ is μ_A -integrable if and only if $f\mathbb{I}_A$ is μ -integrable, in which case $\mu_A(f|_A) = \mu(f\mathbb{I}_A) = \mu(f; A)$.

Proof. First, suppose $f = \mathbb{I}_B$ for some set $B \in \Sigma$. Clearly, $\mu(f\mathbb{I}_A) = \mu(\mathbb{I}_B\mathbb{I}_A) = \mu(\mathbb{I}_{B \cap A}) = \mu(B \cap A)$ and $\mu_A(f|_A) = \mu_A(\mathbb{I}_B|_A) = \mu_A(\mathbb{I}_{B \cap A}) = \mu_A(B \cap A)$. Because $B \cap A \subseteq A$, we have $\mu_A(B \cap A) = \mu(B \cap A)$, which implies $\mu_A(f|_A) = \mu(f\mathbb{I}_A)$. Because $\mu(|f\mathbb{I}_A|) = \mu(f\mathbb{I}_A) = \mu_A(f|_A) = \mu_A(|f|_A)$, we know that $f|_A$ is μ_A -integrable if and only if $f\mathbb{I}_A$ is μ -integrable.

Next, suppose f is a simple function that can be written as $f = \sum_{k=1}^m a_k \mathbb{I}_{A_k}$ for some fixed $a_1, a_2, \dots, a_m \in [0, \infty]$ and $A_1, A_2, \dots, A_m \in \Sigma$. In that case, the integral with respect to μ of the function $f\mathbb{I}_A$ is given by

$$\mu(f\mathbb{I}_A) = \mu\left(\sum_{k=1}^m a_k \mathbb{I}_{A_k} \mathbb{I}_A\right) = \mu\left(\sum_{k=1}^m a_k \mathbb{I}_{A_k \cap A}\right) = \sum_{k=1}^m a_k \mu(A_k \cap A).$$

Furthermore, the integral of the function $f|_A$ with respect to μ_A is given by

$$\mu_A(f|_A) = \mu_A\left(\left(\sum_{k=1}^m a_k \mathbb{I}_{A_k}\right)\Big|_A\right) = \mu_A\left(\sum_{k=1}^m a_k \mathbb{I}_{A_k \cap A}\right) = \sum_{k=1}^m a_k \mu_A(\mathbb{I}_{A_k \cap A}) = \sum_{k=1}^m a_k \mu_A(A_k \cap A).$$

Because $A_k \cap A \subseteq A$ for every $k \leq m$, we have $\mu_A(A_k \cap A) = \mu(A_k \cap A)$, which implies $\mu_A(f|_A) = \mu(f\mathbb{I}_A)$. Because $\mu(|f\mathbb{I}_A|) = \mu(f\mathbb{I}_A) = \mu_A(f|_A) = \mu_A(|f|_A)$, we know that $f|_A$ is μ_A -integrable if and only if $f\mathbb{I}_A$ is μ -integrable.

Next, suppose f is non-negative. For any $n \in \mathbb{N}$, let $f_n = \alpha_n \circ f$, where α_n is the n -th staircase function. Because $(f_n\mathbb{I}_A \mid n \in \mathbb{N})$ is a sequence of Σ -measurable functions such that $f_n\mathbb{I}_A \uparrow f\mathbb{I}_A$, we know that $\mu(f_n\mathbb{I}_A) \uparrow \mu(f\mathbb{I}_A)$. Because $(f_n|_A \mid n \in \mathbb{N})$ is a sequence of Σ_A -measurable functions such that $f_n|_A \uparrow f|_A$, we know that $\mu_A(f_n|_A) \uparrow \mu_A(f|_A)$. For every $n \in \mathbb{N}$, the fact that f_n is a simple function implies $\mu(f_n\mathbb{I}_A) = \mu_A(f_n|_A)$. Therefore, $\mu_A(f_n|_A) \uparrow \mu(f\mathbb{I}_A)$, and $\mu(f_n\mathbb{I}_A) \uparrow \mu_A(f|_A)$, and $\mu(f\mathbb{I}_A) = \mu_A(f|_A)$. Because $\mu(|f\mathbb{I}_A|) = \mu(f\mathbb{I}_A) = \mu_A(f|_A) = \mu_A(|f|_A)$, we know that $f|_A$ is μ_A -integrable if and only if $f\mathbb{I}_A$ is μ -integrable.

Finally, suppose $f : S \rightarrow \mathbb{R}$. By definition,

$$\mu(|f\mathbb{I}_A|) = \mu((f\mathbb{I}_A)^+) + \mu((f\mathbb{I}_A)^-) = \mu(f^+\mathbb{I}_A) + \mu(f^-\mathbb{I}_A) = \mu_A(f^+|_A) + \mu_A(f^-|_A) = \mu_A((f|_A)^+) + \mu_A((f|_A)^-) = \mu_A(|f|_A).$$

Therefore, $f|_A$ is μ_A -integrable if and only if $f\mathbb{I}_A$ is μ -integrable. In that case,

$$\mu(f\mathbb{I}_A) = \mu((f\mathbb{I}_A)^+) - \mu((f\mathbb{I}_A)^-) = \mu(f^+\mathbb{I}_A) - \mu(f^-\mathbb{I}_A) = \mu_A(f^+|_A) - \mu_A(f^-|_A) = \mu_A((f|_A)^+) - \mu_A((f|_A)^-) = \mu_A(f|_A).$$

□

Proposition 5.19. If $(f_n : S \rightarrow [0, \infty] \mid n \in \mathbb{N})$ is a sequence of non-negative Σ -measurable functions,

$$\mu\left(\sum_k f_k\right) = \sum_k \mu(f_k).$$

Proof. For any $n \in \mathbb{N}$, let $g_n = \sum_{k=0}^n f_k$, such that $\mu(g_n) = \sum_{k=0}^n \mu(f_k)$. Clearly, $g_n \geq 0$, $g_n \leq g_{n+1}$, and $\lim_{n \rightarrow \infty} g_n = \sum_k f_k$. Therefore, $g_n \uparrow \sum_k f_k$. By the monotone-convergence theorem, $\mu(g_n) \uparrow \mu(\sum_k f_k)$. □

Definition 5.13. Consider a Σ -measurable function $f : S \rightarrow [0, \infty]$. The function $(f\mu) : \Sigma \rightarrow [0, \infty]$ is given by

$$(f\mu)(A) = \mu(f; A) = \mu(f\mathbb{I}_A) = \int_A f \, d\mu.$$

Proposition 5.20. If $f : S \rightarrow [0, \infty]$ is a Σ -measurable function, then the function $(f\mu)$ is a measure on (S, Σ) .

Proof. Clearly, $(f\mu)(\emptyset) = \mu(f\mathbb{I}_\emptyset) = \mu(\emptyset) = 0$. Consider a sequence $(B_n \in \Sigma \mid n \in \mathbb{N})$ such that $B_n \cap B_m = \emptyset$ for $n \neq m$. Because $(f\mathbb{I}_{B_n} \mid n \in \mathbb{N})$ is a sequence of non-negative Σ -measurable functions,

$$(f\mu)\left(\bigcup_n B_n\right) = \mu(f\mathbb{I}_{\bigcup_n B_n}) = \mu\left(\sum_n f\mathbb{I}_{B_n}\right) = \sum_n \mu(f\mathbb{I}_{B_n}) = \sum_n (f\mu)(B_n).$$

□

Definition 5.14. If λ and μ are measures on (S, Σ) and $f : S \rightarrow [0, \infty]$ is a Σ -measurable function such that $\lambda = (f\mu)$, then f is a version of the Radon-Nikodym derivative $d\lambda/d\mu$ of λ with respect to μ . In that case, we say that $f = d\lambda/d\mu$ almost everywhere.

Proposition 5.21. Consider a Σ -measurable function $f : S \rightarrow [0, \infty]$ and the measure $\lambda = (f\mu)$ on (S, Σ) . If $h : S \rightarrow \mathbb{R}$ is a Σ -measurable function, then h is λ -integrable if and only if hf is μ -integrable. In that case,

$$\int_S h \, d\lambda = \lambda(h) = \mu(hf) = \int_S hf \, d\mu.$$

Proof. First, note that if h is Σ -measurable then hf is also Σ -measurable.

Next, let $h = \mathbb{I}_A$ for some $A \in \Sigma$. In that case, $\mu(hf) = \mu(\mathbb{I}_A f) = \mu(f; A) = \lambda(A) = \lambda(\mathbb{I}_A) = \lambda(h)$. This step is complete since $\mu(|hf|) < \infty$ if and only if $\lambda(|h|) < \infty$.

Next, suppose h is a simple function that can be written as $h = \sum_{k=1}^m a_k \mathbb{I}_{A_k}$ for some fixed $a_1, a_2, \dots, a_m \in [0, \infty]$ and $A_1, A_2, \dots, A_m \in \Sigma$. By the linearity of the integral and considering the previous step,

$$\mu(hf) = \mu\left(\sum_{k=1}^m a_k \mathbb{I}_{A_k} f\right) = \sum_{k=1}^m a_k \mu(\mathbb{I}_{A_k} f) = \sum_{k=1}^m a_k \lambda(\mathbb{I}_{A_k}) = \lambda\left(\sum_{k=1}^m a_k \mathbb{I}_{A_k}\right) = \lambda(h).$$

This step is complete since $\mu(|hf|) < \infty$ if and only if $\lambda(|h|) < \infty$.

Next, suppose h is a non-negative Σ -measurable function. For any $n \in \mathbb{N}$, consider the simple function $h_n = \alpha_n \circ h$, where α_n is the n -th staircase function. Because $h_n \uparrow h$, the monotone-convergence theorem implies that $\lambda(h_n) \uparrow \lambda(h)$. Similarly, because $h_n f \uparrow hf$, the monotone-convergence theorem implies that $\mu(h_n f) \uparrow \mu(hf)$. Because our previous result implies that $\lambda(h_n) = \mu(h_n f)$, the limit when $n \rightarrow \infty$ shows that $\mu(hf) = \lambda(h)$. This step is complete since $\mu(|hf|) < \infty$ if and only if $\lambda(|h|) < \infty$.

Finally, suppose $h : S \rightarrow \mathbb{R}$ is a Σ -measurable function. Recall that $h = h^+ - h^-$, where h^+ and h^- are non-negative Σ -measurable functions. If either $\lambda(|h|) < \infty$ or $\mu(|hf|) < \infty$, then

$$\mu(hf) = \mu((h^+ - h^-)f) = \mu(h^+ f) - \mu(h^- f) = \lambda(h^+) - \lambda(h^-) = \lambda(h) < \infty.$$

Since $\lambda(|h|) = \mu(|hf|) = \infty$ implies that h is not λ -integrable and that hf is not μ -integrable, the proof is complete. $\square$

Proposition 5.22. Consider a Σ -measurable function $f : S \rightarrow [0, \infty]$ and the measure $\lambda = (f\mu)$ on (S, Σ) . For every $A \in \Sigma$, if $\mu(A) = 0$, then $\lambda(A) = 0$.

Proof. Suppose that $\mu(A) = 0$ for some $A \in \Sigma$. The fact that $\{f\mathbb{I}_A \neq 0\} \subseteq A$ implies $\mu(\{f\mathbb{I}_A \neq 0\}) \leq \mu(A) = 0$. Because $f\mathbb{I}_A$ and 0 are Σ -measurable functions such that $\mu(\{f\mathbb{I}_A \neq 0\}) = 0$, we know that $\mu(f\mathbb{I}_A) = \mu(0) = 0$. $\square$

Definition 5.15. If λ and μ are measures on (S, Σ) such that $\mu(A) = 0$ implies $\lambda(A) = 0$ for every $A \in \Sigma$, then λ is absolutely continuous with respect to μ , which is denoted by $\lambda \ll \mu$.

Theorem 5.3 (Radon-Nikodym theorem). If λ and μ are σ -finite measures on (S, Σ) such that $\lambda \ll \mu$, then a version of the Radon-Nikodym derivative $d\lambda/d\mu$ of λ with respect to μ exists. If f and $\tilde{f}$ are such versions, then $\mu(f \neq \tilde{f}) = 0$.

Proposition 5.23 (Radon-Nikodym chain rule). Let λ , μ , and η be σ -finite measures on (S, Σ) such that $\lambda \ll \mu$ and $\mu \ll \eta$. If $f = d\lambda/d\mu$ almost everywhere and $g = d\mu/d\eta$ almost everywhere, then $fg = d\lambda/d\eta$ almost everywhere.

Proof. Note that $fg : S \rightarrow [0, \infty]$ is Σ -measurable. For every $A \in \Sigma$,

$$\lambda(A) = \int_S \mathbb{I}_A d\lambda = \int_S \mathbb{I}_A f d\mu = \int_S (\mathbb{I}_A f) g d\eta = \int_S \mathbb{I}_A (fg) d\eta = \int_A fg d\eta = ((fg)\eta)(A).$$

$\square$

6 Expectation

Definition 6.1. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. The expectation $\mathbb{E}(X)$ of a $\mathbb{P}$ -integrable random variable $X : \Omega \rightarrow \mathbb{R}$ is defined as the integral of X with respect to the probability measure $\mathbb{P}$. Therefore,

$$\mathbb{E}(X) = \mathbb{P}(X) = \int_{\Omega} X d\mathbb{P} = \int_{\Omega} X(\omega) \mathbb{P}(d\omega).$$

Definition 6.2. The expectation $\mathbb{E}(X)$ of a non-negative random variable $X : \Omega \rightarrow [0, \infty]$ is also defined as the integral of X with respect to the probability measure $\mathbb{P}$.

Consider a sequence of random variables $(X_n : \Omega \rightarrow \mathbb{R} \mid n \in \mathbb{N})$ and a random variable $X : \Omega \rightarrow \mathbb{R}$ such that

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} X_n = X\right) = \mathbb{P}\left(\{\omega \in \Omega \mid \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}\right) = 1.$$

The integration results discussed in the previous section can be restated as follows.

Theorem 6.1 (Monotone-convergence theorem). If $X_n \geq 0$ and $X_n \leq X_{n+1}$ for every $n \in \mathbb{N}$, then $\mathbb{E}(X_n) \uparrow \mathbb{E}(X)$.

Lemma 6.1 (Fatou lemma). If $X_n \geq 0$ for every $n \in \mathbb{N}$, then $\mathbb{E}(X) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(X_n)$.

Theorem 6.2 (Dominated convergence theorem). If there is a $\mathbb{P}$ -integrable non-negative function $Y : \Omega \rightarrow [0, \infty]$ such that $|X_n| \leq Y$ for every $n \in \mathbb{N}$, then X is $\mathbb{P}$ -integrable and $\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X)$.

Lemma 6.2 (Scheffé's lemma). If X and X_n are $\mathbb{P}$ -integrable for every $n \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} \mathbb{E}(|X_n - X|) = 0$ if and only if $\lim_{n \rightarrow \infty} \mathbb{E}(|X_n|) = \mathbb{E}(|X|)$.

Theorem 6.3 (Bounded convergence theorem). If there is a $K \in [0, \infty)$ such that $|X_n| \leq K$ for every $n \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X)$ and $\lim_{n \rightarrow \infty} \mathbb{E}(|X_n - X|) = 0$.

Proof. Note that the simple function $Y = K$ is $\mathbb{P}$ -integrable, since $\mathbb{P}(|Y|) = \mathbb{P}(Y) = \mathbb{P}(K\mathbb{I}_\Omega) = K\mathbb{P}(\Omega) = K$. Therefore, X is $\mathbb{P}$ -integrable and $\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X)$. The dominated convergence theorem also guarantees that $\lim_{n \rightarrow \infty} \mathbb{E}(|X_n - X|) = 0$. $\square$

Definition 6.3. The expectation $\mathbb{E}(X; F)$ of the $\mathbb{P}$ -integrable random variable $X : \Omega \rightarrow \mathbb{R}$ over the set $F \in \mathcal{F}$ is defined as

$$\mathbb{E}(X; F) = \mathbb{E}(X\mathbb{I}_F) = \mathbb{P}(X; F) = \mathbb{P}(X\mathbb{I}_F) = \int_F X d\mathbb{P} = \int_F X(\omega) \mathbb{P}(d\omega).$$

Proposition 6.1. Consider a random variable $Z : \Omega \rightarrow \mathbb{R}$ and a $\mathcal{B}(\mathbb{R})$ -measurable non-negative function $g : \mathbb{R} \rightarrow [0, \infty]$ such that $a \leq b$ implies $g(a) \leq g(b)$. Recall that the function $g(Z) : \Omega \rightarrow [0, \infty]$ defined by $g(Z) = g \circ Z$ is also a random variable. For every $c \in \mathbb{R}$, Markov's inequality states that

$$\mathbb{E}(g(Z)) \geq g(c)\mathbb{P}(Z \geq c),$$

since $g(Z) \geq g(Z)\mathbb{I}_{\{Z \geq c\}} \geq g(c)\mathbb{I}_{\{Z \geq c\}}$ implies $\mathbb{E}(g(Z)) \geq \mathbb{E}(g(c)\mathbb{I}_{\{Z \geq c\}}) = g(c)\mathbb{P}(Z \geq c)$.

Proposition 6.2. Consider a non-negative random variable $Z : \Omega \rightarrow [0, \infty]$ and let $g(c) = \max(c, 0)$. For $c \geq 0$, Markov's inequality implies that $\mathbb{E}(Z) \geq c\mathbb{P}(Z \geq c)$.

Proposition 6.3. Consider a random variable $Z : \Omega \rightarrow \mathbb{R}$ and let $g(c) = e^{\theta c}$ for some $\theta > 0$. Markov's inequality implies that $\mathbb{E}(e^{\theta Z}) \geq e^{\theta c}\mathbb{P}(Z \geq c)$.

Proposition 6.4. Consider a non-negative random variable $X : \Omega \rightarrow [0, \infty]$. If $\mathbb{E}(X) < \infty$, then $\mathbb{P}(X < \infty) = 1$. Note that $\infty\mathbb{I}_{\{X=\infty\}} \leq X$, such that $\infty\mathbb{P}(X = \infty) \leq \mathbb{E}(X)$. Therefore, $\mathbb{P}(X = \infty) > 0$ implies $\mathbb{E}[X] = \infty$.

Proposition 6.5. Consider a sequence $(Z_n : \Omega \rightarrow [0, \infty] \mid n \in \mathbb{N})$ of non-negative random variables. In that case,

$$\mathbb{E}\left(\sum_k Z_k\right) = \sum_k \mathbb{E}(Z_k).$$

Proposition 6.6. Consider a sequence $(Z_n : \Omega \rightarrow [0, \infty] \mid n \in \mathbb{N})$ of non-negative random variables such that $\sum_k \mathbb{E}(Z_k) < \infty$. In that case, $\sum_k Z_k < \infty$ almost surely and $\lim_{n \rightarrow \infty} Z_n = 0$ almost surely, where 0 denotes the zero function.

Proof. Because $\mathbb{E}(\sum_k Z_k) < \infty$, we know that $\mathbb{P}(\sum_k Z_k < \infty) = 1$. Because the n -th term test implies that $\{\sum_k Z_k < \infty\} \subseteq \{\lim_{n \rightarrow \infty} Z_n = 0\}$, we know that $1 = \mathbb{P}(\sum_k Z_k < \infty) \leq \mathbb{P}(\lim_{n \rightarrow \infty} Z_n = 0)$. $\square$

Lemma 6.3 (Borel-Cantelli lemma). Consider a sequence of events $(F_n \in \mathcal{F} \mid n \in \mathbb{N})$ such that $\sum_n \mathbb{P}(F_n) < \infty$. Let $(\mathbb{I}_{F_n} \mid n \in \mathbb{N})$ be the corresponding sequence of indicator functions. Because $\mathbb{E}(\mathbb{I}_{F_k}) = \mathbb{P}(F_k)$, we know that $\sum_n \mathbb{E}(\mathbb{I}_{F_n}) < \infty$, which implies $\sum_n \mathbb{I}_{F_n} < \infty$ almost surely. Because $\sum_n \mathbb{I}_{F_n}(\omega)$ is the number of times that the outcome $\omega \in \Omega$ belongs to an event in the sequence, we know that the outcome ω almost surely belongs to a finite number of events in the sequence, which implies that $\mathbb{P}(\limsup_{n \rightarrow \infty} F_n) = 0$.

Definition 6.4. A function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is convex if $\lambda\phi(x) + (1 - \lambda)\phi(y) \geq \phi(\lambda x + (1 - \lambda)y)$, for every $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $\lambda \in [0, 1]$. If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is convex, it is also continuous and therefore $\mathcal{B}(\mathbb{R})$ -measurable.

Important examples of convex functions include $x \mapsto |x|$, $x \mapsto x^2$, and $x \mapsto e^{\theta x}$ for $\theta \in \mathbb{R}$.

Proposition 6.7. If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is a convex function, for every $z \in \mathbb{R}$ there is a function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = ax + b$ for every $x \in \mathbb{R}$ and some $a \in \mathbb{R}$ and $b \in \mathbb{R}$ such that $g(z) = \phi(z)$ and $g(x) \leq \phi(x)$ for every $x \in \mathbb{R}$.

In other words, for every point in the domain of a convex function, there is a linear function that never surpasses the convex function such that the value of the linear function at that point matches the value of the convex function at that point.

Proposition 6.8 (Jensen's inequality). Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, so that $\mathbb{E}(|X|) < \infty$, and a convex function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ such that $\mathbb{E}(\phi(X))$ is well defined. In that case, $\mathbb{E}(\phi(X)) \geq \phi(\mathbb{E}(X))$.

Proof. Consider a function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(\mathbb{E}(X)) = \phi(\mathbb{E}(X))$ and $g(x) = ax + b \leq \phi(x)$ for every $x \in \mathbb{R}$ and some $a, b \in \mathbb{R}$. Clearly $g(X) = g \circ X \leq \phi \circ X = \phi(X)$. Therefore,

$$\mathbb{E}(\phi(X)) \geq \mathbb{E}(g(X)) = \mathbb{E}[aX + b] = a\mathbb{E}(X) + b = g(\mathbb{E}(X)) = \phi(\mathbb{E}(X)).$$

□

Definition 6.5. For every $p \in [1, \infty)$, the set $\mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ contains exactly each random variable $X : \Omega \rightarrow \mathbb{R}$ such that $\mathbb{E}(|X|^p) < \infty$.

Definition 6.6. If $X \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$, the p -norm $\|X\|_p$ of the random variable X is given by $\|X\|_p = \mathbb{E}(|X|^p)^{1/p}$.

Proposition 6.9 (Monotonicity of norm). For every $p \in [1, \infty)$ and $r \in [1, \infty)$ such that $p \leq r$, if $Y \in \mathcal{L}^r(\Omega, \mathcal{F}, \mathbb{P})$ then $Y \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ and $\|Y\|_p \leq \|Y\|_r$.

Proof. For every $n \in \mathbb{N}$, consider the function $X_n = \min(|Y|, n)^p$. Clearly, $0 \leq X_n \leq n^p$, so $0 \leq \mathbb{E}(|X_n|) = \mathbb{E}(X_n) \leq n^p$. Consider also the convex function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ given by $\phi(x) = |x|^{r/p}$ such that $\phi(X_n) = |X_n|^{r/p} = X_n^{r/p}$. Clearly, $0 \leq X_n^{r/p} = \min(|Y|, n)^r \leq n^r$, so $0 \leq \mathbb{E}(|X_n^{r/p}|) = \mathbb{E}(X_n^{r/p}) \leq n^r$. Using Jensen's inequality,

$$\mathbb{E}(X_n^{r/p}) = \mathbb{E}(\phi(X_n)) \geq \phi(\mathbb{E}(X_n)) = |\mathbb{E}(X_n)|^{r/p} = \mathbb{E}(X_n)^{r/p}.$$

Because $X_n^{r/p} \geq 0$ and $X_n^{r/p} \uparrow |Y|^r$, the monotone-convergence theorem guarantees that $\mathbb{E}(X_n^{r/p}) \uparrow \mathbb{E}(|Y|^r)$. Because $X_n \geq 0$ and $X_n \uparrow |Y|^p$, the monotone-convergence theorem guarantees that $\mathbb{E}(X_n) \uparrow \mathbb{E}(|Y|^p)$. By taking the limit of both sides of the previous inequation,

$$\mathbb{E}(|Y|^r) = \lim_{n \rightarrow \infty} \mathbb{E}(X_n^{r/p}) \geq \lim_{n \rightarrow \infty} \mathbb{E}(X_n)^{r/p} = \left(\lim_{n \rightarrow \infty} \mathbb{E}(X_n) \right)^{r/p} = \mathbb{E}(|Y|^p)^{r/p}.$$

By taking the r -th root of both sides of the previous inequation,

$$\infty > \mathbb{E}(|Y|^r)^{1/r} \geq \mathbb{E}(|Y|^p)^{1/p}.$$

□

Proposition 6.10. For every $p \in [1, \infty)$, the set $\mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ is a vector space over the field $\mathbb{R}$.

Proof. First, recall that the set of all functions from Ω to $\mathbb{R}$ is a vector space over the field $\mathbb{R}$ when scalar multiplication and addition are performed pointwise. Because such set includes $\mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$, it is sufficient to show that $\mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ is non-empty and closed under scalar multiplication and addition. Because $0 : \Omega \rightarrow \mathbb{R}$ is a random variable and $\mathbb{E}(|0|^p) = \mathbb{E}(0) = 0$, we know that $0 \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$. If $X \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ and $c \in \mathbb{R}$, then $cX : \Omega \rightarrow \mathbb{R}$ is a random variable and $\mathbb{E}(|cX|^p) = \mathbb{E}(|c|^p |X|^p) = |c|^p \mathbb{E}(|X|^p)$, we know that $cX \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$. Finally, if $X \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$, then

$$|X + Y|^p \leq (|X| + |Y|)^p \leq (2 \max(|X|, |Y|))^p \leq 2^p(|X|^p + |Y|^p),$$

which implies $X + Y \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ since

$$\mathbb{E}(|X + Y|^p) \leq \mathbb{E}(2^p(|X|^p + |Y|^p)) = 2^p \mathbb{E}(|X|^p) + 2^p \mathbb{E}(|Y|^p) < \infty.$$

□

Proposition 6.11 (Schwarz inequality). Consider the random variables $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. In that case, $XY \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathbb{E}(|XY|) \leq \|X\|_2 \|Y\|_2$.

Proof. First, consider the case where $\|X\|_2 \neq 0$ and $\|Y\|_2 \neq 0$. Let $Z = |X|/\|X\|_2$ and $W = |Y|/\|Y\|_2$. Clearly, $\mathbb{E}(Z^2) = \mathbb{E}(|X|^2)/\|X\|_2^2 = 1$. Analogously, $\mathbb{E}(W^2) = 1$. Because $(Z - W)^2 \geq 0$, we know that

$$0 \leq \mathbb{E}((Z - W)^2) = \mathbb{E}(Z^2) + \mathbb{E}(W^2) - \mathbb{E}(2ZW) = 2 - \mathbb{E}(2ZW).$$

Because the previous inequation implies that $\mathbb{E}(ZW) \leq 1$,

$$1 \geq \mathbb{E}(ZW) = \mathbb{E}(|X||Y|/\|X\|_2\|Y\|_2) = \mathbb{E}(|XY|)/\|X\|_2\|Y\|_2.$$

Using the fact that $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$,

$$\mathbb{E}(|XY|) \leq \|X\|_2\|Y\|_2 < \infty.$$

Finally, consider the case where $\|X\|_2 = \mathbb{E}(X^2)^{1/2} = 0$, which will prove analogous to the case where $\|Y\|_2 = 0$. Because X^2 is a non-negative random variable, the fact that $\mathbb{E}(X^2) = 0$ implies that $\mathbb{P}(X^2 > 0) = \mathbb{P}(X \neq 0) = 0$. Therefore, $\mathbb{P}(X = 0) = 1$. Because $\{X = 0\} \subseteq \{XY = 0\}$, we know that $\mathbb{P}(X = 0) \leq \mathbb{P}(XY = 0)$, which implies $\mathbb{P}(XY = 0) = \mathbb{P}(|XY| = 0) = 1$. Because $\{|XY| = 0\}$ happens almost surely, we know that $\mathbb{E}(|XY|) = \mathbb{E}(0) = 0$. Therefore, $XY \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $0 = \mathbb{E}(|XY|) \leq \|X\|_2\|Y\|_2 = 0$. $\square$

Proposition 6.12 (Triangle law). Consider the random variables $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. Because $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ is a vector space over $\mathbb{R}$, we know that $X + Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. In that case, $\|X + Y\|_2 \leq \|X\|_2 + \|Y\|_2$.

Proof. Since $|X + Y| \leq |X| + |Y|$, we know that $|X + Y|^2 \leq (|X| + |Y|)^2 = |X|^2 + 2|X||Y| + |Y|^2$. Therefore,

$$\mathbb{E}(|X + Y|^2) \leq \mathbb{E}(|X|^2) + 2\mathbb{E}(|X||Y|) + \mathbb{E}(|Y|^2) = \mathbb{E}(|X|^2) + 2\mathbb{E}(|XY|) + \mathbb{E}(|Y|^2).$$

Using the Schwarz inequality,

$$\mathbb{E}(|X + Y|^2) \leq \mathbb{E}(|X|^2) + 2\|X\|_2\|Y\|_2 + \mathbb{E}(|Y|^2) = (\|X\|_2 + \|Y\|_2)^2$$

By taking the square root of both sides,

$$\|X + Y\|_2 = \mathbb{E}(|X + Y|^2)^{1/2} \leq \|X\|_2 + \|Y\|_2.$$

$\square$

Definition 6.7. Consider the random variables $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. Recall that $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mu_X = \mathbb{E}(X)$ and $\mu_Y = \mathbb{E}(Y)$. Because $(X - \mu_X) \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $(Y - \mu_Y) \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$, we know that $(X - \mu_X)(Y - \mu_Y) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. The covariance $\text{Cov}(X, Y)$ between X and Y is defined by

$$\text{Cov}(X, Y) = \mathbb{E}((X - \mu_X)(Y - \mu_Y)) = \mathbb{E}(XY) - \mathbb{E}(X\mu_Y) - \mathbb{E}(Y\mu_X) + \mathbb{E}(\mu_X\mu_Y) = \mathbb{E}(XY) - \mu_X\mu_Y.$$

Definition 6.8. Consider the random variable $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. The variance $\text{Var}(X)$ of X is defined by

$$\text{Var}(X) = \text{Cov}(X, X) = \mathbb{E}((X - \mu_X)^2) = \mathbb{E}(X^2) - \mu_X^2.$$

Definition 6.9. Consider the random variables $U \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $V \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. The inner product $\langle U, V \rangle$ between U and V is given by $\langle U, V \rangle = \mathbb{E}(UV)$.

Definition 6.10. If $\|U\|_2 \neq 0$ and $\|V\|_2 \neq 0$, the cosine of the angle θ between U and V is defined as

$$\cos \theta = \frac{\langle U, V \rangle}{\|U\|_2\|V\|_2}.$$

Because $|\langle U, V \rangle| = |\mathbb{E}(UV)| \leq \mathbb{E}(|UV|) \leq \|U\|_2\|V\|_2$, we know that $|\cos \theta| \leq 1$.

Proposition 6.13. Consider the random variables $U, V, W, Z \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. The following are properties of the inner product:

- $\langle U, U \rangle = \mathbb{E}(U^2) = \|U\|_2^2$.
- $\langle U, V \rangle = \mathbb{E}(UV) = \mathbb{E}(VU) = \langle V, U \rangle$.
- $\langle aU, V \rangle = \mathbb{E}(aUV) = a\mathbb{E}(UV) = a\langle U, V \rangle$, for any $a \in \mathbb{R}$.
- $\langle U, aV \rangle = \mathbb{E}(UaV) = a\mathbb{E}(UV) = a\langle U, V \rangle$, for any $a \in \mathbb{R}$.
- $\langle U + V, W \rangle = \mathbb{E}((U + V)W) = \mathbb{E}(UW + VW) = \langle U, W \rangle + \langle V, W \rangle$.

- $\langle U, V + W \rangle = \mathbb{E}(U(V + W)) = \mathbb{E}(UV + UW) = \langle U, V \rangle + \langle U, W \rangle.$
- $\langle U + V, W + Z \rangle = \langle U, W + Z \rangle + \langle V, W + Z \rangle = \langle U, W \rangle + \langle U, Z \rangle + \langle V, W \rangle + \langle V, Z \rangle.$

Definition 6.11. Consider the random variables $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mu_X = \mathbb{E}(X)$ and $\mu_Y = \mathbb{E}(Y)$. The correlation ρ between X and Y is defined as the cosine of the angle between $X - \mu_X$ and $Y - \mu_Y$, which is given by

$$\rho = \frac{\langle X - \mu_X, Y - \mu_Y \rangle}{\|X - \mu_X\|_2 \|Y - \mu_Y\|_2} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

Proposition 6.14. Consider the random variables $U \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $V \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. Because $U + V \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$,

$$\|U + V\|_2^2 = \mathbb{E}(|U + V|^2) = \mathbb{E}((U + V)^2) = \mathbb{E}(U^2) + 2\mathbb{E}(UV) + \mathbb{E}(V^2) = \|U\|_2^2 + \|V\|_2^2 + 2\langle U, V \rangle.$$

Definition 6.12. If $\langle U, V \rangle = 0$, we say that U and V are orthogonal, which is denoted by $U \perp V$. In that case,

$$\|U + V\|_2^2 = \|U\|_2^2 + \|V\|_2^2.$$

Proposition 6.15. Consider the random variables $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. Note that $X + Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and

$$\text{Var}(X + Y) = \mathbb{E}((X + Y)^2) - \mathbb{E}(X + Y)^2 = \mathbb{E}(X^2 + 2XY + Y^2) - (\mathbb{E}(X)^2 + 2\mathbb{E}(X)\mathbb{E}(Y) + \mathbb{E}(Y)^2).$$

By the linearity of expectation and reorganizing terms,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

Therefore, if $\text{Cov}(X, Y) = 0$, then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Proposition 6.16. More generally, if $X_1, \dots, X_n \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$, then

$$\text{Var}\left(\sum_{k=1}^n X_k\right) = \sum_{k=1}^n \text{Var}(X_k) + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{Cov}(X_i, X_j).$$

Proposition 6.17 (Parallelogram law). Consider the random variables $U \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $V \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. In that case,

$$\|U + V\|_2^2 + \|U - V\|_2^2 = 2\|U\|_2^2 + 2\|V\|_2^2.$$

Proof. Using the relationship between the inner product and the 2-norm,

$$\|U + V\|_2^2 + \|U - V\|_2^2 = \langle U + V, U + V \rangle + \langle U - V, U - V \rangle.$$

By the bilinearity of the inner product,

$$\|U + V\|_2^2 + \|U - V\|_2^2 = \langle U, U \rangle + \langle U, V \rangle + \langle V, U \rangle + \langle V, V \rangle + \langle U, U \rangle + \langle U, -V \rangle + \langle -V, U \rangle + \langle -V, -V \rangle.$$

By cancelling terms,

$$\|U + V\|_2^2 + \|U - V\|_2^2 = 2\langle U, U \rangle + 2\langle V, V \rangle = 2\|U\|_2^2 + 2\|V\|_2^2.$$

□

Proposition 6.18. For some $p \in [1, \infty)$, consider a sequence of random variables $(X_n \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P}) \mid n \in \mathbb{N})$ such that

$$\lim_{k \rightarrow \infty} \sup_{r, s \geq k} \|X_r - X_s\|_p = 0.$$

In that case, there is a random variable $X \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ such that

$$\lim_{n \rightarrow \infty} \|X_n - X\|_p = 0.$$

Proof. By definition, for every $\epsilon > 0$ there is an $N \in \mathbb{N}$ such that $k \geq N$ implies $\sup_{r,s \geq k} \|X_r - X_s\|_p < \epsilon$. Therefore, there is a sequence $(k_n \in \mathbb{N} \mid n \in \mathbb{N})$ such that $k_{n+1} \geq k_n$ and $\sup_{r,s \geq k_n} \|X_r - X_s\|_p < 1/2^n$ for every $n \in \mathbb{N}$.

For every $n \in \mathbb{N}$, the monotonicity of the norm implies that

$$\mathbb{E}(|X_{k_{n+1}} - X_{k_n}|) = \|X_{k_{n+1}} - X_{k_n}\|_1 \leq \|X_{k_{n+1}} - X_{k_n}\|_p < \frac{1}{2^n}.$$

Because $|X_{k_{n+1}} - X_{k_n}|$ is a non-negative random variable for every $n \in \mathbb{N}$,

$$\sum_n \mathbb{E}(|X_{k_{n+1}} - X_{k_n}|) = \mathbb{E} \left(\sum_n |X_{k_{n+1}} - X_{k_n}| \right) \leq \sum_n \frac{1}{2^n} < \infty.$$

Because the expectation above is finite,

$$\mathbb{P} \left(\sum_n |X_{k_{n+1}} - X_{k_n}| < \infty \right) = 1.$$

Suppose $\sum_n |X_{k_{n+1}}(\omega) - X_{k_n}(\omega)| < \infty$ for some $\omega \in \Omega$. For every $\epsilon > 0$, the Cauchy test guarantees that there is an $N \in \mathbb{N}$ such that $j > i > N$ implies

$$\left| \sum_{n=i}^j |X_{k_{n+1}}(\omega) - X_{k_n}(\omega)| \right| = \sum_{n=i}^j |X_{k_{n+1}}(\omega) - X_{k_n}(\omega)| < \epsilon.$$

Furthermore, for every $j > i$,

$$|X_{k_j}(\omega) - X_{k_i}(\omega)| = \left| X_{k_j}(\omega) - X_{k_i}(\omega) + \sum_{n=i+1}^{j-1} X_{k_n}(\omega) - \sum_{n=i+1}^{j-1} X_{k_n}(\omega) \right| = \left| \sum_{n=i+1}^j X_{k_n}(\omega) - \sum_{n=i}^{j-1} X_{k_n}(\omega) \right|.$$

By shifting indices and using the triangle inequality, for $j > i > N$,

$$|X_{k_j}(\omega) - X_{k_i}(\omega)| = \left| \sum_{n=i}^{j-1} X_{k_{n+1}}(\omega) - X_{k_n}(\omega) \right| \leq \sum_{n=i}^{j-1} |X_{k_{n+1}}(\omega) - X_{k_n}(\omega)| < \epsilon.$$

For $j = i > N$, note that $|X_{k_j}(\omega) - X_{k_i}(\omega)| = 0 < \epsilon$. Therefore, for every $\epsilon > 0$ there is an $N \in \mathbb{N}$ such that $j > N$ and $i > N$ implies $|X_{k_j}(\omega) - X_{k_i}(\omega)| < \epsilon$, such that $(X_{k_n}(\omega) \mid n \in \mathbb{N})$ is a Cauchy sequence of real numbers.

Because every Cauchy sequence of real numbers converges to a real number, consider the function $\tilde{X} = \limsup_{n \rightarrow \infty} X_{k_n}$ such that $\lim_{n \rightarrow \infty} X_{k_n}(\omega) = \limsup_{n \rightarrow \infty} X_{k_n}(\omega) = \tilde{X}(\omega)$ whenever $(X_{k_n}(\omega) \mid n \in \mathbb{N})$ is a Cauchy sequence of real numbers. Because $\tilde{X}$ may take values outside $\mathbb{R}$ on the set $A = \{\sum_n |X_{k_{n+1}} - X_{k_n}| < \infty\}^c$, which satisfies $\mathbb{P}(A) = 0$, let $X = \tilde{X} \mathbb{I}_{A^c}$, which is a random variable such that $X = \tilde{X}$ almost surely.

Since $\{\sum_n |X_{k_{n+1}} - X_{k_n}| < \infty\} \subseteq \{\lim_{n \rightarrow \infty} X_{k_n} = X\}$,

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} X_{k_n} = X \right) \geq \mathbb{P} \left(\sum_n |X_{k_{n+1}} - X_{k_n}| < \infty \right) = 1.$$

Suppose $\lim_{n \rightarrow \infty} X_{k_n}(\omega) = X(\omega)$ for some $\omega \in \Omega$. For every $r \in \mathbb{N}$,

$$\left| \lim_{n \rightarrow \infty} X_{k_n}(\omega) - X_r(\omega) \right|^p = \lim_{n \rightarrow \infty} |X_{k_n}(\omega) - X_r(\omega)|^p = |X(\omega) - X_r(\omega)|^p.$$

Because $\{\lim_{n \rightarrow \infty} X_{k_n} = X\} \subseteq \{\lim_{n \rightarrow \infty} |X_{k_n} - X_r|^p = |X - X_r|^p\}$ for every $r \in \mathbb{N}$,

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} |X_{k_n} - X_r|^p = |X - X_r|^p \right) \geq \mathbb{P} \left(\lim_{n \rightarrow \infty} X_{k_n} = X \right) = 1.$$

Because $|X_{k_n} - X_r|^p \geq 0$ for every $n \in \mathbb{N}$, by the Fatou lemma,

$$\mathbb{E}(|X - X_r|^p) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(|X_{k_n} - X_r|^p).$$

For any $t \in \mathbb{N}$, suppose $r \geq k_t$ and recall that $k_n \geq k_t$ whenever $n \geq t$. In that case,

$$\mathbb{E}(|X_{k_n} - X_r|^p) = \|X_{k_n} - X_r\|_p^p < \frac{1}{2^{tp}}.$$

For any $\epsilon > 0$, choose $t \in \mathbb{N}$ such that $1/2^{tp} < \epsilon$. In that case, for any $r \geq k_t$,

$$\mathbb{E}(|X - X_r|^p) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(|X_{k_n} - X_r|^p) \leq \frac{1}{2^{tp}} < \epsilon.$$

Because $\mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ is a vector space over the field $\mathbb{R}$, the fact that $(X - X_r) \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ and $X_r \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ implies that $X \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$. The previous inequality also shows that

$$\lim_{r \rightarrow \infty} \mathbb{E}(|X - X_r|^p) = \lim_{r \rightarrow \infty} \|X - X_r\|_p^p = 0.$$

□

Definition 6.13. A vector space $\mathcal{K} \subseteq \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ is said to be complete if for every sequence $(V_n \in \mathcal{K} \mid n \in \mathbb{N})$ such that

$$\lim_{k \rightarrow \infty} \sup_{r, s \geq k} \|V_r - V_s\|_p = 0$$

there is a $V \in \mathcal{K}$ such that

$$\lim_{n \rightarrow \infty} \|V_n - V\|_p = 0.$$

Proposition 6.19. If the vector space $\mathcal{K} \subseteq \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ is complete, then for every $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ there is a so-called version $Y \in \mathcal{K}$ of the orthogonal projection of X onto $\mathcal{K}$ such that $\|X - Y\|_2 = \inf\{\|X - W\|_2 \mid W \in \mathcal{K}\}$ and $X - Y \perp Z$ for every $Z \in \mathcal{K}$. Furthermore, if Y and $\tilde{Y}$ are versions of the orthogonal projection of X onto $\mathcal{K}$, then $\mathbb{P}(Y = \tilde{Y}) = 1$.

Proof. For some $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$, let $\Delta = \inf\{\|X - W\|_2 \mid W \in \mathcal{K}\}$. First, we will show that it is possible to choose a sequence $(Y_n \in \mathcal{K} \mid n \in \mathbb{N})$ such that $\lim_{n \rightarrow \infty} \|X - Y_n\|_2 = \Delta$. Recall that for every $\epsilon > 0$ there is a $W \in \mathcal{K}$ such that $\|X - W\|_2 < \Delta + \epsilon$. Choose Y_n such that $\|X - Y_n\|_2 < \Delta + \frac{1}{n+1}$. For every $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that $n \geq N$ implies that $\|X - Y_n\|_2 < \Delta + \epsilon$, which is equivalent to $|\|X - Y_n\|_2 - \Delta| < \epsilon$ since $\Delta \leq \|X - Y_n\|_2$.

Let $U = X - \frac{1}{2}(Y_r + Y_s)$ and $V = \frac{1}{2}(Y_r - Y_s)$ such that $U + V = X - Y_s$ and $U - V = X - Y_r$. Because $U \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $V \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$, the parallelogram law guarantees that

$$\|X - Y_s\|_2^2 + \|X - Y_r\|_2^2 = 2 \left\| X - \frac{1}{2}(Y_r + Y_s) \right\|_2^2 + 2 \left\| \frac{1}{2}(Y_r - Y_s) \right\|_2^2.$$

Therefore,

$$2 \left\| \frac{1}{2}(Y_r - Y_s) \right\|_2^2 = 2 \left\langle \frac{1}{2}(Y_r - Y_s), \frac{1}{2}(Y_r - Y_s) \right\rangle = \|X - Y_s\|_2^2 + \|X - Y_r\|_2^2 - 2 \left\| X - \frac{1}{2}(Y_r + Y_s) \right\|_2^2.$$

Using properties of the inner product and reorganizing terms,

$$\|Y_r - Y_s\|_2^2 = 2\|X - Y_s\|_2^2 + 2\|X - Y_r\|_2^2 - 4 \left\| X - \frac{1}{2}(Y_r + Y_s) \right\|_2^2.$$

Because $(Y_r + Y_s)/2 \in \mathcal{K}$, we know that $\|X - (Y_r + Y_s)/2\|_2^2 \geq \Delta^2$. Therefore,

$$\|Y_r - Y_s\|_2^2 \leq 2\|X - Y_s\|_2^2 + 2\|X - Y_r\|_2^2 - 4\Delta^2.$$

For every $\epsilon > 0$, since $\lim_{n \rightarrow \infty} \|X - Y_n\|_2^2 = \Delta^2$, there is a k such that $n \geq k$ implies $|\|X - Y_n\|_2^2 - \Delta^2| < \frac{\epsilon}{4}$, which is equivalent to $\|X - Y_n\|_2^2 < \frac{\epsilon}{4} + \Delta^2$. Therefore, whenever $r, s \geq k$,

$$\|Y_r - Y_s\|_2^2 \leq 2\|X - Y_s\|_2^2 + 2\|X - Y_r\|_2^2 - 4\Delta^2 < 2\left(\frac{\epsilon}{4} + \Delta^2\right) + 2\left(\frac{\epsilon}{4} + \Delta^2\right) - 4\Delta^2 = \epsilon,$$

which implies

$$\lim_{k \rightarrow \infty} \sup_{r, s \geq k} \|Y_r - Y_s\|_2 = 0.$$

Because $\mathcal{K}$ is complete, there is an $Y \in \mathcal{K}$ such that

$$\lim_{n \rightarrow \infty} \|Y_n - Y\|_2 = 0.$$

Let $U = X - Y_n$ and $V = Y_n - Y$ such that $U + V = X - Y$. Because $U \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $V \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$,

$$\Delta \leq \|X - Y\|_2 \leq \|X - Y_n\|_2 + \|Y_n - Y\|_2.$$

Using the squeeze theorem when $n \rightarrow \infty$ shows that $\|X - Y\|_2 = \Delta = \inf\{\|X - W\|_2 \mid W \in \mathcal{K}\}$.

For some $Z \in \mathcal{K}$ and $t \in \mathbb{R}$, let $U = X - Y$ and $V = -tZ$ such that $U + V = X - Y - tZ$. Because $U \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $V \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and considering the bilinearity of the inner product,

$$\|X - Y - tZ\|_2^2 = \|X - Y\|_2^2 + \|-tZ\|_2^2 + 2\langle X - Y, -tZ \rangle = \|X - Y\|_2^2 + t^2\|Z\|_2^2 - 2t\langle X - Y, Z \rangle.$$

Because $(Y + tZ) \in \mathcal{K}$, we know that $\|X - Y\|_2^2 \leq \|X - (Y + tZ)\|_2^2$. Therefore, for every $Z \in \mathcal{K}$ and $t \in \mathbb{R}$,

$$t^2\|Z\|_2^2 \geq 2t\langle X - Y, Z \rangle.$$

We will now show that the previous inequation can only be true for every $Z \in \mathcal{K}$ and $t \in \mathbb{R}$ if $\langle X - Y, Z \rangle = 0$ for every $Z \in \mathcal{K}$, which implies $X - Y \perp Z$ for every $Z \in \mathcal{K}$.

In order to find a contradiction, suppose that $\langle X - Y, Z \rangle \neq 0$ for some $Z \in \mathcal{K}$. Because $(X - Y) \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $Z \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$, the Schwarz inequality implies that

$$\|X - Y\|_2\|Z\|_2 \geq \mathbb{E}(|(X - Y)Z|) \geq |\mathbb{E}((X - Y)Z)| \geq 0.$$

Clearly, $|\mathbb{E}((X - Y)Z)| = 0$ when $\|Z\|_2 = 0$, which implies $\mathbb{E}((X - Y)Z) = \langle X - Y, Z \rangle = 0$. Therefore, we can suppose that $\|Z\|_2 > 0$. If $\langle X - Y, Z \rangle > 0$, then choose a $t \in \mathbb{R}$ such that $0 < t < 2\langle X - Y, Z \rangle / \|Z\|_2^2$. If $\langle X - Y, Z \rangle < 0$, then choose a $t \in \mathbb{R}$ such that $2\langle X - Y, Z \rangle / \|Z\|_2^2 < t < 0$. In either case, $t^2\|Z\|_2^2 < 2t\langle X - Y, Z \rangle$, which is a contradiction.

Suppose that Y and $\tilde{Y}$ are versions of the orthogonal projection of X onto $\mathcal{K}$. Because $(\tilde{Y} - Y) \in \mathcal{K}$,

$$\langle X - Y, \tilde{Y} - Y \rangle = \langle X - \tilde{Y}, \tilde{Y} - Y \rangle = 0.$$

By the bilinearity of the inner product,

$$\langle X, \tilde{Y} - Y \rangle + \langle -Y, \tilde{Y} - Y \rangle - \langle X, \tilde{Y} - Y \rangle - \langle -\tilde{Y}, \tilde{Y} - Y \rangle = \langle -Y, \tilde{Y} - Y \rangle - \langle -\tilde{Y}, \tilde{Y} - Y \rangle = \langle \tilde{Y} - Y, \tilde{Y} - Y \rangle = 0$$

Because $\langle \tilde{Y} - Y, \tilde{Y} - Y \rangle = \mathbb{E}((\tilde{Y} - Y)^2) = 0$ and $(\tilde{Y} - Y)^2$ is a non-negative random variable, we know that $\mathbb{P}((\tilde{Y} - Y)^2 \neq 0) = 0$, which implies that $\mathbb{P}(\tilde{Y} = Y) = 1$. $\square$

Proposition 6.20. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable $X : \Omega \rightarrow \mathbb{R}$. Recall that $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L}_X)$ is also a probability triple, where $\mathcal{L}_X : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$ is the law of X given by $\mathcal{L}_X(B) = \mathbb{P}(X^{-1}(B))$ for every $B \in \mathcal{B}(\mathbb{R})$. If $h : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel function, then $(h \circ X) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ if and only if $h \in \mathcal{L}^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L}_X)$. Furthermore, in that case,

$$\int_{\Omega} (h \circ X) d\mathbb{P} = \mathbb{P}(h \circ X) = \mathcal{L}_X(h) = \int_{\mathbb{R}} h d\mathcal{L}_X.$$

Proof. First, suppose $h = \mathbb{I}_B$ for some $B \in \mathcal{B}(\mathbb{R})$. For every $\omega \in \Omega$,

$$(h \circ X)(\omega) = \mathbb{I}_B(X(\omega)) = \mathbb{I}_{X^{-1}(B)}(\omega) = \begin{cases} 1, & \text{if } X(\omega) \in B, \\ 0, & \text{if } X(\omega) \notin B. \end{cases}$$

Therefore, $\mathbb{P}(h \circ X) = \mathbb{P}(\mathbb{I}_{X^{-1}(B)}) = \mathbb{P}(X^{-1}(B)) = \mathcal{L}_X(B) = \mathcal{L}_X(\mathbb{I}_B) = \mathcal{L}_X(h) < \infty$. Because h is $\mathcal{B}(\mathbb{R})$ -measurable and $(h \circ X)$ is $\mathcal{F}$ -measurable, this step is complete.

Next, suppose h is a simple function that can be written as $h = \sum_{k=1}^m a_k \mathbb{I}_{A_k}$ for some fixed $a_1, a_2, \dots, a_m \in [0, \infty]$ and $A_1, A_2, \dots, A_m \in \mathcal{B}(\mathbb{R})$. For every $\omega \in \Omega$,

$$(h \circ X)(\omega) = \sum_{k=1}^m a_k \mathbb{I}_{A_k}(X(\omega)) = \sum_{k=1}^m a_k \mathbb{I}_{X^{-1}(A_k)}(\omega).$$

Therefore, $\mathbb{P}(h \circ X) = \sum_{k=1}^m a_k \mathbb{P}(X^{-1}(A_k)) = \sum_{k=1}^m a_k \mathcal{L}_X(A_k) = \mathcal{L}_X(\sum_{k=1}^m a_k \mathbb{I}_{A_k}) = \mathcal{L}_X(h)$. Because h is $\mathcal{B}(\mathbb{R})$ -measurable and $(h \circ X)$ is $\mathcal{F}$ -measurable, this step is complete since $\mathcal{L}_X(h) < \infty$ if and only if $\mathbb{P}(h \circ X) < \infty$.

Next, suppose h is a non-negative Borel function. For any $n \in \mathbb{N}$, consider the simple function $h_n = \alpha_n \circ h$, where α_n is the n -th staircase function. Because $h_n \uparrow h$, the monotone-convergence theorem implies that $\mathcal{L}_X(h_n) \uparrow \mathcal{L}_X(h)$. Similarly, consider the simple function $\alpha_n \circ (h \circ X) = (\alpha_n \circ h) \circ X = h_n \circ X$. Because $(h_n \circ X) \uparrow (h \circ X)$, the monotone-convergence theorem implies that $\mathbb{P}(h_n \circ X) \uparrow \mathbb{P}(h \circ X)$. Because our previous result implies that $\mathbb{P}(h_n \circ X) = \mathcal{L}_X(h_n)$, the limit when $n \rightarrow \infty$ shows that $\mathbb{P}(h \circ X) = \mathcal{L}_X(h)$. Because h is Borel and $(h \circ X)$ is $\mathcal{F}$ -measurable, this step is complete since $\mathcal{L}_X(h) < \infty$ if and only if $\mathbb{P}(h \circ X) < \infty$.

Finally, suppose h is a Borel function. Recall that $h = h^+ - h^-$, where h^+ and h^- are non-negative Borel functions. Therefore, if either $\mathbb{P}(|h \circ X|) < \infty$ or $\mathcal{L}_X(|h|) < \infty$, then

$$\mathbb{P}(h \circ X) = \mathbb{P}((h \circ X)^+) - \mathbb{P}((h \circ X)^-) = \mathbb{P}(h^+ \circ X) - \mathbb{P}(h^- \circ X) = \mathcal{L}_X(h^+) - \mathcal{L}_X(h^-) = \mathcal{L}_X(h) < \infty,$$

where the second equality follows from associativity. Because h is $\mathcal{B}(\mathbb{R})$ -measurable and $(h \circ X)$ is $\mathcal{F}$ -measurable, this completes the proof, since $\mathbb{P}(|h \circ X|) = \mathcal{L}_X(|h|) = \infty$ implies $(h \circ X) \notin \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $h \notin \mathcal{L}^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L}_X)$. $\square$

Definition 6.14. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. A random variable $X : \Omega \rightarrow \mathbb{R}$ has a probability density function f_X if $f_X : \mathbb{R} \rightarrow [0, \infty]$ is a Borel function such that the law $\mathcal{L}_X$ of X is given by

$$\mathcal{L}_X(B) = \mathbb{P}(X^{-1}(B)) = \text{Leb}(f_X; B) = \text{Leb}(f_X \mathbb{I}_B) = \int_B f_X \, d\text{Leb},$$

for every $B \in \mathcal{B}(\mathbb{R})$, where Leb is the Lebesgue measure on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Proposition 6.21. In that case, since $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \text{Leb})$ is a measure space and $f_X : \mathbb{R} \rightarrow [0, \infty]$ is $\mathcal{B}(\mathbb{R})$ -measurable, recall that the measure $(f_X \text{Leb})$ on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is given by $(f_X \text{Leb})(B) = \text{Leb}(f_X \mathbb{I}_B)$ for every $B \in \mathcal{B}(\mathbb{R})$, so that $\mathcal{L}_X = (f_X \text{Leb})$. Therefore, using the terminology introduced in the previous section, the probability density function f_X is a version of the Radon-Nikodym derivative $d\mathcal{L}_X/d\text{Leb}$ of $\mathcal{L}_X$ with respect to Leb , so that $f_X = d\mathcal{L}_X/d\text{Leb}$ almost everywhere.

Proposition 6.22. Consider a random variable $X : \Omega \rightarrow \mathbb{R}$ that has a probability density function $f_X : \mathbb{R} \rightarrow [0, \infty]$. Furthermore, consider a Borel function $g_X : \mathbb{R} \rightarrow [0, \infty]$ such that $\text{Leb}(\{f_X \neq g_X\}) = 0$. Because these two functions are non-negative and $\text{Leb}(\{f_X \mathbb{I}_B \neq g_X \mathbb{I}_B\}) = 0$, we know that $\text{Leb}(f_X \mathbb{I}_B) = \text{Leb}(g_X \mathbb{I}_B)$, which implies that the random variable X also has a probability density function g_X .

Proposition 6.23. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable $X : \Omega \rightarrow \mathbb{R}$ with a probability density function $f_X : \mathbb{R} \rightarrow [0, \infty]$. If $\mathcal{L}_X = (f_X \text{Leb})$ is the law of X , recall that $f_X = d\mathcal{L}_X/d\text{Leb}$ almost everywhere. If $h : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel function, the fact that $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \text{Leb})$ is a measure space implies that $h \in \mathcal{L}^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L}_X)$ if and only if $hf_X \in \mathcal{L}^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \text{Leb})$. Furthermore, in that case,

$$\int_{\mathbb{R}} h \, d\mathcal{L}_X = \mathcal{L}_X(h) = \text{Leb}(hf_X) = \int_{\mathbb{R}} hf_X \, d\text{Leb}.$$

Definition 6.15. Consider a measure space (S, Σ, μ) . For every $p \in [1, \infty)$, the set $\mathcal{L}^p(S, \Sigma, \mu)$ contains exactly each Σ -measurable function $f : S \rightarrow \mathbb{R}$ such that $\mu(|f|^p) < \infty$. If $f \in \mathcal{L}^p(S, \Sigma, \mu)$, the p -norm $\|f\|_p$ of the function f is given by $\|f\|_p = \mu(|f|^p)^{1/p}$.

Proposition 6.24. Suppose that $p > 1$ and $p^{-1} + q^{-1} = 1$. Furthermore, suppose $f, g \in \mathcal{L}^p(S, \Sigma, \mu)$ and $h \in \mathcal{L}^q(S, \Sigma, \mu)$. Hölder's inequality states that $fh \in \mathcal{L}^1(S, \Sigma, \mu)$ and $\mu(|fh|) \leq \|f\|_p \|h\|_q$. Minkowski's inequality states that $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

Proof. First, note that $fh \in \mathcal{L}^1(S, \Sigma, \mu)$ and $\mu(|fh|) \leq \|f\|_p \|h\|_q$ if and only if $|f||h| \in \mathcal{L}^1(S, \Sigma, \mu)$ and $\mu(|f||h|) \leq \|f\|_p \|h\|_q$. Therefore, we only need to consider the case where f and h are non-negative. In that case, if $\mu(f^p) = 0$, then $0 = \mu(\{f^p > 0\}) = \mu(\{f \neq 0\}) \geq \mu(\{fh \neq 0\})$ and $\mu(fh) = 0$, so that Hölder's inequality is trivial.

Consider the case where f and h are non-negative and $0 < \mu(f^p) < \infty$. Let $\mathbb{P} : \Sigma \rightarrow [0, 1]$ be given by

$$\mathbb{P}(A) = \frac{(f^p \mu)(A)}{\mu(f^p)} = \frac{\mu(f^p; A)}{\mu(f^p)} = \frac{\mu(f^p \mathbb{I}_A)}{\mu(f^p)} = \mu\left(\frac{f^p}{\mu(f^p)} \mathbb{I}_A\right) = \mu\left(\frac{f^p}{\mu(f^p)}; A\right).$$

The function $\mathbb{P}$ is a probability measure on (S, Σ) . Clearly, $\mathbb{P}(S) = 1$ and $\mathbb{P}(\emptyset) = 0$. Because $(f^p \mu)$ is a measure on (S, Σ) , for any sequence $(A_n \in \Sigma \mid n \in \mathbb{N})$ such that $A_n \cap A_m = \emptyset$ for $n \neq m$,

$$\mathbb{P}\left(\bigcup_n A_n\right) = \frac{(f^p \mu)(\bigcup_n A_n)}{\mu(f^p)} = \frac{\sum_n (f^p \mu)(A_n)}{\mu(f^p)} = \sum_n \frac{(f^p \mu)(A_n)}{\mu(f^p)} = \sum_n \mathbb{P}(A_n).$$

Note that the probability measure $\mathbb{P}$ has density $f^p/\mu(f^p)$ relative to μ , so that $d\mathbb{P}/d\mu = f^p/\mu(f^p)$ almost everywhere. Therefore, if $v : S \rightarrow \mathbb{R}$ is a Σ -measurable function, then $v \in \mathcal{L}^1(S, \Sigma, \mathbb{P})$ if and only if $vf^p/\mu(f^p) \in \mathcal{L}^1(S, \Sigma, \mu)$. In that case,

$$\int_S v \, d\mathbb{P} = \mathbb{P}(v) = \mu\left(\frac{vf^p}{\mu(f^p)}\right) = \int_S \frac{vf^p}{\mu(f^p)} \, d\mu.$$

Consider the Σ -measurable function $u : S \rightarrow [0, \infty]$ given by

$$u(s) = \begin{cases} \frac{h(s)}{f(s)^{p-1}}, & \text{if } f(s) > 0, \\ 0, & \text{if } f(s) = 0. \end{cases}$$

By inspecting the pointwise definition of uf^p ,

$$\mathbb{P}(u) = \mu\left(\frac{uf^p}{\mu(f^p)}\right) = \frac{\mu(uf^p)}{\mu(f^p)} = \frac{\mu(hf)}{\mu(f^p)}.$$

Similarly, by inspecting the pointwise definition of $u^q f^p$ and using the fact that $q(p-1) = p$,

$$\mathbb{P}(u^q) = \mu\left(\frac{u^q f^p}{\mu(f^p)}\right) = \frac{\mu(u^q f^p)}{\mu(f^p)} = \frac{\mu(h^q \mathbb{I}_{\{f>0\}})}{\mu(f^p)}.$$

Suppose $\mathbb{P}(u) = \infty$. In that case, $\mathbb{P}(u) = \mathbb{P}(u \mathbb{I}_{\{u<1\}}) + \mathbb{P}(u \mathbb{I}_{\{u \geq 1\}}) = \infty$. The fact that $\mathbb{P}(u \mathbb{I}_{\{u<1\}}) \leq \mathbb{P}(\mathbb{I}_{\{u<1\}}) = \mathbb{P}(\{u < 1\}) \leq 1$ implies that $\mathbb{P}(u \mathbb{I}_{\{u \geq 1\}}) = \infty$. Consequently, $\mathbb{P}(u^q) \geq \mathbb{P}(u^q \mathbb{I}_{\{u \geq 1\}}) \geq \mathbb{P}(u \mathbb{I}_{\{u \geq 1\}}) = \infty$, so that $\mathbb{P}(u^q) \geq \mathbb{P}(u)^q$. In contrast, suppose $\mathbb{P}(u) < \infty$. Consider the convex function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ given by $\phi(x) = |x|^q$. Jensen's inequality also guarantees that $\mathbb{P}(u^q) \geq \mathbb{P}(u)^q$. Therefore,

$$\frac{\mu(h^q \mathbb{I}_{\{f>0\}})}{\mu(f^p)} \geq \frac{\mu(hf)^q}{\mu(f^p)^q}.$$

By multiplying both sides of the previous inequality by $\mu(f^p)^q$,

$$\mu(h^q \mathbb{I}_{\{f>0\}}) \frac{\mu(f^p)^q}{\mu(f^p)} = \mu(h^q \mathbb{I}_{\{f>0\}}) \mu(f^p)^{q-1} \geq \mu(hf)^q.$$

Because $\mu(h^q) \geq \mu(h^q \mathbb{I}_{\{f>0\}})$,

$$\mu(h^q) \mu(f^p)^{q-1} \geq \mu(hf)^q.$$

From the definition of norm and using the fact that $p(q-1) = q$,

$$\|h\|_q^q \|f\|_p^q \geq \mu(hf)^q,$$

which completes the proof of Hölder's inequality.

In order to show Minkowski's inequality, recall that $|f+g| \leq |f| + |g|$. Therefore,

$$|f+g|^p = |f+g||f+g|^{p-1} \leq |f||f+g|^{p-1} + |g||f+g|^{p-1}.$$

By integrating both sides of the previous inequality with respect to μ and employing Hölder's inequality,

$$\mu(|f+g|^p) \leq \mu(|f||f+g|^{p-1}) + \mu(|g||f+g|^{p-1}) \leq \|f\|_p \|f+g|^{p-1}\|_q + \|g\|_p \|f+g|^{p-1}\|_q.$$

Note that $\|f + g\|_q^{p-1} = \mu(|f + g|^{p-1})^{1/q} = \mu(|f + g|^p)^{1/q} < \infty$ because $q(p-1) = p$. Therefore,

$$\mu(|f + g|^p) \leq (\|f\|_p + \|g\|_p)\mu(|f + g|^p)^{1/q}.$$

By dividing both sides of the previous inequality by $\mu(|f + g|^p)^{1/q}$ and using the fact that $p^{-1} = 1 - q^{-1}$,

$$\|f + g\|_p = \mu(|f + g|^p)^{1/p} \leq \|f\|_p + \|g\|_p.$$

□

7 Strong law

Proposition 7.1. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable $Y \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. If X and Y are independent, then $XY \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.

Proof. First, suppose that X and Y are non-negative and let α_n denote the n -th staircase function. For any $n \in \mathbb{N}$, consider the simple function $X_n = \alpha_n \circ X = \sum_{k_x=1}^{m_x} a_{k_x} \mathbb{I}_{A_{k_x}}$, where $a_1, \dots, a_{m_x} \in [0, n]$ are distinct and $A_1, \dots, A_{m_x} \in \mathcal{F}$ partition Ω . Similarly, consider the simple function $Y_n = \alpha_n \circ Y = \sum_{k_y=1}^{m_y} b_{k_y} \mathbb{I}_{B_{k_y}}$, where $b_1, \dots, b_{m_y} \in [0, n]$ are distinct and $B_1, \dots, B_{m_y} \in \mathcal{F}$ partition Ω . In that case,

$$\begin{aligned} \mathbb{E}(X_n) &= \mathbb{E} \left(\sum_{k_x=1}^{m_x} a_{k_x} \mathbb{I}_{A_{k_x}} \right) = \sum_{k_x=1}^{m_x} a_{k_x} \mathbb{P}(A_{k_x}), \\ \mathbb{E}(Y_n) &= \mathbb{E} \left(\sum_{k_y=1}^{m_y} b_{k_y} \mathbb{I}_{B_{k_y}} \right) = \sum_{k_y=1}^{m_y} b_{k_y} \mathbb{P}(B_{k_y}). \end{aligned}$$

Because $X_n \uparrow X$, the monotone-convergence theorem guarantees that $\mathbb{E}(X_n) \uparrow \mathbb{E}(X)$. Similarly, because $Y_n \uparrow Y$, the monotone-convergence theorem guarantees that $\mathbb{E}(Y_n) \uparrow \mathbb{E}(Y)$. Because $\mathbb{E}(X) < \infty$ and $\mathbb{E}(Y) < \infty$, we also know that $\mathbb{E}(X_n)\mathbb{E}(Y_n) \uparrow \mathbb{E}(X)\mathbb{E}(Y)$. By distributing terms and using the fact that $\mathbb{I}_{A_{k_x}} \mathbb{I}_{B_{k_y}} = \mathbb{I}_{A_{k_x} \cap B_{k_y}}$,

$$\mathbb{E}(X_n Y_n) = \mathbb{E} \left[\left(\sum_{k_x=1}^{m_x} a_{k_x} \mathbb{I}_{A_{k_x}} \right) \left(\sum_{k_y=1}^{m_y} b_{k_y} \mathbb{I}_{B_{k_y}} \right) \right] = \mathbb{E} \left(\sum_{k_x=1}^{m_x} \sum_{k_y=1}^{m_y} a_{k_x} b_{k_y} \mathbb{I}_{A_{k_x} \cap B_{k_y}} \right) = \sum_{k_x=1}^{m_x} \sum_{k_y=1}^{m_y} a_{k_x} b_{k_y} \mathbb{P}(A_{k_x} \cap B_{k_y}).$$

Recall that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel function and $Z : \Omega \rightarrow \mathbb{R}$ is a random variable, then

$$\sigma(f \circ Z) = \{(f \circ Z)^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R})\} = \{Z^{-1}(f^{-1}(B)) \mid B \in \mathcal{B}(\mathbb{R})\} \subseteq \{Z^{-1}(A) \mid A \in \mathcal{B}(\mathbb{R})\} = \sigma(Z).$$

Recall that X and Y are independent if and only if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ for every $A \in \sigma(X)$ and $B \in \sigma(Y)$. Therefore, X_n and Y_n are also independent. Because $A_{k_x} = X_n^{-1}(\{a_{k_x}\})$, we know that $A_{k_x} \in \sigma(X_n)$. Because $B_{k_y} = Y_n^{-1}(\{b_{k_y}\})$, we know that $B_{k_y} \in \sigma(Y_n)$. Therefore,

$$\mathbb{E}(X_n Y_n) = \sum_{k_x=1}^{m_x} \sum_{k_y=1}^{m_y} a_{k_x} b_{k_y} \mathbb{P}(A_{k_x}) \mathbb{P}(B_{k_y}) = \left(\sum_{k_x=1}^{m_x} a_{k_x} \mathbb{P}(A_{k_x}) \right) \left(\sum_{k_y=1}^{m_y} b_{k_y} \mathbb{P}(B_{k_y}) \right) = \mathbb{E}(X_n) \mathbb{E}(Y_n).$$

Since $X_n \uparrow X$ and $Y_n \uparrow Y$ imply $X_n Y_n \uparrow XY$, the monotone-convergence theorem guarantees that $\mathbb{E}(X_n Y_n) \uparrow \mathbb{E}(XY)$. Since $\mathbb{E}(X_n Y_n) = \mathbb{E}(X_n) \mathbb{E}(Y_n)$, taking the limit when $n \rightarrow \infty$ shows that $\mathbb{E}(XY) = \mathbb{E}(X) \mathbb{E}(Y) < \infty$, which completes the proof when X and Y are non-negative.

Finally, let $X = X^+ - X^-$, where $X^+ \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $X^- \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ are non-negative. Analogously, let $Y = Y^+ - Y^-$. Because $|X|$ and $|Y|$ are independent non-negative random variables, the previous step guarantees that $\mathbb{E}(|XY|) = \mathbb{E}(|X|) \mathbb{E}(|Y|) < \infty$, so that $XY \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Therefore,

$$\mathbb{E}(XY) = \mathbb{E}((X^+ - X^-)(Y^+ - Y^-)) = \mathbb{E}(X^+ Y^+) - \mathbb{E}(X^+ Y^-) - \mathbb{E}(X^- Y^+) + \mathbb{E}(X^- Y^-).$$

Since X and Y are independent, each pair of variables inside an expectation above is independent. Therefore,

$$\mathbb{E}(XY) = \mathbb{E}(X^+) \mathbb{E}(Y^+) - \mathbb{E}(X^+) \mathbb{E}(Y^-) - \mathbb{E}(X^-) \mathbb{E}(Y^+) + \mathbb{E}(X^-) \mathbb{E}(Y^-) = (\mathbb{E}(X^+) - \mathbb{E}(X^-))(\mathbb{E}(Y^+) - \mathbb{E}(Y^-)),$$

which completes the proof. □

Proposition 7.2. Consider the random variables $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. If X and Y are independent, the previous result guarantees that $\text{Cov}(X, Y) = 0$ and $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Proposition 7.3. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a random variable $X : \Omega \rightarrow \mathbb{R}$, and the random variables $Y_1, \dots, Y_n$, where $n \in \mathbb{N}^+$. Suppose that $X, Y_1, \dots, Y_n$ are independent. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a Borel function and $Z : \Omega \rightarrow \mathbb{R}$ is a random variable given by $Z(\omega) = f(Y_1(\omega), \dots, Y_n(\omega))$, then X and Z are independent.

Proof. First, recall that a previous result establishes that Z is $\sigma(\{Y_1, \dots, Y_n\})$ -measurable, so that

$$\sigma(Z) \subseteq \sigma(\{Y_1, \dots, Y_n\}) = \sigma(\{Y_i^{-1}(B) \mid i \in \{1, \dots, n\}, B \in \mathcal{B}(\mathbb{R})\}) = \sigma\left(\bigcup_{i=1}^n \sigma(Y_i)\right).$$

Therefore, if $\sigma(X)$ and $\sigma(\{Y_1, \dots, Y_n\})$ are independent, then X and Z are independent.

Consider the set $\mathcal{I} = \{\cap_{i=1}^n A_i \mid (A_1, \dots, A_n) \in \sigma(Y_1) \times \dots \times \sigma(Y_n)\}$. If $B \in \mathcal{I}$ and $C \in \mathcal{I}$, then $B = \cap_{i=1}^n A_i$ and $C = \cap_{i=1}^n A'_i$, where $A_i \in \sigma(Y_i)$ and $A'_i \in \sigma(Y_i)$ for every $i \in \{1, \dots, n\}$. Because

$$B \cap C = \left(\bigcap_{i=1}^n A_i\right) \cap \left(\bigcap_{i=1}^n A'_i\right) = \bigcap_{i=1}^n (A_i \cap A'_i)$$

and $(A_i \cap A'_i) \in \sigma(Y_i)$ for every $i \in \{1, \dots, n\}$, we know that $(B \cap C) \in \mathcal{I}$. Therefore, $\mathcal{I}$ is a π -system on Ω .

Let $\mathcal{J} = \sigma(X)$ and note that $\mathcal{J}$ is also a π -system on Ω . Consider a set $(\cap_{i=1}^n A_i) \in \mathcal{I}$, where $A_i \in \sigma(Y_i)$ for every $i \in \{1, \dots, n\}$, and a set $B \in \mathcal{J}$. Since $X, Y_1, \dots, Y_n$ are independent,

$$\mathbb{P}\left(\left(\bigcap_{i=1}^n A_i\right) \cap B\right) = \left(\prod_{i=1}^n \mathbb{P}(A_i)\right) \mathbb{P}(B) = \mathbb{P}\left(\bigcap_{i=1}^n A_i\right) \mathbb{P}(B),$$

which implies that $\mathcal{I}$ and $\mathcal{J}$ are independent. Because $\sigma(\mathcal{I})$ and $\sigma(\mathcal{J})$ are then independent from a previous result and $\sigma(\mathcal{J}) = \sigma(X)$, the proof will be complete if $\sigma(\mathcal{I}) = \sigma(\{Y_1, \dots, Y_n\})$, which we will now show.

Note that $\Omega \in \sigma(Y_i)$ for every $i \in \{1, \dots, n\}$, which implies $\sigma(Y_i) \subseteq \mathcal{I}$ for every $i \in \{1, \dots, n\}$. Therefore, $\cup_{i=1}^n \sigma(Y_i) \subseteq \mathcal{I}$ and $\sigma(\cup_{i=1}^n \sigma(Y_i)) = \sigma(\{Y_1, \dots, Y_n\}) \subseteq \sigma(\mathcal{I})$.

Consider a set $(\cap_{i=1}^n A_i) \in \mathcal{I}$, where $A_i \in \sigma(Y_i)$ for every $i \in \{1, \dots, n\}$. Clearly, $A_i \in \cup_{j=1}^n \sigma(Y_j)$. Because $\sigma(\cup_{j=1}^n \sigma(Y_j)) = \sigma(\{Y_1, \dots, Y_n\})$ is a σ -algebra, we know that $(\cap_{i=1}^n A_i) \in \sigma(\{Y_1, \dots, Y_n\})$, which implies $\mathcal{I} \subseteq \sigma(\{Y_1, \dots, Y_n\})$ and $\sigma(\mathcal{I}) \subseteq \sigma(\{Y_1, \dots, Y_n\})$. □

Theorem 7.1 (Strong law of large numbers for a finite fourth moment). Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence of independent random variables $(X_k : \Omega \rightarrow \mathbb{R} \mid k \in \mathbb{N}^+)$. Furthermore, suppose $\mathbb{E}(X_k) = 0$ and $\mathbb{E}(X_k^4) \leq K$ for some $K \in [0, \infty)$, for every $k \in \mathbb{N}^+$. In that case,

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k = 0\right) = 1.$$

Proof. Consider the random variable $S_n = \sum_{k=1}^n X_k$. From the multinomial theorem,

$$S_n^4 = \left(\sum_{k=1}^n X_k\right)^4 = \sum_{(k_1, \dots, k_n) \in I_4^{(n)}} \frac{4!}{k_1! \dots k_n!} \prod_{t=1}^n X_t^{k_t},$$

where $I_p^{(n)} = \{(k_1, \dots, k_n) \mid k_i \in \{0, \dots, p\} \text{ for every } i \in \{1, \dots, n\} \text{ and } \sum_i k_i = p\}$. By the linearity of expectation,

$$\mathbb{E}(S_n^4) = \sum_{(k_1, \dots, k_n) \in I_4^{(n)}} \frac{4!}{k_1! \dots k_n!} \mathbb{E}\left(\prod_{t=1}^n X_t^{k_t}\right).$$

From the restrictions imposed on $(k_1, \dots, k_n) \in I_4^{(n)}$, the expectation $\mathbb{E}\left(\prod_{t=1}^n X_t^{k_t}\right)$ can be written as either $\mathbb{E}(X_i^4)$, $\mathbb{E}(X_i^3 X_j)$, $\mathbb{E}(X_i^2 X_j^2)$, $\mathbb{E}(X_i^2 X_j X_k)$, or $\mathbb{E}(X_i X_j X_k X_l)$, where $i, j, k, l \in \{1, \dots, n\}$ are distinct indices.

Consider the expectation $\mathbb{E}(X_i^3 X_j)$. Because X_i and X_j are independent, X_i^3 and X_j are independent. By the monotonicity of the norm, $X_i^3 \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $X_j \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Therefore, $\mathbb{E}(X_i^3 X_j) = \mathbb{E}(X_i^3) \mathbb{E}(X_j) = 0$.

Consider the expectation $\mathbb{E}(X_i^2 X_j X_k)$. Because X_i^2, X_j, X_k are independent, $X_i^2 X_j$ and X_k are independent. By the monotonicity of the norm, $X_i^2 \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, $X_j \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, and $X_k \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Due to independence, $X_i^2 X_j \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Therefore, $\mathbb{E}(X_i^2 X_j X_k) = \mathbb{E}(X_i^2 X_j) \mathbb{E}(X_k) = 0$.

Consider the expectation $\mathbb{E}(X_i X_j X_k X_l)$. Because X_i, X_j, X_k, X_l are independent, $X_i X_j X_k$ and X_l are independent. By the monotonicity of the norm, $X_i, X_j, X_k, X_l \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Because X_i and X_j are independent, $X_i X_j \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Because $X_i X_j$ and X_k are independent, $X_i X_j X_k \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Therefore, $\mathbb{E}(X_i X_j X_k X_l) = \mathbb{E}(X_i X_j X_k) \mathbb{E}(X_l) = 0$.

These observations allow rewriting the expectation $\mathbb{E}(S_n^4)$ as

$$\mathbb{E}(S_n^4) = \sum_{i=1}^n \mathbb{E}(X_i^4) + 6 \sum_{i=1}^{n-1} \sum_{j=i+1}^n \mathbb{E}(X_i^2 X_j^2).$$

For every $k \in \mathbb{N}^+$, recall that $\|X_k\|_2 = \mathbb{E}(X_k^2)^{1/2} \leq \mathbb{E}(X_k^4)^{1/4} = \|X_k\|_4$. Therefore, $\mathbb{E}(X_k^2) \leq \mathbb{E}(X_k^4)^{1/2} \leq K^{1/2}$. For every $i \neq j$, X_i^2 and X_j^2 are independent and $X_i^2, X_j^2 \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ by the monotonicity of the norm. Therefore,

$$\mathbb{E}(X_i^2 X_j^2) = \mathbb{E}(X_i^2) \mathbb{E}(X_j^2) \leq \mathbb{E}(X_i^4)^{1/2} \mathbb{E}(X_j^4)^{1/2} \leq K.$$

Consequently,

$$\mathbb{E}(S_n^4) \leq \sum_{i=1}^n K + 6 \sum_{i=1}^{n-1} \sum_{j=i+1}^n K = nK + 3n(n-1)K = K(3n^2 - 2n) \leq 3Kn^2.$$

Because $\mathbb{E}(S_n^4/n^4) \leq 3K/n^2$ for every $n \in \mathbb{N}^+$,

$$\sum_{n=1}^k \mathbb{E} \left(\frac{S_n^4}{n^4} \right) \leq 3K \sum_{n=1}^k \frac{1}{n^2}.$$

Because the summation on the right of the inequality above converges to a real number when $k \rightarrow \infty$,

$$\sum_n \mathbb{E} \left(\frac{S_n^4}{n^4} \right) < \infty.$$

Since S_n^4/n^4 is a non-negative random variable for every $n \in \mathbb{N}^+$, a previous result guarantees that

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{S_n^4}{n^4} = 0 \right) = \mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = 0 \right) = \mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k = 0 \right) = 1.$$

□

Proposition 7.4. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence of independent random variables $(X_k : \Omega \rightarrow \mathbb{R} \mid k \in \mathbb{N}^+)$. Furthermore, suppose $\mathbb{E}(X_k) = \mu$ and $\mathbb{E}(X_k^4) \leq K$ for some $\mu \in \mathbb{R}$ and $K \in [0, \infty)$, for every $k \in \mathbb{N}^+$. As a corollary, the strong law of large numbers for a finite fourth moment guarantees that

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k = \mu \right) = 1.$$

Proof. For every $k \in \mathbb{N}^+$, let $Y_k = X_k - \mu$. By the monotonicity of the norm, $X_k \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, so that $\mathbb{E}(Y_k) = \mathbb{E}(X_k) - \mu = 0$. Furthermore, $(Y_k : \Omega \rightarrow \mathbb{R} \mid k \in \mathbb{N}^+)$ is a sequence of independent random variables, since $\sigma(Y_k) \subseteq \sigma(X_k)$. Using Minkowski's inequality and the fact that $X_k \in \mathcal{L}^4(\Omega, \mathcal{F}, \mathbb{P})$,

$$\infty > \|X_k\|_4 + |\mu| = \|X_k\|_4 + \|-\mu \mathbb{I}_\Omega\|_4 \geq \|X_k - \mu \mathbb{I}_\Omega\|_4 = \|X_k - \mu\|_4 = \|Y_k\|_4.$$

Therefore, $\mathbb{E}(Y_k^4) \leq K'$ for some $K' \in [0, \infty)$. Using the strong law of large numbers for a finite fourth moment,

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n Y_k = 0 \right) = \mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k = \mu \right) = 1$$

□

Proposition 7.5 (Chebyshev's inequality). Consider a random variable $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mu = \mathbb{E}(X)$. For $c \geq 0$,

$$\text{Var}(X) = \mathbb{E}(|X - \mu|^2) \geq c^2 \mathbb{P}(|X - \mu| \geq c),$$

which is a consequence of Markov's inequality.

Example 7.1. Consider the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence of independent and identically distributed random variables $(X_k : \Omega \rightarrow \{0, 1\} \mid k \in \mathbb{N}^+)$. Let $p = \mathbb{E}(X_k) = \mathbb{E}(\mathbb{I}_{\{X_k=1\}}) = \mathbb{P}(X_k = 1)$. Since $X_k^2 = X_k$, $X_k \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $\text{Var}(X_k) = \mathbb{E}(X_k^2) - \mathbb{E}(X_k)^2 = p - p^2$, so that $\text{Var}(X_k) \leq 1/4$.

Let $S_n = \sum_{k=1}^n X_k$. so that $\mathbb{E}(S_n) = \sum_{k=1}^n \mathbb{E}(X_k) = np$. Due to independence,

$$\text{Var}(S_n) = \text{Var}\left(\sum_{k=1}^n X_k\right) = \sum_{k=1}^n \text{Var}(X_k) = \sum_{k=1}^n (p - p^2) = n(p - p^2) \leq \frac{n}{4}.$$

For any $Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $a \in \mathbb{R}$, $\text{Var}(aY) = \mathbb{E}((aY)^2) - \mathbb{E}(aY)^2 = a^2 \mathbb{E}(Y^2) - a^2 \mathbb{E}(Y)^2 = a^2 \text{Var}(Y)$. Therefore, $\mathbb{E}(S_n/n) = p$ and $\text{Var}(S_n/n) \leq 1/4n$. Using Chebyshev's inequality, for any $\delta > 0$,

$$\mathbb{P}\left(\left|\left(\frac{1}{n} \sum_{k=1}^n X_k\right) - p\right| \geq \delta\right) \leq \frac{1}{4n\delta^2}.$$

8 Product measure

Consider a measurable space (S_1, Σ_1) and a measurable space (S_2, Σ_2) . Let $S = S_1 \times S_2$.

Proposition 8.1. Consider the functions $\rho_1 : S \rightarrow S_1$ and $\rho_2 : S \rightarrow S_2$ given by $\rho_1(s_1, s_2) = s_1$ and $\rho_2(s_1, s_2) = s_2$. For $B_1 \in \Sigma_1$ and $B_2 \in \Sigma_2$, let

$$\begin{aligned} \rho_1^{-1}(B_1) &= \{(s_1, s_2) \in S \mid \rho_1(s_1, s_2) \in B_1\} = \{(s_1, s_2) \in S \mid s_1 \in B_1\} = B_1 \times S_2, \\ \rho_2^{-1}(B_2) &= \{(s_1, s_2) \in S \mid \rho_2(s_1, s_2) \in B_2\} = \{(s_1, s_2) \in S \mid s_2 \in B_2\} = S_1 \times B_2. \end{aligned}$$

For $i \in \{1, 2\}$, let $\mathcal{A}_i = \{\rho_i^{-1}(B_i) \mid B_i \in \Sigma_i\}$. In that case, $\mathcal{A}_i$ is a σ -algebra on S .

Proof. First, note that $\rho_i^{-1}(S_i) = S$ and $S_i \in \Sigma_i$. Therefore, $S \in \mathcal{A}_i$. Consider an element $\rho_i^{-1}(B_i) \in \mathcal{A}_i$. Note that $B_i^c \in \Sigma_i$ and $\rho_i^{-1}(B_i^c) = \rho_i^{-1}(B_i)^c$. Therefore, $\rho_i^{-1}(B_i)^c \in \mathcal{A}_i$. Finally, consider a sequence of sets $(\rho_i^{-1}(B_{i,j}) \in \mathcal{A}_i \mid j \in \mathbb{N})$. Note that $\cup_j B_{i,j} \in \Sigma_i$ and $\rho_i^{-1}(\cup_j B_{i,j}) = \cup_j \rho_i^{-1}(B_{i,j})$. Therefore, $\cup_j \rho_i^{-1}(B_{i,j}) \in \mathcal{A}_i$. $\square$

Definition 8.1. Considering the previous result, let $\sigma(\rho_1)$ and $\sigma(\rho_2)$ denote the σ -algebras on S given by

$$\begin{aligned} \sigma(\rho_1) &= \mathcal{A}_1 = \{\rho_1^{-1}(B_1) \mid B_1 \in \Sigma_1\} = \{B_1 \times S_2 \mid B_1 \in \Sigma_1\}, \\ \sigma(\rho_2) &= \mathcal{A}_2 = \{\rho_2^{-1}(B_2) \mid B_2 \in \Sigma_2\} = \{S_1 \times B_2 \mid B_2 \in \Sigma_2\}. \end{aligned}$$

Definition 8.2. The product Σ between the σ -algebras Σ_1 and Σ_2 is a σ -algebra on S denoted by $\Sigma_1 \times \Sigma_2$ but defined by

$$\Sigma = \Sigma_1 \times \Sigma_2 = \sigma(\{\rho_1, \rho_2\}) = \sigma(\sigma(\rho_1) \cup \sigma(\rho_2)),$$

which should not be confused with the Cartesian product between Σ_1 and Σ_2 .

Proposition 8.2. If $\Sigma = \Sigma_1 \times \Sigma_2$ and $\mathcal{I} = \{B_1 \times B_2 \mid B_1 \in \Sigma_1 \text{ and } B_2 \in \Sigma_2\}$, then $\sigma(\mathcal{I}) = \Sigma$.

Proof. For any $B_1 \in \Sigma_1$ and $B_2 \in \Sigma_2$, note that

$$B_1 \times B_2 = (B_1 \cap S_1) \times (S_2 \cap B_2) = (B_1 \times S_2) \cap (S_1 \times B_2).$$

Suppose $B_1 \times B_2 \in \mathcal{I}$ and $B'_1 \times B'_2 \in \mathcal{I}$. In that case, $(B_1 \times B_2) \cap (B'_1 \times B'_2) = (B_1 \cap B'_1) \times (B_2 \cap B'_2)$. Because $(B_1 \cap B'_1) \in \Sigma_1$ and $(B_2 \cap B'_2) \in \Sigma_2$, this implies that $\mathcal{I}$ is a π -system on S .

For any $B_1 \times B_2 \in \mathcal{I}$, we know that $B_1 \times B_2 \in \Sigma$ because $(B_1 \times S_2) \in \sigma(\rho_1)$ and $(S_1 \times B_2) \in \sigma(\rho_2)$. Since Σ is a σ -algebra, $\sigma(\mathcal{I}) \subseteq \Sigma$. For any $B_1 \in \Sigma_1$ and $B_2 \in \Sigma_2$, we know that $B_1 \times S_2 \in \mathcal{I}$ and $S_1 \times B_2 \in \mathcal{I}$. Therefore, $\sigma(\rho_1) \cup \sigma(\rho_2) \subseteq \mathcal{I}$. Because $\sigma(\mathcal{I})$ is a σ -algebra, $\Sigma \subseteq \sigma(\mathcal{I})$. $\square$

Proposition 8.3. Let $\mathcal{I}_2$ be a set of subsets of S_2 such that $S_2 \in \mathcal{I}_2$ and $\sigma(\mathcal{I}_2) = \Sigma_2$. If $\Sigma = \Sigma_1 \times \Sigma_2$ and $\mathcal{I} = \{B_1 \times B_2 \mid B_1 \in \Sigma_1 \text{ and } B_2 \in \mathcal{I}_2\}$, then $\sigma(\mathcal{I}) = \Sigma$.

Proof. For every $B_1 \in \Sigma_1$, let $\mathcal{S}_{B_1} = \{B_2 \in \Sigma_2 \mid B_1 \times B_2 \in \sigma(\mathcal{I})\}$. For every $B_1 \in \Sigma_1$ and $B_2 \in \mathcal{I}_2$, note that $B_1 \times B_2 \in \sigma(\mathcal{I})$, so that $\mathcal{I}_2 \subseteq \mathcal{S}_{B_1}$. Suppose that $\mathcal{S}_{B_1}$ is a σ -algebra on S_2 for every $B_1 \in \Sigma_1$, which we will show soon. In that case, $\Sigma_2 = \sigma(\mathcal{I}_2) \subseteq \mathcal{S}_{B_1}$, so that $\mathcal{S}_{B_1} = \Sigma_2$ for every $B_1 \in \Sigma_1$. In other words, $B_1 \times B_2 \in \sigma(\mathcal{I})$ for every $B_1 \in \Sigma_1$ and $B_2 \in \Sigma_2$. Our previous result then implies that $\Sigma \subseteq \sigma(\mathcal{I})$, so that $\sigma(\mathcal{I}) = \Sigma$.

For every $B_1 \in \Sigma_1$, the following shows that $\mathcal{S}_{B_1}$ is a σ -algebra on S_2 :

- $B_1 \times S_2 \in \sigma(\mathcal{I})$, since $S_2 \in \mathcal{I}_2$.
- If $B_2 \in \Sigma_2$ and $(B_1 \times B_2) \in \sigma(\mathcal{I})$, then $(B_1 \times B_2^c) \in \sigma(\mathcal{I})$, since

$$B_1 \times B_2^c = B_1 \times (S_2 \setminus B_2) = (B_1 \times S_2) \setminus (B_1 \times B_2) = (B_1 \times S_2) \cap (B_1 \times B_2)^c.$$

- If $(B_{2,n} \in \Sigma_2 \mid n \in \mathbb{N})$ is a sequence and $B_1 \times B_{2,n} \in \sigma(\mathcal{I})$ for every $n \in \mathbb{N}$, then $B_1 \times \bigcup_n B_{2,n} \in \sigma(\mathcal{I})$, since

$$B_1 \times \bigcup_n B_{2,n} = \bigcup_n (B_1 \times B_{2,n}).$$

□

Proposition 8.4. Consider a measurable space (S_1, Σ_1) and a measurable space (S_2, Σ_2) . Furthermore, consider the measurable space (S, Σ) , where $S = S_1 \times S_2$ and $\Sigma = \Sigma_1 \times \Sigma_2$. Let $\mathcal{H}$ denote a set that contains exactly each bounded Σ -measurable function $f : S \rightarrow \mathbb{R}$ such that, for every $s_1 \in S_1$, there is a Σ_2 -measurable function $f_{s_1} : S_2 \rightarrow \mathbb{R}$ with $f_{s_1}(s_2) = f(s_1, s_2)$ for every $s_2 \in S_2$ and, for every $s_2 \in S_2$, there is a Σ_1 -measurable function $f_{s_2} : S_1 \rightarrow \mathbb{R}$ with $f_{s_2}(s_1) = f(s_1, s_2)$ for every $s_1 \in S_1$. In that case, $\mathcal{H}$ contains every bounded Σ -measurable function on S , so that $\mathcal{H} = \text{b}\Sigma$.

Proof. Note that the set of bounded Σ -measurable functions $\text{b}\Sigma$ is a vector space over the field $\mathbb{R}$ when scalar multiplication and addition are performed pointwise. Because $\mathcal{H} \subseteq \text{b}\Sigma$, showing that $\mathcal{H}$ is a vector space only requires showing that $\mathcal{H}$ is non-empty and closed under scalar multiplication and addition. For every $s_1 \in S_1$ and $s_2 \in S_2$, let $f = \mathbb{I}_S$, $f_{s_1} = \mathbb{I}_{S_2}$, and $f_{s_2} = \mathbb{I}_{S_1}$, so that $\mathbb{I}_S(s_1, s_2) = \mathbb{I}_{S_2}(s_2) = \mathbb{I}_{S_1}(s_1) = 1$. Clearly, $f \in \mathcal{H}$. Now suppose $f \in \mathcal{H}$ and $a \in \mathbb{R}$. Note that $af \in \text{b}\Sigma$. For every $s_1 \in S_1$ and $s_2 \in S_2$, also note that af_{s_1} is Σ_2 -measurable, af_{s_2} is Σ_1 -measurable, and $(af)(s_1, s_2) = (af_{s_1})(s_2) = (af_{s_2})(s_1)$. Therefore, $af \in \mathcal{H}$. Finally, suppose that $g, h \in \mathcal{H}$. Note that $g + h \in \text{b}\Sigma$. For every $s_1 \in S_1$ and $s_2 \in S_2$, note that $g_{s_1} + h_{s_1}$ is Σ_2 -measurable, $g_{s_2} + h_{s_2}$ is Σ_1 -measurable, and $(g + h)(s_1, s_2) = (g_{s_1} + h_{s_1})(s_2) = (g_{s_2} + h_{s_2})(s_1)$. Therefore, $g + h \in \mathcal{H}$.

Suppose $(f_n \in \mathcal{H} \mid n \in \mathbb{N})$ is a sequence of non-negative functions in $\mathcal{H}$ such that $f_n \uparrow f$, where $f : S \rightarrow [0, \infty)$ is a bounded function. Note that $f \in \text{b}\Sigma$, since f is the limit of a sequence of (bounded) Σ -measurable functions. For every $s_1 \in S_1$ and $s_2 \in S_2$, note that $f_{s_1} = \lim_{n \rightarrow \infty} f_{n, s_1}$ is Σ_2 -measurable, $f_{s_2} = \lim_{n \rightarrow \infty} f_{n, s_2}$ is Σ_1 -measurable, and $f(s_1, s_2) = f_{s_1}(s_2) = f_{s_2}(s_1)$. Therefore, $f \in \mathcal{H}$.

Consider the π -system $\mathcal{I} = \{B_1 \times B_2 \mid B_1 \in \Sigma_1 \text{ and } B_2 \in \Sigma_2\}$ and the indicator function $f = \mathbb{I}_{B_1 \times B_2}$ of a set $B_1 \times B_2 \in \mathcal{I}$. Note that f is a bounded Σ -measurable function, since $B_1 \times B_2 \in \Sigma$. For every $s_1 \in S_1$ and $s_2 \in S_2$, note that $f_{s_1} = \mathbb{I}_{B_1}(s_1)\mathbb{I}_{B_2}$ is Σ_2 -measurable, $f_{s_2} = \mathbb{I}_{B_2}(s_2)\mathbb{I}_{B_1}$ is Σ_1 -measurable, and $f(s_1, s_2) = f_{s_1}(s_2) = f_{s_2}(s_1)$. Therefore, $f \in \mathcal{H}$. Since $\sigma(\mathcal{I}) = \Sigma$, the monotone-class theorem completes the proof.

□

Proposition 8.5. Consider a measure space (S_1, Σ_1, μ_1) , a measure space (S_2, Σ_2, μ_2) , and the measurable space (S, Σ) , where $S = S_1 \times S_2$ and $\Sigma = \Sigma_1 \times \Sigma_2$. Furthermore, suppose μ_1 and μ_2 are finite measures.

For any bounded Σ -measurable function $f : S \rightarrow \mathbb{R}$, let $I_1^f : S_1 \rightarrow \mathbb{R}$ and $I_2^f : S_2 \rightarrow \mathbb{R}$ be given by

$$\begin{aligned} I_1^f(s_1) &= \int_{S_2} f(s_1, s_2) \mu_2(ds_2) = \int_{S_2} f_{s_1}(s_2) \mu_2(ds_2) = \mu_2(f_{s_1}), \\ I_2^f(s_2) &= \int_{S_1} f(s_1, s_2) \mu_1(ds_1) = \int_{S_1} f_{s_2}(s_1) \mu_1(ds_1) = \mu_1(f_{s_2}), \end{aligned}$$

where, for every $s_1 \in S_1$, the function $f_{s_1} : S_2 \rightarrow \mathbb{R}$ is the Σ_2 -measurable function given by $f_{s_1}(s_2) = f(s_1, s_2)$ and, for every $s_2 \in S_2$, the function $f_{s_2} : S_1 \rightarrow \mathbb{R}$ is the Σ_1 -measurable function given by $f_{s_2}(s_1) = f(s_1, s_2)$. Note that

$\mu_2(|f_{s_1}|) < \infty$ because μ_2 is finite and $|f_{s_1}| \in \mathbf{b}\Sigma_2$. Similarly, $\mu_1(|f_{s_2}|) < \infty$ because μ_1 is finite and $|f_{s_2}| \in \mathbf{b}\Sigma_1$. Therefore, I_1^f and I_2^f are bounded.

Let $\mathcal{H}$ denote a set that contains exactly each function $f \in \mathbf{b}\Sigma$ such that $I_1^f \in \mathbf{b}\Sigma_1$ and $I_2^f \in \mathbf{b}\Sigma_2$ and

$$\mu_1(I_1^f) = \int_{S_1} I_1^f(s_1) \mu_1(ds_1) = \int_{S_2} I_2^f(s_2) \mu_2(ds_2) = \mu_2(I_2^f).$$

In that case, $\mathcal{H}$ contains every bounded Σ -measurable function on S , so that $\mathcal{H} = \mathbf{b}\Sigma$.

Proof. Because $\mathcal{H} \subseteq \mathbf{b}\Sigma$, showing that $\mathcal{H}$ is a vector space only requires showing that $\mathcal{H}$ is non-empty and closed under scalar multiplication and addition. For every $s_1 \in S_1$ and $s_2 \in S_2$, let $f = \mathbb{I}_S$, $f_{s_1} = \mathbb{I}_{S_2}$, and $f_{s_2} = \mathbb{I}_{S_1}$, so that $I_1^f(s_1) = \mu_2(\mathbb{I}_{S_2}) = \mu_2(S_2)\mathbb{I}_{S_1}(s_1)$ and $I_2^f(s_2) = \mu_1(\mathbb{I}_{S_1}) = \mu_1(S_1)\mathbb{I}_{S_2}(s_2)$. Because $S_1 \in \Sigma_1$, we have $I_1^f \in \mathbf{b}\Sigma_1$. Similarly, because $S_2 \in \Sigma_2$, we have $I_2^f \in \mathbf{b}\Sigma_2$. In that case, $f \in \mathcal{H}$, since

$$\mu_1(I_1^f) = \int_{S_1} \mu_2(S_2)\mathbb{I}_{S_1}(s_1) \mu_1(ds_1) = \mu_1(S_1)\mu_2(S_2) = \int_{S_2} \mu_1(S_1)\mathbb{I}_{S_2}(s_2) \mu_2(ds_2) = \mu_2(I_2^f).$$

Now suppose that $f \in \mathcal{H}$ and $a \in \mathbb{R}$. Note that $af \in \mathbf{b}\Sigma$. For every $s_1 \in S_1$ and $s_2 \in S_2$, note that $I_1^{af}(s_1) = \mu_2(af_{s_1}) = a\mu_2(f_{s_1}) = aI_1^f(s_1)$ and $I_2^{af}(s_2) = \mu_1(af_{s_2}) = a\mu_1(f_{s_2}) = aI_2^f(s_2)$. Clearly, $I_1^{af} \in \mathbf{b}\Sigma_1$ and $I_2^{af} \in \mathbf{b}\Sigma_2$. Therefore, $af \in \mathcal{H}$, since the fact that $\mu_1(I_1^f) = \mu_2(I_2^f)$ implies

$$\mu_1(I_1^{af}) = \int_{S_1} aI_1^f(s_1) \mu_1(ds_1) = a\mu_1(I_1^f) = a\mu_2(I_2^f) = \int_{S_2} aI_2^f(s_2) \mu_2(ds_2) = \mu_2(I_2^{af}).$$

Finally, suppose that $g, h \in \mathcal{H}$. Note that $g + h \in \mathbf{b}\Sigma$. For every $s_1 \in S_1$ and $s_2 \in S_2$, note that $I_1^{g+h}(s_1) = \mu_2(g_{s_1} + h_{s_1}) = \mu_2(g_{s_1}) + \mu_2(h_{s_1}) = I_1^g(s_1) + I_1^h(s_1)$ and $I_2^{g+h}(s_2) = \mu_1(g_{s_2} + h_{s_2}) = \mu_1(g_{s_2}) + \mu_1(h_{s_2}) = I_2^g(s_2) + I_2^h(s_2)$. Clearly, $I_1^{g+h} \in \mathbf{b}\Sigma_1$ and $I_2^{g+h} \in \mathbf{b}\Sigma_2$. Therefore, $g + h \in \mathcal{H}$, since $\mu_1(I_1^g) = \mu_2(I_2^g)$ and $\mu_1(I_1^h) = \mu_2(I_2^h)$ imply

$$\int_{S_1} [I_1^g(s_1) + I_1^h(s_1)] \mu_1(ds_1) = \mu_1(I_1^g) + \mu_1(I_1^h) = \mu_2(I_2^g) + \mu_2(I_2^h) = \int_{S_2} [I_2^g(s_2) + I_2^h(s_2)] \mu_2(ds_2).$$

Suppose $(f_n \in \mathcal{H} \mid n \in \mathbb{N})$ is a sequence of non-negative functions in $\mathcal{H}$ such that $f_n \uparrow f$, where $f : S \rightarrow [0, \infty)$ is a bounded function. Note that $f \in \mathbf{b}\Sigma$, since f is the limit of a sequence of (bounded) Σ -measurable functions.

For every $s_1 \in S_1$ and $s_2 \in S_2$, note that $f_{n,s_1} \uparrow f_{s_1}$ and $f_{n,s_2} \uparrow f_{s_2}$, so that the monotone-convergence theorem implies that $\mu_2(f_{n,s_1}) \uparrow \mu_2(f_{s_1})$ and $\mu_1(f_{n,s_2}) \uparrow \mu_1(f_{s_2})$. Therefore,

$$\begin{aligned} I_1^f(s_1) &= \mu_2(f_{s_1}) = \lim_{n \rightarrow \infty} \mu_2(f_{n,s_1}) = \lim_{n \rightarrow \infty} I_1^{f_n}(s_1), \\ I_2^f(s_2) &= \mu_1(f_{s_2}) = \lim_{n \rightarrow \infty} \mu_1(f_{n,s_2}) = \lim_{n \rightarrow \infty} I_2^{f_n}(s_2). \end{aligned}$$

Because I_1^f is the limit of (bounded) Σ_1 -measurable functions, $I_1^f \in \mathbf{b}\Sigma_1$. Similarly, because I_2^f is the limit of (bounded) Σ_2 -measurable functions, $I_2^f \in \mathbf{b}\Sigma_2$. Furthermore, $I_1^{f_n} \uparrow I_1^f$ and $I_2^{f_n} \uparrow I_2^f$, since $f_{n+1} \geq f_n$ implies

$$\begin{aligned} I_1^{f_{n+1}}(s_1) &= \mu_2(f_{n+1,s_1}) \geq \mu_2(f_{n,s_1}) = I_1^{f_n}(s_1), \\ I_2^{f_{n+1}}(s_2) &= \mu_1(f_{n+1,s_2}) \geq \mu_1(f_{n,s_2}) = I_2^{f_n}(s_2). \end{aligned}$$

Therefore, $f \in \mathcal{H}$, since the monotone-convergence theorem implies that

$$\mu_1(I_1^f) = \lim_{n \rightarrow \infty} \mu_1(I_1^{f_n}) = \lim_{n \rightarrow \infty} \mu_2(I_2^{f_n}) = \mu_2(I_2^f).$$

Consider the π -system $\mathcal{I} = \{B_1 \times B_2 \mid B_1 \in \Sigma_1 \text{ and } B_2 \in \Sigma_2\}$ and the indicator function $f = \mathbb{I}_{B_1 \times B_2}$ of a set $B_1 \times B_2 \in \mathcal{I}$. Note that f is a bounded Σ -measurable function, since $B_1 \times B_2 \in \Sigma$. For every $s_1 \in S_1$ and $s_2 \in S_2$, note that $I_1^f(s_1) = \mu_2(\mathbb{I}_{B_1}(s_1)\mathbb{I}_{B_2}) = \mathbb{I}_{B_1}(s_1)\mu_2(B_2)$ and $I_2^f(s_2) = \mu_1(\mathbb{I}_{B_2}(s_2)\mathbb{I}_{B_1}) = \mathbb{I}_{B_2}(s_2)\mu_1(B_1)$. Clearly, $I_1^f \in \mathbf{b}\Sigma_1$ and $I_2^f \in \mathbf{b}\Sigma_2$. Therefore, $f \in \mathcal{H}$, since

$$\mu_1(I_1^f) = \mu_1(\mu_2(B_2)\mathbb{I}_{B_1}) = \mu_1(B_1)\mu_2(B_2) = \mu_2(\mu_1(B_1)\mathbb{I}_{B_2}) = \mu_2(I_2^f).$$

Because $\sigma(\mathcal{I}) = \Sigma$, the monotone-class theorem completes the proof. $\square$

Consider a measure space (S_1, Σ_1, μ_1) , a measure space (S_2, Σ_2, μ_2) , and the measurable space (S, Σ) , where $S = S_1 \times S_2$ and $\Sigma = \Sigma_1 \times \Sigma_2$. Furthermore, suppose μ_1 and μ_2 are finite measures.

Definition 8.3. For any $F \in \Sigma$, define $\mu(F)$ by

$$\mu(F) = \mu_1(I_1^{\mathbb{I}_F}) = \int_{S_1} I_1^{\mathbb{I}_F}(s_1) \mu_1(ds_1) = \int_{S_2} I_2^{\mathbb{I}_F}(s_2) \mu_2(ds_2) = \mu_2(I_2^{\mathbb{I}_F}).$$

The function μ is called the product measure of μ_1 and μ_2 and denoted by $\mu = \mu_1 \times \mu_2$.

Proposition 8.6. The function $\mu = \mu_1 \times \mu_2$ is the unique measure on (S, Σ) such that $\mu(B_1 \times B_2) = \mu_1(B_1)\mu_2(B_2)$ for every $B_1 \in \Sigma_1$ and $B_2 \in \Sigma_2$.

Proof. Consider the π -system $\mathcal{I} = \{B_1 \times B_2 \mid B_1 \in \Sigma_1 \text{ and } B_2 \in \Sigma_2\}$, the indicator function $f = \mathbb{I}_{B_1 \times B_2}$ of a set $B_1 \times B_2 \in \mathcal{I}$, and recall that $\mu_1(I_1^f) = \mu_1(B_1)\mu_2(B_2) = \mu_2(I_2^f)$. Therefore, $\mu(\emptyset) = \mu_1(\emptyset)\mu_2(\emptyset) = 0$.

Consider a sequence $(F_n \in \Sigma \mid n \in \mathbb{N})$ such that $F_n \cap F_m = \emptyset$ for $n \neq m$. Furthermore, consider the sequence of non-negative (bounded) Σ -measurable functions $(f_n : S \rightarrow \{0, 1\} \mid n \in \mathbb{N})$ given by

$$f_n = \mathbb{I}_{\bigcup_{k=0}^n F_k} = \sum_{k=0}^n \mathbb{I}_{F_k}.$$

Let $f = \mathbb{I}_{\bigcup_k F_k}$ so that $f_n \uparrow f$. Because f is a bounded function,

$$\mu\left(\bigcup_k F_k\right) = \mu_1(I_1^f) = \lim_{n \rightarrow \infty} \mu_1(I_1^{f_n}) = \lim_{n \rightarrow \infty} \mu_2(I_2^{f_n}) = \mu_2(I_2^f).$$

By the linearity of the integral with respect to μ_2 ,

$$I_1^{f_n}(s_1) = \int_{S_2} \sum_{k=0}^n \mathbb{I}_{F_k}(s_1, s_2) \mu_2(ds_2) = \sum_{k=0}^n \int_{S_2} \mathbb{I}_{F_k}(s_1, s_2) \mu_2(ds_2) = \sum_{k=0}^n I_1^{\mathbb{I}_{F_k}}(s_1).$$

By the linearity of the integral with respect to μ_1 ,

$$\mu\left(\bigcup_k F_k\right) = \lim_{n \rightarrow \infty} \mu_1(I_1^{f_n}) = \lim_{n \rightarrow \infty} \int_{S_1} \sum_{k=0}^n I_1^{\mathbb{I}_{F_k}}(s_1) \mu_1(ds_1) = \lim_{n \rightarrow \infty} \sum_{k=0}^n \int_{S_1} I_1^{\mathbb{I}_{F_k}}(s_1) \mu_1(ds_1) = \sum_k \mu(F_k),$$

which completes the proof that μ is a measure on (S, Σ) . The measure μ is also finite since $\mu(S_1 \times S_2) = \mu_1(S_1)\mu_2(S_2)$.

Notably, μ is the unique measure on (S, Σ) such that $\mu(B_1 \times B_2) = \mu_1(B_1)\mu_2(B_2)$ for every $B_1 \in \Sigma_1$ and $B_2 \in \Sigma_2$, since $\mathcal{I}$ is a π -system on S such that $\sigma(\mathcal{I}) = \Sigma$ and μ is a finite measure on (S, Σ) . $\square$

Proposition 8.7. If $f : S \rightarrow \mathbb{R}$ is a bounded Σ -measurable function, then

$$\mu(f) = \mu_1(I_1^f) = \int_{S_1} I_1^f(s_1) \mu_1(ds_1) = \int_{S_2} I_2^f(s_2) \mu_2(ds_2) = \mu_2(I_2^f).$$

Proof. Let $\mathcal{H}$ denote a set that contains exactly each function $f \in \text{b}\Sigma$ such that $\mu(f) = \mu_1(I_1^f) = \mu_2(I_2^f)$.

Consider the π -system $\mathcal{I} = \{B_1 \times B_2 \mid B_1 \in \Sigma_1 \text{ and } B_2 \in \Sigma_2\}$. Suppose that $f = \mathbb{I}_{B_1 \times B_2}$ is the indicator function of a set $B_1 \times B_2 \in \mathcal{I}$. In that case, $\mu(f) = \mu(B_1 \times B_2) = \mu_1(I_1^f) = \mu_2(I_2^f)$, so that $f \in \mathcal{H}$. In particular, $\mathbb{I}_S \in \mathcal{H}$, since $S_1 \times S_2 \in \mathcal{I}$.

Because $\mathcal{H} \subseteq \text{b}\Sigma$ and $\mathcal{H}$ is non-empty, showing that $\mathcal{H}$ is a vector space only requires showing that $\mathcal{H}$ is closed under scalar multiplication and addition.

Suppose that $f \in \mathcal{H}$ and $a \in \mathbb{R}$. Note that $af \in \text{b}\Sigma$ and $af \in \mathcal{L}^1(S, \Sigma, \mu)$, so that $\mu(af) = a\mu(f)$. Because $f \in \mathcal{H}$, we have $\mu(af) = \mu_1(aI_1^f) = \mu_1(I_1^{af})$ and $\mu(af) = \mu_2(aI_2^f) = \mu_2(I_2^{af})$, so that $af \in \mathcal{H}$.

Now suppose that $g, h \in \mathcal{H}$. Note that $g + h \in \text{b}\Sigma$ and $g + h \in \mathcal{L}^1(S, \Sigma, \mu)$, so that $\mu(g + h) = \mu(g) + \mu(h)$. Because $g, h \in \mathcal{H}$, we have $\mu(g + h) = \mu_1(I_1^g + I_1^h) = \mu_1(I_1^{g+h})$ and $\mu(g + h) = \mu_2(I_2^g + I_2^h) = \mu_2(I_2^{g+h})$, so that $g + h \in \mathcal{H}$.

Finally, suppose $(f_n \in \mathcal{H} \mid n \in \mathbb{N})$ is a sequence of non-negative functions in $\mathcal{H}$ such that $f_n \uparrow f$, where $f : S \rightarrow [0, \infty)$ is a bounded function. By the monotone-convergence theorem, $\mu(f_n) \uparrow \mu(f)$. Since $f_n \in \mathcal{H}$,

$$\mu(f) = \lim_{n \rightarrow \infty} \mu(f_n) = \lim_{n \rightarrow \infty} \mu_1(I_1^{f_n}) = \lim_{n \rightarrow \infty} \mu_2(I_2^{f_n}) = \mu_1(I_1^f) = \mu_2(I_2^f),$$

which implies $f \in \mathcal{H}$. Because $\sigma(\mathcal{I}) = \Sigma$, the monotone-class theorem completes the proof. $\square$

Proposition 8.8. If $f : S \rightarrow [0, \infty]$ is a Σ -measurable function, then

$$\mu(f) = \mu_1(I_1^f) = \int_{S_1} I_1^f(s_1) \mu_1(ds_1) = \int_{S_2} I_2^f(s_2) \mu_2(ds_2) = \mu_2(I_2^f),$$

where the Σ_1 -measurable function $I_1^f : S_1 \rightarrow [0, \infty]$ and the Σ_2 -measurable function $I_2^f : S_2 \rightarrow [0, \infty]$ are given by

$$\begin{aligned} I_1^f(s_1) &= \int_{S_2} f(s_1, s_2) \mu_2(ds_2) = \int_{S_2} f_{s_1}(s_2) \mu_2(ds_2) = \mu_2(f_{s_1}), \\ I_2^f(s_2) &= \int_{S_1} f(s_1, s_2) \mu_1(ds_1) = \int_{S_1} f_{s_2}(s_1) \mu_1(ds_1) = \mu_1(f_{s_2}), \end{aligned}$$

where, for every $s_1 \in S_1$, the function $f_{s_1} : S_2 \rightarrow [0, \infty]$ is the Σ_2 -measurable function given by $f_{s_1}(s_2) = f(s_1, s_2)$ and, for every $s_2 \in S_2$, the function $f_{s_2} : S_1 \rightarrow [0, \infty]$ is the Σ_1 -measurable function given by $f_{s_2}(s_1) = f(s_1, s_2)$.

Proof. For any $n \in \mathbb{N}$, let $f_n = \alpha_n \circ f$, where α_n is the n -th staircase function. Because $f_n : S \rightarrow [0, n]$ is bounded and Σ -measurable, there is a bounded Σ_2 -measurable function $f_{n,s_1} : S_2 \rightarrow [0, n]$ and a bounded Σ_1 -measurable function $f_{n,s_2} : S_1 \rightarrow [0, n]$ such that $f_n(s_1, s_2) = f_{n,s_1}(s_2) = f_{n,s_2}(s_1)$ for every $s_1 \in S_1$ and $s_2 \in S_2$. Since $f_n \uparrow f$, consider the Σ_2 -measurable function $f_{s_1} = \lim_{n \rightarrow \infty} f_{n,s_1}$ and the Σ_1 -measurable function $f_{s_2} = \lim_{n \rightarrow \infty} f_{n,s_2}$. Note that $f(s_1, s_2) = f_{s_1}(s_2) = f_{s_2}(s_1)$.

For every $s_1 \in S_1$ and $s_2 \in S_2$, note that $f_{n,s_1} \uparrow f_{s_1}$ and $f_{n,s_2} \uparrow f_{s_2}$, so that the monotone-convergence theorem implies that $\mu_2(f_{n,s_1}) \uparrow \mu_2(f_{s_1})$ and $\mu_1(f_{n,s_2}) \uparrow \mu_1(f_{s_2})$. Therefore,

$$\begin{aligned} I_1^f(s_1) &= \mu_2(f_{s_1}) = \lim_{n \rightarrow \infty} \mu_2(f_{n,s_1}) = \lim_{n \rightarrow \infty} I_1^{f_n}(s_1), \\ I_2^f(s_2) &= \mu_1(f_{s_2}) = \lim_{n \rightarrow \infty} \mu_1(f_{n,s_2}) = \lim_{n \rightarrow \infty} I_2^{f_n}(s_2). \end{aligned}$$

Since $f_n \in \text{b}\Sigma$, recall that $I_1^{f_n} \in \text{b}\Sigma_1$ and $I_2^{f_n} \in \text{b}\Sigma_2$. Because I_1^f is the limit of Σ_1 -measurable functions, $I_1^f \in \text{m}\Sigma_1$. Similarly, because I_2^f is the limit of Σ_2 -measurable functions, $I_2^f \in \text{m}\Sigma_2$. Furthermore, $I_1^{f_n} \uparrow I_1^f$ and $I_2^{f_n} \uparrow I_2^f$, since $f_{n+1} \geq f_n$ implies

$$\begin{aligned} I_1^{f_{n+1}}(s_1) &= \mu_2(f_{n+1,s_1}) \geq \mu_2(f_{n,s_1}) = I_1^{f_n}(s_1), \\ I_2^{f_{n+1}}(s_2) &= \mu_1(f_{n+1,s_2}) \geq \mu_1(f_{n,s_2}) = I_2^{f_n}(s_2). \end{aligned}$$

Because $f_n \uparrow f$, the monotone-convergence theorem implies that $\mu(f_n) \uparrow \mu(f)$. Because $I_1^{f_n} \uparrow I_1^f$ and $I_2^{f_n} \uparrow I_2^f$, the monotone-convergence theorem implies that $\mu_1(I_1^{f_n}) \uparrow \mu_1(I_1^f)$ and $\mu_2(I_2^{f_n}) \uparrow \mu_2(I_2^f)$. Because $f_n \in \text{b}\Sigma$,

$$\mu(f) = \lim_{n \rightarrow \infty} \mu(f_n) = \lim_{n \rightarrow \infty} \mu_1(I_1^{f_n}) = \mu_1(I_1^f) = \lim_{n \rightarrow \infty} \mu_2(I_2^{f_n}) = \mu_2(I_2^f).$$

$\square$

Consider the measure spaces (S_1, Σ_1, μ_1) and (S_2, Σ_2, μ_2) and suppose that μ_1 and μ_2 are finite measures. Let $(S, \Sigma, \mu) = (S_1, \Sigma_1, \mu_1) \times (S_2, \Sigma_2, \mu_2)$ denote the measure space where $S = S_1 \times S_2$, $\Sigma = \Sigma_1 \times \Sigma_2$, and $\mu = \mu_1 \times \mu_2$.

Theorem 8.1 (Fubini's theorem). Consider a function $f \in \mathcal{L}^1(S, \Sigma, \mu)$, and recall that $f = f^+ - f^-$ and $|f| = f^+ + f^-$, where $f^+ : S \rightarrow [0, \infty]$ and $f^- : S \rightarrow [0, \infty]$ are non-negative Σ -measurable functions. Therefore, for every $s_1 \in S_1$ and $s_2 \in S_2$,

$$\begin{aligned} f(s_1, s_2) &= f^+(s_1, s_2) - f^-(s_1, s_2) = f_{s_1}^+(s_2) - f_{s_1}^-(s_2) = f_{s_2}^+(s_1) - f_{s_2}^-(s_1), \\ |f(s_1, s_2)| &= f^+(s_1, s_2) + f^-(s_1, s_2) = f_{s_1}^+(s_2) + f_{s_1}^-(s_2) = f_{s_2}^+(s_1) + f_{s_2}^-(s_1), \end{aligned}$$

where $f_{s_1}^+ : S_2 \rightarrow [0, \infty]$ and $f_{s_1}^- : S_2 \rightarrow [0, \infty]$ are non-negative Σ_2 -measurable functions and $f_{s_2}^+ : S_1 \rightarrow [0, \infty]$ and $f_{s_2}^- : S_1 \rightarrow [0, \infty]$ are non-negative Σ_1 -measurable functions.

For every $s_1 \in S_1$ and $s_2 \in S_2$, let $f_{s_1} = f_{s_1}^+ - f_{s_1}^-$ and $f_{s_2} = f_{s_2}^+ - f_{s_2}^-$, so that $f(s_1, s_2) = f_{s_1}(s_2) = f_{s_2}(s_1)$. Note that f_{s_1} is Σ_2 -measurable and f_{s_2} is Σ_1 -measurable. Furthermore, $|f_{s_1}| = f_{s_1}^+ + f_{s_1}^-$ and $|f_{s_2}| = f_{s_2}^+ + f_{s_2}^-$.

Finally, let $F_1^f = \{s_1 \in S_1 \mid \mu_2(|f_{s_1}|) < \infty\}$ and $F_2^f = \{s_2 \in S_2 \mid \mu_1(|f_{s_2}|) < \infty\}$. In that case,

$$\mu(f) = \mu_1(I_1^f; F_1^f) = \int_{F_1^f} I_1^f(s_1) \mu_1(ds_1) = \int_{F_2^f} I_2^f(s_2) \mu_2(ds_2) = \mu_2(I_2^f; F_2^f),$$

where $I_1^f : S_1 \rightarrow \mathbb{R}$ and $I_2^f : S_2 \rightarrow \mathbb{R}$ are given by

$$\begin{aligned} I_1^f(s_1) &= \int_{S_2} f(s_1, s_2) \mu_2(ds_2) = \int_{S_2} f_{s_1}(s_2) \mu_2(ds_2) = \mu_2(f_{s_1}), \\ I_2^f(s_2) &= \int_{S_1} f(s_1, s_2) \mu_1(ds_1) = \int_{S_1} f_{s_2}(s_1) \mu_1(ds_1) = \mu_1(f_{s_2}), \end{aligned}$$

for every $s_1 \in F_1^f$ and $s_2 \in F_2^f$.

Proof. Because $|f| : S \rightarrow [0, \infty]$ is a non-negative Σ -measurable function such that $\mu(|f|) < \infty$,

$$\begin{aligned} \mu(|f|) &= \mu_1(I_1^{|f|}) = \mu_1(I_1^{f^+ + f^-}) = \mu_1(I_1^{f^+} + I_1^{f^-}) < \infty, \\ \mu(|f|) &= \mu_2(I_2^{|f|}) = \mu_2(I_2^{f^+ + f^-}) = \mu_2(I_2^{f^+} + I_2^{f^-}) < \infty. \end{aligned}$$

For every $s_1 \in S_1$, note that $I_1^{f^+}(s_1) + I_1^{f^-}(s_1) = \mu_2(f_{s_1}^+) + \mu_2(f_{s_1}^-) = \mu_2(|f_{s_1}|)$. Because $\mu_1(I_1^{f^+} + I_1^{f^-}) < \infty$, we know that $\mu_1(S_1 \setminus F_1^f) = \mu_1(\{s_1 \in S_1 \mid \mu_2(|f_{s_1}|) = \infty\}) = 0$. Similarly, for every $s_2 \in S_2$, note that $I_2^{f^+}(s_2) + I_2^{f^-}(s_2) = \mu_1(f_{s_2}^+) + \mu_1(f_{s_2}^-) = \mu_1(|f_{s_2}|)$. Because $\mu_2(I_2^{f^+} + I_2^{f^-}) < \infty$, we know that $\mu_2(S_2 \setminus F_2^f) = \mu_2(\{s_2 \in S_2 \mid \mu_1(|f_{s_2}|) = \infty\}) = 0$. Therefore, by the linearity of the integral,

$$\begin{aligned} \mu(f) &= \mu(f^+) - \mu(f^-) = \mu_1(I_1^{f^+}) - \mu_1(I_1^{f^-}) = \mu_1(I_1^{f^+} \mathbb{I}_{F_1^f}) - \mu_1(I_1^{f^-} \mathbb{I}_{F_1^f}) = \mu_1((I_1^{f^+} - I_1^{f^-}) \mathbb{I}_{F_1^f}) = \mu_1(I_1^f; F_1^f), \\ \mu(f) &= \mu(f^+) - \mu(f^-) = \mu_2(I_2^{f^+}) - \mu_2(I_2^{f^-}) = \mu_2(I_2^{f^+} \mathbb{I}_{F_2^f}) - \mu_2(I_2^{f^-} \mathbb{I}_{F_2^f}) = \mu_2((I_2^{f^+} - I_2^{f^-}) \mathbb{I}_{F_2^f}) = \mu_2(I_2^f; F_2^f). \end{aligned}$$

□

Proposition 8.9. Fubini's theorem is also valid when μ_1 and μ_2 are σ -finite measures.

Proposition 8.10. Consider the measure space $(S, \Sigma, \mu) = (\Omega, \mathcal{F}, \mathbb{P}) \times ([0, \infty), \mathcal{B}([0, \infty)), \text{Leb})$, where $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability triple. Furthermore, consider a random variable $X : \Omega \rightarrow [0, \infty]$. In that case,

$$\mathbb{E}(X) = \int_{[0, \infty)} \mathbb{P}(X \geq x) \text{Leb}(dx).$$

Proof. First, let $A = \{(\omega, x) \in S \mid x \leq X(\omega)\}$ and $f(\omega, x) = x - X(\omega) = \rho_2(\omega, x) - X(\rho_1(\omega, x))$. Because f is Σ -measurable and $f^{-1}((-\infty, 0]) = A$, we know that $A \in \Sigma$. For every $(\omega, x) \in S$, note that

$$\mathbb{I}_A(\omega, x) = \mathbb{I}_{\{\omega \in \Omega \mid x \leq X(\omega)\}}(\omega) = \mathbb{I}_{\{x \in [0, \infty) \mid x \leq X(\omega)\}}(x).$$

Because $\mathbb{I}_A$ is a bounded Σ -measurable function,

$$\begin{aligned} I_1^{\mathbb{I}_A}(\omega) &= \text{Leb}(\{x \in [0, \infty) \mid x \leq X(\omega)\}) = X(\omega), \\ I_2^{\mathbb{I}_A}(x) &= \mathbb{P}(\{\omega \in \Omega \mid x \leq X(\omega)\}) = \mathbb{P}(X \geq x). \end{aligned}$$

By the definition of the product measure μ ,

$$\mu(A) = \mathbb{P}(I_1^{\mathbb{I}_A}) = \mathbb{E}(X) = \text{Leb}(I_2^{\mathbb{I}_A}) = \int_{[0, \infty)} P(X \geq x) \text{Leb}(dx).$$

□

Definition 8.4. Let $\mathcal{C}$ denote the set of open subsets of $\mathbb{R}^2$. The Borel σ -algebra $\mathcal{B}(\mathbb{R}^2)$ on $\mathbb{R}^2$ is defined as $\mathcal{B}(\mathbb{R}^2) = \sigma(\mathcal{C})$.

Proposition 8.11. Let $\mathcal{B}(\mathbb{R})^2 = \mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})$ denote the product between the Borel σ -algebra $\mathcal{B}(\mathbb{R})$ on $\mathbb{R}$ and itself. In that case, $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R})^2$.

Proof. Because the functions $\rho_1 : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\rho_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $\rho_1(x, y) = x$ and $\rho_2(x, y) = y$ for every $(x, y) \in \mathbb{R}^2$ are continuous, recall that $\rho_1^{-1}(A) \in \mathcal{C}$ and $\rho_2^{-1}(A) \in \mathcal{C}$ for every open set $A \subseteq \mathbb{R}$, so that a previous result guarantees that ρ_1 and ρ_2 are $\mathcal{B}(\mathbb{R}^2)$ -measurable. Therefore, $\sigma(\rho_1) \cup \sigma(\rho_2) \subseteq \mathcal{B}(\mathbb{R}^2)$. Because $\mathcal{B}(\mathbb{R})^2 = \sigma(\sigma(\rho_1) \cup \sigma(\rho_2))$, we know that $\mathcal{B}(\mathbb{R})^2 \subseteq \mathcal{B}(\mathbb{R}^2)$.

Recall that every open subset $C \subseteq \mathbb{R}^2$ can be written as $C = \cup_n (a_n, b_n) \times (c_n, d_n)$, where $a_n \leq b_n$ and $c_n \leq d_n$ for every $n \in \mathbb{N}$. Because $\mathcal{B}(\mathbb{R})$ contains every open interval and $\mathcal{B}(\mathbb{R})^2 = \sigma(\{B_1 \times B_2 \mid B_1, B_2 \in \mathcal{B}(\mathbb{R})\})$, we know that $\mathcal{C} \subseteq \mathcal{B}(\mathbb{R})^2$, so that $\mathcal{B}(\mathbb{R}^2) \subseteq \mathcal{B}(\mathbb{R})^2$. Therefore, $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R})^2$. $\square$

Proposition 8.12. The set $\mathcal{I} = \{(-\infty, x] \times (-\infty, y] \mid x, y \in \mathbb{R}\}$ is a π -system on $\mathbb{R}^2$ such that $\sigma(\mathcal{I}) = \mathcal{B}(\mathbb{R})^2$.

Proof. Let $A_1 = (-\infty, x_1] \times (-\infty, y_1]$ and $A_2 = (-\infty, x_2] \times (-\infty, y_2]$ be elements of $\mathcal{I}$. In that case,

$$A_1 \cap A_2 = ((-\infty, x_1] \cap (-\infty, x_2]) \times ((-\infty, y_1] \cap (-\infty, y_2]) = (-\infty, \min(x_1, x_2)] \times (-\infty, \min(y_1, y_2)],$$

so that $A_1 \cap A_2 \in \mathcal{I}$. Therefore, $\mathcal{I}$ is a π -system.

Because $(-\infty, x] \in \mathcal{B}(\mathbb{R})$ and $(-\infty, y] \in \mathcal{B}(\mathbb{R})$ for every $x, y \in \mathbb{R}$ and $\mathcal{B}(\mathbb{R})^2 = \sigma(\{B_1 \times B_2 \mid B_1, B_2 \in \mathcal{B}(\mathbb{R})\})$, we know that $\mathcal{I} \subseteq \mathcal{B}(\mathbb{R})^2$, so that $\sigma(\mathcal{I}) \subseteq \mathcal{B}(\mathbb{R})^2$.

Note that $(a, b] \times (c, d] \in \sigma(\mathcal{I})$ for every $a \leq b$ and $c \leq d$, since

$$(a, b] \times (c, d] = ((-\infty, b] \times (-\infty, d]) \cap (((-\infty, b] \times (-\infty, c]) \cup ((-\infty, a] \times (-\infty, d]))^c.$$

Also note that $(a, b) \times (c, d] \in \sigma(\mathcal{I})$ for every $a \leq b$ and $c \leq d$, since

$$(a, b) \times (c, d] = \left(\bigcup_{n \in \mathbb{N}^+} (a, b - \epsilon_1 n^{-1}] \right) \times (c, d] = \bigcup_{n \in \mathbb{N}^+} (a, b - \epsilon_1 n^{-1}] \times (c, d],$$

where $\epsilon_1 = (b - a)/2$.

Finally, note that $(a, b) \times (c, d) \in \sigma(\mathcal{I})$ for every $a \leq b$ and $c \leq d$, since

$$(a, b) \times (c, d) = (a, b) \times \bigcup_{n \in \mathbb{N}^+} (c, d - \epsilon_2 n^{-1}] = \bigcup_{n \in \mathbb{N}^+} (a, b) \times (c, d - \epsilon_2 n^{-1}],$$

where $\epsilon_2 = (d - c)/2$.

Because every open set $C \in \mathcal{C}$ can be written as $C = \cup_n (a_n, b_n) \times (c_n, d_n)$, where $a_n \leq b_n$ and $c_n \leq d_n$ for every $n \in \mathbb{N}$, we know that $\mathcal{C} \subseteq \sigma(\mathcal{I})$. Since $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R})^2$, we know that $\mathcal{B}(\mathbb{R})^2 \subseteq \sigma(\mathcal{I})$. $\square$

Proposition 8.13. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$. Let $Z : \Omega \rightarrow \mathbb{R}^2$ be given by $Z(\omega) = (X(\omega), Y(\omega))$. The function Z is $\mathcal{F}/\mathcal{B}(\mathbb{R})^2$ -measurable.

Proof. Let $\rho_1 : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by $\rho_1(x, y) = x$ and $\rho_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by $\rho_2(x, y) = y$. Note that $X = \rho_1 \circ Z$ and $Y = \rho_2 \circ Z$, so that $X^{-1}(B) = (\rho_1 \circ Z)^{-1}(B) = Z^{-1}(\rho_1^{-1}(B))$ and $Y^{-1}(B) = (\rho_2 \circ Z)^{-1}(B) = Z^{-1}(\rho_2^{-1}(B))$ for every $B \in \mathcal{B}(\mathbb{R})$. Because X and Y are $\mathcal{F}$ -measurable, $Z^{-1}(C) \in \mathcal{F}$ for every $C \in (\sigma(\rho_1) \cup \sigma(\rho_2))$.

Note that $\mathcal{E} = \{\Gamma \in \mathcal{B}(\mathbb{R})^2 \mid Z^{-1}(\Gamma) \in \mathcal{F}\}$ is a σ -algebra on $\mathbb{R}^2$. Because $(\sigma(\rho_1) \cup \sigma(\rho_2)) \subseteq \mathcal{B}(\mathbb{R})^2$, we know that $\sigma(\sigma(\rho_1) \cup \sigma(\rho_2)) = \mathcal{B}(\mathbb{R})^2 \subseteq \mathcal{E}$, so that $\mathcal{E} = \mathcal{B}(\mathbb{R})^2$. Therefore, Z is $\mathcal{F}/\mathcal{B}(\mathbb{R})^2$ -measurable. $\square$

Definition 8.5. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$. For any $\Gamma \in \mathcal{B}(\mathbb{R})^2$, the joint law $\mathcal{L}_{X,Y} : \mathcal{B}(\mathbb{R})^2 \rightarrow [0, 1]$ of X and Y is defined by

$$\mathcal{L}_{X,Y}(\Gamma) = \mathbb{P}(\{\omega \in \Omega \mid (X(\omega), Y(\omega)) \in \Gamma\}) = \mathbb{P}((X, Y) \in \Gamma).$$

Proposition 8.14. The function $\mathcal{L}_{X,Y}$ defined above is a probability measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2)$.

Proof. Clearly, $\mathcal{L}_{X,Y}(\mathbb{R}^2) = \mathbb{P}(\Omega) = 1$ and $\mathcal{L}_{X,Y}(\emptyset) = \mathbb{P}(\emptyset) = 0$. Furthermore, for any sequence of sets $(\Gamma_n \in \mathcal{B}(\mathbb{R})^2 \mid n \in \mathbb{N})$ such that $\Gamma_n \cap \Gamma_m = \emptyset$ for $n \neq m$,

$$\mathcal{L}_{X,Y} \left(\bigcup_n \Gamma_n \right) = \mathbb{P} \left(\left\{ \omega \in \Omega \mid (X(\omega), Y(\omega)) \in \bigcup_n \Gamma_n \right\} \right) = \mathbb{P} \left(\bigcup_n \{ \omega \in \Omega \mid (X(\omega), Y(\omega)) \in \Gamma_n \} \right) = \sum_n \mathcal{L}_{X,Y}(\Gamma_n).$$

□

Definition 8.6. The joint distribution $F_{X,Y} : \mathbb{R}^2 \rightarrow [0, 1]$ of X and Y is defined by

$$F_{X,Y}(x, y) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) \leq x \text{ and } Y(\omega) \leq y\}) = \mathbb{P}(X \leq x, Y \leq y) = \mathcal{L}_{X,Y}((-\infty, x] \times (-\infty, y]).$$

Proposition 8.15. Because the π -system $\mathcal{I} = \{(-\infty, x] \times (-\infty, y] \mid x, y \in \mathbb{R}\}$ generates $\mathcal{B}(\mathbb{R})^2$, the joint law $\mathcal{L}_{X,Y}$ of X and Y is the unique measure on the measurable space $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2)$ such that $\mathcal{L}_{X,Y}((-\infty, x] \times (-\infty, y]) = F_{X,Y}(x, y)$ for every $(x, y) \in \mathbb{R}^2$. Therefore, the joint distribution $F_{X,Y}$ completely determines the joint law $\mathcal{L}_{X,Y}$.

Definition 8.7. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$. Consider also the measure space $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2, \text{Leb}^2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \text{Leb})^2$. The random variables X and Y have a joint probability density function $f_{X,Y}$ if $f_{X,Y} : \mathbb{R}^2 \rightarrow [0, \infty]$ is a $\mathcal{B}(\mathbb{R})^2$ -measurable function such that the joint law $\mathcal{L}_{X,Y}$ is given by

$$\mathcal{L}_{X,Y}(\Gamma) = \int_{\Gamma} f_{X,Y}(z) \text{Leb}^2(dz) = \int_{\mathbb{R}^2} \mathbb{I}_{\Gamma}(z) f_{X,Y}(z) \text{Leb}^2(dz).$$

In that case, the joint law $\mathcal{L}_{X,Y}$ has density $f_{X,Y}$ relative to Leb^2 , which is denoted by $d\mathcal{L}_{X,Y}/d\text{Leb}^2 = f_{X,Y}$ almost everywhere. Furthermore, because $\mathbb{I}_{\Gamma} f_{X,Y}$ is a non-negative $\mathcal{B}(\mathbb{R})^2$ -measurable function,

$$\mathcal{L}_{X,Y}(\Gamma) = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_{\Gamma}(x, y) f_{X,Y}(x, y) \text{Leb}(dy) \right] \text{Leb}(dx) = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_{\Gamma}(x, y) f_{X,Y}(x, y) \text{Leb}(dx) \right] \text{Leb}(dy).$$

Proposition 8.16. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$. Note that

$$\begin{aligned} \mathcal{L}_X(B) &= \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) \in B\}) = \mathbb{P}(\{\omega \in \Omega \mid (X(\omega), Y(\omega)) \in (B \times \mathbb{R})\}) = \mathcal{L}_{X,Y}(B \times \mathbb{R}), \\ \mathcal{L}_Y(B) &= \mathbb{P}(Y^{-1}(B)) = \mathbb{P}(\{\omega \in \Omega \mid Y(\omega) \in B\}) = \mathbb{P}(\{\omega \in \Omega \mid (X(\omega), Y(\omega)) \in (\mathbb{R} \times B)\}) = \mathcal{L}_{X,Y}(\mathbb{R} \times B), \end{aligned}$$

for every $B \in \mathcal{B}(\mathbb{R})$, where $\mathcal{L}_X$ is the law of X and $\mathcal{L}_Y$ is the law of Y . Therefore,

$$\begin{aligned} \mathcal{L}_X(B) &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_{B \times \mathbb{R}}(x, y) f_{X,Y}(x, y) \text{Leb}(dy) \right] \text{Leb}(dx) = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_B(x) f_{X,Y}(x, y) \text{Leb}(dy) \right] \text{Leb}(dx), \\ \mathcal{L}_Y(B) &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_{\mathbb{R} \times B}(x, y) f_{X,Y}(x, y) \text{Leb}(dx) \right] \text{Leb}(dy) = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_B(y) f_{X,Y}(x, y) \text{Leb}(dx) \right] \text{Leb}(dy), \end{aligned}$$

for every $B \in \mathcal{B}(\mathbb{R})$. By the linearity of the integral with respect to Leb ,

$$\begin{aligned} \mathcal{L}_X(B) &= \int_{\mathbb{R}} \mathbb{I}_B(x) \left[\int_{\mathbb{R}} f_{X,Y}(x, y) \text{Leb}(dy) \right] \text{Leb}(dx) = \int_{\mathbb{R}} \mathbb{I}_B(x) f_X(x) \text{Leb}(dx) = \int_B f_X(x) \text{Leb}(dx), \\ \mathcal{L}_Y(B) &= \int_{\mathbb{R}} \mathbb{I}_B(y) \left[\int_{\mathbb{R}} f_{X,Y}(x, y) \text{Leb}(dx) \right] \text{Leb}(dy) = \int_{\mathbb{R}} \mathbb{I}_B(y) f_Y(y) \text{Leb}(dy) = \int_B f_Y(y) \text{Leb}(dy), \end{aligned}$$

where $f_X : \mathbb{R} \rightarrow [0, \infty]$ and $f_Y : \mathbb{R} \rightarrow [0, \infty]$ are Borel functions given by

$$\begin{aligned} f_X(x) &= \int_{\mathbb{R}} f_{X,Y}(x, y) \text{Leb}(dy), \\ f_Y(y) &= \int_{\mathbb{R}} f_{X,Y}(x, y) \text{Leb}(dx). \end{aligned}$$

By definition, f_X is a probability density function for X and f_Y is a probability density function for Y .

Proposition 8.17. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$. Let $\mathcal{L}_{X,Y}$ denote the joint law of X and Y , $\mathcal{L}_X$ denote the law of X , $\mathcal{L}_Y$ denote the law of Y , $F_{X,Y}$ denote the joint distribution function of X and Y , F_X denote the distribution function of X , and F_Y denote the distribution function of Y . The following are equivalent: X and Y are independent; $\mathcal{L}_{X,Y} = \mathcal{L}_X \times \mathcal{L}_Y$; and $F_{X,Y} = F_X F_Y$.

Proof. Suppose X and Y are independent. In that case, for every $B_1, B_2 \in \mathcal{B}(\mathbb{R})$,

$$\mathcal{L}_{X,Y}(B_1 \times B_2) = \mathbb{P}(\{\omega \in \Omega \mid (X(\omega), Y(\omega)) \in (B_1 \times B_2)\}) = \mathbb{P}(X^{-1}(B_1) \cap Y^{-1}(B_2)) = \mathcal{L}_X(B_1)\mathcal{L}_Y(B_2).$$

Because $\mathcal{L}_X \times \mathcal{L}_Y$ is the unique measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2)$ such that $(\mathcal{L}_X \times \mathcal{L}_Y)(B_1 \times B_2) = \mathcal{L}_X(B_1)\mathcal{L}_Y(B_2)$ for every $B_1, B_2 \in \mathcal{B}(\mathbb{R})$ and $\mathcal{L}_{X,Y}$ is a measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2)$, we know that $\mathcal{L}_{X,Y} = \mathcal{L}_X \times \mathcal{L}_Y$.

Suppose $\mathcal{L}_{X,Y} = \mathcal{L}_X \times \mathcal{L}_Y$. In that case, for every $x, y \in \mathbb{R}$,

$$F_{X,Y}(x, y) = (\mathcal{L}_X \times \mathcal{L}_Y)((-\infty, x] \times (-\infty, y]) = \mathcal{L}_X((-\infty, x])\mathcal{L}_Y((-\infty, y]) = F_X(x)F_Y(y).$$

Finally, suppose that $F_{X,Y} = F_X F_Y$. In that case, for every $x, y \in \mathbb{R}$,

$$\mathbb{P}(X \leq x, Y \leq y) = F_{X,Y}(x, y) = F_X(x)F_Y(y) = \mathbb{P}(X \leq x)\mathbb{P}(Y \leq y),$$

so that a previous result implies that X and Y are independent, which completes the proof. $\square$

Proposition 8.18. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$. Suppose $f_{X,Y}$ is a joint probability density function for X and Y , f_X is a probability density function for X , and f_Y is a probability density function for Y . Furthermore, let $F = \{(x, y) \in \mathbb{R}^2 \mid f_X(x)f_Y(y) \neq f_{X,Y}(x, y)\}$. In that case, $\text{Leb}^2(F) = 0$ if and only if X and Y are independent random variables.

Proof. Suppose $\text{Leb}^2(F) = 0$. For every $\Gamma \in \mathcal{B}(\mathbb{R})^2$, let $F_\Gamma = \{z \in \mathbb{R}^2 \mid \mathbb{I}_\Gamma(z)f_X(\rho_1(z))f_Y(\rho_2(z)) \neq \mathbb{I}_\Gamma(z)f_{X,Y}(z)\}$, so that $F_\Gamma \subseteq F$. Because $F_\Gamma \subseteq F_{\mathbb{R}^2} = F$, we know that $\text{Leb}^2(F_\Gamma) = 0$. Therefore, because $\mathbb{I}_\Gamma(f_X \circ \rho_1)(f_Y \circ \rho_2)$ and $\mathbb{I}_\Gamma f_{X,Y}$ are non-negative $\mathcal{B}(\mathbb{R})^2$ -measurable functions,

$$\mathcal{L}_{X,Y}(\Gamma) = \int_{\mathbb{R}^2} \mathbb{I}_\Gamma(z)f_{X,Y}(z) \text{Leb}^2(dz) = \int_{\mathbb{R}^2} \mathbb{I}_\Gamma(z)f_X(\rho_1(z))f_Y(\rho_2(z)) \text{Leb}^2(dz).$$

For every $B_1, B_2 \in \mathcal{B}(\mathbb{R})$, since $\mathbb{I}_\Gamma(f_X \circ \rho_1)(f_Y \circ \rho_2)$ is a non-negative $\mathcal{B}(\mathbb{R})^2$ -measurable function,

$$\mathcal{L}_{X,Y}(B_1 \times B_2) = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_{B_1 \times B_2}(x, y)f_X(x)f_Y(y) \text{Leb}(dy) \right] \text{Leb}(dx).$$

Using the fact that $\mathbb{I}_{B_1 \times B_2}(x, y) = \mathbb{I}_{B_1}(x)\mathbb{I}_{B_2}(y)$ and the linearity of the integral with respect to Leb ,

$$\mathcal{L}_{X,Y}(B_1 \times B_2) = \left[\int_{\mathbb{R}} \mathbb{I}_{B_1}(x)f_X(x) \text{Leb}(dx) \right] \left[\int_{\mathbb{R}} \mathbb{I}_{B_2}(y)f_Y(y) \text{Leb}(dy) \right] = \mathcal{L}_X(B_1)\mathcal{L}_Y(B_2).$$

Because $\mathcal{L}_X \times \mathcal{L}_Y$ is the unique measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2)$ such that $(\mathcal{L}_X \times \mathcal{L}_Y)(B_1 \times B_2) = \mathcal{L}_X(B_1)\mathcal{L}_Y(B_2)$ for every $B_1, B_2 \in \mathcal{B}(\mathbb{R})$ and $\mathcal{L}_{X,Y}$ is a measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2)$, we know that X and Y are independent.

Suppose X and Y are independent. Let $f = (f_X \circ \rho_1)(f_Y \circ \rho_2)$. Because f is a $\mathcal{B}(\mathbb{R})^2$ -measurable non-negative function, recall that $(f \text{Leb}^2)$ is a measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2)$ given by

$$(f \text{Leb}^2)(\Gamma) = \int_{\Gamma} f d\text{Leb}^2 = \int_{\mathbb{R}^2} \mathbb{I}_\Gamma(z)f_X(\rho_1(z))f_Y(\rho_2(z)) \text{Leb}^2(dz) = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_\Gamma(x, y)f_X(x)f_Y(y) \text{Leb}(dy) \right] \text{Leb}(dx).$$

By the linearity of the integral with respect to Leb , for every $B_1, B_2 \in \mathcal{B}(\mathbb{R})$,

$$\mathcal{L}_X(B_1)\mathcal{L}_Y(B_2) = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_{B_1 \times B_2}(x, y)f_X(x)f_Y(y) \text{Leb}(dy) \right] \text{Leb}(dx) = (f \text{Leb}^2)(B_1 \times B_2).$$

Because $\mathcal{L}_X \times \mathcal{L}_Y$ is the unique measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2)$ such that $(\mathcal{L}_X \times \mathcal{L}_Y)(B_1 \times B_2) = \mathcal{L}_X(B_1)\mathcal{L}_Y(B_2)$ for every $B_1, B_2 \in \mathcal{B}(\mathbb{R})$ and $(f \text{Leb}^2)$ is a measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R})^2)$, we know that $\mathcal{L}_X \times \mathcal{L}_Y = (f \text{Leb}^2)$. Since X and Y are independent, $\mathcal{L}_{X,Y} = (f \text{Leb}^2)$. Therefore, f is a joint probability density function for X and Y .

Let $F_1 = \{z \in \mathbb{R}^2 \mid f(z) - f_{X,Y}(z) > 0\}$ and $F_2 = \{z \in \mathbb{R}^2 \mid f_{X,Y}(z) - f(z) > 0\}$, so that $F = F_1 \cup F_2$. Since $F_1 \cap F_2 = \emptyset$, we have $\text{Leb}^2(F) = \text{Leb}^2(F_1) + \text{Leb}^2(F_2)$. In order to find a contradiction, suppose $\text{Leb}^2(F) > 0$, so that $\text{Leb}^2(F_1) > 0$ or $\text{Leb}^2(F_2) > 0$. Because $(f - f_{X,Y})\mathbb{I}_{F_1}$ and $(f_{X,Y} - f)\mathbb{I}_{F_2}$ are non-negative $\mathcal{B}(\mathbb{R})^2$ -measurable functions, a previous result then implies that $\text{Leb}^2((f - f_{X,Y})\mathbb{I}_{F_1}) > 0$ or $\text{Leb}^2((f_{X,Y} - f)\mathbb{I}_{F_2}) > 0$. The linearity of the integral with respect to Leb^2 then implies that $\mathcal{L}_{X,Y}(F_1) = \text{Leb}^2(f\mathbb{I}_{F_1}) > \text{Leb}^2(f_{X,Y}\mathbb{I}_{F_1}) = \mathcal{L}_{X,Y}(F_1)$ or $\mathcal{L}_{X,Y}(F_2) = \text{Leb}^2(f_{X,Y}\mathbb{I}_{F_2}) > \text{Leb}^2(f\mathbb{I}_{F_2}) = \mathcal{L}_{X,Y}(F_2)$, which is a contradiction. Therefore, $\text{Leb}^2(F) = 0$. $\square$

The results in this section can be generalized to products between any number of measure spaces.

Theorem 8.2 (Kolmogorov's extension theorem). Consider the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and a sequence of probability measures $(\mathcal{L}_n \mid n \in \mathbb{N})$. Let $\Omega = \prod_n \mathbb{R}$, so that each $\omega \in \Omega$ corresponds to a sequence $(\omega_n \in \mathbb{R} \mid n \in \mathbb{N})$. For every $n \in \mathbb{N}$, let $X_n : \Omega \rightarrow \mathbb{R}$ be given by $X_n(\omega) = \omega_n$. Furthermore, consider the σ -algebra $\mathcal{F}$ on Ω given by $\mathcal{F} = \sigma(\cup_n \sigma(X_n))$. In that case, there is a unique probability measure $\mathbb{P}$ on the measurable space $(\Omega, \mathcal{F})$ such that, for every sequence $(B_n \in \mathcal{B}(\mathbb{R}) \mid n \in \mathbb{N})$,

$$\mathbb{P} \left(\prod_n B_n \right) = \prod_n \mathcal{L}_n(B_n).$$

The measure space $(\Omega, \mathcal{F}, \mathbb{P})$ is denoted by $(\Omega, \mathcal{F}, \mathbb{P}) = \prod_n (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L}_n)$. The sequence $(X_n : \Omega \rightarrow \mathbb{R} \mid n \in \mathbb{N})$ is composed of independent random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ so that $\mathcal{L}_n$ is the law of X_n .

Proposition 8.19. Consider a measurable space $(\tilde{\Omega}, \tilde{\mathcal{F}})$ and a stochastic process $(\tilde{X}_n : \tilde{\Omega} \rightarrow \mathbb{R} \mid n \in \mathbb{N})$. Let $\tilde{X} : \tilde{\Omega} \rightarrow \mathbb{R}^\infty$ be given by $\tilde{X}(\tilde{\omega}) = (\tilde{X}_n(\tilde{\omega}) \mid n \in \mathbb{N})$. For every $n \in \mathbb{N}$, let $X_n : \mathbb{R}^\infty \rightarrow \mathbb{R}$ be given by $X_n(\omega) = \omega_n$ and let $\mathcal{F} = \sigma(\cup_n \sigma(X_n))$. In that case, $\tilde{X}$ is $\tilde{\mathcal{F}}/\mathcal{F}$ -measurable.

Proof. For every $n \in \mathbb{N}$, note that $\tilde{X}_n = X_n \circ \tilde{X}$, so that $\tilde{X}_n^{-1}(B) = \tilde{X}^{-1}(X_n^{-1}(B))$ for every $B \in \mathcal{B}(\mathbb{R})$. Because $\tilde{X}_n$ is $\tilde{\mathcal{F}}$ -measurable for every $n \in \mathbb{N}$, we know that $\tilde{X}^{-1}(C) \in \tilde{\mathcal{F}}$ for every $C \in \cup_n \sigma(X_n)$. Since $(\mathbb{R}^\infty, \mathcal{F})$ is a measurable space, note that $\mathcal{E} = \{F \in \mathcal{F} \mid \tilde{X}^{-1}(F) \in \tilde{\mathcal{F}}\}$ is a σ -algebra on $\mathbb{R}^\infty$. Because $\cup_n \sigma(X_n) \subseteq \mathcal{F}$, we know that $\sigma(\cup_n \sigma(X_n)) = \mathcal{F} \subseteq \mathcal{E}$, so that $\mathcal{E} = \mathcal{F}$. Therefore, $\tilde{X}$ is $\tilde{\mathcal{F}}/\mathcal{F}$ -measurable. $\square$

9 Probability kernels

Consider the measurable spaces (S_1, Σ_1) , (S_2, Σ_2) , and $(S, \Sigma) = (S_1 \times S_2, \Sigma_1 \times \Sigma_2)$.

Definition 9.1. A probability kernel K from S_1 to S_2 is a function $K : S_1 \times S_2 \rightarrow [0, 1]$ such that

- For every $s_1 \in S_1$, the function $K(s_1, \cdot) : S_2 \rightarrow [0, 1]$ is a probability measure on (S_2, Σ_2) ;
- For every $B_2 \in \Sigma_2$, the function $K(\cdot, B_2) : S_1 \rightarrow [0, 1]$ is Σ_1 -measurable.

Proposition 9.1. Consider a π -system $\mathcal{I}$ on S_2 such that $\sigma(\mathcal{I}) = \Sigma_2$. Let $K : S_1 \times S_2 \rightarrow [0, 1]$ be a function such that the function $K(s_1, \cdot) : S_2 \rightarrow [0, 1]$ is a probability measure on (S_2, Σ_2) for every $s_1 \in S_1$. If the function $K(\cdot, B_2) : S_1 \rightarrow [0, 1]$ is Σ_1 -measurable for every $B_2 \in \mathcal{I}$, then K is a probability kernel from S_1 to S_2 .

Proof. Let $\mathcal{D} = \{B_2 \in \Sigma_2 \mid \sigma(K(\cdot, B_2)) \subseteq \Sigma_1\}$. By assumption, $\mathcal{I} \subseteq \mathcal{D}$. Furthermore, $\mathcal{D}$ is a d -system on S_2 :

- $S_2 \in \mathcal{D}$, since $S_2 \in \Sigma_2$ and $K(\cdot, S_2) = 1 = \mathbb{I}_{S_1}$ and $\mathbb{I}_{S_1}$ is Σ_1 -measurable.
- If $B_1, B_2 \in \mathcal{D}$ and $B_1 \subseteq B_2$, then $B_2 \setminus B_1 \in \mathcal{D}$. In order to see this, note that $B_2 \setminus B_1 \in \Sigma_2$ and

$$K(\cdot, B_2 \setminus B_1) = K(\cdot, B_2 \cap B_1^c) = 1 - K(\cdot, B_2^c \cup B_1) = 1 - K(\cdot, B_2^c) - K(\cdot, B_1) = K(\cdot, B_2) - K(\cdot, B_1).$$

Since $K(\cdot, B_2)$ and $K(\cdot, B_1)$ are Σ_1 -measurable, we know that $K(\cdot, B_2 \setminus B_1)$ is Σ_1 -measurable.

- For any sequence $(B_n \in \mathcal{D} \mid n \in \mathbb{N})$, if $B_n \subseteq B_{n+1}$ for every $n \in \mathbb{N}$, then $\cup_n B_n \in \mathcal{D}$. In order to see this, first note that $\cup_n B_n \in \Sigma_2$. By the monotone-convergence property of measure,

$$K(\cdot, \cup_n B_n) = \lim_{n \rightarrow \infty} K(\cdot, B_n).$$

Because $K(\cdot, B_n)$ is Σ_1 -measurable for every $n \in \mathbb{N}$, we know that $K(\cdot, \cup_n B_n)$ is Σ_1 -measurable.

Because $\mathcal{I}$ is a π -system on S_2 and $\mathcal{D}$ is a d -system on S_2 such that $\mathcal{I} \subseteq \mathcal{D}$, Dynkin's lemma shows that $\Sigma_2 \subseteq \mathcal{D}$. Since $\mathcal{D} \subseteq \Sigma_2$, we know that $\mathcal{D} = \Sigma_2$. Therefore, for every $B_2 \in \Sigma_2$, the function $K(\cdot, B_2)$ is Σ_1 -measurable. $\square$

Proposition 9.2. Consider a probability kernel $K : S_1 \times S_2 \rightarrow [0, 1]$ and a Σ -measurable function $f : S \rightarrow [0, \infty]$. The function J_1^f is Σ_1 -measurable, where $J_1^f : S_1 \rightarrow [0, \infty]$ is given by

$$J_1^f(s_1) = \int_{S_2} f(s_1, s_2) K(s_1, ds_2).$$

Proof. Recall that there is a Σ_2 -measurable function $f_{s_1} : S_2 \rightarrow [0, \infty]$ such that $f_{s_1}(s_2) = f(s_1, s_2)$ for every $s_1 \in S_1$ and $s_2 \in S_2$, so that J_1^f is indeed well-defined.

Let $\mathcal{I} = \{B_1 \times B_2 \mid B_1 \in \Sigma_1 \text{ and } B_2 \in \Sigma_2\}$, so that $\sigma(\mathcal{I}) = \Sigma$. For every $B_1 \times B_2 \in \mathcal{I}$,

$$J_1^{\mathbb{I}_{B_1 \times B_2}}(s_1) = \int_{S_2} \mathbb{I}_{B_1 \times B_2}(s_1, s_2) K(s_1, ds_2) = \mathbb{I}_{B_1}(s_1) \int_{S_2} \mathbb{I}_{B_2}(s_2) K(s_1, ds_2) = \mathbb{I}_{B_1}(s_1) K(s_1, B_2).$$

Therefore, for every $B_1 \times B_2 \in \mathcal{I}$, the function $J_1^{\mathbb{I}_{B_1 \times B_2}}$ is Σ_1 -measurable, since $\mathbb{I}_{B_1}$ and $K(\cdot, B_2)$ are Σ_1 -measurable.

Let $\mathcal{D} = \{A \in \Sigma \mid \sigma(J_1^{\mathbb{I}_A}) \subseteq \Sigma_1\}$, so that $\mathcal{I} \subseteq \mathcal{D}$. Note that $\mathcal{D}$ is a d -system on S :

- $S \in \mathcal{D}$, since $S \in \Sigma$ and $S = S_1 \times S_2$ and $J_1^{\mathbb{I}_S}(s_1) = \mathbb{I}_{S_1}(s_1) K(s_1, S_2) = \mathbb{I}_{S_1}(s_1) = 1$ and $\mathbb{I}_{S_1}$ is Σ_1 -measurable.
- If $A_1, A_2 \in \mathcal{D}$ and $A_1 \subseteq A_2$, then $A_2 \setminus A_1 \in \mathcal{D}$. In order to see this, note that $A_2 \setminus A_1 \in \Sigma$ and

$$\mathbb{I}_{A_2 \setminus A_1} = \mathbb{I}_{A_2 \cap A_1^c} = \mathbb{I}_{A_2} \mathbb{I}_{A_1^c} = \mathbb{I}_{A_2} (1 - \mathbb{I}_{A_1}) = \mathbb{I}_{A_2} - \mathbb{I}_{A_1} \mathbb{I}_{A_2} = \mathbb{I}_{A_2} - \mathbb{I}_{A_1 \cap A_2} = \mathbb{I}_{A_2} - \mathbb{I}_{A_1},$$

so that

$$J_1^{\mathbb{I}_{A_2 \setminus A_1}}(s_1) = \int_{S_2} \mathbb{I}_{A_2 \setminus A_1}(s_1, s_2) K(s_1, ds_2) = \int_{S_2} \mathbb{I}_{A_2}(s_1, s_2) K(s_1, ds_2) - \int_{S_2} \mathbb{I}_{A_1}(s_1, s_2) K(s_1, ds_2) = J_1^{\mathbb{I}_{A_2}}(s_1) - J_1^{\mathbb{I}_{A_1}}(s_1).$$

Because $J_1^{\mathbb{I}_{A_2}}$ and $J_1^{\mathbb{I}_{A_1}}$ are Σ_1 -measurable, $J_1^{\mathbb{I}_{A_2 \setminus A_1}}$ is Σ_1 -measurable.

- For any sequence $(A_n \in \mathcal{D} \mid n \in \mathbb{N})$, if $A_n \subseteq A_{n+1}$ for every $n \in \mathbb{N}$, then $\cup_n A_n \in \mathcal{D}$. In order to see this, note that $\cup_n A_n \in \Sigma$ and $\mathbb{I}_{A_n}(s_1, \cdot) \leq \mathbb{I}_{A_{n+1}}(s_1, \cdot)$ for every $n \in \mathbb{N}$ and $s_1 \in S_1$, so that $\mathbb{I}_{A_k}(s_1, \cdot) \uparrow \mathbb{I}_{\cup_n A_n}(s_1, \cdot)$. By the monotone-convergence theorem,

$$J_1^{\mathbb{I}_{\cup_n A_n}}(s_1) = \int_{S_2} \mathbb{I}_{\cup_n A_n}(s_1, s_2) K(s_1, ds_2) = \lim_{n \rightarrow \infty} \int_{S_2} \mathbb{I}_{A_n}(s_1, s_2) K(s_1, ds_2) = \lim_{n \rightarrow \infty} J_1^{\mathbb{I}_{A_n}}(s_1).$$

Because $J_1^{\mathbb{I}_{A_n}}$ is Σ_1 -measurable for every $n \in \mathbb{N}$, $J_1^{\mathbb{I}_{\cup_n A_n}}$ is Σ_1 -measurable.

Because $\mathcal{I}$ is a π -system on S and $\mathcal{D}$ is a d -system on S such that $\mathcal{I} \subseteq \mathcal{D}$, Dynkin's lemma shows that $\Sigma \subseteq \mathcal{D}$. Since $\mathcal{D} \subseteq \Sigma$, we know that $\mathcal{D} = \Sigma$. Therefore, for every $A \in \Sigma$, the function $J_1^{\mathbb{I}_A}$ is Σ_1 -measurable.

Next, suppose $f : S \rightarrow [0, \infty]$ is a simple function that can be written as $f = \sum_{k=1}^m a_k \mathbb{I}_{A_k}$ for some fixed $a_1, a_2, \dots, a_m \in [0, \infty]$ and $A_1, A_2, \dots, A_m \in \Sigma$. In that case,

$$J_1^f(s_1) = \int_{S_2} \sum_{k=1}^m a_k \mathbb{I}_{A_k}(s_1, s_2) K(s_1, ds_2) = \sum_{k=1}^m a_k \int_{S_2} \mathbb{I}_{A_k}(s_1, s_2) K(s_1, ds_2) = \sum_{k=1}^m a_k J_1^{\mathbb{I}_{A_k}}(s_1).$$

Because $J_1^{\mathbb{I}_{A_k}}$ is Σ_1 -measurable for every $A_1, A_2, \dots, A_m \in \Sigma$, the function J_1^f is Σ_1 -measurable.

Finally, consider a Σ -measurable function $f : S \rightarrow [0, \infty]$. For any $n \in \mathbb{N}$, let $f_n = \alpha_n \circ f$, where α_n is the n -th staircase function. For every $n \in \mathbb{N}$, because $f_n : S \rightarrow [0, n]$ is bounded and Σ -measurable, there is a bounded Σ_2 -measurable function $f_{n, s_1} : S_2 \rightarrow [0, n]$ such that $f_n(s_1, s_2) = f_{n, s_1}(s_2)$ for every $s_1 \in S_1$ and $s_2 \in S_2$. Since $f_n \uparrow f$, consider the Σ_2 -measurable function $f_{s_1} = \lim_{n \rightarrow \infty} f_{n, s_1}$ and note that $f(s_1, s_2) = f_{s_1}(s_2)$ for every $s_1 \in S_1$ and $s_2 \in S_2$. Since $f_{n, s_1} \uparrow f_{s_1}$ for every $s_1 \in S_1$, by the monotone-convergence theorem,

$$J_1^f(s_1) = \int_{S_2} f_{s_1}(s_2) K(s_1, ds_2) = \lim_{n \rightarrow \infty} \int_{S_2} f_{n, s_1}(s_2) K(s_1, ds_2) = \lim_{n \rightarrow \infty} J_1^{f_n}(s_1).$$

Because f_n is a simple function for every $n \in \mathbb{N}$, the function $J_1^{f_n}$ is Σ_1 -measurable for every $n \in \mathbb{N}$, so that the function J_1^f is Σ_1 -measurable for every Σ -measurable function $f : S \rightarrow [0, \infty]$. $\square$

Theorem 9.1. Consider a probability kernel $K_2 : S_1 \times S_2 \rightarrow [0, 1]$, a probability measure μ_1 on the measurable space (S_1, Σ_1) , and the function $\mu : \Sigma \rightarrow [0, \infty]$ given by

$$\mu(A) = \int_{S_1} \int_{S_2} \mathbb{I}_A(s_1, s_2) K_2(s_1, ds_2) \mu_1(ds_1).$$

The function μ , which is also denoted by $\mu = \mu_1 \times K_2$, is the only probability measure on (S, Σ) such that

$$\mu(B_1 \times B_2) = \int_{B_1} K_2(s_1, B_2) \mu_1(ds_1)$$

for every $B_1 \in \Sigma_1$ and $B_2 \in \Sigma_2$.

Proof. Note that the function $\mu : \Sigma \rightarrow [0, \infty]$ is given by

$$\mu(A) = \int_{S_1} J_1^{\mathbb{I}_A}(s_1) \mu_1(ds_1),$$

where $J_1^{\mathbb{I}_A} : S_1 \rightarrow [0, \infty]$ is a Σ_1 -measurable function given by $J_1^{\mathbb{I}_A}(s_1) = \int_{S_2} \mathbb{I}_A(s_1, s_2) K_2(s_1, ds_2)$.

Clearly, $\mu(\emptyset) = 0$ and $\mu(S) = 1$. For any sequence $(A_n \in \Sigma \mid n \in \mathbb{N})$ such that $A_n \cap A_m = \emptyset$ for $n \neq m$,

$$\mu\left(\bigcup_n A_n\right) = \int_{S_1} \int_{S_2} \mathbb{I}_{\bigcup_n A_n}(s_1, s_2) K_2(s_1, ds_2) \mu_1(ds_1) = \int_{S_1} \int_{S_2} \sum_n \mathbb{I}_{A_n}(s_1, s_2) K_2(s_1, ds_2) \mu_1(ds_1) = \sum_n \mu(A_n),$$

where the last step relies on the fact that $\mathbb{I}_{A_n} \geq 0$ for every $n \in \mathbb{N}$. Therefore, μ is a probability measure on (S, Σ) .

Finally, let $\mathcal{I} = \{B_1 \times B_2 \mid B_1 \in \Sigma_1 \text{ and } B_2 \in \Sigma_2\}$. Because $\mathcal{I}$ is a π -system on S such that $\sigma(\mathcal{I}) = \Sigma$, μ is the unique probability measure on (S, Σ) such that, for every $B_1 \times B_2 \in \mathcal{I}$,

$$\mu(B_1 \times B_2) = \int_{S_1} J_1^{\mathbb{I}_{B_1 \times B_2}}(s_1) \mu_1(ds_1) = \int_{S_1} \mathbb{I}_{B_1}(s_1) K_2(s_1, B_2) \mu_1(ds_1) = \int_{B_1} K_2(s_1, B_2) \mu_1(ds_1).$$

□

Consider the measurable spaces (S_1, Σ_1) , (S_2, Σ_2) , (S_3, Σ_3) , and $(S, \Sigma) = (S_1 \times S_2 \times S_3, \Sigma_1 \times \Sigma_2 \times \Sigma_3)$.

Theorem 9.2. Consider a probability kernel $K_2 : S_1 \times \Sigma_2 \rightarrow [0, 1]$, a probability kernel $K_3 : (S_1 \times S_2) \times \Sigma_3 \rightarrow [0, 1]$, and a function $K : S_1 \times (\Sigma_2 \times \Sigma_3) \rightarrow [0, 1]$ given by

$$K(s_1, A) = \int_{S_2} \int_{S_3} \mathbb{I}_A(s_2, s_3) K_3(s_1, s_2, ds_3) K_2(s_1, ds_2).$$

The function K , which is also denoted by $K = K_2 \times K_3$, is a probability kernel from S_1 to $S_2 \times S_3$. For every $s_1 \in S_1$, the function $K(s_1, \cdot) : \Sigma_2 \times \Sigma_3 \rightarrow [0, 1]$ is the only probability measure on $(S_2 \times S_3, \Sigma_2 \times \Sigma_3)$ such that

$$K(s_1, B_2 \times B_3) = \int_{B_2} K_3(s_1, s_2, B_3) K_2(s_1, ds_2)$$

for every $B_2 \in \Sigma_2$ and $B_3 \in \Sigma_3$.

Proof. For every $s_1 \in S_1$, the function $K_3(s_1, \cdot) : S_2 \times \Sigma_3 \rightarrow [0, 1]$ is a probability kernel from S_2 to S_3 :

- For every $s_2 \in S_2$, the function $K_3(s_1, s_2, \cdot) : \Sigma_3 \rightarrow [0, 1]$ is a probability measure on (S_3, Σ_3) .
- For every $B_3 \in \Sigma_3$, the function $K_3(s_1, \cdot, B_3) : S_2 \rightarrow [0, 1]$ is Σ_2 -measurable since $K_3(\cdot, B_3) : S_1 \times S_2 \rightarrow [0, 1]$ is a (bounded) $\Sigma_1 \times \Sigma_2$ -measurable function.

For every $s_1 \in S_1$, consider the probability kernel $K_3(s_1, \cdot) : S_2 \times \Sigma_3 \rightarrow [0, 1]$ and the probability measure $K_2(s_1, \cdot) : \Sigma_2 \rightarrow [0, 1]$ on the measurable space (S_2, Σ_2) . By Theorem 9.1, the function $K(s_1, \cdot) : (\Sigma_2 \times \Sigma_3) \rightarrow [0, 1]$ is the only probability measure on $(S_2 \times S_3, \Sigma_2 \times \Sigma_3)$ such that, for every $B_2 \in \Sigma_2$ and $B_3 \in \Sigma_3$,

$$K(s_1, B_2 \times B_3) = \int_{B_2} K_3(s_1, s_2, B_3) K_2(s_1, ds_2).$$

By Proposition 9.1, if the function $K(\cdot, B_2 \times B_3) : S_1 \rightarrow [0, 1]$ is Σ_1 -measurable for every $B_2 \in \Sigma_2$ and $B_3 \in \Sigma_3$, then K is a probability kernel from S_1 to $S_2 \times S_3$.

For every $B_2 \in \Sigma_2$ and $B_3 \in \Sigma_3$, the functions $\mathbb{I}_{S_1 \times B_2} : S_1 \times S_2 \rightarrow [0, 1]$ and $K_3(\cdot, B_3) : S_1 \times S_2 \rightarrow [0, 1]$ are $\Sigma_1 \times \Sigma_2$ -measurable, so that the function $\mathbb{I}_{S_1 \times B_2} K_3(\cdot, B_3)$ is $\Sigma_1 \times \Sigma_2$ -measurable. By Proposition 9.2, the function $K(\cdot, B_2 \times B_3) : S_1 \rightarrow [0, 1]$ is Σ_1 -measurable, since

$$K(s_1, B_2 \times B_3) = \int_{B_2} K_3(s_1, s_2, B_3) K_2(s_1, ds_2) = \int_{S_2} \mathbb{I}_{S_1 \times B_2}(s_1, s_2) K_3(s_1, s_2, B_3) K_2(s_1, ds_2).$$

□

Theorem 9.3 (Ionescu-Tulcea theorem). Consider a sequence $((S_n, \Sigma_n) \mid n \in \mathbb{N}^+)$ of measurable spaces and a probability measure μ_1 on the measurable space (S_1, Σ_1) . For every $n > 1$, let $K_n : \left(\prod_{k=1}^{n-1} S_k\right) \times \Sigma_n \rightarrow [0, 1]$ be a probability kernel. There is a probability triple $(S, \Sigma, \mathbb{P})$ and a sequence of functions $(X_n : S \rightarrow S_n \mid n \in \mathbb{N}^+)$ such that X_n is Σ/Σ_n -measurable for every $n \in \mathbb{N}^+$ and, for every $\Gamma \in \Sigma_1 \times \Sigma_2 \times \cdots \times \Sigma_n$,

$$\mathbb{P}((X_1, X_2, \dots, X_n) \in \Gamma) = (\mu_1 \times (K_2 \times \cdots \times K_n))(\Gamma).$$

10 Conditional expectation

Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable $X : \Omega \rightarrow \mathbb{R}$. For every $\omega \in \Omega$, note that knowing $\mathbb{I}_{\{X=x\}}(\omega)$ for every $x \in \mathbb{R}$ is equivalent to knowing $X(\omega)$. Furthermore, from a previous result,

$$\sigma(X) = \left\{ X^{-1} \left(\bigcup_{x \in B} \{x\} \right) \mid B \in \mathcal{B}(\mathbb{R}) \right\} = \left\{ \bigcup_{x \in B} X^{-1}(\{x\}) \mid B \in \mathcal{B}(\mathbb{R}) \right\} = \left\{ \bigcup_{x \in B} \{X = x\} \mid B \in \mathcal{B}(\mathbb{R}) \right\}.$$

Let $F = \cup_{x \in B} \{X = x\}$ for some $B \in \mathcal{B}(\mathbb{R})$. Because F is a union of disjoint sets, for every $\omega \in \Omega$ we have $\mathbb{I}_F(\omega) = 1$ if and only if $\mathbb{I}_{\{X=x\}}(\omega) = 1$ for exactly one $x \in B$, and $\mathbb{I}_F(\omega) = 0$ otherwise. Finally, note that $\{X = x\} \in \sigma(X)$ for every $x \in \mathbb{R}$. Therefore, for every $\omega \in \Omega$, knowing $\mathbb{I}_{\{X=x\}}(\omega)$ for every $x \in \mathbb{R}$ is also equivalent to knowing $\mathbb{I}_F(\omega)$ for every $F \in \sigma(X)$.

In conclusion, for every $\omega \in \Omega$, knowing $X(\omega)$ is equivalent to knowing $\mathbb{I}_F(\omega)$ for every $F \in \sigma(X)$.

More generally, consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a set of random variables $\{Y_\gamma \mid \gamma \in \mathcal{C}\}$ where $Y_\gamma : \Omega \rightarrow \mathbb{R}$ for every $\gamma \in \mathcal{C}$. Suppose that an unknown outcome $\omega \in \Omega$ results in a known value $Y_\gamma(\omega) \in \mathbb{R}$ for every $\gamma \in \mathcal{C}$. The σ -algebra $\sigma(\{Y_\gamma \mid \gamma \in \mathcal{C}\})$ contains exactly each event $F \in \mathcal{F}$ such that it is possible to state whether $\omega \in F$. In other words, for every $\omega \in \Omega$, knowing $Y_\gamma(\omega) \in \mathbb{R}$ for every $\gamma \in \mathcal{C}$ is equivalent to knowing $\mathbb{I}_F(\omega)$ for every $F \in \sigma(\{Y_\gamma \mid \gamma \in \mathcal{C}\})$.

Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$. Suppose $\sigma(Y) \subseteq \sigma(X)$. For every $\omega \in \Omega$, knowing $X(\omega)$ allows knowing $\mathbb{I}_F(\omega)$ for every $F \in \sigma(Y)$. Therefore, knowing $X(\omega)$ allows knowing $Y(\omega)$.

Proposition 10.1. For every function $Z : \Omega \rightarrow \mathbb{R}$, a function $Y : \Omega \rightarrow \mathbb{R}$ is $\sigma(Z)$ -measurable if and only if there is a Borel function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $Y = f \circ Z$. Furthermore, if $Z_1, Z_2, \dots, Z_n$ are functions from Ω to $\mathbb{R}$, then a function $Y : \Omega \rightarrow \mathbb{R}$ is $\sigma(\{Z_1, Z_2, \dots, Z_n\})$ -measurable if and only if there is a Borel function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $Y(\omega) = f(Z_1(\omega), Z_2(\omega), \dots, Z_n(\omega))$ for every $\omega \in \Omega$.

Definition 10.1. Consider the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a random variable $X : \Omega \rightarrow \mathbb{R}$ such that $\mathbb{E}(|X|) < \infty$, and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. A random variable $Y : \Omega \rightarrow \mathbb{R}$ is called a version of the conditional expectation $\mathbb{E}(X \mid \mathcal{G})$ of X given $\mathcal{G}$ if and only if Y is $\mathcal{G}$ -measurable, $\mathbb{E}(|Y|) < \infty$, and, for every set $G \in \mathcal{G}$,

$$\int_G Y d\mathbb{P} = \int_G X d\mathbb{P}.$$

In that case, we say that $Y = \mathbb{E}(X \mid \mathcal{G})$ almost surely.

Proposition 10.2. Given the definition above, a version Y of the conditional expectation $\mathbb{E}(X \mid \mathcal{G})$ of X given $\mathcal{G}$ always exists. Furthermore, if Y and $\tilde{Y}$ are such versions, then $\mathbb{P}(Y = \tilde{Y}) = 1$.

Proof. First, suppose $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and recall that $\mathcal{L}^2(\Omega, \mathcal{G}, \mathbb{P})$ is a complete vector space. Because $\mathcal{L}^2(\Omega, \mathcal{G}, \mathbb{P}) \subseteq \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$, there is a version $Y \in \mathcal{L}^2(\Omega, \mathcal{G}, \mathbb{P})$ of the orthogonal projection of X onto $\mathcal{L}^2(\Omega, \mathcal{G}, \mathbb{P})$ such that $\|X - Y\|_2 = \inf\{\|X - W\|_2 \mid W \in \mathcal{L}^2(\Omega, \mathcal{G}, \mathbb{P})\}$ and $\mathbb{E}((X - Y)Z) = 0$, for every $Z \in \mathcal{L}^2(\Omega, \mathcal{G}, \mathbb{P})$. Clearly, Y is $\mathcal{G}$ -measurable. By the monotonicity of norm, $\mathbb{E}(|Y|) < \infty$. For every $G \in \mathcal{G}$, we have $\mathbb{I}_G \in \mathcal{L}^2(\Omega, \mathcal{G}, \mathbb{P})$, so that $\mathbb{E}((X - Y)\mathbb{I}_G) = 0$. Therefore, by the linearity of expectation, $\mathbb{E}(X\mathbb{I}_G) = \mathbb{E}(Y\mathbb{I}_G)$, which completes this step.

Suppose that X is a bounded non-negative random variable, so that $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. As an auxiliary step, we will now show that if $Y = \mathbb{E}(X \mid \mathcal{G})$ almost surely, then $\mathbb{P}(Y \geq 0) = 1$. In order to find a contradiction, suppose that $\mathbb{P}(Y \geq 0) < 1$, so that $\mathbb{P}(Y < 0) > 0$. Let $A_n = \{Y < -n^{-1}\} = Y^{-1}((-\infty, -n^{-1}))$, so that $A_n \subseteq A_{n+1}$ and $\cup_n A_n = \{Y < 0\}$. Since $A_n \uparrow \{Y < 0\}$, the monotone-convergence property of measure guarantees that $\mathbb{P}(A_n) \uparrow \mathbb{P}(Y < 0)$. Because we supposed that $\mathbb{P}(Y < 0) > 0$, there is an $n \in \mathbb{N}$ such that $\mathbb{P}(A_n) = \mathbb{P}(Y < -n^{-1}) > 0$. Consider the random variable $Y\mathbb{I}_{A_n}$ given by

$$(Y\mathbb{I}_{A_n})(\omega) = Y(\omega)\mathbb{I}_{A_n}(\omega) = \begin{cases} Y(\omega), & \text{if } Y(\omega) < -n^{-1}, \\ 0, & \text{if } Y(\omega) \geq -n^{-1}. \end{cases}$$

Because $Y\mathbb{I}_{A_n} \leq -n^{-1}\mathbb{I}_{A_n}$, we know that $\mathbb{E}(Y\mathbb{I}_{A_n}) \leq -n^{-1}\mathbb{P}(A_n) < 0$. Because $X \geq 0$, we know that $\mathbb{E}(X\mathbb{I}_{A_n}) \geq 0$. However, $A_n \in \mathcal{G}$, so that $\mathbb{E}(X\mathbb{I}_{A_n}) = \mathbb{E}(Y\mathbb{I}_{A_n})$. Because this is a contradiction, we know that $\mathbb{P}(Y \geq 0) = 1$.

Next, suppose $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ is non-negative. For every $n \in \mathbb{N}$, let $X_n = \alpha_n \circ X$, where α_n is the n -th staircase function, so that $X_n \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. Furthermore, let $Y_n = \mathbb{E}(X_n \mid \mathcal{G})$ almost surely. Because X_n is a bounded non-negative random variable, we know that $\mathbb{P}(Y_n \geq 0) = 1$. For every $n \in \mathbb{N}$ and $G \in \mathcal{G}$, note that

$$\mathbb{E}((Y_{n+1} - Y_n)\mathbb{I}_G) = \mathbb{E}(Y_{n+1}\mathbb{I}_G) - \mathbb{E}(Y_n\mathbb{I}_G) = \mathbb{E}(X_{n+1}\mathbb{I}_G) - \mathbb{E}(X_n\mathbb{I}_G) = \mathbb{E}((X_{n+1} - X_n)\mathbb{I}_G).$$

Because $Y_n \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathbb{P})$ and $Y_{n+1} \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathbb{P})$, we know that $Y_{n+1} - Y_n = \mathbb{E}(X_{n+1} - X_n \mid \mathcal{G})$ almost surely. Because $X_{n+1} - X_n$ is non-negative and bounded for every $n \in \mathbb{N}$, we know that $\mathbb{P}(Y_{n+1} - Y_n \geq 0) = 1$.

Consider the set $A^c = \bigcup_n \{Y_n < 0\} \cup \{Y_{n+1} - Y_n < 0\}$. Note that $A \in \mathcal{G}$ and $\mathbb{P}(A) = 1$, since

$$\mathbb{P}(A^c) = \mathbb{P}\left(\bigcup_n \{Y_n < 0\} \cup \{Y_{n+1} - Y_n < 0\}\right) \leq \sum_n \mathbb{P}(Y_n < 0) + \mathbb{P}(Y_{n+1} - Y_n < 0) = 0.$$

For every $n \in \mathbb{N}$, note that $Y_n \mathbb{I}_A \geq 0$ and $Y_{n+1} \mathbb{I}_A \geq Y_n \mathbb{I}_A$. Let $Y = \limsup_{n \rightarrow \infty} Y_n \mathbb{I}_A$. For every $G \in \mathcal{G}$, because every non-decreasing sequence of real numbers converges (possibly to infinity), we know that $Y_n \mathbb{I}_A \mathbb{I}_G \uparrow Y \mathbb{I}_G$. By the monotone-convergence theorem, we know that $\mathbb{E}(Y_n \mathbb{I}_A \mathbb{I}_G) \uparrow \mathbb{E}(Y \mathbb{I}_G)$.

For every $n \in \mathbb{N}$ and $G \in \mathcal{G}$, we have $(A \cap G) \in \mathcal{G}$ and $\mathbb{P}(X_n \mathbb{I}_G \mathbb{I}_{A^c} \neq 0) = 0$, so that

$$\mathbb{E}(Y_n \mathbb{I}_A \mathbb{I}_G) = \mathbb{E}(Y_n \mathbb{I}_{A \cap G}) = \mathbb{E}(X_n \mathbb{I}_{A \cap G}) = \mathbb{E}(X_n \mathbb{I}_A \mathbb{I}_G) + \mathbb{E}(X_n \mathbb{I}_{A^c} \mathbb{I}_G) = \mathbb{E}(X_n \mathbb{I}_G),$$

which implies $\mathbb{E}(X_n \mathbb{I}_G) \uparrow \mathbb{E}(Y \mathbb{I}_G)$. Since $X_n \mathbb{I}_G \uparrow X \mathbb{I}_G$, we also know that $\mathbb{E}(X_n \mathbb{I}_G) \uparrow \mathbb{E}(X \mathbb{I}_G)$, so that $\mathbb{E}(Y \mathbb{I}_G) = \mathbb{E}(X \mathbb{I}_G)$. Because $\Omega \in \mathcal{G}$, this gives $\mathbb{E}(Y) = \mathbb{E}(X) < \infty$, so that the non-negative $\mathcal{G}$ -measurable function Y is almost surely finite and may be taken to be real valued. Therefore, $Y = \mathbb{E}(X \mid \mathcal{G})$ almost surely.

Finally, suppose $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Let $X = X^+ - X^-$, where $X^+ : \Omega \rightarrow [0, \infty]$ and $X^- : \Omega \rightarrow [0, \infty]$. Let $Y^+ = \mathbb{E}(X^+ \mid \mathcal{G})$ almost surely and $Y^- = \mathbb{E}(X^- \mid \mathcal{G})$ almost surely. For every $G \in \mathcal{G}$,

$$\mathbb{E}(X \mathbb{I}_G) = \mathbb{E}((X^+ - X^-) \mathbb{I}_G) = \mathbb{E}(X^+ \mathbb{I}_G) - \mathbb{E}(X^- \mathbb{I}_G) = \mathbb{E}(Y^+ \mathbb{I}_G) - \mathbb{E}(Y^- \mathbb{I}_G) = \mathbb{E}((Y^+ - Y^-) \mathbb{I}_G),$$

so that $Y^+ - Y^- = \mathbb{E}(X \mid \mathcal{G})$ almost surely.

It remains to show that if $Y = \mathbb{E}(X \mid \mathcal{G})$ almost surely and $\tilde{Y} = \mathbb{E}(X \mid \mathcal{G})$ almost surely then $\mathbb{P}(Y = \tilde{Y}) = 1$. For the purpose of finding a contradiction, suppose that $\mathbb{P}(Y = \tilde{Y}) < 1$, so that $\mathbb{P}(Y \neq \tilde{Y}) > 0$. In that case, $\mathbb{P}(Y > \tilde{Y}) + \mathbb{P}(\tilde{Y} > Y) > 0$, so that $\mathbb{P}(Y > \tilde{Y}) > 0$ or $\mathbb{P}(\tilde{Y} > Y) > 0$. Suppose $\mathbb{P}(Y > \tilde{Y}) > 0$. Let $A_n = \{Y > \tilde{Y} + n^{-1}\} = (Y - \tilde{Y})^{-1}((n^{-1}, \infty))$, so that $A_n \subseteq A_{n+1}$ and $\bigcup_n A_n = \{Y > \tilde{Y}\}$. By the monotone-convergence property of measure, we know that $\mathbb{P}(A_n) \uparrow \mathbb{P}(Y > \tilde{Y})$. Because $\mathbb{P}(Y > \tilde{Y}) > 0$, there is an $n \in \mathbb{N}$ such that $\mathbb{P}(A_n) = \mathbb{P}(Y > \tilde{Y} + n^{-1}) > 0$. Note that $(Y - \tilde{Y}) \mathbb{I}_{A_n} \geq n^{-1} \mathbb{I}_{A_n}$, since

$$(Y - \tilde{Y})(\omega) \mathbb{I}_{A_n}(\omega) = \begin{cases} (Y - \tilde{Y})(\omega), & \text{if } (Y - \tilde{Y})(\omega) > n^{-1}, \\ 0, & \text{if } (Y - \tilde{Y})(\omega) \leq n^{-1}. \end{cases}$$

Therefore, $\mathbb{E}((Y - \tilde{Y}) \mathbb{I}_{A_n}) \geq \mathbb{E}(n^{-1} \mathbb{I}_{A_n}) = n^{-1} \mathbb{P}(A_n) > 0$. However, for every $G \in \mathcal{G}$, note that $\mathbb{E}(Y \mathbb{I}_G) = \mathbb{E}(\tilde{Y} \mathbb{I}_G)$, so that $\mathbb{E}((Y - \tilde{Y}) \mathbb{I}_G) = 0$. Because $A_n \in \mathcal{G}$, we arrived at a contradiction. An analogous contradiction is found by supposing that $\mathbb{P}(\tilde{Y} > Y) > 0$. Therefore, $\mathbb{P}(Y = \tilde{Y}) = 1$. $\square$

Definition 10.2. Consider the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a random variable $X : \Omega \rightarrow \mathbb{R}$ such that $\mathbb{E}(|X|) < \infty$, and a random variable $Z : \Omega \rightarrow \mathbb{R}$. A random variable $Y : \Omega \rightarrow \mathbb{R}$ is called a version of the conditional expectation $\mathbb{E}(X \mid Z)$ of X given Z if and only if it is a version of the conditional expectation $\mathbb{E}(X \mid \sigma(Z))$ of X given $\sigma(Z)$. An analogous definition applies when Z is a set of random variables.

Suppose $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $Z : \Omega \rightarrow \mathbb{R}$ are random variables and let $Y = \mathbb{E}(X \mid Z)$ almost surely. Recall that for every $W \in \mathcal{L}^2(\Omega, \sigma(Z), \mathbb{P})$ there is a Borel function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $W = f \circ Z$ and that $\mathbb{E}((X - Y)^2) \leq \mathbb{E}((X - W)^2)$. In this sense, if $g : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel function such that $Y = g \circ Z$, then $Y(\omega) = g(Z(\omega))$ is almost surely the best prediction about $X(\omega)$ that can be made given $Z(\omega)$.

The next three examples illustrate the definition of conditional expectation.

Proposition 10.3. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $X : \Omega \rightarrow \mathcal{X}$ and $Z : \Omega \rightarrow \mathcal{Z}$, where $\mathcal{X} = \{x_1, \dots, x_m\}$ and $\mathcal{Z} = \{z_1, \dots, z_n\}$. Furthermore, suppose $\mathbb{P}(Z = z) > 0$ for every $z \in \mathcal{Z}$.

Let $\mathcal{P}(\mathcal{Z})$ denote the set of all subsets of $\mathcal{Z}$ and consider the $\mathcal{P}(\mathcal{Z})$ -measurable function $E : \mathcal{Z} \rightarrow \mathbb{R}$ given by

$$E(z) = \sum_i x_i \frac{\mathbb{P}(X = x_i, Z = z)}{\mathbb{P}(Z = z)}.$$

In that case, $Y = E \circ Z$ is a $\sigma(Z)$ -measurable function such that

$$\int_G Y d\mathbb{P} = \int_G X d\mathbb{P},$$

for every $G \in \sigma(Z)$, so that $Y = \mathbb{E}(X \mid Z)$ almost surely.

Proof. For every $B \in \mathcal{B}(\mathbb{R})$, recall that $Y^{-1}(B) = Z^{-1}(E^{-1}(B))$. Because $E^{-1}(B) \in \mathcal{P}(\mathcal{Z})$ and $\mathcal{P}(\mathcal{Z}) \subseteq \mathcal{B}(\mathbb{R})$, we know that $Y^{-1}(B) \in \sigma(Z)$. Therefore, Y is $\sigma(Z)$ -measurable.

Because Y is a bounded $\mathcal{F}$ -measurable function and $\{Z = z\} \in \mathcal{F}$ for every $z \in \mathcal{Z}$,

$$\int_{\{Z=z\}} Y d\mathbb{P} = \int_{\Omega} \mathbb{I}_{\{Z=z\}}(\omega) E(Z(\omega)) \mathbb{P}(d\omega) = \int_{\Omega} \mathbb{I}_{\{Z=z\}}(\omega) E(z) \mathbb{P}(d\omega) = E(z) \mathbb{P}(Z = z) = \sum_i x_i \mathbb{P}(X = x_i, Z = z).$$

By the definition of the integral of a simple function with respect to $\mathbb{P}$,

$$\int_{\{Z=z\}} Y d\mathbb{P} = \int_{\Omega} \left(\sum_i x_i \mathbb{I}_{\{X=x_i, Z=z\}} \right) d\mathbb{P} = \int_{\Omega} \left(\mathbb{I}_{\{Z=z\}} \sum_i x_i \mathbb{I}_{\{X=x_i\}} \right) d\mathbb{P} = \int_{\Omega} \mathbb{I}_{\{Z=z\}} X d\mathbb{P} = \int_{\{Z=z\}} X d\mathbb{P}.$$

Because $Z(\omega) \in \mathcal{Z}$ for every $\omega \in \Omega$ and $\mathcal{P}(\mathcal{Z}) \subseteq \mathcal{B}(\mathbb{R})$,

$$\sigma(Z) = \left\{ \bigcup_{z \in B} \{Z = z\} \mid B \in \mathcal{B}(\mathbb{R}) \right\} = \left\{ \bigcup_{z \in B} \{Z = z\} \mid B \in \mathcal{P}(\mathcal{Z}) \right\}.$$

Let $G = \bigcup_{z \in B} \{Z = z\}$ for some $B \in \mathcal{P}(\mathcal{Z})$. For every $\omega \in \Omega$, note that $\mathbb{I}_G(\omega) = \sum_{z \in B} \mathbb{I}_{\{Z=z\}}(\omega)$, since G is a union of disjoint sets. Therefore, because Y is a bounded $\mathcal{F}$ -measurable function and $G \in \mathcal{F}$,

$$\int_G Y d\mathbb{P} = \int_{\Omega} \sum_{z \in B} \mathbb{I}_{\{Z=z\}}(\omega) Y(\omega) \mathbb{P}(d\omega) = \sum_{z \in B} \int_{\Omega} \mathbb{I}_{\{Z=z\}}(\omega) Y(\omega) \mathbb{P}(d\omega) = \sum_{z \in B} \int_{\Omega} \mathbb{I}_{\{Z=z\}}(\omega) X(\omega) \mathbb{P}(d\omega).$$

By the linearity of the integral with respect to $\mathbb{P}$ and the fact that $\mathbb{I}_G(\omega) = \sum_{z \in B} \mathbb{I}_{\{Z=z\}}(\omega)$,

$$\int_G Y d\mathbb{P} = \int_{\Omega} \mathbb{I}_G(\omega) X(\omega) \mathbb{P}(d\omega) = \int_G X d\mathbb{P}.$$

□

Proposition 10.4. Consider the probability triple $(\Omega, \mathcal{F}, \mathbb{P}) = ([0, 1], \mathcal{B}([0, 1]), \text{Leb}) \times ([0, 1], \mathcal{B}([0, 1]), \text{Leb})$ and the bounded random variables $X : \Omega \rightarrow \mathbb{R}$ and $Z : \Omega \rightarrow [0, 1]$, where $Z(a, b) = a$. Furthermore, consider the bounded $\mathcal{B}([0, 1])$ -measurable function $I_1^X : [0, 1] \rightarrow \mathbb{R}$ given by

$$I_1^X(a) = \int_{[0, 1]} X(a, b) \text{Leb}(db).$$

In that case, $Y = I_1^X \circ Z$ is a $\sigma(Z)$ -measurable function such that

$$\int_G Y d\mathbb{P} = \int_G X d\mathbb{P},$$

for every $G \in \sigma(Z)$, so that $Y = \mathbb{E}(X \mid Z)$ almost surely.

Proof. Recall that $\sigma(Z) = \{A \times [0, 1] \mid A \in \mathcal{B}([0, 1])\}$. For every $B \in \mathcal{B}(\mathbb{R})$, note that $Y^{-1}(B) = Z^{-1}((I_1^X)^{-1}(B))$. Because $(I_1^X)^{-1}(B) \in \mathcal{B}([0, 1])$, we know that Y is $\sigma(Z)$ -measurable.

Let $G = A \times [0, 1]$ for some $A \in \mathcal{B}([0, 1])$. Because Y is a bounded $\mathcal{F}$ -measurable function and $G \in \mathcal{F}$,

$$\int_G Y d\mathbb{P} = \int_{[0, 1]} \left[\int_{[0, 1]} \mathbb{I}_{A \times [0, 1]}(a, b) Y(a, b) \text{Leb}(db) \right] \text{Leb}(da) = \int_{[0, 1]} \left[\int_{[0, 1]} \mathbb{I}_A(a) I_1^X(a) \text{Leb}(db) \right] \text{Leb}(da).$$

By the linearity of the integral with respect to Leb and using the fact that $\text{Leb}([0, 1]) = 1$,

$$\int_G Y d\mathbb{P} = \left[\int_{[0, 1]} \text{Leb}(db) \right] \left[\int_{[0, 1]} \mathbb{I}_A(a) I_1^X(a) \text{Leb}(da) \right] = \int_{[0, 1]} \mathbb{I}_A(a) \left[\int_{[0, 1]} X(a, b) \text{Leb}(db) \right] \text{Leb}(da).$$

Therefore, using the fact that $\mathbb{I}_A(a) = \mathbb{I}_{A \times [0, 1]}(a, b) = \mathbb{I}_G(a, b)$,

$$\int_G Y d\mathbb{P} = \int_{[0, 1]} \left[\int_{[0, 1]} \mathbb{I}_G(a, b) X(a, b) \text{Leb}(db) \right] \text{Leb}(da) = \int_G X d\mathbb{P}.$$

□

Proposition 10.5. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $X : \Omega \rightarrow \mathbb{R}$ and $Z : \Omega \rightarrow \mathbb{R}$. Suppose that $f_{X,Z} : \mathbb{R}^2 \rightarrow [0, \infty]$ is a joint probability density function for X and Z . Let $f_X : \mathbb{R} \rightarrow [0, \infty]$ be a probability density function for X and $f_Z : \mathbb{R} \rightarrow [0, \infty]$ be a probability density function for Z such that

$$\begin{aligned} f_X(x) &= \int_{\mathbb{R}} f_{X,Z}(x, z) \text{Leb}(dz), \\ f_Z(z) &= \int_{\mathbb{R}} f_{X,Z}(x, z) \text{Leb}(dx). \end{aligned}$$

Furthermore, consider the elementary conditional probability density function $f_{X|Z} : \mathbb{R}^2 \rightarrow [0, \infty]$ given by

$$f_{X|Z}(x, z) = \begin{cases} 0, & \text{if } f_Z(z) = 0, \\ f_{X,Z}(x, z)/f_Z(z), & \text{if } 0 < f_Z(z) < \infty, \\ 0, & \text{if } f_Z(z) = \infty. \end{cases}$$

Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a Borel function such that $\mathbb{E}(|h \circ X|) < \infty$, so that

$$\mathbb{E}(h \circ X) = \int_{\Omega} (h \circ X) d\mathbb{P} = \int_{\mathbb{R}} h d\mathcal{L}_X = \int_{\mathbb{R}} h(x) f_X(x) \text{Leb}(dx),$$

where $\mathcal{L}_X$ is the law of X . Finally, consider the function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$g(z) = \begin{cases} 0, & \text{if } z \notin F_2^g, \\ \int_{\mathbb{R}} h(x) f_{X|Z}(x, z) \text{Leb}(dx), & \text{if } z \in F_2^g, \end{cases}$$

where $F_2^g = \{z \in \mathbb{R} \mid \int_{\mathbb{R}} |h(x) f_{X|Z}(x, z)| \text{Leb}(dx) < \infty\}$.

In that case, $Y = g \circ Z$ is a $\sigma(Z)$ -measurable function such that $\mathbb{E}(|Y|) < \infty$ and

$$\int_G Y d\mathbb{P} = \int_G (h \circ X) d\mathbb{P}$$

for every $G \in \sigma(Z)$, so that $Y = \mathbb{E}((h \circ X) \mid Z)$ almost surely.

Proof. First, we will show that $(h \circ \rho_1) f_{X|Z}$ is $\mathcal{B}(\mathbb{R})^2$ -measurable. Let $A_1 = \{z \in \mathbb{R} \mid f_Z(z) > 0\} \cap \{z \in \mathbb{R} \mid f_Z(z) < \infty\}$. Because f_Z is Borel, we know that $\mathbb{R} \times A_1 \in \mathcal{B}(\mathbb{R})^2$. Furthermore, note that

$$f_{X|Z}(x, z) = \mathbb{I}_{\mathbb{R} \times A_1}(x, z) \frac{f_{X,Z}(x, z)}{f_Z(\rho_2(x, z)) + \mathbb{I}_{\mathbb{R} \times A_1^c}(x, z)}.$$

Because the function $u : (0, \infty] \rightarrow [0, \infty]$ given by $u(r) = 1/r$ is Borel, we know that $f_{X|Z}$ is $\mathcal{B}(\mathbb{R})^2$ -measurable. Because h is Borel, we also know that $(h \circ \rho_1) f_{X|Z}$ is $\mathcal{B}(\mathbb{R})^2$ -measurable.

We will now show that g is Borel. Because $|(h \circ \rho_1) f_{X|Z}|$ is non-negative and $\mathcal{B}(\mathbb{R})^2$ -measurable, we know that the function $I_2 : \mathbb{R} \rightarrow [0, \infty]$ given by $I_2(z) = \int_{\mathbb{R}} |h(x) f_{X|Z}(x, z)| \text{Leb}(dx)$ is Borel, so that $F_2^g \in \mathcal{B}(\mathbb{R})$. Furthermore,

$$g(z) = \mathbb{I}_{F_2^g}(z) \int_{\mathbb{R}} ((h \circ \rho_1) f_{X|Z})^+(x, z) \text{Leb}(dx) - \mathbb{I}_{F_2^g}(z) \int_{\mathbb{R}} ((h \circ \rho_1) f_{X|Z})^-(x, z) \text{Leb}(dx).$$

Since $((h \circ \rho_1) f_{X|Z})^+$ and $((h \circ \rho_1) f_{X|Z})^-$ are non-negative and $\mathcal{B}(\mathbb{R})^2$ -measurable, we know that g is Borel, which also implies that $Y = g \circ Z$ is a $\sigma(Z)$ -measurable function.

We will now show that $\mathbb{E}(|Y|) < \infty$. Because $|g(z)| \leq I_2(z)$ for every $z \in \mathbb{R}$,

$$|g(z)| f_Z(z) \leq I_2(z) f_Z(z) = \int_{\mathbb{R}} |h(x) f_{X|Z}(x, z)| f_Z(z) \text{Leb}(dx) = \int_{\mathbb{R}} |h(x)| \mathbb{I}_{A_1}(z) f_{X,Z}(x, z) \text{Leb}(dx).$$

Because $|g| f_Z$ and $I_2 f_Z$ are non-negative and Borel,

$$\int_{\mathbb{R}} |g(z)| f_Z(z) \text{Leb}(dz) \leq \int_{\mathbb{R}} \left[\int_{\mathbb{R}} |h(x)| \mathbb{I}_{A_1}(z) f_{X,Z}(x, z) \text{Leb}(dx) \right] \text{Leb}(dz).$$

Because a previous result for probability density functions extends to joint probability density functions,

$$\int_{\mathbb{R}} |g(z)| f_Z(z) \text{Leb}(dz) \leq \int_{\mathbb{R}^2} |h \circ \rho_1| (\mathbb{I}_{A_1} \circ \rho_2) f_{X,Z} d\text{Leb}^2 = \mathbb{E}(|h \circ X| \mathbb{I}_{Z^{-1}(A_1)}) < \infty,$$

since $(\mathbb{I}_{A_1} \circ Z) = \mathbb{I}_{Z^{-1}(A_1)}$. Because $\text{Leb}(|g|f_Z) = \mathbb{E}(|g \circ Z|)$, we know that $Y \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$.

Let $\mathcal{L}_{X,Z} : \mathcal{B}(\mathbb{R})^2 \rightarrow [0, 1]$ denote the joint law of X and Z .

We will now show that $\mathcal{L}_{X,Z}(\mathbb{I}_{\mathbb{R} \times A_1^c}) = 0$. Because a previous result for laws extends to joint laws,

$$\int_{\mathbb{R}^2} \mathbb{I}_{\mathbb{R} \times A_1^c} d\mathcal{L}_{X,Z} = \int_{\mathbb{R}^2} \mathbb{I}_{\mathbb{R} \times A_1^c} f_{X,Z} d\text{Leb}^2 = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_{A_1^c}(z) f_{X,Z}(x, z) \text{Leb}(dx) \right] \text{Leb}(dz).$$

By rearranging terms,

$$\int_{\mathbb{R}^2} \mathbb{I}_{\mathbb{R} \times A_1^c} d\mathcal{L}_{X,Z} = \int_{\mathbb{R}} \mathbb{I}_{A_1^c}(z) \left[\int_{\mathbb{R}} f_{X,Z}(x, z) \text{Leb}(dx) \right] \text{Leb}(dz) = \int_{\mathbb{R}} \mathbb{I}_{A_1^c}(z) f_Z(z) \text{Leb}(dz).$$

Because $A_1^c = \{f_Z = 0\} \cup \{f_Z = \infty\}$ is a union of disjoint sets, we know that $\mathbb{I}_{A_1^c} = \mathbb{I}_{\{f_Z=0\}} + \mathbb{I}_{\{f_Z=\infty\}}$. Therefore,

$$\int_{\mathbb{R}^2} \mathbb{I}_{\mathbb{R} \times A_1^c} d\mathcal{L}_{X,Z} = \int_{\mathbb{R}} \mathbb{I}_{\{f_Z=0\}}(z) f_Z(z) \text{Leb}(dz) + \int_{\mathbb{R}} \mathbb{I}_{\{f_Z=\infty\}}(z) f_Z(z) \text{Leb}(dz) = 0,$$

since $\mathbb{I}_{\{f_Z=0\}} f_Z = 0$ and $\text{Leb}(f_Z) < \infty$.

Let $A_2 = \{z \in \mathbb{R} \mid \int_{\mathbb{R}} |h(x)| f_{X,Z}(x, z) \text{Leb}(dx) < \infty\}$, so that $A_2 \in \mathcal{B}(\mathbb{R})$. We will now show that $\mathcal{L}_{X,Z}(\mathbb{I}_{\mathbb{R} \times A_2^c}) = 0$. From a previous result about probability density functions,

$$\mathbb{E}(|h \circ X|) = \int_{\mathbb{R}} |h(x)| f_X(x) \text{Leb}(dx) = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} |h(x)| f_{X,Z}(x, z) \text{Leb}(dz) \right] \text{Leb}(dx) = \int_{\mathbb{R}^2} |h \circ \rho_1| f_{X,Z} d\text{Leb}^2.$$

Because $\mathbb{E}(|h \circ X|) < \infty$, we know that $\text{Leb}(A_2^c) = 0$. Because a previous result about laws extends to joint laws,

$$\int_{\mathbb{R}^2} \mathbb{I}_{\mathbb{R} \times A_2^c} d\mathcal{L}_{X,Z} = \int_{\mathbb{R}^2} \mathbb{I}_{\mathbb{R} \times A_2^c} f_{X,Z} d\text{Leb}^2 = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \mathbb{I}_{A_2^c}(z) f_{X,Z}(x, z) \text{Leb}(dx) \right] \text{Leb}(dz).$$

By rearranging terms and the using fact that $\text{Leb}(\mathbb{I}_{A_2^c}) = 0$ implies $\text{Leb}(\{\mathbb{I}_{A_2^c} f_Z > 0\}) \leq \text{Leb}(\{\mathbb{I}_{A_2^c} > 0\}) = 0$,

$$\int_{\mathbb{R}^2} \mathbb{I}_{\mathbb{R} \times A_2^c} d\mathcal{L}_{X,Z} = \int_{\mathbb{R}} \mathbb{I}_{A_2^c}(z) f_Z(z) \text{Leb}(dz) = 0.$$

Finally, we will show that $\mathbb{E}(Y \mathbb{I}_G) = \mathbb{E}((h \circ X) \mathbb{I}_G)$ for every $G \in \sigma(Z)$. Note that, for every $G \in \sigma(Z)$,

$$\mathbb{I}_G(\omega) = \mathbb{I}_{Z^{-1}(B)}(\omega) = (\mathbb{I}_B \circ Z)(\omega) = \begin{cases} 1, & \text{if } Z(\omega) \in B, \\ 0, & \text{if } Z(\omega) \notin B, \end{cases}$$

for some $B \in \mathcal{B}(\mathbb{R})$. Let $S = (\mathbb{R} \times A_1) \cap (\mathbb{R} \times A_2)$, so that $S^c = (\mathbb{R} \times A_1^c) \cup (\mathbb{R} \times A_2^c)$ and $\mathcal{L}_{X,Z}(\mathbb{I}_{S^c}) = 0$. Note that

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\Omega} (h \circ X) (\mathbb{I}_B \circ Z) d\mathbb{P} = \int_{\mathbb{R}^2} (h \circ \rho_1) (\mathbb{I}_B \circ \rho_2) d\mathcal{L}_{X,Z} = \int_{\mathbb{R}^2} (h \circ \rho_1) (\mathbb{I}_B \circ \rho_2) \mathbb{I}_S d\mathcal{L}_{X,Z},$$

since $(h \circ \rho_1) (\mathbb{I}_B \circ \rho_2)$ and $(h \circ \rho_1) (\mathbb{I}_B \circ \rho_2) \mathbb{I}_S$ are $\mathcal{L}_{X,Z}$ -integrable and equal almost everywhere.

Because a previous result for probability density functions extends to joint probability density functions,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\mathbb{R}^2} (h \circ \rho_1) (\mathbb{I}_B \circ \rho_2) \mathbb{I}_S f_{X,Z} d\text{Leb}^2.$$

Because $\mathbb{I}_S(x, z) = \mathbb{I}_{A_1}(z) \mathbb{I}_{A_2}(z)$ for every $(x, z) \in \mathbb{R}^2$,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_F \left[\int_{\mathbb{R}} h(x) \mathbb{I}_B(z) \mathbb{I}_{A_1}(z) \mathbb{I}_{A_2}(z) f_{X,Z}(x, z) \text{Leb}(dx) \right] \text{Leb}(dz),$$

where $F = \{z \in \mathbb{R} \mid \int_{\mathbb{R}} |h(x)| \mathbb{I}_B(z) \mathbb{I}_{A_1}(z) \mathbb{I}_{A_2}(z) f_{X,Z}(x, z) \text{Leb}(dx) < \infty\}$.

Because $A_2 \subseteq F$, we know that $\mathbb{I}_F \mathbb{I}_{A_2} = \mathbb{I}_{A_2}$. Therefore,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} h(x) \mathbb{I}_B(z) \mathbb{I}_{A_1}(z) \mathbb{I}_{A_2}(z) f_{X,Z}(x, z) \text{Leb}(dx) \right] \text{Leb}(dz).$$

Because $f_{X,Z}(x, z) \mathbb{I}_{A_1}(z) = f_{X|Z}(x, z) f_Z(z) \mathbb{I}_{A_1}(z)$ for every $(x, z) \in \mathbb{R}^2$,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\mathbb{R}} \left[\int_{\mathbb{R}} h(x) \mathbb{I}_B(z) \mathbb{I}_{A_1}(z) \mathbb{I}_{A_2}(z) f_{X|Z}(x, z) f_Z(z) \text{Leb}(dx) \right] \text{Leb}(dz).$$

By rearranging terms,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\mathbb{R}} \mathbb{I}_B(z) f_Z(z) \mathbb{I}_{A_1 \cap A_2}(z) \left[\int_{\mathbb{R}} h(x) f_{X|Z}(x, z) \text{Leb}(dx) \right] \text{Leb}(dz).$$

For any $z \in (A_1 \cap A_2)$, by the linearity of the integral with respect to Leb ,

$$\mathbb{I}_{A_1}(z) \int_{\mathbb{R}} |h(x)| f_{X,Z}(x, z) \text{Leb}(dx) = f_Z(z) \int_{\mathbb{R}} |h(x)| f_{X|Z}(x, z) \text{Leb}(dx) < \infty.$$

Because $f_Z(z) > 0$, we know that $\int_{\mathbb{R}} |h(x)| f_{X|Z}(x, z) \text{Leb}(dx) < \infty$, so that $z \in F_2^g$.

Because $(A_1 \cap A_2) \subseteq F_2^g$ implies $\mathbb{I}_{A_1 \cap A_2} = \mathbb{I}_{A_1 \cap A_2} \mathbb{I}_{F_2^g}$,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\mathbb{R}} \mathbb{I}_B(z) f_Z(z) \mathbb{I}_{A_1 \cap A_2}(z) \mathbb{I}_{F_2^g}(z) \left[\int_{\mathbb{R}} h(x) f_{X|Z}(x, z) \text{Leb}(dx) \right] \text{Leb}(dz).$$

By the definition of g ,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\mathbb{R}} \mathbb{I}_B(z) f_Z(z) \mathbb{I}_{A_1 \cap A_2}(z) g(z) \text{Leb}(dz).$$

By once again applying results about probability density functions and joint laws,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\Omega} (\mathbb{I}_B \circ Z)(\mathbb{I}_{A_1 \cap A_2} \circ Z)(g \circ Z) d\mathbb{P} = \int_{\mathbb{R}^2} (\mathbb{I}_B \circ \rho_2)(\mathbb{I}_{A_1 \cap A_2} \circ \rho_2)(g \circ \rho_2) d\mathcal{L}_{X,Z}.$$

Because $\mathbb{I}_S(x, z) = \mathbb{I}_{A_1}(z) \mathbb{I}_{A_2}(z)$ for every $(x, z) \in \mathbb{R}^2$,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\mathbb{R}^2} (g \circ \rho_2)(\mathbb{I}_B \circ \rho_2) \mathbb{I}_S d\mathcal{L}_{X,Z}.$$

Because $(g \circ \rho_2)(\mathbb{I}_B \circ \rho_2)$ and $(g \circ \rho_2)(\mathbb{I}_B \circ \rho_2) \mathbb{I}_S$ are $\mathcal{L}_{X,Z}$ -integrable functions that are equal almost everywhere,

$$\int_{\Omega} (h \circ X) \mathbb{I}_G d\mathbb{P} = \int_{\mathbb{R}^2} (g \circ \rho_2)(\mathbb{I}_B \circ \rho_2) d\mathcal{L}_{X,Z} = \int_{\Omega} (g \circ Z)(\mathbb{I}_B \circ Z) d\mathbb{P} = \int_{\Omega} Y \mathbb{I}_G d\mathbb{P}.$$

□

Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. For the remainder of this text, we let $\mathbb{E}(X | \mathcal{G})$ denote an arbitrary version of the conditional expectation of X given $\mathcal{G}$.

Proposition 10.6. Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. Note that $\mathbb{E}(\mathbb{E}(X | \mathcal{G})) = \mathbb{E}(\mathbb{E}(X | \mathcal{G}) \mathbb{I}_{\Omega}) = \mathbb{E}(X \mathbb{I}_{\Omega}) = \mathbb{E}(X)$.

Proposition 10.7. Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. Note that if X is $\mathcal{G}$ -measurable, then $X = \mathbb{E}(X | \mathcal{G})$ almost surely.

Proposition 10.8. Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and let $Y = \mathbb{E}(X) \mathbb{I}_{\Omega}$. In that case, $Y = \mathbb{E}(X | \{\emptyset, \Omega\})$ almost surely.

Proof. For every $B \in \mathcal{B}(\mathbb{R})$, we have $Y^{-1}(B) = \emptyset$ if $\mathbb{E}(X) \notin B$ and $Y^{-1}(B) = \Omega$ if $\mathbb{E}(X) \in B$. Furthermore, $\mathbb{E}(|Y|) = \mathbb{E}(|\mathbb{E}(X) \mathbb{I}_{\Omega}|) = |\mathbb{E}(X)| \leq \mathbb{E}(|X|) < \infty$. Therefore, $Y \in \mathcal{L}^1(\Omega, \{\emptyset, \Omega\}, \mathbb{P})$. Finally, $\mathbb{E}(Y \mathbb{I}_{\Omega}) = \mathbb{E}(\mathbb{E}(X) \mathbb{I}_{\Omega} \mathbb{I}_{\Omega}) = \mathbb{E}(X \mathbb{I}_{\Omega})$ and $\mathbb{E}(Y \mathbb{I}_{\emptyset}) = 0 = \mathbb{E}(X \mathbb{I}_{\emptyset})$. □

Proposition 10.9. Consider the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a random variable $X : \Omega \rightarrow \mathbb{R}$, and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. If $X = 0$ almost surely, then $0 = \mathbb{E}(X | \mathcal{G})$ almost surely, where 0 denotes the zero function.

Proof. Clearly, $0 \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathbb{P})$. For every $G \in \mathcal{G}$, because $\mathbb{P}(X\mathbb{I}_G = 0) = 1$, we know that $\mathbb{E}(X\mathbb{I}_G) = 0 = \mathbb{E}(0\mathbb{I}_G)$. $\square$

Proposition 10.10. Consider the random variables $X_1 \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $X_2 \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. In that case, $a_1\mathbb{E}(X_1 | \mathcal{G}) + a_2\mathbb{E}(X_2 | \mathcal{G}) = \mathbb{E}(a_1X_1 + a_2X_2 | \mathcal{G})$ almost surely for every $a_1, a_2 \in \mathbb{R}$.

Proof. Because $\mathcal{L}^1(\Omega, \mathcal{G}, \mathbb{P})$ is a vector space, we know that $a_1\mathbb{E}(X_1 | \mathcal{G}) + a_2\mathbb{E}(X_2 | \mathcal{G}) \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathbb{P})$. For every $G \in \mathcal{G}$,

$$\mathbb{E}((a_1\mathbb{E}(X_1 | \mathcal{G}) + a_2\mathbb{E}(X_2 | \mathcal{G}))\mathbb{I}_G) = a_1\mathbb{E}(\mathbb{E}(X_1 | \mathcal{G})\mathbb{I}_G) + a_2\mathbb{E}(\mathbb{E}(X_2 | \mathcal{G})\mathbb{I}_G).$$

From the definition of a version of the conditional expectation,

$$\mathbb{E}((a_1\mathbb{E}(X_1 | \mathcal{G}) + a_2\mathbb{E}(X_2 | \mathcal{G}))\mathbb{I}_G) = a_1\mathbb{E}(X_1\mathbb{I}_G) + a_2\mathbb{E}(X_2\mathbb{I}_G) = \mathbb{E}((a_1X_1 + a_2X_2)\mathbb{I}_G).$$

$\square$

Proposition 10.11. Consider the random variables $X_1 \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $X_2 \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. If $X_1 = X_2$ almost surely, then $\mathbb{E}(X_1 | \mathcal{G}) = \mathbb{E}(X_2 | \mathcal{G})$ almost surely.

Proof. Because $\mathbb{P}(X_1 - X_2 = 0) = 1$, we know that $\mathbb{P}(\mathbb{E}(X_1 - X_2 | \mathcal{G}) = 0) = 1$. Therefore, by linearity, $\mathbb{P}(\mathbb{E}(X_1 | \mathcal{G}) = \mathbb{E}(X_2 | \mathcal{G})) = 1$. $\square$

Proposition 10.12. Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. If $X \geq 0$, then $\mathbb{P}(\mathbb{E}(X | \mathcal{G}) \geq 0) = 1$.

Proof. In order to find a contradiction, suppose that $\mathbb{P}(\mathbb{E}(X | \mathcal{G}) \geq 0) < 1$, so that $\mathbb{P}(\mathbb{E}(X | \mathcal{G}) < 0) > 0$. Let $A_n = \{\mathbb{E}(X | \mathcal{G}) < -n^{-1}\} = \mathbb{E}(X | \mathcal{G})^{-1}((-\infty, -n^{-1}))$, so that $A_n \subseteq A_{n+1}$ and $\cup_n A_n = \{\mathbb{E}(X | \mathcal{G}) < 0\}$. Since $A_n \uparrow \{\mathbb{E}(X | \mathcal{G}) < 0\}$, the monotone-convergence property of measure guarantees that $\mathbb{P}(A_n) \uparrow \mathbb{P}(\mathbb{E}(X | \mathcal{G}) < 0)$. Because we supposed that $\mathbb{P}(\mathbb{E}(X | \mathcal{G}) < 0) > 0$, there is an $n \in \mathbb{N}$ such that $\mathbb{P}(A_n) = \mathbb{P}(\mathbb{E}(X | \mathcal{G}) < -n^{-1}) > 0$. Consider the random variable $\mathbb{E}(X | \mathcal{G})\mathbb{I}_{A_n}$ given by

$$(\mathbb{E}(X | \mathcal{G})\mathbb{I}_{A_n})(\omega) = \mathbb{E}(X | \mathcal{G})(\omega)\mathbb{I}_{A_n}(\omega) = \begin{cases} \mathbb{E}(X | \mathcal{G})(\omega), & \text{if } \mathbb{E}(X | \mathcal{G})(\omega) < -n^{-1}, \\ 0, & \text{if } \mathbb{E}(X | \mathcal{G})(\omega) \geq -n^{-1}. \end{cases}$$

Because $\mathbb{E}(X | \mathcal{G})\mathbb{I}_{A_n} \leq -n^{-1}\mathbb{I}_{A_n}$, we know that $\mathbb{E}(\mathbb{E}(X | \mathcal{G})\mathbb{I}_{A_n}) \leq -n^{-1}\mathbb{P}(A_n) < 0$. Because $X \geq 0$, we know that $\mathbb{E}(X\mathbb{I}_{A_n}) \geq 0$. However, $A_n \in \mathcal{G}$, so that $\mathbb{E}(X\mathbb{I}_{A_n}) = \mathbb{E}(\mathbb{E}(X | \mathcal{G})\mathbb{I}_{A_n})$. Because this is a contradiction, we know that $\mathbb{P}(\mathbb{E}(X | \mathcal{G}) \geq 0) = 1$. $\square$

Proposition 10.13. Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. In that case, $|\mathbb{E}(X | \mathcal{G})| \leq \mathbb{E}(|X| | \mathcal{G})$ almost surely.

Proof. By the linearity of conditional expectation,

$$\begin{aligned} \mathbb{P}(|\mathbb{E}(X | \mathcal{G})| = |\mathbb{E}(X^+ - X^- | \mathcal{G})| = |\mathbb{E}(X^+ | \mathcal{G}) - \mathbb{E}(X^- | \mathcal{G})|) &= 1, \\ \mathbb{P}(\mathbb{E}(|X| | \mathcal{G}) = \mathbb{E}(X^+ + X^- | \mathcal{G}) = \mathbb{E}(X^+ | \mathcal{G}) + \mathbb{E}(X^- | \mathcal{G})) &= 1. \end{aligned}$$

By the triangle inequality, $|\mathbb{E}(X^+ | \mathcal{G}) - \mathbb{E}(X^- | \mathcal{G})| \leq |\mathbb{E}(X^+ | \mathcal{G})| + |\mathbb{E}(X^- | \mathcal{G})|$.

Because $\mathbb{P}(|\mathbb{E}(X^+ | \mathcal{G})| = \mathbb{E}(X^+ | \mathcal{G})) = 1$ and $\mathbb{P}(|\mathbb{E}(X^- | \mathcal{G})| = \mathbb{E}(X^- | \mathcal{G})) = 1$,

$$\mathbb{P}(|\mathbb{E}(X | \mathcal{G})| \leq |\mathbb{E}(X^+ | \mathcal{G})| + |\mathbb{E}(X^- | \mathcal{G})| = \mathbb{E}(X^+ | \mathcal{G}) + \mathbb{E}(X^- | \mathcal{G}) = \mathbb{E}(|X| | \mathcal{G})) = 1.$$

$\square$

Theorem 10.1 (Conditional monotone-convergence theorem). Consider a sequence of non-negative random variables $(X_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P}) \mid n \in \mathbb{N})$, a non-negative random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. If $X_n \uparrow X$, then $\mathbb{P}(\mathbb{E}(X_n | \mathcal{G})\mathbb{I}_A \uparrow \mathbb{E}(X | \mathcal{G})) = 1$, where $A \in \mathcal{G}$ is a set such that $\mathbb{P}(A) = 1$.

Proof. Because X_n is a non-negative random variable, $\mathbb{P}(\mathbb{E}(X_n | \mathcal{G}) \geq 0) = 1$. For every $n \in \mathbb{N}$, because $X_{n+1} - X_n$ is non-negative and $\mathbb{E}(X_{n+1} | \mathcal{G}) - \mathbb{E}(X_n | \mathcal{G}) = \mathbb{E}(X_{n+1} - X_n | \mathcal{G})$ almost surely, $\mathbb{P}(\mathbb{E}(X_{n+1} | \mathcal{G}) - \mathbb{E}(X_n | \mathcal{G}) \geq 0) = 1$.

Let $A^c = \bigcup_n \{\mathbb{E}(X_n | \mathcal{G}) < 0\} \cup \{\mathbb{E}(X_{n+1} | \mathcal{G}) - \mathbb{E}(X_n | \mathcal{G}) < 0\}$. Note that $A \in \mathcal{G}$ and $\mathbb{P}(A) = 1$, since

$$\mathbb{P}(A^c) \leq \sum_n \mathbb{P}(\mathbb{E}(X_n | \mathcal{G}) < 0) + \mathbb{P}(\mathbb{E}(X_{n+1} | \mathcal{G}) - \mathbb{E}(X_n | \mathcal{G}) < 0) = 0.$$

For every $n \in \mathbb{N}$, note that $\mathbb{E}(X_n | \mathcal{G})\mathbb{I}_A \geq 0$ and $\mathbb{E}(X_{n+1} | \mathcal{G})\mathbb{I}_A \geq \mathbb{E}(X_n | \mathcal{G})\mathbb{I}_A$.

Let $Y = \limsup_{n \rightarrow \infty} \mathbb{E}(X_n | \mathcal{G})\mathbb{I}_A$. For every $G \in \mathcal{G}$, because every non-decreasing sequence of real numbers converges (possibly to infinity), we know that $\mathbb{E}(X_n | \mathcal{G})\mathbb{I}_A\mathbb{I}_G \uparrow Y\mathbb{I}_G$, which also implies $\mathbb{E}(X_n | \mathcal{G})\mathbb{I}_A \uparrow Y$. By the monotone-convergence theorem, we know that $\mathbb{E}(\mathbb{E}(X_n | \mathcal{G})\mathbb{I}_A\mathbb{I}_G) \uparrow \mathbb{E}(Y\mathbb{I}_G)$.

For every $n \in \mathbb{N}$ and $G \in \mathcal{G}$, we have $(A \cap G) \in \mathcal{G}$ and $\mathbb{P}(X_n\mathbb{I}_G\mathbb{I}_{A^c} \neq 0) = 0$, so that

$$\mathbb{E}(\mathbb{E}(X_n | \mathcal{G})\mathbb{I}_A\mathbb{I}_G) = \mathbb{E}(\mathbb{E}(X_n | \mathcal{G})\mathbb{I}_{A \cap G}) = \mathbb{E}(X_n\mathbb{I}_{A \cap G}) = \mathbb{E}(X_n\mathbb{I}_A\mathbb{I}_G) + \mathbb{E}(X_n\mathbb{I}_{A^c}\mathbb{I}_G) = \mathbb{E}(X_n\mathbb{I}_G),$$

which implies $\mathbb{E}(X_n\mathbb{I}_G) \uparrow \mathbb{E}(Y\mathbb{I}_G)$. Since $X_n\mathbb{I}_G \uparrow X\mathbb{I}_G$, we also know that $\mathbb{E}(X_n\mathbb{I}_G) \uparrow \mathbb{E}(X\mathbb{I}_G)$, so that $\mathbb{E}(Y\mathbb{I}_G) = \mathbb{E}(X\mathbb{I}_G)$. Because $\Omega \in \mathcal{G}$, this gives $\mathbb{E}(Y) = \mathbb{E}(X) < \infty$, so that the non-negative $\mathcal{G}$ -measurable function Y is almost surely finite and may be taken to be real valued. Therefore, $Y = \mathbb{E}(X | \mathcal{G})$ almost surely. $\square$

Lemma 10.1 (Conditional Fatou lemma). Consider a sequence of non-negative random variables $(X_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P}) \mid n \in \mathbb{N})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. If $\mathbb{E}(\liminf_{n \rightarrow \infty} X_n) < \infty$, then

$$\mathbb{P}\left(\mathbb{E}\left(\liminf_{n \rightarrow \infty} X_n \mid \mathcal{G}\right) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(X_n \mid \mathcal{G})\right) = 1.$$

Proof. For any $m \in \mathbb{N}$, consider the function $Z_m = \inf_{n \geq m} X_n$, such that

$$\liminf_{n \rightarrow \infty} X_n = \lim_{m \rightarrow \infty} \inf_{n \geq m} X_n = \lim_{m \rightarrow \infty} Z_m.$$

Because $Z_m \leq Z_{m+1}$ for every $m \in \mathbb{N}$, we have $Z_m \uparrow \liminf_{n \rightarrow \infty} X_n$. Furthermore, $Z_m \geq 0$ and $Z_m \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ for every $m \in \mathbb{N}$. Therefore, by the conditional monotone-convergence theorem,

$$\mathbb{P}\left(\mathbb{E}(Z_m \mid \mathcal{G})\mathbb{I}_A \uparrow \mathbb{E}\left(\liminf_{n \rightarrow \infty} X_n \mid \mathcal{G}\right)\right) = 1,$$

where $A \in \mathcal{G}$ and $\mathbb{P}(A) = 1$.

For any $n \geq m$, note that $X_n \geq Z_m$. Therefore, $\mathbb{P}(\mathbb{E}(X_n - Z_m \mid \mathcal{G}) \geq 0) = 1$ and $\mathbb{P}(\mathbb{E}(X_n \mid \mathcal{G}) \geq \mathbb{E}(Z_m \mid \mathcal{G})) = 1$. Furthermore, for every $m \in \mathbb{N}$, because $\mathbb{P}(A^c) = 0$,

$$\mathbb{P}\left(\inf_{n \geq m} \mathbb{E}(X_n \mid \mathcal{G}) \geq \mathbb{E}(Z_m \mid \mathcal{G})\mathbb{I}_A\right) = 1.$$

By taking the limit of both sides of the previous inequation when $m \rightarrow \infty$,

$$\mathbb{P}\left(\liminf_{n \rightarrow \infty} \mathbb{E}(X_n \mid \mathcal{G}) \geq \mathbb{E}\left(\liminf_{n \rightarrow \infty} X_n \mid \mathcal{G}\right)\right) = 1.$$

$\square$

Lemma 10.2 (Reverse conditional Fatou lemma). Consider a sequence of non-negative random variables $(X_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P}) \mid n \in \mathbb{N})$, a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, and a non-negative random variable $Y \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ such that $X_n \leq Y$ for every $n \in \mathbb{N}$. In that case,

$$\mathbb{P}\left(\mathbb{E}\left(\limsup_{n \rightarrow \infty} X_n \mid \mathcal{G}\right) \geq \limsup_{n \rightarrow \infty} \mathbb{E}(X_n \mid \mathcal{G})\right) = 1.$$

Proof. Because $X_n \leq Y$ for every $n \in \mathbb{N}$, we know that $\mathbb{E}(\limsup_{n \rightarrow \infty} X_n) \leq \mathbb{E}(Y) < \infty$.

For every $n \in \mathbb{N}$, consider the non-negative function $Z_n = Y - X_n$, so that $Z_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. From the conditional Fatou lemma, since $\mathbb{E}(\liminf_{n \rightarrow \infty} Z_n) < \infty$,

$$\mathbb{P}\left(\mathbb{E}\left(\liminf_{n \rightarrow \infty} Y - X_n \mid \mathcal{G}\right) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(Y - X_n \mid \mathcal{G})\right) = 1.$$

For every $n \in \mathbb{N}$, by moving constants outside the corresponding limits and linearity,

$$\mathbb{P} \left(\mathbb{E}(Y | \mathcal{G}) + \mathbb{E} \left(\liminf_{n \rightarrow \infty} -X_n | \mathcal{G} \right) \leq \mathbb{E}(Y | \mathcal{G}) + \liminf_{n \rightarrow \infty} -\mathbb{E}(X_n | \mathcal{G}) \right) = 1.$$

By the relationship between limit inferior and limit superior and linearity,

$$\mathbb{P} \left(\mathbb{E}(Y | \mathcal{G}) - \mathbb{E} \left(\limsup_{n \rightarrow \infty} X_n | \mathcal{G} \right) \leq \mathbb{E}(Y | \mathcal{G}) - \limsup_{n \rightarrow \infty} \mathbb{E}(X_n | \mathcal{G}) \right) = 1.$$

The proof is completed by reorganizing terms in the inequation above. $\square$

Theorem 10.2 (Conditional dominated convergence theorem). Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a sequence of random variables $(X_n | n \in \mathbb{N})$, a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, a random variable X , and a non-negative random variable $V \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ such that $|X_n| \leq V$ for every $n \in \mathbb{N}$. If $\mathbb{P}(\lim_{n \rightarrow \infty} X_n = X) = 1$, then $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \mathbb{E}(X_n | \mathcal{G}) \mathbb{I}_C = \mathbb{E}(X | \mathcal{G}) \right) = 1.$$

where $C \in \mathcal{G}$ is a set such that $\mathbb{P}(C) = 1$.

Proof. Because $|X_n| \leq V$ for every $n \in \mathbb{N}$, we know that $\mathbb{E}(|X_n|) \leq \mathbb{E}(V) < \infty$, which implies that $X_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Because the function $|\cdot|$ is continuous, we know that $\mathbb{P}(\lim_{n \rightarrow \infty} |X_n| = |X|) = 1$. Because $\mathbb{P}(\lim_{n \rightarrow \infty} |X_n| \leq V) = 1$, we know that $\mathbb{P}(|X| \leq V) = 1$. Because $\mathbb{P}(|X| \neq |X| \mathbb{I}_{\{|X| \leq V\}}) = 0$, we know that $\mathbb{E}(|X|) = \mathbb{E}(|X| \mathbb{I}_{\{|X| \leq V\}}) \leq \mathbb{E}(V) < \infty$, so that $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$.

Since $\mathbb{P}(|X_n| \leq V) = 1$ and $\mathbb{P}(|X| \leq V) = 1$, we have $\mathbb{P}(|X_n| + |X| \leq 2V) = 1$. By the triangle inequality,

$$|X_n - X| = |X_n + (-X)| \leq |X_n| + |X|,$$

which implies that $\mathbb{P}(|X_n - X| \leq 2V) = 1$.

Let $A = \{|X| \leq V\}$, which does not depend on $n \in \mathbb{N}$ and satisfies $\mathbb{P}(A) = 1$, so that $\mathbb{P}(|X_n - X| \mathbb{I}_A = |X_n - X| \mathbb{I}_A) = 1$ and $\mathbb{E}(|X_n - X|) = \mathbb{E}(|X_n - X| \mathbb{I}_A)$ for every $n \in \mathbb{N}$. Because $|X_n - X| \mathbb{I}_A$ is an $\mathcal{F}$ -measurable function and $|X_n - X| \mathbb{I}_A \leq (|X_n| + |X|) \mathbb{I}_A \leq 2V$ for every $n \in \mathbb{N}$, where $2V : \Omega \rightarrow [0, \infty]$ is an $\mathcal{F}$ -measurable function such that $\mathbb{E}(2V) = 2\mathbb{E}(V) < \infty$, the reverse conditional Fatou lemma states that

$$\mathbb{P} \left(\mathbb{E} \left(\limsup_{n \rightarrow \infty} |X_n - X| \mathbb{I}_A | \mathcal{G} \right) \geq \limsup_{n \rightarrow \infty} \mathbb{E}(|X_n - X| \mathbb{I}_A | \mathcal{G}) \right) = 1.$$

Since $|\cdot|$ is continuous, we have $\mathbb{P}(\lim_{n \rightarrow \infty} |X_n - X| \mathbb{I}_A = 0) = 1$, where 0 is the zero function. Therefore,

$$\mathbb{P} \left(\limsup_{n \rightarrow \infty} |X_n - X| \mathbb{I}_A = \liminf_{n \rightarrow \infty} |X_n - X| \mathbb{I}_A = \lim_{n \rightarrow \infty} |X_n - X| \mathbb{I}_A = 0 \right) = 1.$$

Because each of the random variables above is almost surely equal to zero,

$$\mathbb{P} \left(\mathbb{E} \left(\limsup_{n \rightarrow \infty} |X_n - X| \mathbb{I}_A | \mathcal{G} \right) = \mathbb{E} \left(\liminf_{n \rightarrow \infty} |X_n - X| \mathbb{I}_A | \mathcal{G} \right) = \mathbb{E} \left(\lim_{n \rightarrow \infty} |X_n - X| \mathbb{I}_A | \mathcal{G} \right) = 0 \right) = 1.$$

Since $(X_n - X) \mathbb{I}_A \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ for every $n \in \mathbb{N}$, we have $\mathbb{P}(|\mathbb{E}((X_n - X) \mathbb{I}_A | \mathcal{G})| \leq \mathbb{E}(|X_n - X| \mathbb{I}_A | \mathcal{G})) = 1$. By taking the limit superior of both sides of the previous inequation and employing the previous results,

$$\mathbb{P} \left(0 \leq \limsup_{n \rightarrow \infty} |\mathbb{E}((X_n - X) \mathbb{I}_A | \mathcal{G})| \leq \limsup_{n \rightarrow \infty} \mathbb{E}(|X_n - X| \mathbb{I}_A | \mathcal{G}) \leq \mathbb{E} \left(\limsup_{n \rightarrow \infty} |X_n - X| \mathbb{I}_A | \mathcal{G} \right) = 0 \right) = 1.$$

Therefore, by the relationship between limits,

$$\mathbb{P} \left(\liminf_{n \rightarrow \infty} \mathbb{E}((X_n - X) \mathbb{I}_A | \mathcal{G}) = \limsup_{n \rightarrow \infty} \mathbb{E}((X_n - X) \mathbb{I}_A | \mathcal{G}) = 0 \right) = 1.$$

Because $\mathbb{P}((X_n - X) \mathbb{I}_A = (X_n - X)) = 1$ implies $\mathbb{P}(\mathbb{E}((X_n - X) \mathbb{I}_A | \mathcal{G}) = \mathbb{E}(X_n - X | \mathcal{G})) = 1$.

$$\mathbb{P} \left(\liminf_{n \rightarrow \infty} \mathbb{E}(X_n - X | \mathcal{G}) = \limsup_{n \rightarrow \infty} \mathbb{E}(X_n - X | \mathcal{G}) = 0 \right) = 1.$$

By the linearity of conditional expectation,

$$\mathbb{P} \left(\liminf_{n \rightarrow \infty} \mathbb{E}(X_n | \mathcal{G}) = \limsup_{n \rightarrow \infty} \mathbb{E}(X_n | \mathcal{G}) = \mathbb{E}(X | \mathcal{G}) \right) = 1.$$

Let $C = \{\omega \in \Omega \mid \lim_{n \rightarrow \infty} \mathbb{E}(X_n | \mathcal{G})(\omega) \text{ exists in } \mathbb{R}\}$. Because $\mathbb{E}(X_n | \mathcal{G})$ is $\mathcal{G}$ -measurable for every $n \in \mathbb{N}$, recall that $C \in \mathcal{G}$. Because $\mathbb{E}(|\mathbb{E}(X | \mathcal{G})|) < \infty$, recall that $\mathbb{P}(|\mathbb{E}(X | \mathcal{G})| < \infty) = 1$, so that $\mathbb{P}(C) = 1$. Furthermore,

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \mathbb{E}(X_n | \mathcal{G}) \mathbb{I}_C = \mathbb{E}(X | \mathcal{G}) \right) = 1.$$

□

Proposition 10.14 (Conditional Jensen's inequality). Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, and a convex function $\phi : \mathbb{R} \rightarrow \mathbb{R}$. If $(\phi \circ X) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, then $\mathbb{P}((\phi \circ \mathbb{E}(X | \mathcal{G})) \leq \mathbb{E}((\phi \circ X) | \mathcal{G})) = 1$.

Proof. Because ϕ is a convex function, it is possible to show that there is a sequence $((a_n, b_n) \in \mathbb{R}^2 \mid n \in \mathbb{N})$ such that $\phi(x) = \sup_n a_n x + b_n$ for every $x \in \mathbb{R}$. Therefore, $\phi(x) \geq a_n x + b_n$ for every $x \in \mathbb{R}$ and $n \in \mathbb{N}$. Furthermore, if $(\phi \circ X) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, then $(\phi \circ X) - a_n X - b_n \geq 0$ for every $n \in \mathbb{N}$ and

$$\mathbb{P}(\mathbb{E}((\phi \circ X) - a_n X - b_n | \mathcal{G}) \geq 0) = 1.$$

For every $n \in \mathbb{N}$, by the linearity of conditional expectation,

$$\mathbb{P}(\mathbb{E}((\phi \circ X) | \mathcal{G}) \geq a_n \mathbb{E}(X | \mathcal{G}) + b_n) = 1.$$

By taking the supremum of both sides of the previous inequation,

$$\mathbb{P} \left(\mathbb{E}((\phi \circ X) | \mathcal{G}) \geq \sup_n a_n \mathbb{E}(X | \mathcal{G}) + b_n = (\phi \circ \mathbb{E}(X | \mathcal{G})) \right) = 1.$$

□

Proposition 10.15. Consider a random variable $X \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$, where $p \in [1, \infty)$, and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. In that case, $\|\mathbb{E}(X | \mathcal{G})\|_p \leq \|X\|_p$.

Proof. From the monotonicity of norm, we know that $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Consider the convex function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ given by $\phi(x) = |x|^p$, so that $(\phi \circ X) = |X|^p$. Because $\mathbb{E}(|X|^p) < \infty$, we know that $|X|^p \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. From the conditional Jensen's inequality, $\mathbb{P}(|\mathbb{E}(X | \mathcal{G})|^p \leq \mathbb{E}(|X|^p | \mathcal{G})) = 1$. Let $A = \{|\mathbb{E}(X | \mathcal{G})|^p \leq \mathbb{E}(|X|^p | \mathcal{G})\}$.

Because $|\mathbb{E}(X | \mathcal{G})|^p$ is non-negative and $\mathcal{G}$ -measurable and $\mathbb{E}(|X|^p | \mathcal{G}) \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathbb{P})$,

$$\mathbb{E}(|\mathbb{E}(X | \mathcal{G})|^p) = \mathbb{E}(|\mathbb{E}(X | \mathcal{G})|^p \mathbb{I}_A) \leq \mathbb{E}(\mathbb{E}(|X|^p | \mathcal{G}) \mathbb{I}_A) = \mathbb{E}(\mathbb{E}(|X|^p | \mathcal{G})) = \mathbb{E}(|X|^p).$$

□

Proposition 10.16 (Tower property). Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, and a σ -algebra $\mathcal{H} \subseteq \mathcal{G}$. In that case, $\mathbb{E}(\mathbb{E}(X | \mathcal{G}) | \mathcal{H}) = \mathbb{E}(X | \mathcal{H})$ almost surely.

Proof. Because $\mathbb{E}(X | \mathcal{G}) \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathbb{P})$, we know that $\mathbb{E}(\mathbb{E}(X | \mathcal{G}) | \mathcal{H}) \in \mathcal{L}^1(\Omega, \mathcal{H}, \mathbb{P})$. For every $H \in \mathcal{H}$, since $H \in \mathcal{G}$,

$$\int_{\Omega} \mathbb{E}(\mathbb{E}(X | \mathcal{G}) | \mathcal{H}) \mathbb{I}_H d\mathbb{P} = \int_{\Omega} \mathbb{E}(X | \mathcal{G}) \mathbb{I}_H d\mathbb{P} = \int_{\Omega} X \mathbb{I}_H d\mathbb{P}.$$

□

For the remainder of this text, we let $\mathbb{E}(X | \mathcal{G} | \mathcal{H})$ denote $\mathbb{E}(\mathbb{E}(X | \mathcal{G}) | \mathcal{H})$.

Proposition 10.17 (Taking out what is known). Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, and a $\mathcal{G}$ -measurable random variable $Z : \Omega \rightarrow \mathbb{R}$. If $\mathbb{E}(|ZX|) < \infty$, then $\mathbb{E}(ZX | \mathcal{G}) = Z\mathbb{E}(X | \mathcal{G})$ almost surely.

Proof. We will start by assuming that $X \geq 0$.

First, suppose that $Z = \mathbb{I}_A$, where $A \in \mathcal{G}$. For every $G \in \mathcal{G}$, since $ZX \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $A \cap G \in \mathcal{G}$,

$$\mathbb{E}(ZX \mathbb{I}_G) = \mathbb{E}(X \mathbb{I}_{A \cap G}) = \mathbb{E}(\mathbb{E}(X | \mathcal{G}) \mathbb{I}_{A \cap G}) = \mathbb{E}(Z \mathbb{E}(X | \mathcal{G}) \mathbb{I}_G).$$

Because $Z \mathbb{E}(X | \mathcal{G})$ is $\mathcal{G}$ -measurable and $\mathbb{E}(Z \mathbb{E}(X | \mathcal{G})) = \mathbb{E}(ZX) < \infty$, we know that $Z \mathbb{E}(X | \mathcal{G}) = \mathbb{E}(ZX | \mathcal{G})$ almost surely.

Next, suppose that Z is a simple function that can be written as $Z = \sum_{k=1}^m a_k \mathbb{I}_{A_k}$ for some fixed $a_1, a_2, \dots, a_m \in [0, \infty]$ and $A_1, A_2, \dots, A_m \in \mathcal{G}$. By the linearity of the conditional expectation and the previous step,

$$\mathbb{P} \left(\mathbb{E}(ZX | \mathcal{G}) = \mathbb{E} \left(\sum_{k=1}^m a_k \mathbb{I}_{A_k} X | \mathcal{G} \right) = \sum_{k=1}^m a_k \mathbb{E}(\mathbb{I}_{A_k} X | \mathcal{G}) = \sum_{k=1}^m a_k \mathbb{I}_{A_k} \mathbb{E}(X | \mathcal{G}) = Z \mathbb{E}(X | \mathcal{G}) \right) = 1,$$

where we also used the fact that $\mathbb{E}(\mathbb{I}_{A_k} X) \leq \mathbb{E}(X) < \infty$.

Next, suppose that Z is a non-negative $\mathcal{G}$ -measurable function. For any $n \in \mathbb{N}$, consider the simple function $Z_n = \alpha_n \circ Z$, where α_n is the n -th staircase function.

For every $G \in \mathcal{G}$, since $Z_n \uparrow Z$ and $X \mathbb{I}_G \geq 0$, note that $Z_n X \mathbb{I}_G \uparrow ZX \mathbb{I}_G$. For every $G \in \mathcal{G}$, since $Z_n \uparrow Z$ and $|\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G \geq 0$, note that $Z_n |\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G \uparrow Z |\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G$. Therefore, by the monotone-convergence theorem, we know that $\mathbb{E}(Z_n X \mathbb{I}_G) \uparrow \mathbb{E}(ZX \mathbb{I}_G)$ and $\mathbb{E}(Z_n |\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G) \uparrow \mathbb{E}(Z |\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G)$.

Because Z_n is a simple $\mathcal{G}$ -measurable function and $\mathbb{E}(Z_n X) \leq \mathbb{E}(ZX) < \infty$, note that $\mathbb{E}(Z_n X | \mathcal{G}) = Z_n \mathbb{E}(X | \mathcal{G})$ almost surely. Because $Z_n \mathbb{E}(X | \mathcal{G}) = Z_n |\mathbb{E}(X | \mathcal{G})|$ almost surely, $\mathbb{E}(Z_n X \mathbb{I}_G) = \mathbb{E}(Z_n |\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G)$ for every $G \in \mathcal{G}$. Therefore, the previous result implies that $\mathbb{E}(ZX \mathbb{I}_G) = \mathbb{E}(Z |\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G)$ for every $G \in \mathcal{G}$, so that $Z |\mathbb{E}(X | \mathcal{G})| = \mathbb{E}(ZX | \mathcal{G})$ almost surely. Because $Z |\mathbb{E}(X | \mathcal{G})| = Z \mathbb{E}(X | \mathcal{G})$ almost surely, this step is complete.

Next, suppose that Z is a $\mathcal{G}$ -measurable function. Recall that $Z = Z^+ - Z^-$, where Z^+ and Z^- are non-negative $\mathcal{G}$ -measurable functions. By the linearity of the conditional expectation and the previous step,

$$\mathbb{P}(\mathbb{E}(ZX | \mathcal{G}) = \mathbb{E}(Z^+ X | \mathcal{G}) - \mathbb{E}(Z^- X | \mathcal{G}) = Z^+ \mathbb{E}(X | \mathcal{G}) - Z^- \mathbb{E}(X | \mathcal{G}) = Z \mathbb{E}(X | \mathcal{G})) = 1,$$

where we have also used the fact that $\mathbb{E}(Z^+ X) + \mathbb{E}(Z^- X) = \mathbb{E}((Z^+ + Z^-)X) = \mathbb{E}(|ZX|) < \infty$.

Finally, suppose that $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Recall that $X = X^+ - X^-$, where X^+ and X^- are non-negative $\mathcal{F}$ -measurable functions. By the linearity of the conditional expectation,

$$\mathbb{P}(\mathbb{E}(ZX | \mathcal{G}) = \mathbb{E}(ZX^+ | \mathcal{G}) - \mathbb{E}(ZX^- | \mathcal{G}) = Z \mathbb{E}(X^+ | \mathcal{G}) - Z \mathbb{E}(X^- | \mathcal{G}) = Z \mathbb{E}(X | \mathcal{G})) = 1,$$

where we have also used the fact that $\mathbb{E}(|Z|X^+) + \mathbb{E}(|Z|X^-) = \mathbb{E}(|Z|(X^+ + X^-)) = \mathbb{E}(|ZX|) < \infty$. □

Proposition 10.18 (Role of independence). Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, and a σ -algebra $\mathcal{H} \subseteq \mathcal{F}$. If $\mathcal{H}$ and $\sigma(\sigma(X) \cup \mathcal{G})$ are independent, then $\mathbb{E}(X | \sigma(\mathcal{G} \cup \mathcal{H})) = \mathbb{E}(X | \mathcal{G})$ almost surely.

Proof. We will start by assuming that $X \geq 0$.

For every $G \in \mathcal{G}$, note that $|\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G$ is $\mathcal{G}$ -measurable. Consider the Borel function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(a, b) = ab$. Since $(X \mathbb{I}_G)(\omega) = f(X(\omega), |\mathbb{E}(X | \mathcal{G})|(\omega))$ for every $\omega \in \Omega$, we also know that $X \mathbb{I}_G$ is $\sigma(\sigma(X) \cup \mathcal{G})$ -measurable.

For every $G \in \mathcal{G}$ and $H \in \mathcal{H}$, we know that $X \mathbb{I}_G$ and $\mathbb{I}_H$ are independent, since $\mathbb{I}_H$ is $\mathcal{H}$ -measurable. We also know that $|\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G$ and $\mathbb{I}_H$ are independent, since $\mathcal{G} \subseteq \sigma(\sigma(X) \cup \mathcal{G})$.

For every $G \in \mathcal{G}$ and $H \in \mathcal{H}$, because $X \mathbb{I}_G \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, $|\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$, and $\mathbb{I}_H \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$,

$$\mathbb{E}(X; G \cap H) = \mathbb{E}(X \mathbb{I}_G \mathbb{I}_H) = \mathbb{E}(X \mathbb{I}_G) \mathbb{E}(\mathbb{I}_H) = \mathbb{E}(|\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G) \mathbb{E}(\mathbb{I}_H) = \mathbb{E}(|\mathbb{E}(X | \mathcal{G})| \mathbb{I}_G \mathbb{I}_H) = \mathbb{E}(|\mathbb{E}(X | \mathcal{G})|; G \cap H).$$

Consider the set $\mathcal{I} = \{G \cap H \mid G \in \mathcal{G} \text{ and } H \in \mathcal{H}\}$. Suppose that $(G_1 \cap H_1) \in \mathcal{I}$ and $(G_2 \cap H_2) \in \mathcal{I}$, and note that $(G_1 \cap H_1) \cap (G_2 \cap H_2) = (G_1 \cap G_2) \cap (H_1 \cap H_2)$. Because $(G_1 \cap G_2) \in \mathcal{G}$ and $(H_1 \cap H_2) \in \mathcal{H}$, we know that $((G_1 \cap H_1) \cap (G_2 \cap H_2)) \in \mathcal{I}$, so that $\mathcal{I}$ is a π -system.

Since $\Omega \in \mathcal{G}$, we know that $\mathcal{H} \subseteq \mathcal{I}$. Since $\Omega \in \mathcal{H}$, we know that $\mathcal{G} \subseteq \mathcal{I}$. Therefore, $\mathcal{G} \cup \mathcal{H} \subseteq \mathcal{I}$, so that $\sigma(\mathcal{G} \cup \mathcal{H}) \subseteq \sigma(\mathcal{I})$. For every $G \in \mathcal{G}$ and $H \in \mathcal{H}$, we know that $(G \cap H) \in \sigma(\mathcal{G} \cup \mathcal{H})$. Therefore $\mathcal{I} \subseteq \sigma(\mathcal{G} \cup \mathcal{H})$, so that $\sigma(\mathcal{I}) \subseteq \sigma(\mathcal{G} \cup \mathcal{H})$. In conclusion, $\sigma(\mathcal{I}) = \sigma(\mathcal{G} \cup \mathcal{H})$.

Consider the measure $(X\mathbb{P}) : \mathcal{F} \rightarrow [0, \infty]$ given by $(X\mathbb{P})(A) = \mathbb{E}(X; A)$ and the measure $(|\mathbb{E}(X | \mathcal{G})| \mathbb{P}) : \mathcal{F} \rightarrow [0, \infty]$ given by $(|\mathbb{E}(X | \mathcal{G})| \mathbb{P})(A) = \mathbb{E}(|\mathbb{E}(X | \mathcal{G})|; A)$. For every $I \in \mathcal{I}$, we know that $(X\mathbb{P})(I) = (|\mathbb{E}(X | \mathcal{G})| \mathbb{P})(I)$. In particular, we know that $(X\mathbb{P})(\Omega) = \mathbb{E}(X) = (|\mathbb{E}(X | \mathcal{G})| \mathbb{P})(\Omega) < \infty$. Therefore, from a previous result, we know that $\mathbb{E}(X \mathbb{I}_A) = \mathbb{E}(|\mathbb{E}(X | \mathcal{G})| \mathbb{I}_A)$ for every $A \in \sigma(\mathcal{G} \cup \mathcal{H})$. Because $|\mathbb{E}(X | \mathcal{G})|$ is $\sigma(\mathcal{G} \cup \mathcal{H})$ -measurable and

$\mathbb{E}(|\mathbb{E}(X | \mathcal{G})|) = \mathbb{E}(X) < \infty$, we know that $|\mathbb{E}(X | \mathcal{G})| = \mathbb{E}(X | \sigma(\mathcal{G} \cup \mathcal{H}))$ almost surely. Since $|\mathbb{E}(X | \mathcal{G})| = \mathbb{E}(X | \mathcal{G})$ almost surely, this step is complete.

Finally, suppose $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Recall that $X = X^+ - X^-$, where $X^+ \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $X^- \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ are non-negative. By the linearity of the conditional expectation,

$$\mathbb{P}(\mathbb{E}(X | \sigma(\mathcal{G} \cup \mathcal{H})) = \mathbb{E}(X^+ | \sigma(\mathcal{G} \cup \mathcal{H})) - \mathbb{E}(X^- | \sigma(\mathcal{G} \cup \mathcal{H})) = \mathbb{E}(X^+ | \mathcal{G}) - \mathbb{E}(X^- | \mathcal{G}) = \mathbb{E}(X | \mathcal{G}) = 1,$$

where we used the fact that $\sigma(\sigma(X^+) \cup \mathcal{G}) \subseteq \sigma(\sigma(X) \cup \mathcal{G})$ and $\sigma(\sigma(X^-) \cup \mathcal{G}) \subseteq \sigma(\sigma(X) \cup \mathcal{G})$. $\square$

Proposition 10.19. Consider a random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{H} \subseteq \mathcal{F}$. If $\mathcal{H}$ and $\sigma(X)$ are independent, then $\mathbb{E}(X | \mathcal{H}) = \mathbb{E}(X)$ almost surely.

Proof. Let $\mathcal{G} = \{\emptyset, \Omega\}$. Using the previous result, we know that $\mathbb{E}(X | \mathcal{H}) = \mathbb{E}(X | \mathcal{G})$ almost surely. Based on a previous result, we know that $\mathbb{E}(X) = \mathbb{E}(X | \mathcal{G})$ almost surely. $\square$

Definition 10.3. Consider the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. For every $F \in \mathcal{F}$, we let $\mathbb{P}(F | \mathcal{G})$ denote a version of the conditional expectation $\mathbb{E}(\mathbb{I}_F | \mathcal{G})$ of $\mathbb{I}_F$ given $\mathcal{G}$, so that $\mathbb{P}(F | \mathcal{G}) = \mathbb{E}(\mathbb{I}_F | \mathcal{G})$ almost surely. Note that $\mathbb{P}(F | \{\emptyset, \Omega\}) = \mathbb{E}(\mathbb{I}_F | \{\emptyset, \Omega\}) = \mathbb{E}(\mathbb{I}_F) = \mathbb{P}(F)$ almost surely.

Proposition 10.20. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the random variables $\mathbb{I}_F : \Omega \rightarrow \{0, 1\}$ and $Z : \Omega \rightarrow \mathcal{Z}$, where $F \in \mathcal{F}$ and $\mathcal{Z} = \{z_1, \dots, z_n\}$. Furthermore, suppose $\mathbb{P}(Z = z) > 0$ for every $z \in \mathcal{Z}$. Recall that if $E : \mathcal{Z} \rightarrow [0, 1]$ is given by

$$E(z) = \frac{\mathbb{P}(\mathbb{I}_F = 1, Z = z)}{\mathbb{P}(Z = z)} = \frac{\mathbb{P}(F \cap \{Z = z\})}{\mathbb{P}(Z = z)},$$

then $E \circ Z = \mathbb{E}(\mathbb{I}_F | Z) = \mathbb{P}(F | Z)$ almost surely.

Proposition 10.21. Consider a sequence of events $(F_n \in \mathcal{F} | n \in \mathbb{N})$ such that $F_n \cap F_m = \emptyset$ for every $n \neq m$. In that case, $\mathbb{P}(\bigcup_n F_n | \mathcal{G}) = \sum_n \mathbb{I}_A \mathbb{P}(F_n | \mathcal{G})$ almost surely, where $A \in \mathcal{G}$ is a set such that $\mathbb{P}(A) = 1$.

Proof. For every $k \in \mathbb{N}$, by the linearity of conditional expectation,

$$\mathbb{P}\left(\mathbb{P}\left(\bigcup_{i=0}^k F_i | \mathcal{G}\right)\right) = \mathbb{E}\left(\mathbb{I}_{\bigcup_{i=0}^k F_i} | \mathcal{G}\right) = \mathbb{E}\left(\sum_{i=0}^k \mathbb{I}_{F_i} | \mathcal{G}\right) = \sum_{i=0}^k \mathbb{E}(\mathbb{I}_{F_i} | \mathcal{G}) = \sum_{i=0}^k \mathbb{P}(F_i | \mathcal{G}) = 1.$$

Because $\mathbb{I}_{\bigcup_{i=0}^k F_i} \uparrow \mathbb{I}_{\bigcup_n F_n}$ with respect to k , by the conditional monotone-convergence theorem,

$$\mathbb{P}\left(\sum_n \mathbb{I}_A \mathbb{P}(F_n | \mathcal{G})\right) = \lim_{k \rightarrow \infty} \sum_{i=0}^k \mathbb{I}_A \mathbb{P}(F_i | \mathcal{G}) = \lim_{k \rightarrow \infty} \mathbb{E}\left(\mathbb{I}_{\bigcup_{i=0}^k F_i} | \mathcal{G}\right) \mathbb{I}_A = \mathbb{E}\left(\mathbb{I}_{\bigcup_n F_n} | \mathcal{G}\right) \mathbb{I}_A = \mathbb{P}\left(\bigcup_n F_n | \mathcal{G}\right) \mathbb{I}_A = 1,$$

where $A \in \mathcal{G}$ is a set such that $\mathbb{P}(A) = 1$. $\square$

Definition 10.4. Consider the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. A function $\mathbb{P}_{\mathcal{G}} : \Omega \times \mathcal{F} \rightarrow [0, 1]$ is called a regular conditional probability given $\mathcal{G}$ if

- There is a set $A \in \mathcal{F}$ such that $\mathbb{P}(A) = 1$ and, for every $\omega \in A$, the function $\mathbb{P}_{\mathcal{G}}(\omega, \cdot) : \mathcal{F} \rightarrow [0, 1]$ is a probability measure on $(\Omega, \mathcal{F})$.
- For every $F \in \mathcal{F}$, the function $\mathbb{P}_{\mathcal{G}}(\cdot, F) : \Omega \rightarrow [0, 1]$ is a version of the conditional expectation $\mathbb{E}(\mathbb{I}_F | \mathcal{G})$ of $\mathbb{I}_F$ given $\mathcal{G}$, so that $\mathbb{P}_{\mathcal{G}}(\cdot, F) = \mathbb{P}(F | \mathcal{G}) = \mathbb{E}(\mathbb{I}_F | \mathcal{G})$ almost surely.

It can be shown that a regular conditional probability given $\mathcal{G}$ exists under very permissive assumptions.

Proposition 10.22. Consider the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a bounded Borel function $h : \mathbb{R}^n \rightarrow \mathbb{R}$, and the independent random variables $X_1, X_2, \dots, X_n$. Let $h(X_1, X_2, \dots, X_n) : \Omega \rightarrow \mathbb{R}$ be given by

$$h(X_1, X_2, \dots, X_n)(\omega) = h(X_1(\omega), X_2(\omega), \dots, X_n(\omega)).$$

Furthermore, for every $x_1 \in \mathbb{R}$, let $h(x_1, X_2, \dots, X_n) : \Omega \rightarrow \mathbb{R}$ be given by

$$h(x_1, X_2, \dots, X_n)(\omega) = h(x_1, X_2(\omega), \dots, X_n(\omega)).$$

Finally, let $\gamma : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$\gamma(x_1) = \mathbb{E}(h(x_1, X_2, \dots, X_n)).$$

In that case, $\gamma(X_1) = \mathbb{E}(h(X_1, X_2, \dots, X_n) \mid X_1)$ almost surely, where $\gamma(X_1) = \gamma \circ X_1$.

Proof. For every $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, let $h_{x_1} : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ be given by $h_{x_1}(x_2, \dots, x_n) = h(x_1, x_2, \dots, x_n)$, and recall that h_{x_1} is a bounded Borel function. Furthermore, recall that the function $Z : \Omega \rightarrow \mathbb{R}^n$ given by $Z(\omega) = (X_1(\omega), X_2(\omega), \dots, X_n(\omega))$ is $\mathcal{F}/\mathcal{B}(\mathbb{R})^n$ -measurable and that the function $Y : \Omega \rightarrow \mathbb{R}^{n-1}$ given by $Y(\omega) = (X_2(\omega), \dots, X_n(\omega))$ is $\mathcal{F}/\mathcal{B}(\mathbb{R})^{n-1}$ -measurable.

For every $x_1 \in \mathbb{R}$, note that $h(X_1, X_2, \dots, X_n) = h \circ Z$ and $h(x_1, X_2, \dots, X_n) = h_{x_1} \circ Y$. Because h and h_{x_1} are Borel, for every $B \in \mathcal{B}(\mathbb{R})$, we know that $Z^{-1}(h^{-1}(B)) \in \mathcal{F}$ and $Y^{-1}(h_{x_1}^{-1}(B)) \in \mathcal{F}$. Because h and h_{x_1} are bounded, $h(X_1, X_2, \dots, X_n) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and $h(x_1, X_2, \dots, X_n) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$.

For every $k \in \{1, \dots, n\}$, let $\mathcal{L}_k : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$ denote the law of X_k . Because the random variables $X_1, X_2, \dots, X_n$ are independent, recall that the joint law of $X_i, X_{i+1}, \dots, X_n$ is given by $\mathcal{L}_i \times \mathcal{L}_{i+1} \times \dots \times \mathcal{L}_n$.

For every $x_1 \in \mathbb{R}$, because a previous result for laws extends to joint laws,

$$\gamma(x_1) = \int_{\Omega} h(x_1, X_2, \dots, X_n) d\mathbb{P} = \int_{\Omega} (h_{x_1} \circ Y) d\mathbb{P} = \int_{\mathbb{R}^{n-1}} h_{x_1} d(\mathcal{L}_2 \times \dots \times \mathcal{L}_n).$$

Because h_{x_1} is a bounded Borel function,

$$\gamma(x_1) = \int_{\mathbb{R}} \dots \int_{\mathbb{R}} h(x_1, x_2, \dots, x_n) \mathcal{L}_n(dx_n) \dots \mathcal{L}_2(dx_2),$$

which also implies that γ is $\mathcal{B}(\mathbb{R})$ -measurable, so that $\gamma(X_1)$ is $\sigma(X_1)$ -measurable.

For every $B \in \mathcal{B}(\mathbb{R})$, recall that $\mathbb{I}_{X_1^{-1}(B)} = \mathbb{I}_B(X_1)$. Therefore, for every $X_1^{-1}(B) \in \sigma(X_1)$,

$$\int_{\Omega} h(X_1, X_2, \dots, X_n) \mathbb{I}_{X_1^{-1}(B)} d\mathbb{P} = \int_{\mathbb{R}^n} h \mathbb{I}_B(\rho_1) d(\mathcal{L}_1 \times \dots \times \mathcal{L}_n).$$

Because $h \mathbb{I}_B(\rho_1)$ is bounded Borel function,

$$\int_{\Omega} h(X_1, X_2, \dots, X_n) \mathbb{I}_{X_1^{-1}(B)} d\mathbb{P} = \int_{\mathbb{R}} \mathbb{I}_B(x_1) \left[\int_{\mathbb{R}} \dots \int_{\mathbb{R}} h(x_1, x_2, \dots, x_n) \mathcal{L}_n(dx_n) \dots \mathcal{L}_2(dx_2) \right] \mathcal{L}_1(dx_1).$$

Using the previous expression for $\gamma(x_1)$ and a previous result for laws,

$$\int_{\Omega} h(X_1, X_2, \dots, X_n) \mathbb{I}_{X_1^{-1}(B)} d\mathbb{P} = \int_{\mathbb{R}} \mathbb{I}_B(x_1) \gamma(x_1) \mathcal{L}_1(dx_1) = \int_{\Omega} \gamma(X_1) \mathbb{I}_{X_1^{-1}(B)} d\mathbb{P}.$$

Because $\mathbb{E}(\gamma(X_1)) = \mathbb{E}(h(X_1, X_2, \dots, X_n)) < \infty$, the proof is complete. $\square$

Proposition 10.23. Consider a measurable space $(\Omega, \mathcal{F})$ and the sequence of σ -algebras $(\mathcal{F}_n \subseteq \mathcal{F} \mid n \in \mathbb{N}^+)$. For every $n \in \mathbb{N}^+$, let $\mathcal{I}_n = \{\cap_{i=1}^n F_i \mid F_i \in \mathcal{F}_i \text{ for every } i \in \{1, \dots, n\}\}$. In that case, $\mathcal{I} = \cup_n \mathcal{I}_n$ is a π -system on Ω such that $\sigma(\mathcal{I}) = \sigma(\mathcal{F}_1, \mathcal{F}_2, \dots)$, where $\sigma(\mathcal{F}_1, \mathcal{F}_2, \dots) = \sigma(\{F_1, F_2, \dots\}) = \sigma(\cup_n \mathcal{F}_n)$.

Proof. For some $n \in \mathbb{N}^+$, consider the sets $B \in \mathcal{I}_n$ and $C \in \mathcal{I}_n$ such that $B = \cap_{i=1}^n F_i$ and $C = \cap_{i=1}^n F'_i$, where $F_i \in \mathcal{F}_i$ and $F'_i \in \mathcal{F}_i$ for every $i \in \{1, \dots, n\}$. In that case,

$$B \cap C = \left(\bigcap_{i=1}^n F_i \right) \cap \left(\bigcap_{i=1}^n F'_i \right) = \bigcap_{i=1}^n (F_i \cap F'_i).$$

Because $(F_i \cap F'_i) \in \mathcal{F}_i$ for every $i \in \{1, \dots, n\}$, we know that $(B \cap C) \in \mathcal{I}_n$. Therefore, $\mathcal{I}_n$ is a π -system on Ω . Because $\Omega \in \mathcal{F}_n$ for every $n \in \mathbb{N}^+$, we know that $\mathcal{I}_n \subseteq \mathcal{I}_{n+1}$. Therefore, $\mathcal{I} = \cup_n \mathcal{I}_n$ is also a π -system on Ω .

Since $\Omega \in \mathcal{F}_n$ for every $n \in \mathbb{N}^+$, we also know that $\mathcal{F}_n \subseteq \mathcal{I}$ for every $n \in \mathbb{N}^+$. Therefore, $\cup_n \mathcal{F}_n \subseteq \mathcal{I}$ and $\sigma(\cup_n \mathcal{F}_n) \subseteq \sigma(\mathcal{I})$. Consider a set $(\cap_{i=1}^m F_i) \in \mathcal{I}$, where $m \in \mathbb{N}^+$ and $F_i \in \mathcal{F}_i$ for every $i \in \{1, \dots, m\}$. Clearly, $F_i \in \cup_n \mathcal{F}_n$ for every $i \in \{1, \dots, m\}$. Because $\sigma(\cup_n \mathcal{F}_n)$ is a σ -algebra, we know that $(\cap_{i=1}^m F_i) \in \sigma(\cup_n \mathcal{F}_n)$, which implies $\mathcal{I} \subseteq \sigma(\cup_n \mathcal{F}_n)$ and $\sigma(\mathcal{I}) \subseteq \sigma(\cup_n \mathcal{F}_n)$. $\square$

Proposition 10.24. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and the sequence of independent σ -algebras $(\mathcal{F}_n \subseteq \mathcal{F} \mid n \in \mathbb{N}^+)$. In that case, $\sigma(\mathcal{F}_1, \dots, \mathcal{F}_k)$ and $\sigma(\mathcal{F}_{k+1}, \mathcal{F}_{k+2}, \dots)$ are independent for every $k \in \mathbb{N}^+$.

Proof. From the previous proof, we know that $\mathcal{I} = \{\cap_{i=1}^k F_i \mid F_i \in \mathcal{F}_i \text{ for every } i \in \{1, \dots, k\}\}$ is a π -system on Ω such that $\sigma(\mathcal{I}) = \sigma(\mathcal{F}_1, \dots, \mathcal{F}_k)$. We also know that $\mathcal{J} = \cup_n \{\cap_{i=k+1}^{k+n} F_i \mid F_i \in \mathcal{F}_i \text{ for every } i \in \{k+1, \dots, k+n\}\}$ is a π -system on Ω such that $\sigma(\mathcal{J}) = \sigma(\mathcal{F}_{k+1}, \mathcal{F}_{k+2}, \dots)$.

Consider a set $(\cap_{i=1}^k F_i) \in \mathcal{I}$, where $F_i \in \mathcal{F}_i$ for every $i \in \{1, \dots, k\}$, and a set $(\cap_{i=k+1}^{k+n} F_i) \in \mathcal{J}$, where $n \in \mathbb{N}^+$ and $F_i \in \mathcal{F}_i$ for every $i \in \{k+1, \dots, k+n\}$. Because $\mathcal{F}_1, \dots, \mathcal{F}_{k+n}$ are independent,

$$\mathbb{P}\left(\left(\bigcap_{i=1}^k F_i\right) \cap \left(\bigcap_{i=k+1}^{k+n} F_i\right)\right) = \left(\prod_{i=1}^k \mathbb{P}(F_i)\right) \left(\prod_{i=k+1}^{k+n} \mathbb{P}(F_i)\right) = \mathbb{P}\left(\bigcap_{i=1}^k F_i\right) \mathbb{P}\left(\bigcap_{i=k+1}^{k+n} F_i\right),$$

which implies that $\mathcal{I}$ and $\mathcal{J}$ are independent. Because $\sigma(\mathcal{I})$ and $\sigma(\mathcal{J})$ are then independent, the proof is complete. $\square$

Proposition 10.25. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence of independent identically distributed random variables $(X_n : \Omega \rightarrow \mathbb{R} \mid n \in \mathbb{N}^+)$, each of which has the same law $\mathcal{L}_X$ as the random variable $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Let $S_n : \Omega \rightarrow \mathbb{R}$ be a random variable given by $S_n = X_1 + \dots + X_n$. In that case,

$$\mathbb{E}(X_k \mid S_n) = \mathbb{E}(X_k \mid S_n, S_{n+1}, \dots) = \frac{S_n}{n}$$

almost surely, where $n \in \mathbb{N}^+$ and $k \in \{1, \dots, n\}$.

Proof. We will start by showing that $\sigma(S_n, S_{n+1}, \dots) = \sigma(S_n, X_{n+1}, X_{n+2}, \dots)$ for every $n \in \mathbb{N}^+$. For every $i \in \mathbb{N}^+$, note that $S_{n+i} = S_n + X_{n+1} + \dots + X_{n+i}$, so that $\sigma(S_{n+i}) \subseteq \sigma(S_n, X_{n+1}, X_{n+2}, \dots)$. Therefore, $\sigma(S_n, S_{n+1}, \dots) \subseteq \sigma(S_n, X_{n+1}, X_{n+2}, \dots)$. For every $i \in \mathbb{N}^+$, note that $X_{n+i} = S_{n+i} - S_{n+i-1}$, so that $\sigma(X_{n+i}) \subseteq \sigma(S_n, S_{n+1}, \dots)$. Therefore, $\sigma(S_n, X_{n+1}, X_{n+2}, \dots) \subseteq \sigma(S_n, S_{n+1}, \dots)$.

Next, we will show that $\sigma(S_n, X_k)$ and $\sigma(X_{n+1}, X_{n+2}, \dots)$ are independent for every $n \in \mathbb{N}^+$ and $k \in \{1, \dots, n\}$. Note that $\sigma(S_n) \subseteq \sigma(X_1, \dots, X_n)$. Therefore, $\sigma(S_n, X_k) \subseteq \sigma(X_1, \dots, X_n)$. From a previous result, we know that $\sigma(X_1, \dots, X_n)$ and $\sigma(X_{n+1}, X_{n+2}, \dots)$ are independent, so that $\sigma(S_n, X_k)$ and $\sigma(X_{n+1}, X_{n+2}, \dots)$ are independent.

By considering this independence, for every $n \in \mathbb{N}^+$ and $k \in \{1, \dots, n\}$,

$$\mathbb{E}(X_k \mid S_n, S_{n+1}, \dots) = \mathbb{E}(X_k \mid S_n, X_{n+1}, X_{n+2}, \dots) = \mathbb{E}(X_k \mid S_n)$$

almost surely.

For every $n \in \mathbb{N}^+$, recall that $\mathbb{I}_{S_n^{-1}(B)} = \mathbb{I}_B(S_n)$ for all $B \in \mathcal{B}(\mathbb{R})$. Since $X_k \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ for every $k \in \{1, \dots, n\}$,

$$\int_{\Omega} X_k \mathbb{I}_{S_n^{-1}(B)} d\mathbb{P} = \int_{\Omega} X_k \mathbb{I}_B(S_n) d\mathbb{P} = \int_{\Omega} f_B(X_k, X_1, \dots, X_{k-1}, X_{k+1}, \dots, X_n) d\mathbb{P},$$

where $f_B : \mathbb{R}^n \rightarrow \mathbb{R}$ is a Borel function given by $f_B(x_1, \dots, x_n) = x_1 \mathbb{I}_B(x_1 + \dots + x_n)$.

Because a previous result for laws extends to joint laws and $X_1, \dots, X_n$ are independent,

$$\int_{\Omega} X_k \mathbb{I}_{S_n^{-1}(B)} d\mathbb{P} = \int_{\mathbb{R}^n} f_B d\mathcal{L}_{X_k, X_1, \dots, X_{k-1}, X_{k+1}, \dots, X_n} = \int_{\mathbb{R}^n} f_B d\mathcal{L}_X^n.$$

Therefore, for every $n \in \mathbb{N}^+$, $B \in \mathcal{B}(\mathbb{R})$, $S_n^{-1}(B) \in \sigma(S_n)$, and $i, j \in \{1, \dots, n\}$,

$$\int_{\Omega} \mathbb{E}(X_i \mid S_n) \mathbb{I}_{S_n^{-1}(B)} d\mathbb{P} = \int_{\Omega} X_i \mathbb{I}_{S_n^{-1}(B)} d\mathbb{P} = \int_{\mathbb{R}^n} f_B d\mathcal{L}_X^n = \int_{\Omega} X_j \mathbb{I}_{S_n^{-1}(B)} d\mathbb{P} = \int_{\Omega} \mathbb{E}(X_j \mid S_n) \mathbb{I}_{S_n^{-1}(B)} d\mathbb{P},$$

so that $\mathbb{E}(X_i \mid S_n) = \mathbb{E}(X_j \mid S_n)$ almost surely.

Finally, for every $n \in \mathbb{N}^+$ and $k \in \{1, \dots, n\}$,

$$n\mathbb{E}(X_k \mid S_n) = \sum_{i=1}^n \mathbb{E}(X_k \mid S_n) = \sum_{i=1}^n \mathbb{E}(X_i \mid S_n) = \mathbb{E}\left(\sum_{i=1}^n X_i \mid S_n\right) = \mathbb{E}(S_n \mid S_n) = S_n$$

almost surely, so that $\mathbb{E}(X_k \mid S_n) = S_n/n$ almost surely. $\square$

Proposition 10.26. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, the random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{N}$, and a Borel function $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\mathbb{E}(|h(X)|) < \infty$. Consider also a sequence of probability measures $(P_n \mid n \in \mathbb{N})$ on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. If $\mathbb{P}(X \in B \mid Y) = P_Y(B)$ almost surely for every $B \in \mathcal{B}(\mathbb{R})$, then almost surely

$$\mathbb{E}(h(X) \mid Y) = \sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} h(x) P_y(dx) = \int_{\mathbb{R}} h(x) P_Y(dx).$$

Proof. First, suppose that $h = \mathbb{I}_B$ for some $B \in \mathcal{B}(\mathbb{R})$. Almost surely,

$$\mathbb{E}(\mathbb{I}_B(X) \mid Y) = \mathbb{E}(\mathbb{I}_{\{X \in B\}} \mid Y) = P_Y(B) = \sum_y \mathbb{I}_{\{Y=y\}} P_y(B) = \sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} \mathbb{I}_B(x) P_y(dx).$$

Next, suppose that h is a simple function that can be written as $h = \sum_{k=1}^m a_k \mathbb{I}_{A_k}$ for some fixed $a_1, a_2, \dots, a_m \in [0, \infty]$ and $A_1, A_2, \dots, A_m \in \mathcal{B}(\mathbb{R})$. If $\mathbb{E}(h(X)) < \infty$, almost surely,

$$\mathbb{E}\left(\sum_{k=1}^m a_k \mathbb{I}_{A_k}(X) \mid Y\right) = \sum_{k=1}^m a_k \sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} \mathbb{I}_{A_k}(x) P_y(dx) = \sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} \sum_{k=1}^m a_k \mathbb{I}_{A_k}(x) P_y(dx).$$

Next, suppose that h is a non-negative Borel function such that $\mathbb{E}(h(X)) < \infty$. For any $n \in \mathbb{N}$, consider the simple function $h_n = \alpha_n \circ h$, where α_n is the n -th staircase function. Almost surely, since $h_n(X) \uparrow h(X)$,

$$\mathbb{E}(h(X) \mid Y) = \mathbb{E}\left(\lim_{n \rightarrow \infty} h_n(X) \mid Y\right) = \lim_{n \rightarrow \infty} \mathbb{E}(h_n(X) \mid Y) = \lim_{n \rightarrow \infty} \sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} h_n(x) P_y(dx).$$

Since $h_n \uparrow h$, by the monotone-convergence theorem, almost surely,

$$\mathbb{E}(h(X) \mid Y) = \sum_y \mathbb{I}_{\{Y=y\}} \lim_{n \rightarrow \infty} \int_{\mathbb{R}} h_n(x) P_y(dx) = \sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} h(x) P_y(dx).$$

Finally, suppose that $h = h^+ - h^-$ is a Borel function such that $\mathbb{E}(|h(X)|) < \infty$. Recall that $\mathbb{E}(h^+(X)) < \infty$ and $\mathbb{E}(h^-(X)) < \infty$. Therefore, almost surely,

$$\mathbb{E}(h(X) \mid Y) = \left(\sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} h^+(x) P_y(dx) \right) - \left(\sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} h^-(x) P_y(dx) \right).$$

By the linearity of the integral, almost surely,

$$\mathbb{E}(h(X) \mid Y) = \sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} (h^+(x) - h^-(x)) P_y(dx) = \sum_y \mathbb{I}_{\{Y=y\}} \int_{\mathbb{R}} h(x) P_y(dx).$$

□

11 Martingales

Definition 11.1. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. A filtration $(\mathcal{F}_n)_n$ is a sequence $(\mathcal{F}_n \subseteq \mathcal{F} \mid n \in \mathbb{N})$ of σ -algebras such that $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$ for every $n \in \mathbb{N}$. In that case, we let $\mathcal{F}_\infty = \sigma(\mathcal{F}_0, \mathcal{F}_1, \dots) = \sigma(\cup_n \mathcal{F}_n)$.

Definition 11.2. A filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$ is composed of a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ and a filtration $(\mathcal{F}_n)_n$.

Intuitively, at a given time $n \in \mathbb{N}$, for every $\omega \in \Omega$, recall that knowing $\mathbb{I}_{F_n}(\omega)$ for every $F_n \in \mathcal{F}_n$ allows knowing $Z_n(\omega)$ for every $\mathcal{F}_n$ -measurable random variable Z_n .

For any set $\mathcal{C}$, recall that a set (or sequence) of random variables $Y = (Y_\gamma \mid \gamma \in \mathcal{C})$ on a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a stochastic process (parameterized by $\mathcal{C}$).

Definition 11.3. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. The natural filtration $(\mathcal{F}_n)_n$ of the stochastic process $(W_n \mid n \in \mathbb{N})$ is given by $\mathcal{F}_n = \sigma(W_0, \dots, W_n)$ for every $n \in \mathbb{N}$.

Intuitively, at a given time $n \in \mathbb{N}$, for every $\omega \in \Omega$, recall that knowing $\mathbb{I}_{F_n}(\omega)$ for every $F_n \in \sigma(W_0, \dots, W_n)$ is equivalent to knowing $W_0(\omega), \dots, W_n(\omega)$.

Definition 11.4. Consider a filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$. A stochastic process $(X_n \mid n \in \mathbb{N})$ is called adapted (to the filtration $(\mathcal{F}_n)_n$) if X_n is $\mathcal{F}_n$ -measurable for every $n \in \mathbb{N}$.

Note that if $(\mathcal{F}_n)_n$ is the natural filtration of the stochastic process $(W_n \mid n \in \mathbb{N})$, then there is a Borel function $f_n : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ such that $X_n = f_n(W_0, \dots, W_n)$.

Consider a filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$.

Definition 11.5. A stochastic process $(X_n \mid n \in \mathbb{N})$ is called a martingale if $(X_n \mid n \in \mathbb{N})$ is adapted; $\mathbb{E}(|X_n|) < \infty$ for every $n \in \mathbb{N}$; and $\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) = X_{n-1}$ almost surely for every $n \in \mathbb{N}^+$.

Definition 11.6. A stochastic process $(X_n \mid n \in \mathbb{N})$ is called a supermartingale if $(X_n \mid n \in \mathbb{N})$ is adapted; $\mathbb{E}(|X_n|) < \infty$ for every $n \in \mathbb{N}$; and $\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) \leq X_{n-1}$ almost surely for every $n \in \mathbb{N}^+$.

Definition 11.7. A stochastic process $(X_n \mid n \in \mathbb{N})$ is called a submartingale if $(X_n \mid n \in \mathbb{N})$ is adapted; $\mathbb{E}(|X_n|) < \infty$ for every $n \in \mathbb{N}$; and $\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) \geq X_{n-1}$ almost surely for every $n \in \mathbb{N}^+$.

Proposition 11.1. Consider an adapted stochastic process $(X_n \mid n \in \mathbb{N})$ and suppose that $\mathbb{E}(|X_n|) < \infty$ for every $n \in \mathbb{N}$. For every $n \in \mathbb{N}^+$, note that $\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) = X_{n-1}$ almost surely if and only if $\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) \leq X_{n-1} \leq \mathbb{E}(X_n \mid \mathcal{F}_{n-1})$ almost surely. Therefore, $(X_n \mid n \in \mathbb{N})$ is a martingale if and only if $(X_n \mid n \in \mathbb{N})$ is a supermartingale and a submartingale.

Proposition 11.2. If $(X_n \mid n \in \mathbb{N})$ is a supermartingale, then $(-X_n \mid n \in \mathbb{N})$ is adapted; $\mathbb{E}(|-X_n|) = \mathbb{E}(|X_n|) < \infty$ for every $n \in \mathbb{N}$; and $\mathbb{E}(-X_n \mid \mathcal{F}_{n-1}) \geq -X_{n-1}$ almost surely for every $n \in \mathbb{N}^+$. Therefore, $(-X_n \mid n \in \mathbb{N})$ is a submartingale.

Proposition 11.3. If $(X_n \mid n \in \mathbb{N})$ is a submartingale, then $(-X_n \mid n \in \mathbb{N})$ is adapted; $\mathbb{E}(|-X_n|) = \mathbb{E}(|X_n|) < \infty$ for every $n \in \mathbb{N}$; and $\mathbb{E}(-X_n \mid \mathcal{F}_{n-1}) \leq -X_{n-1}$ almost surely for every $n \in \mathbb{N}^+$. Therefore, $(-X_n \mid n \in \mathbb{N})$ is a supermartingale.

Proposition 11.4. Consider an adapted stochastic process $(X_n \mid n \in \mathbb{N})$ and suppose that $\mathbb{E}(|X_n|) < \infty$ for every $n \in \mathbb{N}$. Furthermore, consider the stochastic process $(X_n - X_0 \mid n \in \mathbb{N})$. Because $X_n - X_0$ is $\mathcal{F}_n$ -measurable for every $n \in \mathbb{N}$, we know that $(X_n - X_0 \mid n \in \mathbb{N})$ is adapted. Because $\mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ is a vector space, we know that $\mathbb{E}(|X_n - X_0|) < \infty$ for every $n \in \mathbb{N}$. By the linearity of conditional expectation,

$$\mathbb{E}(X_n - X_0 \mid \mathcal{F}_{n-1}) = \mathbb{E}(X_n \mid \mathcal{F}_{n-1}) - \mathbb{E}(X_0 \mid \mathcal{F}_{n-1}) = \mathbb{E}(X_n \mid \mathcal{F}_{n-1}) - X_0$$

almost surely for every $n \in \mathbb{N}^+$. Therefore:

- For every $n \in \mathbb{N}^+$, $\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) = X_{n-1}$ almost surely if and only if $\mathbb{E}(X_n - X_0 \mid \mathcal{F}_{n-1}) = X_{n-1} - X_0$ almost surely. Therefore, $(X_n \mid n \in \mathbb{N})$ is a martingale if and only if $(X_n - X_0 \mid n \in \mathbb{N})$ is a martingale.
- For every $n \in \mathbb{N}^+$, $\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) \leq X_{n-1}$ almost surely if and only if $\mathbb{E}(X_n - X_0 \mid \mathcal{F}_{n-1}) \leq X_{n-1} - X_0$ almost surely. Therefore, $(X_n \mid n \in \mathbb{N})$ is a supermartingale if and only if $(X_n - X_0 \mid n \in \mathbb{N})$ is a supermartingale.
- For every $n \in \mathbb{N}^+$, $\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) \geq X_{n-1}$ almost surely if and only if $\mathbb{E}(X_n - X_0 \mid \mathcal{F}_{n-1}) \geq X_{n-1} - X_0$ almost surely. Therefore, $(X_n \mid n \in \mathbb{N})$ is a submartingale if and only if $(X_n - X_0 \mid n \in \mathbb{N})$ is a submartingale.

Consequently, it is common to assume that a stochastic process $(X_n \mid n \in \mathbb{N})$ has $X_0 = 0$ and $\mathcal{F}_0 = \{\emptyset, \Omega\}$.

Proposition 11.5. If $(X_n \mid n \in \mathbb{N})$ is a martingale, $n \in \mathbb{N}^+$, and $m < n$, then

$$\mathbb{E}(X_n \mid \mathcal{F}_m) = \mathbb{E}(X_n \mid \mathcal{F}_{n-1} \mid \mathcal{F}_m) = \mathbb{E}(\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) \mid \mathcal{F}_m) = \mathbb{E}(X_{n-1} \mid \mathcal{F}_m)$$

almost surely. Therefore, almost surely,

$$\mathbb{E}(X_n \mid \mathcal{F}_m) = \mathbb{E}(X_{n-1} \mid \mathcal{F}_m) = \dots = \mathbb{E}(X_{m+1} \mid \mathcal{F}_m) = \mathbb{E}(X_m \mid \mathcal{F}_m) = X_m.$$

Proposition 11.6. If $(X_n \mid n \in \mathbb{N})$ is a supermartingale, $n \in \mathbb{N}^+$, and $m < n$, then

$$\mathbb{E}(X_n \mid \mathcal{F}_m) = \mathbb{E}(X_n \mid \mathcal{F}_{n-1} \mid \mathcal{F}_m) = \mathbb{E}(\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) \mid \mathcal{F}_m) \leq \mathbb{E}(X_{n-1} \mid \mathcal{F}_m)$$

almost surely. Therefore, almost surely,

$$\mathbb{E}(X_n \mid \mathcal{F}_m) \leq \mathbb{E}(X_{n-1} \mid \mathcal{F}_m) \leq \dots \leq \mathbb{E}(X_{m+1} \mid \mathcal{F}_m) \leq \mathbb{E}(X_m \mid \mathcal{F}_m) = X_m.$$

Proposition 11.7. If $(X_n \mid n \in \mathbb{N})$ is a submartingale, $n \in \mathbb{N}^+$, and $m < n$, then

$$\mathbb{E}(X_n \mid \mathcal{F}_m) = \mathbb{E}(X_n \mid \mathcal{F}_{n-1} \mid \mathcal{F}_m) = \mathbb{E}(\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) \mid \mathcal{F}_m) \geq \mathbb{E}(X_{n-1} \mid \mathcal{F}_m)$$

almost surely. Therefore, almost surely,

$$\mathbb{E}(X_n \mid \mathcal{F}_m) \geq \mathbb{E}(X_{n-1} \mid \mathcal{F}_m) \geq \dots \geq \mathbb{E}(X_{m+1} \mid \mathcal{F}_m) \geq \mathbb{E}(X_m \mid \mathcal{F}_m) = X_m.$$

The next three examples illustrate the definition of martingales.

Proposition 11.8. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a sequence of independent random variables $(X_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P}) \mid n \in \mathbb{N}^+)$, and suppose that $\mathbb{E}(X_n) = 0$ for every $n \in \mathbb{N}^+$. Let $S_n = X_1 + \dots + X_n$ for every $n \in \mathbb{N}^+$ and $S_0 = 0$. In that case, $(S_n \mid n \in \mathbb{N})$ is a martingale.

Proof. Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ for every $n \in \mathbb{N}^+$ and $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Clearly, $(S_n \mid n \in \mathbb{N})$ is adapted to the filtration $(\mathcal{F}_n)_n$. Because $\mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ is a vector space, $S_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ for every $n \in \mathbb{N}$. For every $n \in \mathbb{N}^+$,

$$\mathbb{E}(S_n \mid \mathcal{F}_{n-1}) = \mathbb{E}(S_{n-1} + X_n \mid \mathcal{F}_{n-1}) = \mathbb{E}(S_{n-1} \mid \mathcal{F}_{n-1}) + \mathbb{E}(X_n \mid \mathcal{F}_{n-1}) = S_{n-1} + \mathbb{E}(X_n) = S_{n-1}$$

almost surely, where we used the fact that $\sigma(X_n)$ is independent of $\mathcal{F}_{n-1}$ for every $n \in \mathbb{N}^+$. □

Proposition 11.9. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a sequence of independent random variables $(X_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P}) \mid n \in \mathbb{N}^+)$, and suppose that $\mathbb{E}(X_n) = 1$ for every $n \in \mathbb{N}^+$. Let $M_n = X_1 \cdots X_n$ for every $n \in \mathbb{N}^+$ and $M_0 = 1$. In that case, $(M_n \mid n \in \mathbb{N})$ is a martingale.

Proof. Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ for every $n \in \mathbb{N}^+$ and $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Clearly, $(M_n \mid n \in \mathbb{N})$ is adapted to the filtration $(\mathcal{F}_n)_n$. Because $X_1, \dots, X_n$ are independent, $M_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ for every $n \in \mathbb{N}$. For every $n \in \mathbb{N}^+$,

$$\mathbb{E}(M_n \mid \mathcal{F}_{n-1}) = \mathbb{E}(M_{n-1}X_n \mid \mathcal{F}_{n-1}) = M_{n-1}\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) = M_{n-1}\mathbb{E}(X_n) = M_{n-1}$$

almost surely, where we used the fact that $\sigma(X_n)$ is independent of $\mathcal{F}_{n-1}$ for every $n \in \mathbb{N}^+$. □

Proposition 11.10. Consider a filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$ and a random variable $\xi \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Let $M_n = \mathbb{E}(\xi \mid \mathcal{F}_n)$ almost surely for every $n \in \mathbb{N}$. In that case, $(M_n \mid n \in \mathbb{N})$ is a martingale.

Proof. Clearly, $(M_n \in \mathcal{L}^1(\Omega, \mathcal{F}_n, \mathbb{P}) \mid n \in \mathbb{N})$ is adapted to the filtration $(\mathcal{F}_n)_n$. For every $n \in \mathbb{N}^+$,

$$\mathbb{E}(M_n \mid \mathcal{F}_{n-1}) = \mathbb{E}(\mathbb{E}(\xi \mid \mathcal{F}_n) \mid \mathcal{F}_{n-1}) = \mathbb{E}(\xi \mid \mathcal{F}_n \mid \mathcal{F}_{n-1}) = \mathbb{E}(\xi \mid \mathcal{F}_{n-1}) = M_{n-1}$$

almost surely. □

Consider a filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$.

Definition 11.8. A stochastic process $(C_n \mid n \in \mathbb{N})$ is called previsible if C_n is $\mathcal{F}_{n-1}$ measurable for every $n \in \mathbb{N}^+$.

Note that if $(\mathcal{F}_n)_n$ is the natural filtration of the stochastic process $(W_n \mid n \in \mathbb{N})$, then there is a Borel function $g_n : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $C_n = g_n(W_0, \dots, W_{n-1})$ for every $n \in \mathbb{N}^+$.

Definition 11.9. The martingale transform $(C \bullet X)$ of an adapted process $X = (X_n \mid n \in \mathbb{N})$ by a previsible process $C = (C_n \mid n \in \mathbb{N})$ is the adapted process $((C \bullet X)_n \mid n \in \mathbb{N})$, where $(C \bullet X)_0 = 0$ and

$$(C \bullet X)_n = \sum_{k=1}^n C_k(X_k - X_{k-1})$$

for every $n \in \mathbb{N}^+$.

Note that $(C \bullet X)_n = (C \bullet X)_{n-1} + C_n(X_n - X_{n-1})$ for every $n \in \mathbb{N}^+$.

The following example illustrates the definition of martingale transform.

Example 11.1. For every $\omega \in \Omega$, suppose that $X_n(\omega) - X_{n-1}(\omega)$ represents the profit per unit stake in round $n \in \mathbb{N}^+$ of a game. In that case, $C_n(\omega)$ can be interpreted as the amount staked in round $n \in \mathbb{N}^+$ by a particular gambling strategy C . For every $n \in \mathbb{N}^+$ and $\omega \in \Omega$, the amount staked $C_n(\omega)$ may rely on knowledge about $\mathbb{I}_{\mathcal{F}_{n-1}}(\omega)$ for every $\mathcal{F}_{n-1} \in \mathcal{F}_{n-1}$, which includes at the very least knowledge about $X_0(\omega), \dots, X_{n-1}(\omega)$ and $C_0(\omega), \dots, C_{n-1}(\omega)$. Finally, in this setting, $(C \bullet X)_n(\omega)$ represents the profit after $n \in \mathbb{N}^+$ rounds. Note that:

- If $(X_n \mid n \in \mathbb{N})$ is a martingale, then $\mathbb{E}(X_n - X_{n-1} \mid \mathcal{F}_{n-1}) = \mathbb{E}(X_n \mid \mathcal{F}_{n-1}) - X_{n-1} = 0$ almost surely for every $n \in \mathbb{N}^+$.
- If $(X_n \mid n \in \mathbb{N})$ is a supermartingale, then $\mathbb{E}(X_n - X_{n-1} \mid \mathcal{F}_{n-1}) = \mathbb{E}(X_n \mid \mathcal{F}_{n-1}) - X_{n-1} \leq 0$ almost surely for every $n \in \mathbb{N}^+$.
- If $(X_n \mid n \in \mathbb{N})$ is a submartingale, then $\mathbb{E}(X_n - X_{n-1} \mid \mathcal{F}_{n-1}) = \mathbb{E}(X_n \mid \mathcal{F}_{n-1}) - X_{n-1} \geq 0$ almost surely for every $n \in \mathbb{N}^+$.

Proposition 11.11. Consider an adapted process $X = (X_n \mid n \in \mathbb{N})$ and a previsible process $C = (C_n \mid n \in \mathbb{N})$. If $C_n \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and $X_n \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ for every $n \in \mathbb{N}$, then $C_n(X_n - X_{n-1}) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ for every $n \in \mathbb{N}^+$.

Proof. Since $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ is a vector space, $(X_n - X_{n-1}) \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ for every $n \in \mathbb{N}^+$. By the Schwarz inequality, $C_n(X_n - X_{n-1}) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. $\square$

Proposition 11.12. Consider an adapted process $X = (X_n \mid n \in \mathbb{N})$ and a previsible process $C = (C_n \mid n \in \mathbb{N})$. If $|C_n| \leq K$ and $\mathbb{E}(|X_n|) < \infty$ for every $n \in \mathbb{N}$ and some $K \in [0, \infty)$, then $C_n(X_n - X_{n-1}) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ for every $n \in \mathbb{N}^+$.

Proof. Since $|C_n||X_n - X_{n-1}| \leq K|X_n - X_{n-1}|$ for every $n \in \mathbb{N}^+$, we know that $\mathbb{E}(|C_n(X_n - X_{n-1})|) \leq K\mathbb{E}(|X_n - X_{n-1}|)$. Because $\mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ is a vector space, we know that $C_n(X_n - X_{n-1}) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. $\square$

Proposition 11.13. Consider an adapted process $X = (X_n \in \mathcal{L}^1(\Omega, \mathcal{F}_n, \mathbb{P}) \mid n \in \mathbb{N})$ and a previsible process $C = (C_n \mid n \in \mathbb{N})$. Furthermore, suppose that $C_n(X_n - X_{n-1}) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ for every $n \in \mathbb{N}^+$.

First, recall that $(C \bullet X)$ is adapted. Because $(C \bullet X)_0 = 0$ and $(C \bullet X)_n = (C \bullet X)_{n-1} + C_n(X_n - X_{n-1})$ for every $n \in \mathbb{N}^+$, we know that $(C \bullet X)_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ for every $n \in \mathbb{N}$. Finally, for every $n \in \mathbb{N}^+$,

$$\mathbb{E}((C \bullet X)_n \mid \mathcal{F}_{n-1}) = \mathbb{E}((C \bullet X)_{n-1} + C_n(X_n - X_{n-1}) \mid \mathcal{F}_{n-1}) = (C \bullet X)_{n-1} + C_n\mathbb{E}(X_n - X_{n-1} \mid \mathcal{F}_{n-1})$$

almost surely. Therefore:

- If $(X_n \mid n \in \mathbb{N})$ is a martingale, then, $\mathbb{E}((C \bullet X)_n \mid \mathcal{F}_{n-1}) = (C \bullet X)_{n-1}$ almost surely for every $n \in \mathbb{N}^+$, so that $(C \bullet X)$ is a martingale.
- If $(X_n \mid n \in \mathbb{N})$ is a supermartingale and C is non-negative, then $\mathbb{E}((C \bullet X)_n \mid \mathcal{F}_{n-1}) \leq (C \bullet X)_{n-1}$ almost surely for every $n \in \mathbb{N}^+$, so that $(C \bullet X)$ is a supermartingale.
- If $(X_n \mid n \in \mathbb{N})$ is a submartingale and C is non-negative, then $\mathbb{E}((C \bullet X)_n \mid \mathcal{F}_{n-1}) \geq (C \bullet X)_{n-1}$ almost surely for every $n \in \mathbb{N}^+$, so that $(C \bullet X)$ is a submartingale.

Consider a filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$.

Definition 11.10. A function $T : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ is called a stopping time if $\{T \leq n\} \in \mathcal{F}_n$ for every $n \in \mathbb{N} \cup \{\infty\}$.

Intuitively, for every $\omega \in \Omega$ and $n \in \mathbb{N} \cup \{\infty\}$, knowing $\mathbb{I}_{F_n}(\omega)$ for every $F_n \in \mathcal{F}_n$ allows knowing whether $T(\omega) \leq n$.

Proposition 11.14. The function $T : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ is a stopping time if and only if $\{T = n\} \in \mathcal{F}_n$ for every $n \in \mathbb{N} \cup \{\infty\}$.

Proof. If T is a stopping time, then $\{T \leq n\} \in \mathcal{F}_n$ and $\{T \leq n-1\}^c \in \mathcal{F}_n$ for every $n \in \mathbb{N}$. Because $\{T = n\} = \{T \leq n\} \cap \{T > n-1\}$, we know that $\{T = n\} \in \mathcal{F}_n$. Furthermore, $\{T = \infty\} = \bigcap_n \{T \leq n\}^c$, so that $\{T = \infty\} \in \mathcal{F}_\infty$.

If $\{T = n\} \in \mathcal{F}_n$ for every $n \in \mathbb{N} \cup \{\infty\}$, the fact that $\{T \leq n\} = \bigcup_{k \leq n} \{T = k\}$ and $\{T = k\} \in \mathcal{F}_n$ for every $k \leq n$ implies that $\{T \leq n\} \in \mathcal{F}_n$. $\square$

The following example illustrates the definition of stopping time.

Proposition 11.15. Consider an adapted process $(A_n \mid n \in \mathbb{N})$ and a set $B \in \mathcal{B}(\mathbb{R})$. Let the function $T : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ be given by $T(\omega) = \inf\{n \in \mathbb{N} \mid A_n(\omega) \in B\}$, so that $T(\omega) = \inf \emptyset = \infty$ if $A_n(\omega) \notin B$ for every $n \in \mathbb{N}$. For every $n \in \mathbb{N}$,

$$\{T \leq n\} = \{\omega \in \Omega \mid A_k(\omega) \in B \text{ for some } k \leq n\} = \bigcup_{k \leq n} \{\omega \in \Omega \mid A_k(\omega) \in B\} = \bigcup_{k \leq n} A_k^{-1}(B).$$

Because A_k is $\mathcal{F}_n$ -measurable for every $k \leq n$ and $\{T \leq \infty\} \in \mathcal{F}_\infty$, we know that T is a stopping time.

Proposition 11.16. Consider an adapted process $(X_n \mid n \in \mathbb{N})$ and a stopping time T . For some $a \in \mathbb{R}$, consider the set A given by

$$A = \{\omega \in \Omega \mid T(\omega) < \infty \text{ and } X_{T(\omega)}(\omega) \leq a\}.$$

In that case, $A \in \mathcal{F}_\infty$.

Proof. By definition,

$$A = \bigcup_{k \in \mathbb{N}} \{\omega \in \Omega \mid T(\omega) = k \text{ and } X_k(\omega) \leq a\} = \bigcup_{k \in \mathbb{N}} \{T = k\} \cap \{X_k \leq a\}.$$

Because $\{T = k\} \cap \{X_k \leq a\} \in \mathcal{F}_k$ for every $k \in \mathbb{N}$, we know that $A \in \mathcal{F}_\infty$. $\square$

Definition 11.11. Consider an adapted process $X = (X_n \mid n \in \mathbb{N})$ and a stopping time T . The stopped process X^T is the adapted process $(X_n^T \mid n \in \mathbb{N})$. For every $n \in \mathbb{N}$, the random variable $X_n^T : \Omega \rightarrow \mathbb{R}$ is given by

$$X_n^T(\omega) = X_{\min(T(\omega), n)}(\omega) = \begin{cases} X_n(\omega), & \text{if } n \leq T(\omega), \\ X_{T(\omega)}(\omega), & \text{if } n > T(\omega). \end{cases}$$

Proposition 11.17. The stochastic process X^T defined above is indeed adapted to the filtration $(\mathcal{F}_n)_n$.

Proof. For every $n \in \mathbb{N}$ and $a \in \mathbb{R}$,

$$\{X_n^T \leq a\} = \{\omega \in \Omega \mid n \leq T(\omega) \text{ and } X_n(\omega) \leq a\} \cup \{\omega \in \Omega \mid n > T(\omega) \text{ and } X_{T(\omega)}(\omega) \leq a\}.$$

Let A_1 denote the first set on the right side of the previous equation and A_2 denote the second set.

Because $\{n \leq T\} = \{n-1 \geq T\}^c \in \mathcal{F}_n$ and $\{X_n \leq a\} \in \mathcal{F}_n$, we know that $A_1 \in \mathcal{F}_n$. Regarding A_2 , note that

$$A_2 = \{n > T\} \cap \{\omega \in \Omega \mid T(\omega) < \infty \text{ and } X_{T(\omega)}(\omega) \leq a\}.$$

Using a previous result,

$$A_2 = \{n > T\} \cap \bigcup_{k \in \mathbb{N}} \{T = k\} \cap \{X_k \leq a\} = \bigcup_{k \in \mathbb{N}} \{n > T\} \cap \{T = k\} \cap \{X_k \leq a\} = \bigcup_{k < n} \{T = k\} \cap \{X_k \leq a\}.$$

Because $\{T = k\} \cap \{X_k \leq a\} \in \mathcal{F}_n$ for every $k < n$, we know that $A_2 \in \mathcal{F}_n$. Therefore, $\{X_n^T \leq a\} \in \mathcal{F}_n$ for every $n \in \mathbb{N}$ and $a \in \mathbb{R}$, so that X_n^T is $\mathcal{F}_n$ -measurable. $\square$

Proposition 11.18. Consider the adapted process $X = (X_n \mid n \in \mathbb{N})$, the stopping time T , and the process $C = (C_n \mid n \in \mathbb{N})$, where $C_n = \mathbb{I}_{\{n \leq T\}}$ for every $n \in \mathbb{N}$. Note that C is previsible, since $\{n \leq T\} = \{n-1 \geq T\}^c$ and $\{n-1 \geq T\}^c \in \mathcal{F}_{n-1}$ for every $n \in \mathbb{N}^+$, which implies that $\mathbb{I}_{\{n \leq T\}}$ is $\mathcal{F}_{n-1}$ -measurable.

Now consider the martingale transform $(C \bullet X) = ((C \bullet X)_n \mid n \in \mathbb{N})$, so that $(C \bullet X)_0 = 0$ and

$$(C \bullet X)_n(\omega) = \sum_{k=1}^n \mathbb{I}_{\{k \leq T\}}(\omega) (X_k(\omega) - X_{k-1}(\omega)) = \sum_{k=1}^{\min(T(\omega), n)} X_k(\omega) - X_{k-1}(\omega)$$

for every $n \in \mathbb{N}^+$ and $\omega \in \Omega$. By reorganizing terms,

$$(C \bullet X)_n(\omega) = \sum_{k=1}^{\min(T(\omega), n)} X_k(\omega) - \sum_{k=0}^{\min(T(\omega), n)-1} X_k(\omega) = X_{\min(T(\omega), n)}(\omega) - X_0(\omega)$$

for every $n \in \mathbb{N}^+$ and $\omega \in \Omega$. Therefore, $(C \bullet X)_n = X_n^T - X_0 = X_n^T - X_0^T$ for every $n \in \mathbb{N}$.

Proposition 11.19. When combined with previous results, the result above implies the following:

- If $(X_n \mid n \in \mathbb{N})$ is a martingale and T is a stopping time, then $\mathbb{E}(X_n^T - X_0^T \mid \mathcal{F}_{n-1}) = X_{n-1}^T - X_0^T$ almost surely for every $n \in \mathbb{N}^+$, so that the stopped process X^T is a martingale.

- If $(X_n \mid n \in \mathbb{N})$ is a supermartingale and T is a stopping time, then $\mathbb{E}(X_n^T - X_0^T \mid \mathcal{F}_{n-1}) \leq X_{n-1}^T - X_0^T$ almost surely for every $n \in \mathbb{N}^+$, so that the stopped process X^T is a supermartingale.
- If $(X_n \mid n \in \mathbb{N})$ is a submartingale and T is a stopping time, then $\mathbb{E}(X_n^T - X_0^T \mid \mathcal{F}_{n-1}) \geq X_{n-1}^T - X_0^T$ almost surely for every $n \in \mathbb{N}^+$, so that the stopped process X^T is a submartingale.

Definition 11.12. Consider an adapted process $X = (X_n \mid n \in \mathbb{N})$ and a stopping time T . The function $X_T : \Omega \rightarrow \mathbb{R}$ is given by

$$X_T(\omega) = \begin{cases} X_{T(\omega)}(\omega), & \text{if } T(\omega) < \infty, \\ 0, & \text{if } T(\omega) = \infty. \end{cases}$$

Proposition 11.20. The function X_T defined above is $\mathcal{F}_\infty$ -measurable.

Proof. For every $a \in \mathbb{R}$,

$$\{X_T \leq a\} = \{\omega \in \Omega \mid T(\omega) < \infty \text{ and } X_{T(\omega)}(\omega) \leq a\} \cup \{\omega \in \Omega \mid T(\omega) = \infty \text{ and } 0 \leq a\}.$$

Let A_1 denote the first set on the right side of the previous equation and A_2 denote the second set. We have already shown that $A_1 \in \mathcal{F}_\infty$. If $a \geq 0$, then $A_2 = \{T = \infty\}$. Otherwise, if $a < 0$, then $A_2 = \emptyset$. In either case, $A_2 \in \mathcal{F}_\infty$. Therefore, $\{X_T \leq a\} \in \mathcal{F}_\infty$ for every $a \in \mathbb{R}$, so that X_T is $\mathcal{F}_\infty$ -measurable. $\square$

Theorem 11.1 (Doob's optional-stopping theorem). Consider a supermartingale $X = (X_n \mid n \in \mathbb{N})$, a stopping time T , and suppose at least one of the following:

1. The stopping time T is bounded, so that $T \leq N$ for some $N \in \mathbb{N}$.
2. The stopping time T is almost surely finite, so that $\mathbb{P}(T < \infty) = 1$, and the stochastic process X is bounded, so that $|X_n| \leq K$ for every $n \in \mathbb{N}$ and some $K \in [0, \infty)$.
3. The stopping time T has finite expectation, so that $\mathbb{E}(T) < \infty$, and the stochastic process X has bounded increments, so that $|X_n - X_{n-1}| \leq K$ for every $n \in \mathbb{N}^+$ and some $K \in [0, \infty)$.

In that case, $\mathbb{E}(|X_T|) < \infty$ and $\mathbb{E}(X_T) \leq \mathbb{E}(X_0)$.

Proof. First, recall that the stopped process X^T is a supermartingale. Therefore,

$$\mathbb{E}(X_n^T) \mathbb{I}_\Omega = \mathbb{E}(X_n^T \mid \{\emptyset, \Omega\}) = \mathbb{E}(\mathbb{E}(X_n^T \mid \mathcal{F}_{n-1}) \mid \{\emptyset, \Omega\}) \leq \mathbb{E}(X_{n-1}^T \mid \{\emptyset, \Omega\}) = \mathbb{E}(X_{n-1}^T) \mathbb{I}_\Omega,$$

almost surely for every $n \in \mathbb{N}^+$, which implies that $\mathbb{E}(X_n^T) \leq \mathbb{E}(X_{n-1}^T) \leq \dots \leq \mathbb{E}(X_1^T) \leq \mathbb{E}(X_0^T)$ for every $n \in \mathbb{N}^+$.

Suppose that the stopping time T is bounded, so that $T \leq N$ for some $N \in \mathbb{N}$. In that case, for every $\omega \in \Omega$,

$$X_T(\omega) = X_{T(\omega)}(\omega) = X_{\min(T(\omega), N)}(\omega) = X_N^T(\omega).$$

Because $X_T = X_N^T$, we know that $\mathbb{E}(|X_T|) < \infty$. From the previous result, $\mathbb{E}(X_T) = \mathbb{E}(X_N^T) \leq \mathbb{E}(X_0^T) = \mathbb{E}(X_0)$.

Suppose that the stopping time T is almost surely finite, so that $\mathbb{P}(T < \infty) = 1$, and the stochastic process X is bounded, so that $|X_n| \leq K$ for every $n \in \mathbb{N}$ and some $K \in [0, \infty)$. Because $\mathbb{P}(T < \infty) = 1$, we know that $\mathbb{P}(\lim_{n \rightarrow \infty} X_n^T = X_T) = 1$. Therefore, by the bounded convergence theorem, we know that $\mathbb{E}(|X_T|) < \infty$ and $\lim_{n \rightarrow \infty} \mathbb{E}(X_n^T) = \mathbb{E}(X_T)$. Because $\mathbb{E}(X_n^T) \leq \mathbb{E}(X_0)$ for every $n \in \mathbb{N}^+$, we know that $\mathbb{E}(X_T) \leq \mathbb{E}(X_0)$.

Finally, suppose that the stopping time T has finite expectation, so that $\mathbb{E}(T) < \infty$, and the stochastic process X has bounded increments, so that $|X_n - X_{n-1}| \leq K$ for every $n \in \mathbb{N}^+$ and some $K \in [0, \infty)$.

Because $\mathbb{E}(T) < \infty$ implies $\mathbb{P}(T < \infty) = 1$, we know that $\mathbb{P}(\lim_{n \rightarrow \infty} X_n^T = X_T) = 1$. Therefore,

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} X_n^T - X_0 = X_T - X_0\right) = 1.$$

Note that $|X_n^T - X_0| \leq KT$ for every $n \in \mathbb{N}$, since

$$|X_n^T(\omega) - X_0(\omega)| = \left| \sum_{k=1}^{\min(T(\omega), n)} X_k(\omega) - X_{k-1}(\omega) \right| \leq \sum_{k=1}^{\min(T(\omega), n)} |X_k(\omega) - X_{k-1}(\omega)| \leq \sum_{k=1}^{\min(T(\omega), n)} K \leq KT(\omega).$$

Because $\mathbb{E}(KT) = K\mathbb{E}(T) < \infty$, the dominated convergence theorem guarantees that $\mathbb{E}(|X_T - X_0|) < \infty$ and

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n^T - X_0) = \mathbb{E}(X_T - X_0),$$

so that $\lim_{n \rightarrow \infty} \mathbb{E}(X_n^T) = \mathbb{E}(X_T)$. Because $\mathbb{E}(X_n^T) \leq \mathbb{E}(X_0)$ for every $n \in \mathbb{N}^+$, we know that $\mathbb{E}(X_T) \leq \mathbb{E}(X_0)$. $\square$

Proposition 11.21. Consider a martingale $X = (X_n \mid n \in \mathbb{N})$, a stopping time T , and suppose at least one of the following:

1. The stopping time T is bounded, so that $T \leq N$ for some $N \in \mathbb{N}$.
2. The stopping time T is almost surely finite, so that $\mathbb{P}(T < \infty) = 1$, and the stochastic process X is bounded, so that $|X_n| \leq K$ for every $n \in \mathbb{N}$ and some $K \in [0, \infty)$.
3. The stopping time T has finite expectation, so that $\mathbb{E}(T) < \infty$, and the stochastic process X has bounded increments, so that $|X_n - X_{n-1}| \leq K$ for every $n \in \mathbb{N}^+$ and some $K \in [0, \infty)$.

In that case, $\mathbb{E}(|X_T|) < \infty$ and $\mathbb{E}(X_T) = \mathbb{E}(X_0)$.

Proof. Because X is a supermartingale, we know that $\mathbb{E}(|X_T|) < \infty$ and $\mathbb{E}(X_T) \leq \mathbb{E}(X_0)$. Because X is a submartingale, we know that $-X = (-X_n \mid n \in \mathbb{N})$ is a supermartingale, so that $\mathbb{E}(|(-X)_T|) < \infty$ and $\mathbb{E}((-X)_T) \leq \mathbb{E}(-X_0)$. Since $(-X)_T = -X_T$, we know that $\mathbb{E}(X_T) \geq \mathbb{E}(X_0)$, which implies $\mathbb{E}(X_T) = \mathbb{E}(X_0)$. $\square$

Proposition 11.22. Consider a martingale $M = (M_n \mid n \in \mathbb{N})$ that has bounded increments, so that $|M_n - M_{n-1}| \leq K_1$ for every $n \in \mathbb{N}^+$ and some $K_1 \in [0, \infty)$. Consider also a previsible process $C = (C_n \mid n \in \mathbb{N})$ that is bounded, so that $|C_n| \leq K_2$ for every $n \in \mathbb{N}$ and some $K_2 \in [0, \infty)$. Finally, consider a stopping time T with finite expectation, so that $\mathbb{E}(T) < \infty$. In that case, $\mathbb{E}((C \bullet M)_T) = 0$.

Proof. Note that $|C_n(M_n - M_{n-1})| = |C_n||M_n - M_{n-1}| \leq K_1 K_2$ for every $n \in \mathbb{N}^+$, so that $\mathbb{E}(|C_n(M_n - M_{n-1})|) < \infty$. Therefore, using a previous result, we know that $(C \bullet M)$ is a martingale.

Because $|(C \bullet M)_n - (C \bullet M)_{n-1}| = |C_n(M_n - M_{n-1})| \leq K_1 K_2$ for every $n \in \mathbb{N}^+$, we know that $(C \bullet M)$ has bounded increments. Therefore, using a previous result, we know that $\mathbb{E}(|(C \bullet M)_T|) < \infty$ and

$$\mathbb{E}((C \bullet M)_T) = \mathbb{E}((C \bullet M)_0) = \mathbb{E}(0) = 0.$$

$\square$

Proposition 11.23. Consider a supermartingale $X = (X_n \mid n \in \mathbb{N})$ and a stopping time T . Furthermore, suppose that $X_n \geq 0$ for every $n \in \mathbb{N}$ and that $\mathbb{P}(T < \infty) = 1$. In that case, $\mathbb{E}(X_T) \leq \mathbb{E}(X_0)$.

Proof. Because $\mathbb{P}(T < \infty) = 1$, we have $\mathbb{P}(\lim_{n \rightarrow \infty} X_n^T = X_T) = 1$. By the Fatou lemma, $\mathbb{E}(X_T) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(X_n^T)$. Because $\mathbb{E}(X_n^T) \leq \mathbb{E}(X_0)$ for every $n \in \mathbb{N}$, we know that $\mathbb{E}(X_T) \leq \mathbb{E}(X_0)$. $\square$

Proposition 11.24. For a random variable $T : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$,

$$\mathbb{E}(T) = \sum_{t=1}^{\infty} \mathbb{P}(T \geq t).$$

Furthermore, if $\mathbb{P}(T < \infty) = 1$, then

$$\mathbb{E}(T) = \sum_{t=1}^{\infty} t\mathbb{P}(T = t).$$

Proof. For every $\omega \in \Omega$,

$$T(\omega) = \sum_{t=1}^{\infty} \mathbb{I}_{\{T \geq t\}}(\omega),$$

where both sides are infinite when $T(\omega) = \infty$. Because $\mathbb{I}_{\{T \geq t\}}$ is a non-negative random variable for every $t \in \mathbb{N}^+$, the countable additivity of the integral guarantees that

$$\mathbb{E}(T) = \sum_{t=1}^{\infty} \mathbb{E}(\mathbb{I}_{\{T \geq t\}}) = \sum_{t=1}^{\infty} \mathbb{P}(T \geq t).$$

Now suppose that $\mathbb{P}(T < \infty) = 1$ and consider the random variable $\tilde{T} : \Omega \rightarrow \mathbb{N}$ given by

$$\tilde{T}(\omega) = (T \mathbb{I}_{\{T < \infty\}})(\omega) = \sum_{t=1}^{\infty} t \mathbb{I}_{\{T=t\}}(\omega) = \begin{cases} T(\omega), & \text{if } T(\omega) < \infty, \\ 0, & \text{if } T(\omega) = \infty. \end{cases}$$

Because $\mathbb{P}(\tilde{T} \neq T) \leq \mathbb{P}(T = \infty) = 0$, we know that $\mathbb{E}(T) = \mathbb{E}(\tilde{T})$. Because $t \mathbb{I}_{\{T=t\}}$ is a non-negative random variable for every $t \in \mathbb{N}^+$, the countable additivity of the integral guarantees that

$$\mathbb{E}(T) = \mathbb{E}(\tilde{T}) = \sum_{t=1}^{\infty} t \mathbb{E}(\mathbb{I}_{\{T=t\}}) = \sum_{t=1}^{\infty} t \mathbb{P}(T = t).$$

□

Proposition 11.25. Suppose that T is a stopping time and that for some $N \in \mathbb{N}^+$ and some $\epsilon > 0$

$$\mathbb{P}(T \leq n + N \mid \mathcal{F}_n) = \mathbb{E}(\mathbb{I}_{\{T \leq n+N\}} \mid \mathcal{F}_n) > \epsilon$$

almost surely for every $n \in \mathbb{N}$. In that case, $\mathbb{E}(T) < \infty$.

Proof. For every $k \in \mathbb{N}^+$,

$$\mathbb{P}(T > kN) = \mathbb{P}(\{T > kN\} \cap \{T > (k-1)N\}) = \mathbb{E}(\mathbb{I}_{\{T > kN\}} \mathbb{I}_{\{T > (k-1)N\}}) = \mathbb{E}(\mathbb{E}(\mathbb{I}_{\{T > kN\}} \mathbb{I}_{\{T > (k-1)N\}} \mid \mathcal{F}_{(k-1)N})).$$

Because $\{T \leq (k-1)N\}^c \in \mathcal{F}_{(k-1)N}$, we know that $\mathbb{I}_{\{T > (k-1)N\}}$ is $\mathcal{F}_{(k-1)N}$ -measurable. Therefore,

$$\mathbb{P}(T > kN) = \mathbb{E}(\mathbb{I}_{\{T > (k-1)N\}} \mathbb{E}(\mathbb{I}_{\{T > kN\}} \mid \mathcal{F}_{(k-1)N})).$$

Let $n = (k-1)N$, so that $n + N = kN$. From our assumption, $\mathbb{E}(\mathbb{I}_{\{T \leq kN\}} \mid \mathcal{F}_{(k-1)N}) > \epsilon$ almost surely. Therefore, $\mathbb{E}(\mathbb{I}_{\{T > kN\}} \mid \mathcal{F}_{(k-1)N}) < 1 - \epsilon$ almost surely, so that

$$\mathbb{P}(T > kN) \leq (1 - \epsilon) \mathbb{E}(\mathbb{I}_{\{T > (k-1)N\}}) = (1 - \epsilon) \mathbb{P}(T > (k-1)N).$$

If $k = 1$, then

$$\mathbb{P}(T > N) = 1 - \mathbb{P}(T \leq N) = 1 - \mathbb{E}(\mathbb{I}_{\{T \leq N\}}) = 1 - \mathbb{E}(\mathbb{E}(\mathbb{I}_{\{T \leq N\}} \mid \mathcal{F}_0)) \leq 1 - \epsilon.$$

By induction, for every $k \in \mathbb{N}^+$,

$$\mathbb{P}(T > kN) \leq (1 - \epsilon)^k.$$

Because each $t \in \mathbb{N}$ can be uniquely written as $t = kN + i$ for some $k \in \mathbb{N}$ and $i \in \{0, \dots, N-1\}$,

$$\mathbb{E}(T) = \sum_{t=1}^{\infty} \mathbb{P}(T \geq t) = \sum_{t=0}^{\infty} \mathbb{P}(T > t) = \sum_{k=0}^{\infty} \sum_{i=0}^{N-1} \mathbb{P}(T > kN + i).$$

Because $\{T > kN + i\} \subseteq \{T > kN\}$ for every $k \in \mathbb{N}$ and $i \in \{0, \dots, N-1\}$,

$$\mathbb{E}(T) \leq \sum_{k=0}^{\infty} \sum_{i=0}^{N-1} \mathbb{P}(T > kN) = N \sum_{k=0}^{\infty} \mathbb{P}(T > kN) \leq N \sum_{k=0}^{\infty} (1 - \epsilon)^k = \frac{N}{\epsilon} < \infty.$$

□

Proposition 11.26. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a sequence of independent random variables $(X_n : \Omega \rightarrow \{-1, 1\} \mid n \in \mathbb{N}^+)$, and a random variable $X : \Omega \rightarrow \{-1, 1\}$. Suppose that $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = 1/2$ and that X_n has the same distribution as X for every $n \in \mathbb{N}^+$. Furthermore, let $S_0 = 0$ and $S_n = X_1 + \dots + X_n$ for every $n \in \mathbb{N}^+$. Finally, let $T : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ be given by $T(\omega) = \inf\{n \in \mathbb{N} \mid S_n(\omega) = 1\}$. In that case, $\mathbb{P}(T < \infty) = 1$.

Proof. Consider the filtration $(\mathcal{F}_n)_n$ where $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ for every $n \in \mathbb{N}^+$. Because S_n is $\mathcal{F}_n$ -measurable for every $n \in \mathbb{N}$, the process $(S_n \mid n \in \mathbb{N})$ is adapted, so that T is a stopping time.

For every $n \in \mathbb{N}^+$ and $\theta \in (0, \infty)$, note that $e^{\theta X_n} = e^{\theta} \mathbb{I}_{\{X_n=1\}} + e^{-\theta} \mathbb{I}_{\{X_n=-1\}}$. Therefore,

$$\mathbb{E}(e^{\theta X_n}) = e^{\theta} \mathbb{P}(X_n = 1) + e^{-\theta} \mathbb{P}(X_n = -1) = \frac{e^{\theta} + e^{-\theta}}{2}.$$

For every $n \in \mathbb{N}^+$ and $\theta \in (0, \infty)$, let $W_n = (2/(e^{\theta} + e^{-\theta}))e^{\theta X_n}$, so that

$$\mathbb{E}(W_n) = \mathbb{E}\left(\frac{2}{e^{\theta} + e^{-\theta}}e^{\theta X_n}\right) = \frac{2}{e^{\theta} + e^{-\theta}}\mathbb{E}(e^{\theta X_n}) = 1.$$

Because W_n is $\sigma(X_n)$ -measurable for every $n \in \mathbb{N}^+$, the sequence $(W_n \mid n \in \mathbb{N}^+)$ is composed of independent random variables. Now consider the stochastic process $M = (M_n \mid n \in \mathbb{N})$ where $M_0 = 1$ and

$$M_n = W_1 \cdots W_n = \prod_{i=1}^n \frac{2}{e^{\theta} + e^{-\theta}} e^{\theta X_i} = \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^n e^{\theta S_n}$$

for every $n \in \mathbb{N}^+$. From a previous result, we know that M is a martingale.

Let $Y_n = (2/(e^{\theta} + e^{-\theta}))^n$ and $Z_n = e^{\theta S_n}$ for every $n \in \mathbb{N}$, so that $M_n = Y_n Z_n$. Since $Y = (Y_n \mid n \in \mathbb{N})$ and $Z = (Z_n \mid n \in \mathbb{N})$ are adapted, the stopped processes $Y^T = (Y_n^T \mid n \in \mathbb{N})$ and $Z^T = (Z_n^T \mid n \in \mathbb{N})$ are given by

$$Y_n^T(\omega) = Y_{\min(T(\omega), n)}(\omega) = \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^{\min(T(\omega), n)} \quad \text{and} \quad Z_n^T(\omega) = Z_{\min(T(\omega), n)}(\omega) = e^{\theta S_{\min(T(\omega), n)}(\omega)}$$

for every $n \in \mathbb{N}$ and $\omega \in \Omega$. Furthermore, the stopped process $M^T = (M_n^T \mid n \in \mathbb{N})$ is given by

$$M_n^T(\omega) = M_{\min(T(\omega), n)}(\omega) = Y_{\min(T(\omega), n)}(\omega) Z_{\min(T(\omega), n)}(\omega) = Y_n^T(\omega) Z_n^T(\omega)$$

for every $n \in \mathbb{N}$ and $\omega \in \Omega$. Since M^T is a martingale, $\mathbb{E}(M_n^T) = \mathbb{E}(M_0^T) = 1$ for every $n \in \mathbb{N}$.

For every $n \in \mathbb{N}$ and $\omega \in \Omega$, note that $S_{\min(T(\omega), n)}(\omega) \leq 1$. Since $\theta \in (0, \infty)$, note that $Z_n^T \leq e^{\theta}$ and $Y_n^T \leq 1$, so that $M_n^T \leq e^{\theta}$. For every $\omega \in \Omega$, consider the limit

$$\lim_{n \rightarrow \infty} M_n^T(\omega) = \lim_{n \rightarrow \infty} \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^{\min(T(\omega), n)} e^{\theta S_{\min(T(\omega), n)}(\omega)}.$$

First, suppose that $T(\omega) < \infty$. In that case,

$$\lim_{n \rightarrow \infty} M_n^T(\omega) = \lim_{n \rightarrow \infty} \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^{T(\omega)} e^{\theta S_{T(\omega)}(\omega)} = \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^{T(\omega)} e^{\theta},$$

since $S_{T(\omega)}(\omega) = 1$.

Second, suppose that $T(\omega) = \infty$. In that case, we know that $S_n(\omega) < 1$ for every $n \in \mathbb{N}$, so that

$$0 \leq \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^n e^{\theta S_n(\omega)} \leq \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^n e^{\theta}$$

for every $n \in \mathbb{N}$ and $\theta \in (0, \infty)$. Because $0 < (2/(e^{\theta} + e^{-\theta})) < 1$ for every $\theta \in (0, \infty)$,

$$\lim_{n \rightarrow \infty} \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^n e^{\theta} = e^{\theta} \lim_{n \rightarrow \infty} \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^n = 0.$$

Therefore, when $T(\omega) = \infty$,

$$\lim_{n \rightarrow \infty} M_n^T(\omega) = \lim_{n \rightarrow \infty} \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^n e^{\theta S_n(\omega)} = 0.$$

For every $\theta \in (0, \infty)$, let $Y_{\theta, T} : \Omega \rightarrow [0, \infty)$ denote a random variable given by

$$Y_{\theta, T}(\omega) = \begin{cases} \left(\frac{2}{e^{\theta} + e^{-\theta}}\right)^{T(\omega)}, & \text{if } T(\omega) < \infty, \\ 0, & \text{if } T(\omega) = \infty. \end{cases}$$

Using the previous result,

$$\lim_{n \rightarrow \infty} M_n^T = e^\theta Y_{\theta, T}.$$

Since $|M_n^T| \leq e^\theta$ for every $n \in \mathbb{N}$, by the bounded convergence theorem,

$$\lim_{n \rightarrow \infty} \mathbb{E}(M_n^T) = 1 = e^\theta \mathbb{E}(Y_{\theta, T}),$$

so that $\mathbb{E}(Y_{\theta, T}) = 1/e^\theta$.

Finally, consider a sequence $(\theta_n \in (0, \infty) \mid n \in \mathbb{N})$ so that $\theta_n \downarrow 0$. In that case,

$$\lim_{n \rightarrow \infty} Y_{\theta_n, T}(\omega) = \mathbb{I}_{\{T < \infty\}}(\omega) = \begin{cases} 1, & \text{if } T(\omega) < \infty, \\ 0, & \text{if } T(\omega) = \infty. \end{cases}$$

Since $|Y_{\theta_n, T}| \leq 1$ for every $n \in \mathbb{N}$, by the bounded convergence theorem,

$$\lim_{n \rightarrow \infty} \mathbb{E}(Y_{\theta_n, T}) = \lim_{n \rightarrow \infty} \frac{1}{e^{\theta_n}} = 1 = \mathbb{E}(\mathbb{I}_{\{T < \infty\}}) = \mathbb{P}(T < \infty).$$

□

Proposition 11.27. Consider a measurable space $(\Omega, \mathcal{F})$, a set $E \subseteq \mathbb{N}$, a stochastic process $(Z_n : \Omega \rightarrow E \mid n \in \mathbb{N})$, and let $\mathcal{F}_n = \sigma(Z_0, \dots, Z_n)$ for every $n \in \mathbb{N}$. Furthermore, for every $n \in \mathbb{N}$, let $\mathcal{G}_n$ be given by

$$\mathcal{G}_n = \left\{ \bigcup_{i \in A} \{Z_0 = i_0, \dots, Z_n = i_n\} \mid A \in \mathcal{P}(E^{n+1}) \right\},$$

where $i = (i_0, \dots, i_n)$ and $\mathcal{P}(E^{n+1})$ is the set of all subsets of E^{n+1} . In that case, $\mathcal{F}_n = \mathcal{G}_n$.

Proof. For some $n \in \mathbb{N}$, consider a set given by

$$\bigcup_{i \in A} \{Z_0 = i_0, \dots, Z_n = i_n\} = \bigcup_{i \in A} \bigcap_{k=0}^n \{Z_k = i_k\}$$

for some $A \in \mathcal{P}(E^{n+1})$. For every $k \in \mathbb{N}$, recall that

$$\sigma(Z_k) = \left\{ \bigcup_{i_k \in A_k} \{Z_k = i_k\} \mid A_k \in \mathcal{P}(E) \right\}.$$

The set A is countable, since it is a subset of the countable set E^{n+1} , which is a finite Cartesian product between countable sets. Because $\{Z_k = i_k\} \in \mathcal{F}_n$ for every $k \in \{0, \dots, n\}$ and $i_k \in E$, we know that $\mathcal{G}_n \subseteq \mathcal{F}_n$.

For some $n \in \mathbb{N}$, let $A = A_0 \times \dots \times A_n$, where $A_k \in \mathcal{P}(E)$ for every $k \in \{0, \dots, n\}$. In that case,

$$\bigcup_{i \in A} \bigcap_{k=0}^n \{Z_k = i_k\} = \bigcup_{i_0 \in A_0} \dots \bigcup_{i_n \in A_n} \bigcap_{k=0}^n \{Z_k = i_k\} = \left(\bigcup_{i_0 \in A_0} \{Z_0 = i_0\} \right) \cap \dots \cap \left(\bigcup_{i_n \in A_n} \{Z_n = i_n\} \right).$$

Since $E \in \mathcal{P}(E)$, note that $\sigma(Z_k) \subseteq \mathcal{G}_n$ for every $k \in \{0, \dots, n\}$. Because $\mathcal{F}_n = \sigma(\cup_{k=0}^n \sigma(Z_k))$ and $\mathcal{G}_n \subseteq \mathcal{F}_n$, showing that $\mathcal{F}_n = \mathcal{G}_n$ now only requires showing that $\mathcal{G}_n$ is a σ -algebra on Ω .

For some $n \in \mathbb{N}$, let $A = E^{n+1}$. Using the previous result, we know that $\Omega \in \mathcal{G}_n$.

For some $n \in \mathbb{N}$, consider a sequence $(G_{n,m} \in \mathcal{G}_n \mid m \in \mathbb{N})$ where

$$G_{n,m} = \bigcup_{i \in A_m} \{Z_0 = i_0, \dots, Z_n = i_n\}$$

for some sequence $(A_m \in \mathcal{P}(E^{n+1}) \mid m \in \mathbb{N})$. Clearly,

$$\bigcup_m G_{n,m} = \bigcup_m \bigcup_{i \in A_m} \{Z_0 = i_0, \dots, Z_n = i_n\} = \bigcup_{i \in A} \{Z_0 = i_0, \dots, Z_n = i_n\},$$

where $A = \cup_m A_m$. Because $A \in \mathcal{P}(E^{n+1})$, we know that $\cup_m G_{n,m} \in \mathcal{G}_n$.

For some $n \in \mathbb{N}$ and every $A \in \mathcal{P}(E^{n+1})$, note that $A^c \in \mathcal{P}(E^{n+1})$ and $A \cup A^c = E^{n+1}$, so that

$$\left\{ \bigcup_{i \in A} \{Z_0 = i_0, \dots, Z_n = i_n\} \right\} \cup \left\{ \bigcup_{i \in A^c} \{Z_0 = i_0, \dots, Z_n = i_n\} \right\} = \bigcup_{i \in E^{n+1}} \{Z_0 = i_0, \dots, Z_n = i_n\} = \Omega.$$

Since the leftmost sets above are disjoint, if $G_n \in \mathcal{G}_n$, then $G_n^c \in \mathcal{G}_n$, so that $\mathcal{G}_n$ is a σ -algebra on Ω . $\square$

Proposition 11.28. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a random variable $Z : \Omega \rightarrow \mathbb{N}$, and a non-negative function $h : \mathbb{N} \rightarrow [0, \infty]$. In that case,

$$\mathbb{E}(h(Z)) = \sum_{z \in \mathbb{N}} h(z) \mathbb{P}(Z = z).$$

Proof. For every $B \in \mathcal{B}(\mathbb{R})$, note that $h^{-1}(B) \in \mathcal{P}(\mathbb{N})$, where $\mathcal{P}(\mathbb{N})$ is the set of all subsets of $\mathbb{N}$. Because $\mathcal{P}(\mathbb{N}) \subseteq \mathcal{B}(\mathbb{R})$ and $Z^{-1}(h^{-1}(B)) \in \mathcal{F}$, we know that $h(Z)$ is a random variable. For every $\omega \in \Omega$, note that

$$h(Z)(\omega) = h(Z(\omega)) = \sum_{z \in \mathbb{N}} h(z) \mathbb{I}_{\{Z=z\}}(\omega).$$

Since $h(z) \mathbb{I}_{\{Z=z\}}$ is a non-negative random variable for every $z \in \mathbb{N}$,

$$\mathbb{E}(h(Z)) = \mathbb{E} \left(\sum_{z \in \mathbb{N}} h(z) \mathbb{I}_{\{Z=z\}} \right) = \sum_{z \in \mathbb{N}} h(z) \mathbb{E}(\mathbb{I}_{\{Z=z\}}) = \sum_{z \in \mathbb{N}} h(z) \mathbb{P}(Z = z).$$

$\square$

Proposition 11.29. Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a random variable $Z : \Omega \rightarrow \mathbb{N}$, and a function $h : \mathbb{N} \rightarrow \mathbb{R}$. If $\mathbb{E}(|h(Z)|) < \infty$, then

$$\mathbb{E}(h(Z)) = \sum_{z \in \mathbb{N}} h(z) \mathbb{P}(Z = z).$$

Proof. Note that $h(Z) : \Omega \rightarrow \mathbb{R}$ is a random variable and let $h = h^+ - h^-$. Using the previous result,

$$\mathbb{E}(h(Z)) = \mathbb{E}(h^+(Z)) - \mathbb{E}(h^-(Z)) = \sum_{z \in \mathbb{N}} h^+(z) \mathbb{P}(Z = z) - \sum_{z \in \mathbb{N}} h^-(z) \mathbb{P}(Z = z) = \sum_{z \in \mathbb{N}} h(z) \mathbb{P}(Z = z).$$

$\square$

Consider a probability triple $(\Omega, \mathcal{F}, \mathbb{P})$, a set $E \subseteq \mathbb{N}$, a stochastic process $Z = (Z_n : \Omega \rightarrow E \mid n \in \mathbb{N})$, and let $\mathcal{F}_n = \sigma(Z_0, \dots, Z_n)$ for every $n \in \mathbb{N}$.

Let P be a stochastic matrix whose (i, j) -th element is given by $p_{i,j} \geq 0$ and suppose that $\sum_{k \in E} p_{i,k} = 1$ for every $i \in E$. Let μ be a probability measure on the measurable space $(E, \mathcal{P}(E))$, where $\mathcal{P}(E)$ is the set of all subsets of E , and let μ_i denote $\mu(\{i\})$ for every $i \in E$.

Finally, suppose that Z is a time-homogeneous Markov chain on E with initial distribution μ and 1-step transition matrix P . Recall that, for every $n \in \mathbb{N}^+$ and $i_0, i_1, \dots, i_n \in E$,

$$\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \mu_{i_0} p_{i_0, i_1} \dots p_{i_{n-1}, i_n} = \mu_{i_0} \prod_{k=1}^n p_{i_{k-1}, i_k}.$$

Definition 11.13. For every $n \in \mathbb{N}$ and $i_{n+1} \in E$, let $p(Z_n, i_{n+1}) : \Omega \rightarrow [0, 1]$ be given by $p(Z_n, i_{n+1})(\omega) = p_{Z_n(\omega), i_{n+1}}$.

We will now show three propositions regarding such time-homogeneous Markov chain.

Proposition 11.30. First, for every $n \in \mathbb{N}$, $i_{n+1} \in E$, $p(Z_n, i_{n+1}) = \mathbb{E}(\mathbb{I}_{\{Z_{n+1}=i_{n+1}\}} \mid \mathcal{F}_n) = \mathbb{P}(Z_{n+1} = i_{n+1} \mid \mathcal{F}_n)$ almost surely.

Proof. For every $c \in \mathbb{R}$, note that

$$\{p(Z_n, i_{n+1}) \leq c\} = \bigcup_{i_n \in E} \{\omega \in \Omega \mid Z_n(\omega) = i_n \text{ and } p_{i_n, i_{n+1}} \leq c\} = \bigcup_{i_n \in E} \{Z_n = i_n\} \cap \{p_{i_n, i_{n+1}} \leq c\}.$$

Since $\{p_{i_n, i_{n+1}} \leq c\} \in \{\emptyset, \Omega\}$ for every $i_n \in E$, we know that $\{p(Z_n, i_{n+1}) \leq c\} \in \sigma(Z_n)$, so that $p(Z_n, i_{n+1})$ is $\sigma(Z_n)$ -measurable. Because $|p(Z_n, i_{n+1})| \leq 1$, we know that $\mathbb{E}(|p(Z_n, i_{n+1})|) \leq 1$.

For every $\omega \in \Omega$, $n \in \mathbb{N}$, and $i_0, \dots, i_{n+1} \in E$, note that

$$(p(Z_n, i_{n+1}) \mathbb{I}_{\{Z_0=i_0, \dots, Z_n=i_n\}})(\omega) = p_{i_n, i_{n+1}} \mathbb{I}_{\{Z_0=i_0, \dots, Z_n=i_n\}}(\omega) = \begin{cases} p_{Z_n(\omega), i_{n+1}}, & \text{if } Z_k(\omega) = i_k \text{ for } k \in \{0, \dots, n\}, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, for every $n \in \mathbb{N}$, and $i_0, \dots, i_{n+1} \in E$,

$$\mathbb{E}(p(Z_n, i_{n+1}) \mathbb{I}_{\{Z_0=i_0, \dots, Z_n=i_n\}}) = p_{i_n, i_{n+1}} \mathbb{E}(\mathbb{I}_{\{Z_0=i_0, \dots, Z_n=i_n\}}) = p_{i_n, i_{n+1}} \mathbb{P}(Z_0 = i_0, \dots, Z_n = i_n).$$

Clearly, for every $n \in \mathbb{N}$, and $i_0, \dots, i_{n+1} \in E$,

$$\mathbb{E}(\mathbb{I}_{\{Z_{n+1}=i_{n+1}\}} \mathbb{I}_{\{Z_0=i_0, \dots, Z_n=i_n\}}) = \mathbb{P}(Z_0 = i_0, \dots, Z_{n+1} = i_{n+1}) = p_{i_n, i_{n+1}} \mathbb{P}(Z_0 = i_0, \dots, Z_n = i_n).$$

For every $n \in \mathbb{N}$, recall that every set $F_n \in \mathcal{F}_n$ can be written as

$$F_n = \bigcup_{i \in A} \{Z_0 = i_0, \dots, Z_n = i_n\},$$

where $i = (i_0, \dots, i_n)$ and $A \in \mathcal{P}(E^{n+1})$ is a countable set. Because F_n is a union of disjoint sets,

$$\mathbb{I}_{F_n} = \sum_{i \in A} \mathbb{I}_{\{Z_0=i_0, \dots, Z_n=i_n\}}.$$

Therefore, for every $n \in \mathbb{N}$ and $F_n \in \mathcal{F}_n$, since $p(Z_n, i_{n+1}) \geq 0$ for every $i_{n+1} \in E$,

$$\mathbb{E}(p(Z_n, i_{n+1}) \mathbb{I}_{F_n}) = \sum_{i \in A} \mathbb{E}(p(Z_n, i_{n+1}) \mathbb{I}_{\{Z_0=i_0, \dots, Z_n=i_n\}}) = \sum_{i \in A} p_{i_n, i_{n+1}} \mathbb{P}(Z_0 = i_0, \dots, Z_n = i_n),$$

where $i = (i_0, \dots, i_n)$ and $A \in \mathcal{P}(E^{n+1})$ is a countable set. Using our previous observation,

$$\mathbb{E}(p(Z_n, i_{n+1}) \mathbb{I}_{F_n}) = \mathbb{E}\left(\mathbb{I}_{\{Z_{n+1}=i_{n+1}\}} \sum_{i \in A} \mathbb{I}_{\{Z_0=i_0, \dots, Z_n=i_n\}}\right) = \mathbb{E}(\mathbb{I}_{\{Z_{n+1}=i_{n+1}\}} \mathbb{I}_{F_n}),$$

so that $p(Z_n, i_{n+1}) = \mathbb{E}(\mathbb{I}_{\{Z_{n+1}=i_{n+1}\}} \mid \mathcal{F}_n) = \mathbb{P}(Z_{n+1} = i_{n+1} \mid \mathcal{F}_n)$ almost surely. $\square$

For every $n \in \mathbb{N}$ and $i_{n+1} \in E$, note that $p(Z_n, i_{n+1})$ is $\sigma(Z_n)$ -measurable and $\sigma(Z_n) \subseteq \mathcal{F}_n$. Therefore, we also know that $p(Z_n, i_{n+1}) = \mathbb{E}(\mathbb{I}_{\{Z_{n+1}=i_{n+1}\}} \mid Z_n) = \mathbb{P}(Z_{n+1} = i_{n+1} \mid Z_n)$ almost surely.

Proposition 11.31. Second, for every $n \in \mathbb{N}$ and $i_{n+1} \in E$,

$$\mathbb{P}(Z_{n+1} = i_{n+1}) = \sum_{i_n \in E} p_{i_n, i_{n+1}} \mathbb{P}(Z_n = i_n).$$

Proof. For every $n \in \mathbb{N}$ and $i_{n+1} \in E$, using a property of conditional expectations,

$$\mathbb{P}(Z_{n+1} = i_{n+1}) = \mathbb{E}(\mathbb{I}_{\{Z_{n+1}=i_{n+1}\}}) = \mathbb{E}(\mathbb{E}(\mathbb{I}_{\{Z_{n+1}=i_{n+1}\}} \mid \mathcal{F}_n)) = \mathbb{E}(p(Z_n, i_{n+1})) = \sum_{i_n \in E} p_{i_n, i_{n+1}} \mathbb{P}(Z_n = i_n),$$

where we have noted that $p(Z_n, i_{n+1}) = f_{i_{n+1}}(Z_n)$ if $f_{i_{n+1}} : \mathbb{N} \rightarrow [0, 1]$ is given by $f_{i_{n+1}}(i_n) = p_{i_n, i_{n+1}}$. $\square$

Proposition 11.32. Third, consider a function $h : E \rightarrow [0, \infty]$ and let $Ph : E \rightarrow [0, \infty]$ be given by

$$(Ph)(i) = \sum_{j \in E} p_{i,j} h(j).$$

Furthermore, suppose that $Ph \leq h$ (so that h is P -superharmonic) and that $\sum_{i \in E} \mu_i h(i) < \infty$.

In that case, $(h(Z_n) : \Omega \rightarrow [0, \infty] \mid n \in \mathbb{N})$ is a supermartingale adapted to the filtration $(\mathcal{F}_n)_n$.

Proof. For every $B \in \mathcal{B}(\mathbb{R})$, note that $h^{-1}(B) \in \mathcal{P}(E)$. For every $n \in \mathbb{N}$, because $\mathcal{P}(E) \subseteq \mathcal{B}(\mathbb{R})$ and $Z_n^{-1}(h^{-1}(B)) \in \sigma(Z_n)$, we know that $h(Z_n)$ is $\mathcal{F}_n$ -measurable. Therefore, the stochastic process $(h(Z_n) \mid n \in \mathbb{N})$ is adapted.

We will use induction to show that $\mathbb{E}(h(Z_n)) < \infty$ for every $n \in \mathbb{N}$. Using our assumption and a previous result,

$$\mathbb{E}(h(Z_0)) = \sum_{i_0 \in E} h(i_0) \mathbb{P}(Z_0 = i_0) = \sum_{i_0 \in E} \mu_{i_0} h(i_0) < \infty.$$

Suppose that $\mathbb{E}(h(Z_n)) < \infty$ for some $n \in \mathbb{N}$. Using a previous result,

$$\mathbb{E}(h(Z_{n+1})) = \sum_{i_{n+1} \in E} h(i_{n+1}) \mathbb{P}(Z_{n+1} = i_{n+1}) = \sum_{i_{n+1} \in E} h(i_{n+1}) \sum_{i_n \in E} p_{i_n, i_{n+1}} \mathbb{P}(Z_n = i_n).$$

By rearranging terms, since h is P -superharmonic,

$$\mathbb{E}(h(Z_{n+1})) = \sum_{i_n \in E} \mathbb{P}(Z_n = i_n) \sum_{i_{n+1} \in E} p_{i_n, i_{n+1}} h(i_{n+1}) \leq \sum_{i_n \in E} \mathbb{P}(Z_n = i_n) h(i_n) = \mathbb{E}(h(Z_n)) < \infty,$$

which completes the inductive step.

It remains to show that $\mathbb{E}(h(Z_{n+1}) \mid \mathcal{F}_n) \leq h(Z_n)$ almost surely for every $n \in \mathbb{N}$.

For every $n \in \mathbb{N}$, note that $h(Z_{n+1}) : \Omega \rightarrow [0, \infty]$ is given by

$$h(Z_{n+1})(\omega) = h(Z_{n+1}(\omega)) = \sum_{i_{n+1} \in E} h(i_{n+1}) \mathbb{I}_{\{Z_{n+1}=i_{n+1}\}}(\omega).$$

Therefore, if $(X_m : \Omega \rightarrow [0, \infty] \mid m \in \mathbb{N})$ is a sequence of random variables given by

$$X_m(\omega) = \sum_{i_{n+1} \in E \mid i_{n+1} \leq m} h(i_{n+1}) \mathbb{I}_{\{Z_{n+1}=i_{n+1}\}}(\omega),$$

then $X_m \uparrow h(Z_{n+1})$. Since $\mathbb{E}(h(Z_{n+1})) < \infty$, by the conditional monotone-convergence theorem,

$$\mathbb{E}(h(Z_{n+1}) \mid \mathcal{F}_n) = \lim_{m \rightarrow \infty} \mathbb{I}_A \sum_{i_{n+1} \in E \mid i_{n+1} \leq m} h(i_{n+1}) \mathbb{E}(\mathbb{I}_{\{Z_{n+1}=i_{n+1}\}} \mid \mathcal{F}_n) = \sum_{i_{n+1} \in E} h(i_{n+1}) p(Z_n, i_{n+1}) \mathbb{I}_A$$

almost surely, where $A \in \mathcal{F}_n$ is a set such that $\mathbb{P}(A) = 1$.

For every $n \in \mathbb{N}$, note that $(Ph)(Z_n) : \Omega \rightarrow [0, \infty]$ is given by

$$(Ph)(Z_n)(\omega) = (Ph)(Z_n(\omega)) = \sum_{i_{n+1} \in E} p_{Z_n(\omega), i_{n+1}} h(i_{n+1}) = \sum_{i_{n+1} \in E} p(Z_n, i_{n+1})(\omega) h(i_{n+1}).$$

Therefore, for every $n \in \mathbb{N}$, because h is P -superharmonic,

$$\mathbb{E}(h(Z_{n+1}) \mid \mathcal{F}_n) = (Ph)(Z_n) \mathbb{I}_A = (Ph)(Z_n) \leq h(Z_n)$$

almost surely, so that $(h(Z_n) : \Omega \rightarrow [0, \infty] \mid n \in \mathbb{N})$ is a supermartingale adapted to the filtration $(\mathcal{F}_n)_n$. □

Proposition 11.33. Consider a set $E \subseteq \mathbb{R}$, a measurable space $(\tilde{\Omega}, \tilde{\mathcal{F}})$, and a stochastic process $(\tilde{Z}_n : \tilde{\Omega} \rightarrow E \mid n \in \mathbb{N})$. Let $\tilde{Z} : \tilde{\Omega} \rightarrow E^\infty$ be given by $\tilde{Z}(\tilde{\omega}) = (\tilde{Z}_n(\tilde{\omega}) \mid n \in \mathbb{N})$. For every $n \in \mathbb{N}$, let $Z_n : E^\infty \rightarrow E$ be given by $Z_n(\omega) = \omega_n$ and let $\mathcal{F} = \sigma(\cup_n \sigma(Z_n))$. In that case, $\tilde{Z}$ is $\tilde{\mathcal{F}}/\mathcal{F}$ -measurable.

Proof. For every $n \in \mathbb{N}$, note that $\tilde{Z}_n = Z_n \circ \tilde{Z}$, so that $\tilde{Z}_n^{-1}(B) = \tilde{Z}^{-1}(Z_n^{-1}(B))$ for every $B \in \mathcal{B}(\mathbb{R})$. Because $\tilde{Z}_n$ is $\tilde{\mathcal{F}}$ -measurable for every $n \in \mathbb{N}$, we know that $\tilde{Z}^{-1}(C) \in \tilde{\mathcal{F}}$ for every $C \in \cup_n \sigma(Z_n)$.

Since $(E^\infty, \mathcal{F})$ is a measurable space, note that $\mathcal{E} = \{F \in \mathcal{F} \mid \tilde{Z}^{-1}(F) \in \tilde{\mathcal{F}}\}$ is a σ -algebra on E^∞ . Because $\cup_n \sigma(Z_n) \subseteq \mathcal{F}$, we know that $\sigma(\cup_n \sigma(Z_n)) = \mathcal{F} \subseteq \mathcal{E}$, so that $\mathcal{E} = \mathcal{F}$. Therefore, $\tilde{Z}$ is $\tilde{\mathcal{F}}/\mathcal{F}$ -measurable. □

Proposition 11.34 (Existence of the canonical model). Consider a set $E \subseteq \mathbb{N}$ and a stochastic matrix P whose (i, j) -th element is given by $p_{i,j} \geq 0$ and suppose that $\sum_{k \in E} p_{i,k} = 1$ for every $i \in E$.

Let $\Omega = E^\infty$, so that every $\omega \in \Omega$ is a sequence $\omega = (\omega_n \in E \mid n \in \mathbb{N})$. For every $n \in \mathbb{N}$, consider the function $Z_n : \Omega \rightarrow E$ given by $Z_n(\omega) = \omega_n$ and let $\mathcal{F}_n = \sigma(Z_0, \dots, Z_n)$. Furthermore, let $\mathcal{F} = \sigma(Z_0, Z_1, \dots) = \sigma(\cup_n \mathcal{F}_n)$.

In that case, for every probability measure μ on the measurable space $(E, \mathcal{P}(E))$ there is a unique probability measure $\mathbb{P}^\mu$ on the measurable space $(\Omega, \mathcal{F})$ such that, for every $n \in \mathbb{N}^+$ and $i_0, i_1, \dots, i_n \in E$,

$$\mathbb{P}^\mu(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \mu_{i_0} p_{i_0, i_1} \cdots p_{i_{n-1}, i_n} = \mu_{i_0} \prod_{k=1}^n p_{i_{k-1}, i_k}.$$

The probability triple $(\Omega, \mathcal{F}, \mathbb{P}^\mu)$ is called the canonical model for the time-homogeneous Markov chain $Z = (Z_n : \Omega \rightarrow E \mid n \in \mathbb{N})$ on E with initial distribution μ and 1-step transition matrix P .

Proof. We have already shown that there is a probability triple $(\tilde{\Omega}^\mu, \tilde{\mathcal{F}}^\mu, \tilde{\mathbb{P}}^\mu)$ carrying the stochastic process $(\tilde{Z}_n^\mu : \tilde{\Omega}^\mu \rightarrow E \mid n \in \mathbb{N})$ such that, for every $n \in \mathbb{N}^+$ and $i_0, i_1, \dots, i_n \in E$,

$$\tilde{\mathbb{P}}^\mu(\tilde{Z}_0^\mu = i_0, \tilde{Z}_1^\mu = i_1, \dots, \tilde{Z}_n^\mu = i_n) = \mu_{i_0} p_{i_0, i_1} \cdots p_{i_{n-1}, i_n}.$$

Consider the function $\tilde{Z}^\mu : \tilde{\Omega}^\mu \rightarrow \Omega$ given by $\tilde{Z}^\mu(\tilde{\omega}) = (\tilde{Z}_n^\mu(\tilde{\omega}) \mid n \in \mathbb{N})$. Because $\tilde{Z}^\mu$ is $\tilde{\mathcal{F}}^\mu/\mathcal{F}$ -measurable, the function $\mathbb{P}^\mu : \mathcal{F} \rightarrow [0, 1]$ defined by

$$\mathbb{P}^\mu(F) = \tilde{\mathbb{P}}^\mu((\tilde{Z}^\mu)^{-1}(F)) = \tilde{\mathbb{P}}^\mu(\{\tilde{\omega} \in \tilde{\Omega}^\mu \mid \tilde{Z}^\mu(\tilde{\omega}) \in F\})$$

is a probability measure on the measurable space $(\Omega, \mathcal{F})$. Clearly, $\mathbb{P}^\mu(\Omega) = \tilde{\mathbb{P}}^\mu((\tilde{Z}^\mu)^{-1}(\Omega)) = \tilde{\mathbb{P}}^\mu(\tilde{\Omega}^\mu) = 1$ and $\mathbb{P}^\mu(\emptyset) = \tilde{\mathbb{P}}^\mu((\tilde{Z}^\mu)^{-1}(\emptyset)) = \tilde{\mathbb{P}}^\mu(\emptyset) = 0$. For any sequence of sets $(F_n \in \mathcal{F} \mid n \in \mathbb{N})$ such that $F_n \cap F_m = \emptyset$ for $n \neq m$,

$$\mathbb{P}^\mu\left(\bigcup_n F_n\right) = \tilde{\mathbb{P}}^\mu\left((\tilde{Z}^\mu)^{-1}\left(\bigcup_n F_n\right)\right) = \tilde{\mathbb{P}}^\mu\left(\bigcup_n (\tilde{Z}^\mu)^{-1}(F_n)\right) = \sum_n \tilde{\mathbb{P}}^\mu((\tilde{Z}^\mu)^{-1}(F_n)) = \sum_n \mathbb{P}^\mu(F_n),$$

where we have used the fact that $(\tilde{Z}^\mu)^{-1}(F_n) \cap (\tilde{Z}^\mu)^{-1}(F_m) = (\tilde{Z}^\mu)^{-1}(F_n \cap F_m) = \emptyset$ for $n \neq m$.

For every $n \in \mathbb{N}^+$ and $i_0, i_1, \dots, i_n \in E$,

$$\mathbb{P}^\mu(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \tilde{\mathbb{P}}^\mu(\{\tilde{\omega} \in \tilde{\Omega}^\mu \mid \tilde{Z}^\mu(\tilde{\omega}) \in \{\omega \in \Omega \mid Z_0(\omega) = i_0, Z_1(\omega) = i_1, \dots, Z_n(\omega) = i_n\}\}).$$

Because $\Omega = E^\infty$ and $Z_m(\omega) = \omega_m$ for every $m \in \mathbb{N}$,

$$\mathbb{P}^\mu(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \tilde{\mathbb{P}}^\mu(\{\tilde{\omega} \in \tilde{\Omega}^\mu \mid \tilde{Z}_0^\mu(\tilde{\omega}) = i_0, \tilde{Z}_1^\mu(\tilde{\omega}) = i_1, \dots, \tilde{Z}_n^\mu(\tilde{\omega}) = i_n\}) = \mu_{i_0} p_{i_0, i_1} \cdots p_{i_{n-1}, i_n},$$

so that a probability measure on $(\Omega, \mathcal{F})$ with the desired properties exists.

Naturally, any two desired probability measures on $(\Omega, \mathcal{F})$ must agree on the π -system $\mathcal{I} \subseteq \mathcal{F}$ given by

$$\mathcal{I} = \{\emptyset\} \cup \{\{Z_0 = i_0, \dots, Z_n = i_n\} \mid n \in \mathbb{N} \text{ and } i_0, \dots, i_n \in E\}.$$

Therefore, if $\sigma(\mathcal{I}) = \mathcal{F}$, then $\mathbb{P}^\mu$ will be the unique probability measure on $(\Omega, \mathcal{F})$ with the desired properties.

First, we will show that $\mathcal{I}$ is indeed a π -system on Ω . Clearly, $I \cap \emptyset = \emptyset$ and $\emptyset \in \mathcal{I}$ for every $I \in \mathcal{I}$. For some $n \in \mathbb{N}$ and $i_0, \dots, i_n \in E$, let $I_1 = \{Z_0 = i_0, \dots, Z_n = i_n\}$. For some $m \geq n$ and $j_0, \dots, j_m \in E$, let $I_2 = \{Z_0 = j_0, \dots, Z_m = j_m\}$. In that case,

$$I_1 \cap I_2 = \{\omega \in \Omega \mid Z_0(\omega) = i_0 = j_0, \dots, Z_n(\omega) = i_n = j_n, Z_n(\omega) = j_n, \dots, Z_m(\omega) = j_m\},$$

so that

$$I_1 \cap I_2 = \begin{cases} I_2, & \text{if } i_k = j_k \text{ for every } k \in \{0, \dots, n\}, \\ \emptyset, & \text{if } i_k \neq j_k \text{ for some } k \in \{0, \dots, n\}. \end{cases}$$

Therefore, $I_1 \cap I_2 \in \mathcal{I}$, so that $\mathcal{I}$ is a π -system on Ω .

Finally, we will show that $\mathcal{F} \subseteq \sigma(\mathcal{I})$. For every $n \in \mathbb{N}$, recall that the σ -algebra $\mathcal{F}_n$ is given by

$$\mathcal{F}_n = \left\{ \bigcup_{i \in A} \{Z_0 = i_0, \dots, Z_n = i_n\} \mid A \in \mathcal{P}(E^{n+1}) \right\},$$

where $i = (i_0, \dots, i_n)$ and A is a countable set. For every $n \in \mathbb{N}$, because each $F_n \in \mathcal{F}_n$ is a countable union of elements of $\mathcal{I}$, we know that $F_n \in \sigma(\mathcal{I})$. Therefore, $\cup_n \mathcal{F}_n \subseteq \sigma(\mathcal{I})$, so that $\mathcal{F} = \sigma(\cup_n \mathcal{F}_n) \subseteq \sigma(\mathcal{I})$. $\square$

Consider a set $E \subseteq \mathbb{N}$. Let P be a stochastic matrix whose (i, j) -th element is given by $p_{i,j} \geq 0$ and suppose that $\sum_{k \in E} p_{i,k} = 1$ for every $i \in E$.

Let $(\Omega, \mathcal{F}, \mathbb{P}^\mu)$ denote the canonical model for the time-homogeneous Markov chain $Z = (Z_n : \Omega \rightarrow E \mid n \in \mathbb{N})$ on E with initial distribution μ and 1-step transition matrix P , where μ is a probability measure on the measurable space $(E, \mathcal{P}(E))$. For every $i \in E$, let $\mathbb{P}^i = \mathbb{P}^\mu$ if $\mu(\{i\}) = 1$. Furthermore, consider the filtration $(\mathcal{F}_n)_n$, where $\mathcal{F}_n = \sigma(Z_0, \dots, Z_n)$ for every $n \in \mathbb{N}$. Finally, for every $j \in E$, consider the stopping time $T_j : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ given by $T_j(\omega) = \inf\{n \in \mathbb{N}^+ \mid Z_n(\omega) = j\}$.

Definition 11.14. The stochastic matrix P is called irreducible recurrent if and only if $\mathbb{P}^i(T_j < \infty) = 1$ for every $i, j \in E$.

Definition 11.15. For every function $h : E \rightarrow [0, \infty]$, let $Ph : E \rightarrow [0, \infty]$ be given by

$$(Ph)(i) = \sum_{j \in E} p_{i,j} h(j).$$

The function h is called finite non-negative P -superharmonic if and only if $Ph \leq h < \infty$.

Proposition 11.35. The stochastic matrix P is irreducible recurrent if and only if every finite non-negative P -superharmonic function is constant.

Proof. First, suppose that the stochastic matrix P is irreducible recurrent. Consider a finite non-negative P -superharmonic function $h : E \rightarrow [0, \infty]$ and the stochastic process $h(Z) = (h(Z_n) : \Omega \rightarrow [0, \infty] \mid n \in \mathbb{N})$. Because $h < \infty$, for every $i \in E$, a previous result guarantees that $h(Z)$ is a supermartingale on the filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P}^i)$.

For every $j \in E$, recall that the random variable $h(Z)_{T_j} : \Omega \rightarrow [0, \infty]$ is given by

$$h(Z)_{T_j}(\omega) = \begin{cases} h(j), & \text{if } T_j(\omega) < \infty, \\ 0, & \text{if } T_j(\omega) = \infty. \end{cases}$$

Because $h(Z_n) \geq 0$ for every $n \in \mathbb{N}$ and $\mathbb{P}^i(T_j < \infty) = 1$ for every $i, j \in E$, a previous result guarantees that $\mathbb{E}^i(h(Z)_{T_j}) \leq \mathbb{E}^i(h(Z_0))$ for every $i, j \in E$. This implies that h is constant because, for every $i, j \in E$,

$$h(j) = \mathbb{E}^i(h(Z)_{T_j}) \leq \mathbb{E}^i(h(Z_0)) = \sum_{i_0 \in E} h(i_0) \mathbb{P}^i(Z_0 = i_0) = h(i).$$

Suppose that every finite non-negative P -superharmonic function is constant. For every $i, j \in E$, note that

$$\mathbb{P}^i(T_j = 1) = \mathbb{P}^i(Z_1 = j) = \sum_{k \in E} p_{i,k} \mathbb{P}^i(Z_0 = k) = p_{i,j}.$$

For every $n \in \mathbb{N}^+$ and $i, j \in E$, we will now show that

$$\mathbb{P}^i(T_j = n + 1) = \sum_{k \neq j} p_{i,k} \mathbb{P}^k(T_j = n).$$

For $n = 1$ and every $i, j \in E$,

$$\mathbb{P}^i(T_j = 2) = \mathbb{P}^i(Z_1 \neq j, Z_2 = j) = \mathbb{P}^i\left(\bigcup_{k \neq j} \{Z_1 = k\} \cap \{Z_2 = j\}\right) = \sum_{k \neq j} \mathbb{P}^i(Z_1 = k, Z_2 = j).$$

Since $\mathbb{P}^i(Z_0 = i) = 1$,

$$\mathbb{P}^i(T_j = 2) = \sum_{k \neq j} \mathbb{P}^i(Z_0 = i, Z_1 = k, Z_2 = j) = \sum_{k \neq j} p_{i,k} p_{k,j} = \sum_{k \neq j} p_{i,k} \mathbb{P}^k(T_j = 1).$$

For every $n \geq 2$ and $i, j \in E$,

$$\mathbb{P}^i(T_j = n + 1) = \mathbb{P}^i(Z_1 \neq j, \dots, Z_n \neq j, Z_{n+1} = j) = \mathbb{P}^i\left(\bigcup_{k_0 \neq j} \dots \bigcup_{k_{n-1} \neq j} \{Z_1 = k_0, \dots, Z_n = k_{n-1}, Z_{n+1} = j\}\right).$$

Since $\mathbb{P}^i(Z_0 = i) = 1$,

$$\mathbb{P}^i(T_j = n + 1) = \sum_{k_0 \neq j} \cdots \sum_{k_{n-1} \neq j} \mathbb{P}^i(Z_0 = i, Z_1 = k_0, \dots, Z_n = k_{n-1}, Z_{n+1} = j).$$

From the definition of $\mathbb{P}^i$,

$$\mathbb{P}^i(T_j = n + 1) = \sum_{k_0 \neq j} p_{i,k_0} \sum_{k_1 \neq j} \cdots \sum_{k_{n-1} \neq j} p_{k_0,k_1} \cdots p_{k_{n-2},k_{n-1}} p_{k_{n-1},j}.$$

From the definition of $\mathbb{P}^{k_0}$,

$$\mathbb{P}^i(T_j = n + 1) = \sum_{k_0 \neq j} p_{i,k_0} \mathbb{P}^{k_0} \left(\bigcup_{k_1 \neq j} \cdots \bigcup_{k_{n-1} \neq j} \{Z_1 = k_1, \dots, Z_{n-1} = k_{n-1}, Z_n = j\} \right) = \sum_{k \neq j} p_{i,k} \mathbb{P}^k(T_j = n),$$

which completes this step.

For every $i, j \in E$, we will now provide a recursive expression for $\mathbb{P}^i(T_j < \infty)$. First, note that

$$\mathbb{P}^i(T_j < \infty) = \mathbb{P}^i \left(\bigcup_{n \in \mathbb{N}^+} \{T_j = n\} \right) = \sum_{n \in \mathbb{N}^+} \mathbb{P}^i(T_j = n) = \mathbb{P}^i(T_j = 1) + \sum_{n \in \mathbb{N}^+} \mathbb{P}^i(T_j = n + 1).$$

Using the previous results,

$$\mathbb{P}^i(T_j < \infty) = p_{i,j} + \sum_{n \in \mathbb{N}^+} \sum_{k \neq j} p_{i,k} \mathbb{P}^k(T_j = n) = p_{i,j} + \sum_{k \neq j} p_{i,k} \sum_{n \in \mathbb{N}^+} \mathbb{P}^k(T_j = n).$$

Therefore,

$$\mathbb{P}^i(T_j < \infty) = p_{i,j} + \sum_{k \neq j} p_{i,k} \mathbb{P}^k \left(\bigcup_{n \in \mathbb{N}^+} \{T_j = n\} \right) = p_{i,j} + \sum_{k \neq j} p_{i,k} \mathbb{P}^k(T_j < \infty),$$

which completes this step.

For every $i, j \in E$, let $h_j(i) = \mathbb{P}^i(T_j < \infty)$. Since $\mathbb{P}^j(T_j < \infty) \leq 1$,

$$h_j(i) = p_{i,j} + \sum_{k \neq j} p_{i,k} h_j(k) \geq \sum_{k \in E} p_{i,k} h_j(k) = (Ph_j)(i),$$

so that $h_j : E \rightarrow [0, \infty]$ is finite non-negative P -superharmonic.

By assumption, for every $j \in E$ there is a constant ρ_j such that $h_j(i) = \rho_j$ for every $i \in E$. For every $i, j \in E$,

$$\rho_j = h_j(i) = p_{i,j} + \sum_{k \neq j} p_{i,k} \rho_j = p_{i,j} + \rho_j(1 - p_{i,j}).$$

By reordering terms, we know that $\rho_j p_{i,j} = p_{i,j}$ for every $i, j \in E$.

In order to complete the proof, we will show that for every $j \in E$ there is an $i \in E$ such that $p_{i,j} > 0$, which implies that $\rho_j = 1$. For every $i, j \in E$, let $f_j(i) = 1$ if $i = j$ and $f_j(i) = 0$ if $i \neq j$, so that

$$(Pf_j)(i) = \sum_{k \in E} p_{i,k} f_j(k) = p_{i,j}.$$

If there is a $j \in E$ such that $p_{i,j} = 0$ for every $i \in E$, then $f_j(i) \geq (Pf_j)(i) = 0$, so that $f_j : E \rightarrow [0, \infty]$ is a finite non-negative P -superharmonic function. Because f_j is not constant, such $j \in E$ does not exist.

Because $\rho_j = 1$ for every $j \in E$ and $\rho_j = h_j(i) = \mathbb{P}^i(T_j < \infty)$ for every $i \in E$, the proof is complete. $\square$

12 Martingale convergence

Consider a filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$.

Proposition 12.1. Consider an adapted process $(X_n \mid n \in \mathbb{N})$, a stopping time $S : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$, and a set $B \in \mathcal{B}(\mathbb{R})$. Let the function $T : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ be given by $T(\omega) = \inf\{n > S(\omega) \mid X_n(\omega) \in B\}$, where $\inf \emptyset = \infty$. The function T is a stopping time.

Proof. For every $n \in \mathbb{N}$,

$$\{T \leq n\} = \{\omega \in \Omega \mid X_k(\omega) \in B \text{ for some } k \leq n \text{ such that } k > S(\omega)\} = \bigcup_{k \leq n} X_k^{-1}(B) \cap \{S \leq k-1\}.$$

For every $k \leq n$, we know that X_k is $\mathcal{F}_n$ -measurable and $\{S \leq k-1\} \in \mathcal{F}_n$, so that $\{T \leq n\} \in \mathcal{F}_n$. Because $\{T \leq \infty\} \in \mathcal{F}_\infty$, we know that T is a stopping time. $\square$

Definition 12.1. Consider an adapted process $(X_n \mid n \in \mathbb{N})$. Let $T_0(\omega) = -1$ for every $\omega \in \Omega$. For some $a, b \in \mathbb{R}$ such that $a < b$ and every $i \in \mathbb{N}^+$, let $S_i : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ and $T_i : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ be given by

$$\begin{aligned} S_i(\omega) &= \inf\{n > T_{i-1}(\omega) \mid X_n(\omega) < a\}, \\ T_i(\omega) &= \inf\{n > S_i(\omega) \mid X_n(\omega) > b\}. \end{aligned}$$

For every $n \in \mathbb{N}^+$, the number of upcrossings $U_n[a, b]$ of $[a, b]$ by time n is defined by

$$U_n[a, b] = \sup_{i \in \mathbb{N}^+} i \mathbb{I}_{\{T_i \leq n\}}.$$

For every $\omega \in \Omega$, $U_n[a, b](\omega)$ is the number of times that $X_0(\omega), \dots, X_n(\omega)$ goes from below a to above b .

Proposition 12.2. For every $i \in \mathbb{N}^+$, the functions S_i and T_i are stopping times.

Proof. From a previous result, S_1 is a stopping time, so that T_1 is also a stopping time. If T_i is a stopping time for some $i \in \mathbb{N}^+$, then S_{i+1} and T_{i+1} are stopping times. $\square$

Proposition 12.3. Consider an adapted process $(X_n \mid n \in \mathbb{N})$. For every $a, b \in \mathbb{R}$ such that $a < b$ and every $n \in \mathbb{N}^+$, the number of upcrossings $U_n[a, b]$ is an $\mathcal{F}_n$ -measurable bounded non-negative function.

Proof. Because T_i is a stopping time for every $i \in \mathbb{N}^+$, we know that $i \mathbb{I}_{\{T_i \leq n\}}$ is non-negative and $\mathcal{F}_n$ -measurable. Therefore, $U_n[a, b]$ is non-negative and $\mathcal{F}_n$ -measurable. For every $\omega \in \Omega$ and $i \in \mathbb{N}^+$ such that $T_i(\omega) \leq n$, note that $0 \leq S_1(\omega) < T_1(\omega) < \dots < S_i(\omega) < T_i(\omega) \leq n$, so that $T_i(\omega) \geq 2i - 1$. Therefore,

$$U_n[a, b] = \sup_{i \in \mathbb{N}^+} i \mathbb{I}_{\{2i-1 \leq T_i \leq n\}} = \sup_{i \leq \lfloor \frac{n+1}{2} \rfloor} i \mathbb{I}_{\{T_i \leq n\}},$$

which implies that $U_n[a, b] \leq \lfloor \frac{n+1}{2} \rfloor$. $\square$

Lemma 12.1 (Doob's upcrossing lemma). Consider a supermartingale $X = (X_n \mid n \in \mathbb{N})$. For every $a, b \in \mathbb{R}$ such that $a < b$ and $n \in \mathbb{N}^+$,

$$(b - a) \mathbb{E}(U_n[a, b]) \leq \mathbb{E}(\max(a - X_n, 0)).$$

Proof. Consider the stochastic process $C = (C_n : \Omega \rightarrow \{0, 1\} \mid n \in \mathbb{N})$, where $C_0 = 0$ and

$$C_n = \sup_{i \in \mathbb{N}^+} \mathbb{I}_{\{S_i \leq n-1 < T_i\}}$$

for every $n \in \mathbb{N}^+$. Because S_i and T_i are stopping times for every $i \in \mathbb{N}^+$, we know that $\mathbb{I}_{\{S_i \leq n-1\}}$ and $\mathbb{I}_{\{T_i > n-1\}} = \mathbb{I}_{\{T_i \leq n-1\}}^c$ are $\mathcal{F}_{n-1}$ -measurable for every $n \in \mathbb{N}^+$. Therefore, C is previsible.

Consider the martingale transform $(C \bullet X) = ((C \bullet X)_n \mid n \in \mathbb{N})$, where $(C \bullet X)_0 = 0$ and

$$(C \bullet X)_n = \sum_{k=1}^n C_k (X_k - X_{k-1})$$

for every $n \in \mathbb{N}^+$. Because C is bounded and non-negative, $(C \bullet X)$ is a supermartingale.

For every $\omega \in \Omega$ and $k \in \mathbb{N}^+$, note that $C_k(\omega) = 1$ if and only if $(k-1) \in [S_i(\omega), T_i(\omega))$ for some $i \in \mathbb{N}^+$. Therefore, for every $\omega \in \Omega$ and $n \in \mathbb{N}^+$,

$$(C \bullet X)_n(\omega) = \left[\sum_{i=1}^{U_n[a,b](\omega)} \sum_{k=S_i(\omega)+1}^{T_i(\omega)} X_k(\omega) - X_{k-1}(\omega) \right] + \left[\sum_{k=S_{U_n[a,b](\omega)+1}(\omega)+1}^n X_k(\omega) - X_{k-1}(\omega) \right].$$

By rewriting the sums of differences,

$$(C \bullet X)_n = \left[\sum_{i=1}^{U_n[a,b]} X_{T_i} - X_{S_i} \right] + [X_n - X_{S_{U_n[a,b]+1}}] \mathbb{I}_{\{S_{U_n[a,b]+1} < n\}},$$

where $X_\infty = 0$. For every $\omega \in \Omega$, if $i \leq U_n[a,b](\omega)$, then $X_{T_i}(\omega) - X_{S_i}(\omega) > (b-a)$. Therefore,

$$(C \bullet X)_n \geq (b-a)U_n[a,b] + [X_n - X_{S_{U_n[a,b]+1}}] \mathbb{I}_{\{S_{U_n[a,b]+1} < n\}}.$$

Let $L = [X_n - X_{S_{U_n[a,b]+1}}] \mathbb{I}_{\{S_{U_n[a,b]+1} < n\}}$. For every $\omega \in \Omega$, if $S_{U_n[a,b](\omega)+1}(\omega) \geq n$, then $L(\omega) = 0$. Now suppose $S_{U_n[a,b](\omega)+1}(\omega) < n$. If $X_n(\omega) \geq X_{S_{U_n[a,b](\omega)+1}(\omega)}(\omega)$, then $L(\omega) \geq 0$. If $X_n(\omega) < X_{S_{U_n[a,b](\omega)+1}(\omega)}(\omega)$, then $X_n(\omega) < a$ and $-L(\omega) = |X_n(\omega) - X_{S_{U_n[a,b](\omega)+1}(\omega)}(\omega)| < |X_n(\omega) - a| = \max(a - X_n(\omega), 0)$. In every case, $L(\omega) \geq -\max(a - X_n(\omega), 0)$. Therefore,

$$(C \bullet X)_n \geq (b-a)U_n[a,b] - \max(a - X_n, 0).$$

Because $(C \bullet X)$ is a supermartingale and $(C \bullet X)_0 = 0$,

$$0 \geq (b-a)\mathbb{E}(U_n[a,b]) - \mathbb{E}(\max(a - X_n, 0)).$$

□

Definition 12.2. A stochastic process $(X_n \mid n \in \mathbb{N})$ is bounded in $\mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ if

$$\sup_n \mathbb{E}(|X_n|) < \infty.$$

Proposition 12.4. Consider a supermartingale $X = (X_n \mid n \in \mathbb{N})$ bounded in $\mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. For every $a, b \in \mathbb{R}$ such that $a < b$,

$$\mathbb{P}(U_\infty[a,b] = \infty) = 0,$$

where $U_\infty[a,b] = \lim_{n \rightarrow \infty} U_n[a,b]$.

Proof. Note that $U_\infty[a,b] : \Omega \rightarrow [0, \infty]$ is well defined because $U_{n+1}[a,b] \geq U_n[a,b]$ for every $n \in \mathbb{N}^+$.

For every $n \in \mathbb{N}^+$, by Doob's upcrossing lemma,

$$(b-a)\mathbb{E}(U_n[a,b]) \leq \mathbb{E}(\max(a - X_n, 0)) = \mathbb{E}((X_n - a)^-) = \mathbb{E}(|X_n - a|) - \mathbb{E}((X_n - a)^+) \leq \mathbb{E}(|X_n - a|).$$

For every $n \in \mathbb{N}^+$, by the triangle inequality and because X is bounded in $\mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$,

$$(b-a)\mathbb{E}(U_n[a,b]) \leq \mathbb{E}(|X_n|) + |a| \leq \sup_k \mathbb{E}(|X_k|) + |a|.$$

By the monotone-convergence theorem, since $U_n[a,b] \uparrow U_\infty[a,b]$,

$$(b-a)\mathbb{E}(U_\infty[a,b]) = \lim_{n \rightarrow \infty} (b-a)\mathbb{E}(U_n[a,b]) \leq \sup_k \mathbb{E}(|X_k|) + |a| < \infty,$$

so that $\mathbb{E}(U_\infty[a,b]) < \infty$, which implies $\mathbb{P}(U_\infty[a,b] = \infty) = 0$. □

Theorem 12.1 (Doob's forward convergence theorem). Consider a supermartingale $X = (X_n \mid n \in \mathbb{N})$ bounded in $\mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. If $X_\infty = \limsup_{n \rightarrow \infty} X_n$, then $\lim_{n \rightarrow \infty} X_n = X_\infty$ almost surely and $|X_\infty| < \infty$ almost surely.

Proof. Let $\Lambda = \{\omega \in \Omega \mid \lim_{n \rightarrow \infty} X_n(\omega) \text{ does not exist in } [-\infty, \infty]\}$. In that case,

$$\Lambda = \{\omega \in \Omega \mid \liminf_{n \rightarrow \infty} X_n(\omega) < \limsup_{n \rightarrow \infty} X_n(\omega)\}.$$

For every $\omega \in \Omega$, $\liminf_{n \rightarrow \infty} X_n(\omega) < \limsup_{n \rightarrow \infty} X_n(\omega)$ if and only if $\liminf_{n \rightarrow \infty} X_n(\omega) < a < b < \limsup_{n \rightarrow \infty} X_n(\omega)$ for some rationals $a, b \in \mathbb{Q}$. Therefore,

$$\Lambda = \bigcup_{a, b \in \mathbb{Q} \mid a < b} \{\omega \in \Omega \mid \liminf_{n \rightarrow \infty} X_n(\omega) < a < b < \limsup_{n \rightarrow \infty} X_n(\omega)\}.$$

For every $\omega \in \Omega$, if $\liminf_{n \rightarrow \infty} X_n(\omega) < a$, then $X_n(\omega) < a$ for infinitely many $n \in \mathbb{N}$. Similarly, if $b < \limsup_{n \rightarrow \infty} X_n(\omega)$, then $X_n(\omega) > b$ for infinitely many $n \in \mathbb{N}$. Therefore,

$$\Lambda \subseteq \bigcup_{a, b \in \mathbb{Q} \mid a < b} \{\omega \in \Omega \mid U_\infty[a, b](\omega) = \infty\}.$$

Because the set of rational numbers $\mathbb{Q}$ is countable and by a previous result,

$$\mathbb{P}(\Lambda) \leq \sum_{a, b \in \mathbb{Q} \mid a < b} \mathbb{P}(U_\infty[a, b] = \infty) = 0.$$

Therefore, almost surely,

$$\lim_{n \rightarrow \infty} X_n = \liminf_{n \rightarrow \infty} X_n = \limsup_{n \rightarrow \infty} X_n = X_\infty.$$

Because $|X_\infty| = \lim_{n \rightarrow \infty} |X_n| = \liminf_{n \rightarrow \infty} |X_n|$ almost surely and by the Fatou lemma,

$$\mathbb{E}(|X_\infty|) = \mathbb{E}\left(\liminf_{n \rightarrow \infty} |X_n|\right) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(|X_n|) \leq \sup_n \mathbb{E}(|X_n|) < \infty.$$

Therefore, $\mathbb{P}(|X_\infty| = \infty) = 0$, so that $|X_\infty| < \infty$ almost surely. $\square$

Proposition 12.5. Consider a non-negative supermartingale $X = (X_n : \Omega \rightarrow [0, \infty] \mid n \in \mathbb{N})$. If $X_\infty = \limsup_{n \rightarrow \infty} X_n$, then $\lim_{n \rightarrow \infty} X_n = X_\infty$ almost surely and $|X_\infty| < \infty$ almost surely.

Proof. For every $n \in \mathbb{N}^+$, we know that $\mathbb{E}(X_0) \geq \mathbb{E}(X_n) = \mathbb{E}(|X_n|)$. Therefore, $\sup_n \mathbb{E}(|X_n|) \leq \mathbb{E}(X_0) < \infty$, so that the supermartingale X is bounded in $\mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. $\square$

Acknowledgements

I would like to thank Daniel Valesin for his guidance and the ideas behind many proofs found in these notes.

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